The topoi of the Longue Marche
Multi-Galois theory, mixed topoi, hole filling, and higher structure

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Abstract

We prove corrected recognition and reconstruction theorems for the multi-Galois, mixed, stratified, and higher topoi in Grothendieck's manuscripts. Exact cohomological defects, sharp flag bounds, sphere Postnikov invariants, and arithmetic inertia counterexamples identify where the provisional claims fail. A separate argument proves pinned etale reconstruction over subfields of finitely generated extensions of $\mathbb{Q}^{\mathrm{ab}}$. The paper retains two unresolved obligations: a further base-field test and one literal cross-return audit. Every numbered statement has a status and a manuscript locus in the accompanying registers. No priority is claimed.

Notation and manuscript coordinates

Here	In the manuscript	Meaning; introducing passage
βC, βG	$\widehat{C}$, $\mathrm{Ens}(G)$	Discrete groupoid action topos; LM §1, scans 6–11.
$\mathrm{LC}(\mathcal{E})$	$\mathcal{E}_{\mathrm{lc}}$, $\mathcal{E}_{[1]}$	Locally constant envelope; LM §2, scans 12–15.
$\Pi_n\mathcal{X}$, $\mathcal{X}_{[n]}$	$\mathcal{E}_{[i]}$	Shape truncation and its classifying stage; LM §2, scan 13.
$B_{\mathrm{cont}}G$, SLC	Pro-multigalois variants	Continuous actions and their generated topos; LM §3, scan 16.
$\mathcal{B}(X,S)$	$B_{X,U}$, $B_{U,X}$, $B_{X;S}$	Mixed Artin gluing; LM §14, scans 123–127.
G, H_s, Q_s, I_s	$\mathcal{E}, \mathcal{E}_i, \Gamma_i, \Pi_i$	Global, local, residue and inertia groups; LM §14, scans 122–123.
L_i, T, Λ_i	L_i, T_π, Λ_i	Peripheral lattice, orientation and loop; LM §12, scans 103–109; §14, scans 130–132.
Γ, Σ	Γ, Σ	Operators; LM §12, scan 109; §16, scans 172–178.
$A_0, \widetilde{A}_0; \omega, R$	$A_0, \widetilde{A}_0; \omega, R$	Sphere symmetry, cover, orientation and gerbe; LM §20, scans 233–241.
$\mathfrak{V}, \mathfrak{V}^*$	$\mathcal{V}_{D',Y}, \mathcal{V}^*_{D',Y}$	Full neighbourhood pro-topoi; MPL 147, scans 143–150.
P_S, L, T^J	$P_{D'^*}, X^{*\infty}, T^J$	Normal directions, resolved boundary, tame lattice; MPL 147, scans 143–152; MPL 149, scans 132–133.
D_r, D_r^*	$\mathcal{D}_r, \mathcal{D}_r^*$	Local germ flags and strict flags; MPL 147, scans 184–190.

LM denotes La Longue Marche (cote 140-2 here; §53 uses cote 140-4); MPL denotes the Montpellier cote. The full conventions distinguish ordinary topoi, infinity-topoi, and pro-objects. Auxiliary symbols are defined where introduced; no change of level is implicit.

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Conventions and manuscript coordinates

We distinguish a classifying topos of a discrete groupoid, a topos of continuous actions of a profinite groupoid, a pro-object of groupoids, and an inverse limit of their underlying groups. All profinite actions on sets are continuous for the discrete topology on the set. Cohomology of a topos has abelian coefficients. “Torsion sheaf” never means nonabelian cohomology in unbounded degrees. An expression with $\widehat{\mathbb{Z}}$ denotes continuous derived inverse-limit cohomology of its finite quotients, with the inverse-limit issue stated where used.

A geometric morphism $f : \mathcal{E} \to \mathcal{F}$ has inverse image $f^* : \mathcal{F} \to \mathcal{E}$. Pseudopullbacks and pseudopushouts retain invertible commutation data. A lax gluing category retains arbitrary

specialization maps. Neither construction is interchangeable with its strict categorical counterpart. The disjoint sum $\coprod_i \mathcal{E}_i$ of topoi has underlying category $\prod_i \mathcal{E}_i$; it is not their product in the two-category of topoi. Our diagrams use the former notation for disjoint strata.

Our notation	Manuscript notation	Introduction and meaning
βG	$\mathrm{Ens}(G)$	LM §1, scans 6–11: discrete action topos; a continuous-action variant is expressly marked.
$\mathcal{B}(X, S)$	$B_{X,U}$, $B_{U,X}$, $B_{X;S}$	LM §14, scans 123–127; §16, scans 163–168: the mixed Artin gluing.
G, H_s, Q_s, I_s	$\mathcal{E}$, $\mathcal{E}_i$, Γ_i, Π_i	LM §14, scans 122–123: global, local, residue and inertia groups.
$X^{*\infty}$	$\mathcal{V}^*_{\Theta,X}$	Cote 147, scan 150: punctured-neighborhood pro-object, with its precise realization specified.

We cite LM by section, manuscript folio when marked, and scan. Scan numbers refer to the attached archive images, not to pages of this paper. Readings remain provisional; mathematical consistency is evidence, not a declaration of palaeographic certainty.

1 The discrete multi-Galois envelope and its pro-variants

Throughout this section a topos is a Grothendieck topos in a fixed universe, and choice is available in the ambient category of sets. We write $\beta C = \mathrm{Fun}(C^{\mathrm{op}}, \mathbf{Set})$ for a small groupoid C. A locally constant object may have different sets of values on different members of a covering family. In particular, the cardinality need not be the same on different connected components. Points have the variance used in the manuscript: the category of points is opposite to the category of inverse-image fibre functors. All statements about fundamental groupoids below use this convention.

1.1 The assertion that all objects are locally constant

A.1 (Proved; manuscript 1.1, $b \Leftrightarrow b' \Rightarrow a$). *Manuscript: LM §1, folio 1, scan 6 (1.1 and Definition 1.2).* For every small groupoid C, a choice of one object in each component gives

$$\beta C \simeq \prod_{i \in \pi_0 C} \mathbf{Set}^{G_i}, \qquad G_i = \mathrm{Aut}_C(c_i).$$

Every object of βC is locally constant. Conversely, every such product is the presheaf topos of the disjoint union of the one-object groupoids BG_i. These assertions include the empty groupoid and the degenerate topos. These, and only these, are the multi-Galois topoi of Definition 1.2.

Proof. Restriction to the chosen objects is an equivalence of groupoids and therefore induces an equivalence of presheaf categories. In $\mathbf{Set}^{G_i}$ the regular G_i-set is an epimorphic cover of the terminal object, and its slice topos is $\mathbf{Set}$: an equivariant map $Y \to G_i$ is recovered from its fibre over the identity. The pullback of any G_i-set to this slice is constant. Doing this in every component proves local constancy. The converse construction is literal. No finiteness condition on the groups occurs in the manuscript's definition. □

A.2 (Refuted; manuscript 1.1, $a \Rightarrow b$ without additional hypotheses). *Manuscript: LM §1, folio 1, scan 6 (1.1(a) and the marginal doubt).* Let $B = \mathrm{RO}(\mathbb{R})$ be the complete atomless Boolean algebra of regular open subsets of the real line. Every object of $\mathrm{Sh}(B)$ is locally constant, but this nondegenerate topos has no points. It is therefore not βC for any groupoid C.

Proof. We give the general Boolean-algebra argument, so that the example does not depend on a spatial interpretation of B. Every sheaf on a locale is covered by subterminal objects. In a Boolean topos, well-order such a covering family and subtract from the image of each member the union of the preceding images. The resulting disjoint family still covers the object. Consequently every $X \in \mathrm{Sh}(B)$ is a coproduct $\coprod_{i<\kappa} b_i$ of elements $b_i \in B$ for some ordinal κ.

Here is the elementary Boolean-valued step used to trivialize this coproduct. Regard it as the Boolean-valued subset A of the ground ordinal κ whose membership values are $\|i \in A\| = b_i$. The inherited ordering is a well-order. Its order type is an ordinal at most κ, so

$$1 = \bigvee_{\alpha \le \kappa} \|A \text{ is order-isomorphic to } \check{\alpha}\|.$$

This equality is valid in the Boolean-valued interpretation of the usual theorem that a subset of an ordinal has order type at most that ordinal. The bounded quantifier over $\kappa \mathbin{\check{+}} 1$ is, by definition, the displayed join. Refine that join to a disjoint partition of 1. On each member, choose a witness to the corresponding isomorphism; the witness exists locally by the definition of the existential Boolean value, and a further disjoint refinement glues its witnesses. In sheaf language this is an isomorphism $X|_b \simeq \Delta\alpha|_b$. Thus the resulting partition trivializes X. This argument uses only the usual Boolean-valued interpretation of bounded sets and its mixing rule, not a set-generic ultrafilter. The equivalence of that interpretation with sheaves on B is described in Makkai and Reyes (1977, Theorem 4.2.1 and Proposition 4.3.4).

A point of the locale B would be a frame homomorphism $p : B \to 2$. Since B is Boolean, p preserves complements; since it preserves all joins, it therefore preserves all meets. The element

$$a = \bigwedge \{b \in B : p(b) = 1\}$$

satisfies $p(a) = 1$ and is an atom: for $0 < c \le a$, the possibility $p(c) = 0$ would imply $a \le \neg c$, a contradiction. The algebra $\mathrm{RO}(\mathbb{R})$ is atomless because every nonempty regular open contains two disjoint nonempty regular opens. Hence there is no point. A nonempty groupoid presheaf topos has evaluation points, whereas the empty groupoid gives the degenerate topos. This proves the asserted failure. □

A.3 (Proved; the weakest of the manuscript's two proposed repairs). *Manuscript: LM §1, folio 1, scan 6 (1.1 and the proposed extra hypotheses).* For a Grothendieck topos $\mathcal{E}$ all of whose objects are locally constant, the following are equivalent:

$$\mathcal{E} \simeq \beta C \text{ for a groupoid } C; \qquad \mathcal{E} \text{ is locally connected;} \qquad \mathcal{E} \text{ has enough points.}$$

Thus either proposed extra hypothesis suffices separately; their conjunction is unnecessary. Local connectedness is also necessary, so it is an exact repair within the class under discussion.

Proof. A locally constant object has decidable equality, because that property is local and holds for constant sets. If every object is decidable, Ω is decidable; equality of a truth value with truth is decidable, which is precisely excluded middle. Thus $\mathcal{E}$ is Boolean.

Suppose first that $\mathcal{E}$ has enough points. For each X, the points of the slice $\mathcal{E}/X$ separate its subterminal objects. Hence the complete Boolean algebra $\mathrm{Sub}(X)$ has enough frame points. The argument in A.2 shows that every such frame point comes from an atom. Separation now says that every element is a join of atoms. The atoms of $\mathrm{Sub}(X)$ are disjoint, so X is the coproduct of its atomic subobjects. This proves local connectedness.

Suppose next that $\mathcal{E}$ is locally connected. Since it is Boolean, its nonzero connected objects are atoms. There is only a set of isomorphism classes of atoms: choose a small generating family; every atom is an epimorphic image of one generator, and each generator has only a set of quotients. Choose representatives A_i of all atoms and form $A = \coprod_i A_i$. By hypothesis A is locally constant. Refine a trivializing cover of A into connected objects U_j. Each U_j is an atom. In $\mathcal{E}/U_j$, every $A_i \times U_j$ is a constant object: it is a subobject of the constant object $A \times U_j$, and subobjects of constant sets over an atom are constant subsets. Therefore every $X \times U_j$ is constant, since every X is a coproduct of atoms.

We claim $\mathcal{E}/U_j \simeq \mathbf{Set}$. For $f : Y \to U_j$, the two objects $Y \times U_j$ and $U_j \times U_j$, viewed over the second factor, are constant. A map between constant objects over the connected object U_j is a single set map. Pull back $f \times U_j$ along the diagonal section of $U_j \times U_j$; its fibre is constant and is precisely $Y \to U_j$. Thus every object of the slice is constant, and the constant functor is fully faithful because U_j is connected.

Each U_j is consequently connected and small projective: $\mathrm{Hom}(U_j, -)$ is the inverse image of the point obtained from the slice, and preserves all colimits. The U_j generate $\mathcal{E}$: after pullback to the covering family, every object is a coproduct of copies of its member, giving an epimorphic family of maps from the U_j. A morphism between two such connected projectives is an isomorphism. Indeed its image is a nonzero subobject of an atom, hence the whole atom, and projectivity splits this epimorphism; its section has nonzero image in an atom, hence is an isomorphism. The small full category C of these generators is therefore a groupoid. The usual density presentation of every object as a colimit of representables identifies $\mathcal{E}$ with βC. Equivalently, the explicit nerve functor $X \mapsto \mathrm{Hom}(-, X)|_C$ has a coend left adjoint, and both unit and counit are isomorphisms because the generators are small projective and conservative.

Finally βC is locally connected and has enough evaluation points. This proves all implications. □

A.4 (Proved; explicit exactness condition 1.1(c)). *Manuscript: LM §1, folio 1, scan 6 (1.1(c)).* A locally small category $\mathcal{E}$ is a discrete multi-Galois topos exactly when it satisfies the following axioms: (i) finite limits and all small colimits exist; (ii) small coproducts are disjoint and stable under pullback; (iii) every arrow factors as an effective epimorphism followed by a monomorphism, these factorizations are stable under pullback, and every internal equivalence relation is the kernel pair of its quotient; (iv) $\mathcal{E}$ has a small generating family; (v) it has a small generating family of noninitial connected projective atoms. Equivalently, in (v) one may require small projective generators whose full subcategory is a groupoid.

Proof. The first four are Giraud's exactness axioms, in the formulation as an infinitary pretopos with a set of generators (Makkai and Reyes, 1977, Proposition 3.4.8). They make $\mathcal{E}$ a Grothendieck topos. For an atom P, connectedness gives preservation of arbitrary coproducts by $\mathrm{Hom}(P, -)$. Projectivity gives lifting through effective epimorphisms. To see preservation of coequalizers, write a coequalizer as the quotient by the equivalence relation generated by the given parallel pair. An equality after quotient is locally a finite chain of generating identifications. Because P is an atom, an epimorphic family of its subobjects contains the whole P; the finite chain and each lifted step may therefore be chosen on all of P. This proves that the quotient of the Hom-set is the Hom-set of the quotient. Thus each P is small projective. All maps between such generators are isomorphisms by the splitting argument in A.3, and density gives $\mathcal{E} \simeq \beta C$. Conversely the representables of a groupoid presheaf category have precisely these properties; limits, coproducts and effective quotients are pointwise. This supplies a genuine exactness formulation without introducing a preferred fibre functor. □

1.2 All the assertions of §1

A.5 (Proved; propositions 1.3 and 1.4). *Manuscript: LM §1, folio 1, scans 6–7 (1.3–1.4).* For $S \in \beta C$, let $\int_C S$ be its category of elements. It is a groupoid and

$$(\beta C)/S \simeq \beta\left(\int_C S\right).$$

Arbitrary sums of multi-Galois topoi, whose underlying categories are products, are multi-Galois. The category of points of βC is a groupoid equivalent to C, and the canonical evaluation comparison with presheaves on that point groupoid is an equivalence.

Proof. The slice equivalence sends $T \to S$ to its fibres $T(c)_s$; inverses are fibrewise disjoint unions. Every arrow of the category of elements is invertible because arrows of C are. The sum assertion follows by taking a disjoint union of groupoids.

For completeness we classify all fibre functors, rather than only the evident evaluations. A colimit-preserving functor $F : \beta C \to \mathbf{Set}$ is determined by $P(c) = F(yc)$ and the coend formula. Left exactness says precisely that the category of elements of P is filtering in the relevant variance. For a groupoid, filtering says nonempty and that any two objects have a unique comparison after transport; equivalently, P is a torsor in one component of C. Choose an element of this torsor. Transport identifies it with the representable torsor at its vertex, hence identifies F with evaluation at that vertex. A transformation between two such functors is, by Yoneda, induced by exactly one arrow of C, and is invertible. Evaluation thus identifies the entire point groupoid, not merely a specified collection of points. The final comparison is the Yoneda evaluation equivalence transported along this identification. □

A.6 (Proved; proposition 1.5, all five conditions and the N.B.). *Manuscript: LM §1, folios 1–2, scans 7–9 (1.5 and Definition 1.6).* For $S \in \beta C$ the following are equivalent: S is nonempty and simply connected in the manuscript's sense; S is connected and projective; $\mathrm{Hom}(S, -)$ is a fibre functor; $(\beta C)/S \simeq \mathbf{Set}$; and S is representable. Definition 1.6 calls an object with these equivalent properties a universal cover. In the third condition it suffices to preserve coproducts and coequalizers, and, because the functor is left exact, it suffices to preserve coproducts and effective quotients.

Proof. Work in one component $\mathbf{Set}^G$. A connected object is a transitive G-set G/H. Its slice is $\mathbf{Set}^H$. Every object of that slice is locally constant by A.1, so simple connectedness means that every H-set is trivial. The regular H-set shows this happens exactly when $H = 1$. The epimorphism $G \to G/H$ admits a G-equivariant section exactly when $H = 1$, so projectivity has the same criterion. For $H = 1$, G is the representable object and its Hom-functor is the underlying-set functor, hence a fibre functor; conversely a representable fibre functor preserves epimorphisms and coproducts and therefore makes S projective and connected. The slice statement is the same $H = 1$ criterion. Objects meeting more than one component are not connected, and the initial object does not preserve the empty coproduct, so no exceptional case is suppressed.

Every colimit is a coequalizer of a pair between coproducts. In an exact category each coequalizer is the effective quotient of the equivalence relation generated by the pair. The finite-limit preservation of a representable functor and the atom/projectivity argument in A.4 justify the equivalent quotient formulation. This proves the N.B. with "quotient" interpreted as an effective quotient, rather than as an arbitrary image operation. □

A.7 (Proved; propositions 1.7–1.10). *Manuscript: LM §1, folios 2–3, scans 9–10 (1.7–1.10, including Definition 1.9).* The full subcategory of simply connected objects in a multi-Galois topos is equivalent to its point groupoid, by sending S to the fibre functor it represents with the point variance fixed above. For a specified point p, the pairs (S, s) with S simply connected and $s \in p^*S$ form a contractible groupoid. These are the pointed universal covers of Definition 1.9. Thus a pointed universal cover is unique up to unique isomorphism; an unpointed universal cover generally has automorphisms.

Proof. A.5 represents every fibre functor, and A.6 identifies its representing objects. Yoneda proves full faithfulness. If S represents p^*, the Yoneda element corresponding to id_S is universal. Any other representing pair is carried to it by a unique representing isomorphism. In $\mathbf{Set}^G$ the unpointed cover has automorphism group G^{op}, which explains exactly why the pointing is needed for the uniqueness assertion. □

A.8 (Proved; propositions 1.11 and the missing 1.12). *Manuscript: LM §1, folio 3, scan 11 (1.11–1.12).* For small groupoids C, D, composition on points induces an equivalence

$$\mathrm{Hom}_{\mathrm{Top}}(\beta C, \beta D) \simeq \mathrm{Fun}(C, D).$$

The right side is a groupoid, with natural isomorphisms as arrows. For connected pointed topoi $\mathbf{Set}^G$ and $\mathbf{Set}^H$ it is the action groupoid

$$\big[\mathrm{Hom}_{\mathrm{Grp}}(G, H)/H\big], \qquad h : \varphi \Rightarrow \psi \iff \psi(g) = h\varphi(g)h^{-1}.$$

Its automorphism group at φ is $C_H(\varphi(G))$. Forgetting the chosen comparison of base points therefore produces conjugacy, not equality, of homomorphisms.

Proof. A functor $u : C \to D$ gives restriction $u^* : \beta D \to \beta C$, which preserves all limits and colimits and has right Kan extension as right adjoint. Conversely compose the inverse image of a geometric morphism with every evaluation at $c \in C$. A.5 makes the resulting fibre functors a functor $C \to D$, unique up to natural isomorphism, and pointwise evaluation identifies the original inverse image with restriction. The same argument on transformations gives the asserted Hom-category equivalence. For one-object groupoids, a functor is a homomorphism, and its naturality equation is exactly the displayed conjugacy equation. All homomorphisms are allowed; no injectivity or centralizer hypothesis is involved at this stage. □

1.3 The locally constant envelope without points

A.9 (Proved; theorem 2.1 and definition 2.2). *Manuscript: LM §2, folio 4, scans 12–13 (2.1–2.2).* Suppose $\mathcal{E}$ is locally connected and has a small generating family of nonempty connected objects U_i such that every locally constant object of $\mathcal{E}/U_i$ is constant. Then $\mathrm{LC}(\mathcal{E}) \simeq \beta C$ for a small groupoid C, and its inclusion in $\mathcal{E}$ preserves finite limits and all small colimits. No hypothesis on points is needed. The same conclusion holds under the weaker, exact splitting hypothesis that a covering family of connected objects U_i trivializes every locally constant object of $\mathcal{E}$.

Proof. The generating hypothesis supplies an epimorphic family $U_i \to 1$, and the assumed simple connectedness trivializes every locally constant object after pullback to U_i. Let $U = \coprod_i U_i$. Form the simplicial set $K_n = \pi_0(U^{n+1})$, where products are over 1. Let C be the groupoid presented by vertices K_0, edges K_1, and the identity and triangle relations from K_2. Thus $C = \Pi_1 |K_\bullet|$.

Effective descent for the cover $U \to 1$ identifies objects of $\mathcal{E}$ with objects on its pieces plus cocycle data. For an object constant on each connected U_i, those data are sets A_i and, on each connected component of $U_i \times U_j$, a bijection between the corresponding sets. The cocycle

condition on every component of a triple product gives exactly the triangle relation defining C. A map of such objects is exactly a natural transformation. This proves an equivalence between the objects split by this cover and βC. By the splitting hypothesis these are precisely all locally constant objects.

Finite limits are locally finite limits of constant sets. For an arbitrary diagram of locally constant objects, restriction to each U_i is a diagram of constant objects and all its maps are constant set maps because U_i is connected. The constant functor to $\mathcal{E}/U_i$ preserves colimits, and pullback to U_i does likewise. Hence the colimit in $\mathcal{E}$ is again split by the cover and agrees with the colimit in βC. The inclusion is therefore left exact and cocontinuous, and gives the inverse image of the canonical geometric morphism $q : \mathcal{E} \to \mathcal{E}_{[1]}$. Existence of its right adjoint follows from the adjoint functor theorem for Grothendieck topoi, or from the small presentation just constructed. □

A.10 (Proved; the first phenomenon in 2.3). *Manuscript: LM §2, folio 5, scan 14 (2.3).* There is a locally connected, locally simply connected and simply connected topos having two distinct morphisms between the same two points. In particular, restriction of point morphisms to locally constant objects need not be faithful even when there are enough points.

Proof. Let D have objects $0, 1, 2$, two arrows $a, b : 0 \to 1$, an arrow $c : 1 \to 2$, and one arrow $d : 0 \to 2$, with $ca = cb = d$, and identities. Let $\mathcal{E} = \mathbf{Set}^{D^{\mathrm{op}}}$. Representables are connected; their slice categories are presheaves on D/x, which has a terminal object and therefore contractible nerve. Locally constant presheaves are exactly functors inverting all arrows, hence representations of the groupoid completion. The terminal object 2 makes this completion the terminal groupoid, so $\mathcal{E}$ is simply connected. The same argument in each D/x proves local simple connectedness. Evaluation points at 0 and 1 have two point morphisms induced by a, b; Yoneda distinguishes them. Their restrictions to the locally constant category **Set** coincide. The finite category and its localization relations are checked in the attached program `code/foundations/foundations_checks.py`. □

A.11 (Proved; a pointless locally simply connected example). *Manuscript: LM §2, folio 5, scans 14–15 (2.3).* There exists a nondegenerate, locally connected, locally simply connected topos with no points and with locally constant envelope **Set**. Consequently a universal cover need not be induced by any point, as anticipated in 2.3; the displayed assertion that β is always an equivalence is false without a point-existence hypothesis.

Proof. We give an explicit locale. For a closed subset $F \subset \mathbb{R}^3$, write $P(F)$ for its perfect kernel, supplied by the Cantor–Bendixson theorem. Set

$$j(V) = \mathbb{R}^3 \setminus P(\mathbb{R}^3 \setminus V) \quad (V \text{ open}), \qquad L = \{V : j(V) = V\}.$$

The operation j is a nucleus. It is increasing, inflationary and idempotent. Moreover $P(F \cup G) = P(F) \cup P(G)$ for closed F, G: each difference from its perfect kernel is countable, while a nonempty relatively open subset of a perfect Polish space is uncountable. Hence j preserves binary intersections. The frame L, with joins computed by applying j to ordinary unions, is the locale of coperfect opens. Its top and bottom remain distinct. It has no points: a point of L would compose with the frame quotient to a point $x \in \mathbb{R}^3$, but $j(\mathbb{R}^3 \setminus \{x\}) = \mathbb{R}^3$ contradicts evaluation at x.

We verify the local connectivity and covering claims rather than inferring them from the name of this locale. Ordinary open balls are j-fixed and form a basis of L. If a family of j-fixed opens covers a ball B in L, its ordinary union is $B \setminus D$, where D is closed in B with empty perfect kernel. The Cantor–Bendixson theorem makes this single omitted set D countable. In dimension at least two, removal of a closed countable subset does not disconnect a connected

open set: polygonal paths can be moved off the countably many forbidden points, or joined through a generic intermediate point. A partition in the locale of a connected ordinary open would therefore partition a connected open minus a closed countable set, and must be trivial. The components of a j-fixed open are themselves j-fixed: an isolated point in a component's complement could not lie in another open component, and outside the original open it would be an isolated point of that open's perfect complement. These are both impossible. Its ordinary component decomposition is consequently also its locale component decomposition. In particular the balls are connected, proving local connectedness.

Let F be locally constant on the slice locale over a ball B. Refine a trivializing cover by subterminal opens, and choose their j-fixed representatives V_i. Their union is $V = B \setminus D$ for one closed countable D. A morphism between constant sheaves over any $V_i \cap V_j$ is a family of set maps on its connected components. The preceding component comparison identifies these maps with the ordinary constant-sheaf maps on that intersection. The descent data for F are therefore the descent data of an ordinary covering space of V.

The space $B \setminus D$ is simply connected. Indeed a loop has compact image separated from the relatively closed set D, and can first be replaced, through loops avoiding D, by a polygonal loop. Choose a cone vertex in the convex ball outside the countable union of affine planes determined by one polygonal edge and one point of D, also outside D. The cone on the polygon then avoids D. This contracts the loop. Path connectedness is proved by the same generic-intermediate-point argument. Thus the ordinary covering is constant. Restrict its trivialization to the original V_i; the component comparison turns these into locale trivializations compatible on every overlap, and effective descent glues them over the locale cover of B. This does not require the ordinary union V itself to be j-fixed. Every ball slice is simply connected, and the identical argument with the convex space $\mathbb{R}^3$ proves global simple connectedness. Hence $\mathcal{E} = \mathrm{Sh}(L)$ has the required properties. Its terminal object is the universal cover in $\mathrm{LC}(\mathcal{E}) = \mathbf{Set}$, and no point can define it. □

A.12 (Proved; exact criteria for α and β in 2.3). *Manuscript: §2; folio 5; scans 14-15.*

Let $q : \mathcal{E} \to \beta C$ be the envelope of A.9. Let C_{pts} have the actual points of $\mathcal{E}$ as objects and the morphisms between their images in C as arrows. Then $\alpha : \mathrm{Pt}(\mathcal{E}) \to C_{\mathrm{pts}}$ is always essentially surjective, and $\beta : C_{\mathrm{pts}} \to C$ is always fully faithful. The latter is an equivalence exactly when every connected component of $\mathcal{E}$ has a point.

There is also a precise neighbourhood criterion for fullness and faithfulness. Write N_p for the category of pointed objects (V, v) with $v \in p^*V$, and let (S_p, s_p) be the universal pointed locally constant cover determined by p. In fibre-functor variance,

$$\mathrm{Nat}(p^*, r^*) \simeq \lim_{(V,v)\in N_p} r^*V \longrightarrow r^*S_p \simeq \mathrm{Nat}(p^*|_{\mathrm{LC}}, r^*|_{\mathrm{LC}}).$$

The functor α is faithful exactly when all these displayed maps are injective; it is full exactly when they are surjective. This criterion does not assume a spatial or a Boolean topos.

Proof. The first two assertions follow directly from the definition of C_{pts}. Components of C correspond to components of the terminal object of $\mathcal{E}$, because constant sets detect exactly their coproduct decomposition. A point in a component maps to its component of C; every object of that groupoid component is isomorphic to its image. Conversely an object of C hit up to isomorphism supplies such a point. This proves the criterion for β. Enough points implies this condition but is stronger than necessary; A.11 shows that it cannot simply be removed.

For the formula, every functor $p^* : \mathcal{E} \to \mathbf{Set}$ is the colimit, over the opposite of its category of elements, of the represented functors $\mathrm{Hom}(V, -)$. A natural transformation out of this colimit is a compatible family of elements of r^*V. This proves the first equivalence (a small cofinal presentation may be used). On locally constant objects, Yoneda at S_p gives the second. The

restriction map is evaluation at (S_p, s_p). Injectivity and surjectivity are exactly faithfulness and fullness, with no missing coherence condition because the displayed limit includes every map of pointed objects. For comparison, a Hausdorff interval gives a faithful but nonfull α, since distinct spatial points have no specialization arrows while its intrinsic fundamental groupoid is connected; A.10 supplies nonfaithfulness. □

1.4 Faithful groupoids over a base

A.13 (Proved; §4, discrete induction). *Manuscript: LM §4, folios 6–7, scans 16–18.* A geometric morphism $\beta D \to \beta C$ is a slice morphism if and only if its corresponding functor $p : D \to C$ is faithful. The bicategory of faithful groupoids over C has morphisms (f, α) with $\alpha : p_2 f \Rightarrow p_1$ and 2-morphisms $\eta : f \Rightarrow g$ satisfying

$$\alpha = \beta \circ (p_2 \eta).$$

Every Hom-category is a thin groupoid. Its quotient by isomorphism of 1-morphisms is equivalent to βC.

Proof. For a presheaf S, the category of elements projects faithfully to C, and A.5 identifies its topos with the slice over S. Conversely, for faithful p define

$$S(c) = \pi_0\{(d, \gamma) : d \in D,\ \gamma : c \xrightarrow{\sim} p(d)\},$$

where an arrow $(d, \gamma) \to (d', \gamma')$ is $h : d \to d'$ with $p(h)\gamma = \gamma'$. Each such comma groupoid is thin by faithfulness. Precomposition defines a presheaf on C. The functor $D \to \int_C S$ taking d to $(p(d), [d, 1])$ is fully faithful: equality of the associated classes gives a unique lift of the required arrow, again by faithfulness. It is essentially surjective by the very definition of the comma classes. Thus p is equivalent over C to a category-of-elements projection and gives a slice.

For the asserted 2-category, the compatibility equation determines each component $p_2\eta_d$ uniquely. Faithfulness of p_2 makes η_d unique when it exists. All its components are invertible, so a Hom-category is an equivalence relation in the categorical sense. The preceding construction sends (f, α) to a natural map of the associated presheaves. Conversely every presheaf map induces a functor between the categories of elements over C. The fibrewise construction proves these operations inverse after quotienting by the stated unique 2-isomorphisms. This makes precise the manuscript's marginal request. □

A.14 (Refuted; corrected statement: the profinite qualification on scan 18). *Manuscript: LM §4, folio 7, scan 18 (application to fields).* For profinite groups, faithful continuous homomorphisms do not suffice to characterize slice morphisms. A continuous injection $H \hookrightarrow G$ gives a connected slice $B_{\mathrm{cont}}H \to B_{\mathrm{cont}}G$ exactly when its image is open. Étale schemes over a field k correspond to discrete continuous Γ_k-sets, and hence to faithful action groupoids over Γ_k with open stabilizers; finite étale schemes correspond to finite such sets.

Proof. An object of $B_{\mathrm{cont}}G$ is a discrete set with continuous action, so a transitive object is G/H with H open. Its slice is $B_{\mathrm{cont}}H$, by taking the fibre of an equivariant map over the identity coset. Conversely any connected slice has precisely this form. The inclusion $1 \hookrightarrow \widehat{\mathbb{Z}}$ is faithful, but the regular coset set $\widehat{\mathbb{Z}}/1$ is not discrete continuous; the resulting point is not a slice morphism. Equivalently, a slice realizing it would provide a simply connected transitive continuous action, whose stabilizer would have to be trivial and open, impossible in this infinite profinite group.

For a field, a connected étale scheme is the spectrum of a finite separable extension. A general étale scheme is their disjoint union. Sending it to its k_s-points gives a discrete continuous Galois set; the inverse sends an orbit Γ_k/H to $\mathrm{Spec}(k_s^H)$. Open subgroups have finite index, and this construction is inverse on morphisms by the elementary Galois correspondence. Thus the topology and openness condition are essential to the application, not optional embellishments. □

1.5 The missing §3: pro-multi-Galois and profinite topoi

A.15 (Proved; definitions and recognition). *Manuscript: LM §3, folio 6, scan 16.* Call a topos *pro-multi-Galois* if it is locally connected and generated under colimits by locally constant objects. Call it *componentwise profinite multi-Galois* if it is locally connected and generated by finite locally constant objects, allowing a discrete, possibly infinite set of connected components. These definitions concern topoi, rather than formal pro-objects. A pro-multi-Galois topos is equivalently the classifying topos of a 0-strict pro-groupoid, that is, of a cofiltered system of groupoids whose transition maps have connected nonempty homotopy fibres. Its componentwise profinite analogue is equivalently a product of continuous-action topoi of profinite groups. These assertions are the corrected pro-analogues of 1.1: "generated by" replaces "all objects are" locally constant.

Proof. We spell out the passage from covering data to the pro-object. If $U \to 1$ is a covering object of a locally connected topos, decompose U into connected components. The argument of A.9 identifies the full subcategory $\mathrm{Split}(U)$ of objects trivialized on these components with βC_U, using the groupoid of the componentwise Čech descent data. The inclusion preserves finite limits and all colimits. A refinement $V \to U$ gives a fully faithful restriction functor $\beta C_U \to \beta C_V$. The corresponding functor $C_V \to C_U$ is bijective on connected components and surjective on the automorphism groups after choosing vertices: for a functor of groupoids these are exactly the conditions that restriction of set-valued functors be fully faithful. Equivalently its homotopy fibres are nonempty and connected. Thus the splitting groupoids give a strict fundamental pro-groupoid. Refinements and their isomorphisms must be kept coherently; a chosen cofiltered presentation of the associated pro-object removes this bookkeeping without discarding it.

There is a small cofinal supply of splitting covers. Fix a small generating set for the original topos. Refine every trivializing cover by its members; since a member has only one map to 1, repetitions can be removed. Hence the subfamilies of a fixed set of generators suffice. For each such cover the connected objects of $\mathrm{Split}(U)$ form a set of isomorphism classes, being quotients of its representables. Their union over the chosen covers is still a set.

Let $\mathrm{SLC}(\mathcal{E})$ be the full subcategory of sums of connected locally constant objects. We verify its exactness without asserting that arbitrary subobjects in $\mathcal{E}$ are locally constant (that assertion would be false, already for open subsets of a connected space). The needed lemma is that a *locally constant* subobject of a fixed locally constant object A is split by every connected cover splitting A. On a connected trivializing member, membership of each element of the constant fibre is a locally constant subterminal. Such a subterminal is complemented and therefore is 0 or 1. This proves the lemma; it also proves that arbitrary unions of locally constant subobjects of this fixed A are locally constant. A map between locally constant objects has locally constant image, since a common connected trivializing cover turns it into maps of sets. The same is true of its kernel pair and quotient by a locally constant equivalence relation.

Finite limits of sums of locally constant objects follow by distributing products over sums and applying the preceding observation on connected summands. For a coequalizer of maps between such sums, decompose its target as $\coprod_b Y_b$ with each Y_b connected and locally constant. Each connected source summand maps into a pair Y_b, Y_c; its image is a locally constant relation in $Y_b \times Y_c$. The equivalence relation generated by all these images is obtained from finite composites and arbitrary unions. Finite composites are locally constant by the image and fibre-product assertions; unions remain locally constant on each $Y_b \times Y_c$ by the lemma. Denote the resulting pieces by R_{bc}. If R_{bc} is nonempty, its projections onto Y_b, Y_c are epimorphic: their images are nonempty locally constant subobjects of connected locally constant objects. Thus the images of Y_b and Y_c in the quotient coincide. Choose one b for each component of this relation on the index set. The quotient is the sum of Y_b/R_{bb} over these choices, and each such quotient is locally constant. Hence all colimits and finite limits are computed in $\mathcal{E}$. The small family already constructed generates this category; the exactness axioms are inherited for its own internal

equivalence relations. This proves it is a Grothendieck topos.

Its generating site is the filtered union of the splitting sites, with their canonical topologies. The inverse-limit theorem for topoi identifies it with the bicategorical inverse limit of βC_U (Dubuc, 2004, Theorem 1.1.1 and Propositions 5.1.2–5.1.4). It is locally connected and is generated by its locally constant objects. Conversely such a topos equals its SLC category, giving the claimed recognition. Equivalently this is the $n = 1$ case of the strict-pro-groupoid recognition in Hoyois (2018, Corollary 3.20).

For the profinite version, restrict the generating category to finite locally constant objects. Its fibre functors on each component are finite Galois fibre functors. Finite products and the finite permutation action provide cofinal finite Galois objects. Their automorphism groups form a surjective inverse system of finite groups, whose compact inverse limit is a profinite group G. The finite Galois correspondence identifies the finite objects with finite continuous G-sets. Every continuous G-set is a disjoint union of finite orbits, so passing to sums identifies the full topos with $B_{\mathrm{cont}}G$. Products over components finish the unpointed statement. The existence of fibre functors here can also be obtained from the coherent-to-pos point theorem; the choices disappear on passing to the groupoid of all fibre functors. □

A.16 (Proved; the corrected analogue of 2.1). *Manuscript: LM §3, folio 6, scan 16; §2, folio 4, scan 12.* For every locally connected Grothendieck topos there is a canonical geometric morphism

$$q : \mathcal{E} \longrightarrow \mathrm{SLC}(\mathcal{E}) \simeq \beta\Pi_1\mathcal{E}, \qquad q^* \text{ the full inclusion},$$

and, for a strict presentation $\Pi_1\mathcal{E} = (C_i)$,

$$\mathrm{LC}(\mathcal{E}) \simeq 2\text{–}\operatorname*{colim}_i \mathrm{Fun}(C_i^{\mathrm{op}}, \mathbf{Set}). \tag{A.16.1}$$

The right side consists of representations split at one stage. The topos $\beta\Pi_1\mathcal{E}$ consists of their filtered colimits and can be strictly larger. It is the inverse limit of the *topoi* βC_i, not the category of actions of the abstract inverse-limit group with its topology forgotten. Under A.9's splitting hypothesis the system is essentially constant and A.16 reduces to the manuscript's theorem.

Proof. Every locally constant object is split by a cover and hence lies in one $\mathrm{Split}(U)$. Any two splitting covers have a common refinement, and each inclusion is full. This proves (A.16.1), on both objects and morphisms. Every sum of connected locally constant objects is the filtered union of its finite subsums; a finite sum has a common splitting cover. Conversely a filtered colimit of locally constant objects lies in $\mathrm{SLC}(\mathcal{E})$ by A.15. This identifies the two descriptions of the enlargement. Its inclusion is left exact and cocontinuous, and the generator argument supplies its right adjoint, hence q.

The construction is intrinsic: a map to a discrete groupoid classifying topos is a torsor, and a torsor is locally constant and therefore lies in a splitting category. The coherent system (C_i) thus pro-represents the functor of groupoid-valued torsors; it is uniquely the fundamental pro-groupoid. This agrees with the 1-truncation of the shape constructed in E.1. The full-faithfulness and enlargement assertions are also the $n = 0$ case of Hoyois (2018, Theorem 3.13). Under a cover splitting all locally constant objects, its own splitting category already contains all the others, so the pro-object is constant. □

A.17 (Refuted; corrected statement: pro-analogues of 1.4 and 1.7). *Manuscript: LM §3, folio 6, scan 16; §1, folios 1–3, scans 7–10.* For an infinite profinite group G, the ordinary category of points alone does not reconstruct $B_{\mathrm{cont}}G$, and an ordinary universal-cover object need not exist. The correct replacements are the profinite (or, in the prodiscrete case, localic) groupoid

of fibre functors, and pro-representing universal covers. In the connected pointed profinite case the canonical fibre functor is pro-represented by

$$\widetilde{1} = (G/N)_{N \lhd_o G}, \qquad p^*X = \operatorname*{colim}_{N \lhd_o G} \mathrm{Hom}_G(G/N, X). \tag{A.17.1}$$

It is represented by an object of $B_{\mathrm{cont}}G$ exactly when G is discrete, hence finite. For a prodiscrete group replace the finite quotients by its discrete quotient system; without spatial hypotheses the inverse limit belongs in locales.

Proof. For every x in a continuous G-set its stabilizer is open. In a profinite group it contains an open normal subgroup, so the right side of (A.17.1) is the union of the fixed-point sets X^N, namely X. Compatibility is the identity on each such element. This proves pro-representability directly. If a representing object existed, it would be connected projective. A transitive continuous action is G/H with H open. If $H \neq 1$, choose an open subgroup $K \subset H$ omitting one nontrivial element. The epimorphism $G/K \to G/H$ has no equivariant section; consequently no such G/H is projective. The only remaining possibility is $H = 1$, which is allowed exactly when G is discrete.

The natural automorphism group of the fibre functor is G: its compatible permutations of the finite regular quotients recover $\lim_N G/N$. Equip it with the topology whose basic subgroups fix specified elements of finite objects. This is exactly the original profinite topology. Natural isomorphisms between two fibre functors form a profinite torsor, and their compositions are continuous. Evaluating objects on this enriched groupoid reconstructs continuous actions, component by component. Forgetting the topology loses the restriction that stabilizers be open. Concretely, the ordinary point groupoids of $B_{\mathrm{cont}}G$ and $B_{\mathrm{disc}}(G_{\mathrm{abstract}})$ are both equivalent to the one-object groupoid of the abstract group G, whereas the second admits the regular action and the first does not. The preceding projectivity test distinguishes the topoi categorically.

In a general prodiscrete Galois topos the same construction uses localic spaces of natural isomorphisms. Their inverse-limit descriptions follow by restriction to the discrete splitting categories, and the pointwise evaluation equivalence is Dubuc (2004, Theorems 3.3.5–3.3.6). One must not assert that all ordinary points of a connected prodiscrete Galois topos are isomorphic: Dubuc's Example 3.5.2, referring to SGA 4 IV 7.2.6(d), explicitly records the failure. Nor does the pointless example A.11 contradict existence of points for a Galois topos: its Galois envelope is **Set**, while the original topos is not generated by its locally constant objects. □

A.18 (Proved; analogue of 1.11 and its scope). *Manuscript: LM §3, folio 6, scan 16; §7, folios 12–13, scans 28–30.* For connected profinite Galois topoi, choosing points gives

$$\mathrm{Hom}_{\mathrm{Top}}(B_{\mathrm{cont}}G, B_{\mathrm{cont}}H) \simeq [\mathrm{Hom}_{\mathrm{cont}}(G, H)/H]. \tag{A.18.1}$$

For the unpointed and prodiscrete theory the correct version is the groupoid of coherent maps between the *intrinsic fundamental pro-groupoids* of the topoi, or equivalently of continuous localic torsors; an ordinary Hom-set of inverse-limit groups is not a replacement. This is not a full-faithfulness assertion for arbitrary strict pro-presentations of their classifying topoi; A.21 gives a counterexample even with connected transition fibres. The same intrinsic statement is valid componentwise and hence applies to the diamonds of groupoids in §7.

Proof. A geometric inverse image takes finite locally constant objects to finite locally constant objects. Composing it with the chosen source fibre functor gives a target fibre functor. After choosing an isomorphism with the selected target point, the induced automorphisms define $G \to H$. It is continuous because the stabilizer of every element of every finite H-set pulls back to an open subgroup of G. Conversely continuous restriction is left exact and cocontinuous on all discrete continuous actions and has a right adjoint; it is a geometric inverse image. These

constructions are mutually inverse since all actions are sums of finite orbits. A change of the chosen point isomorphism conjugates the homomorphism, and natural transformations are precisely the elements implementing that conjugacy, proving (A.18.1).

Without choices, pull back the compatible torsors defining the target fundamental pro-groupoid. This gives a coherent morphism into it. Conversely the torsor adjunction of E.1–E.2 gives

$$\mathrm{Hom}_{\mathrm{Top}}(\mathcal{E},\mathcal{F}) \simeq \mathrm{Map}_{\mathrm{Pro}(\mathbf{Gpd})}(\Pi_1\mathcal{E},\Pi_1\mathcal{F})$$

for pro-multi-Galois topoi: A.16 identifies $\mathcal{F}$ with the ordinary classifying topos of its own $\Pi_1\mathcal{F}$. Equivalently the coherent system defines inverse images on the intrinsic splitting categories and extends to the generated topos. The compatibility transformations also descend, so this is an equivalence of Hom-groupoids, not merely a bijection on isomorphism classes. In localic language these data are principal bibundles; selecting points trivializes the source-side torsor only when the required isomorphism of points exists. This explains the qualification in A.17. In §7 the objects $\Pi_U, \Pi_D, \Pi_{D^*}, \Pi_e$ are therefore retained as profinite groupoids, with the comparison $\alpha : \varphi\rho \simeq \psi\sigma$ of scan 30. No equality of chosen representatives or inverse-limit abstract groups is required. □

A.19 (Proved; the scheme range, including nonnormal schemes). *Manuscript: §3; folio 6; scan 16; §7; folios 12-13; scans 28-30.*

For a locally noetherian scheme X, its small étale topos is locally connected. Its pro-multi-Galois envelope is A.16's topos generated by all étale locally constant sets. Its componentwise profinite envelope is generated by finite étale covers and is the product of the continuous-action topoi of the usual étale fundamental groups. These two envelopes need not coincide. They do coincide for geometrically unibranch locally noetherian schemes, in particular for smooth curves and fields, which are the cases used in §7. Even there, the subcategory of locally constant objects need not itself be the whole envelope.

Proof. Every étale object over a locally noetherian scheme is locally noetherian and locally connected. Its connected components give a decomposition of its representable sheaf into connected objects, proving local connectedness of the topos. Finite locally constant sheaves are represented by finite étale schemes; the fibre functor at a geometric point identifies these with finite continuous $\pi_1(X)$-sets (Stacks Project, Theorem 58.6.2, Tag 0BND). Thus A.15 gives the asserted profinite envelope, without assuming that all étale sheaves are locally constant.

For the possible distinction, consider a rational nodal curve over an algebraically closed field. The infinite chain of copies of its normalization, with the second preimage of the node on the nth copy glued to the first on the $(n+1)$st, is a connected étale covering with fibre $\mathbb{Z}$. It trivializes over an étale neighbourhood splitting the two branches and over the smooth complement. Deck translation acts transitively with infinite orbit. Consequently its locally constant sheaf cannot be a continuous action of a profinite group, all of whose orbits on a discrete set are finite. This is the discrete loop part of the nodal example in Bhatt and Scholze (2015, Example 7.4.9). Their Lemma 7.4.5 identifies étale locally constant sets with those pro-étale actions having a common open kernel, and Lemma 7.4.10 proves that for geometrically unibranch schemes every connected geometric covering is finite étale. Applied to an étale locally constant sheaf, the latter says that all its connected components are finite étale. Hence its generated topos is the profinite one, proving the coincidence asserted above.

For the last qualification let $G = \mathbb{Z}_p$ and

$$X = \coprod_{n\geq 1} G/p^nG \in B_{\mathrm{cont}}G.$$

An action in $B_{\mathrm{cont}}G$ is locally constant exactly when an open subgroup acts trivially on the *whole* set: if it is trivialized over a covering action, pull back to any transitive orbit G/H of that

cover, obtaining a trivial restricted H-action; conversely G/H supplies the trivializing cover. For the displayed X the kernel is $\bigcap_n p^n G = 0$, which is not open. Thus X is continuous but not locally constant. This exhibits the precise difference between the two sides of (A.16.1) and the limit topos, already in a connected profinite example. □

A.20 (Refuted; reconstruction from finite tests). *Manuscript: LM §1, folio 3, scan 11; §3, folio 6, scan 16; PATHS E.13 test.* An isomorphism of effects on all finite covers, or even isomorphism after precomposition by every finitely generated discrete group, need not identify morphisms of discrete multi-Galois topoi. Let $G = \mathrm{Sym}_{\mathrm{fin}}(\mathbb{N})$ and let $s : G \to G$ shift the support by one, fixing 0. The endomorphisms of βG induced by s and 1 are not isomorphic, have isomorphic restrictions to every finitely generated subgroup, and induce the same endomorphism of the profinite completion $\widehat{G} = C_2$.

Proof. The shift is not surjective, whereas conjugation is; A.8 therefore separates the two morphisms. Every finite set of permutations is supported on a finite set F. Choose a finite permutation h whose restriction to F is the injection $i \mapsto i+1$. Such an extension exists because a bijection between two finite subsets extends to a permutation of their union. Then $hgh^{-1} = s(g)$ for all permutations supported on F, in particular for the subgroup generated by the given generators. The choices of h are not globally compatible, which is exactly why the tests do not assemble into a natural isomorphism.

The infinite alternating subgroup A_{fin} is simple. Indeed let $g \neq 1$ belong to a normal subgroup, choose a with $g(a) = b \neq a$, and choose c, d outside its support. For $t = (a\ c\ d)$ one has

$$[g, t] = (b\ c\ d)(a\ d\ c) = (a\ b\ c).$$

Thus the normal subgroup contains a 3-cycle. Any two 3-cycles are conjugate by an even finite permutation: adjust the parity of a chosen conjugator by a transposition on two fresh letters. Since 3-cycles generate A_{fin}, this proves simplicity. Any map of A_{fin} to a finite group is therefore trivial, since its kernel cannot be trivial. Thus every finite quotient of G factors through sign; sign itself is a quotient, and s preserves it. This proves the completion assertion. The finite-support conjugation and the displayed commutator are checked in the attached program, while the argument proves them for arbitrary supports. This verifies the PATHS E.13 inference rather than using it as an axiom. Finite tests are complete for *continuous profinite* morphisms by A.18, where compactness supplies compatible conjugators; they are incomplete for arbitrary discrete groups. □

A.21 (Proved; pro-object, limit, and the ind cautions). *Manuscript: LM §3, folio 6, scan 16; pro/ind qualifications in the task.* The pro-object used in §3, its classifying topos, and an inverse-limit group have different universal properties. For diagrams (G_i) and (H_j) the coherent pro-Hom is

$$\mathrm{Map}_{\mathrm{Pro}(\mathbf{Gpd})}((G_i), (H_j)) = \lim_j \operatorname*{colim}_i \mathrm{Map}_{\mathbf{Gpd}}(G_i, H_j), \tag{A.21.1}$$

with limits and colimits taken homotopy coherently. The classifying topos is a bicategorical inverse limit of topoi. For profinite groups, the topological inverse limit with its compact topology gives the same continuous-action topos. For unrestricted systems, passing to an abstract inverse limit loses information.

Proof. Formula (A.21.1) is the defining pro-universal property. The topos statement was proved by the filtered-site construction in A.15. For finite groups with surjective transition maps, compactness makes every projection of the inverse limit surjective; an open stabilizer contains the kernel of a projection after passage to a suitable finite stage. Thus every finite orbit descends

to a stage, and all continuous actions are sums of these orbits. This proves the claimed agreement while retaining the topology.

For a direct warning take the progroup

$$\mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \cdots .$$

Its ordinary inverse limit is zero: an integer divisible by every power of two is zero. The pro-object is not zero, since its maps to the constant group $\mathbb{Z}$ form $\operatorname{colim}(\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \cdots) \cong \mathbb{Z}[1/2]$, which is nonzero. These transition maps are not 0-connected, so this example also shows why the strictness qualification in A.15 cannot be suppressed.

Even strictness alone does not make arbitrary pro-presentations intrinsic. Fix a prime p and let

$$C_n = \coprod_{k \geq 1} B(\mathbb{Z}/p^{\min(n,k)}), \qquad D = \coprod_{k \geq 1} B(\mathbb{Z}/p^k), \tag{A.21.2}$$

with reduction maps $C_{n+1} \to C_n$. Each transition has connected nonempty homotopy fibres. In topoi the inverse limit of the βC_n is βD: all maps are over the fixed discrete component topos, so the connected inverse limit is computed componentwise, and its kth component is eventually the constant topos $\beta(\mathbb{Z}/p^k)$. Nevertheless (C_n) is not the constant pro-groupoid D. A pro-map $(C_n) \to D$ must factor through one stage C_n. The source vertex groups at that stage have exponent at most p^n, while D has groups of unbounded exponent. Its composite with any map $D \to (C_n)$ therefore cannot be an equivalence on all components; a permutation of components cannot fix this bound. In particular the canonical $D \to (C_n)$ is not invertible. This example uses infinitely many components, so it does not contradict the compact finite-groupoid correspondence. It proves that A.15 is an essential-image recognition theorem and that A.18 must use intrinsic fundamental pro-groupoids, not arbitrary strict presentations having the same limit topos.

The ind-results CATHOT E.4 and PATHS D.10–D.13 concern another category, formed by quotienting functors by zigzags of natural transformations. Neither that quotient nor an ordinary union enters (A.21.1). No claim about its limits is needed in the present proofs. An actual filtered union of groupoids remains a groupoid, since inverses already occur at finite stages; its classifying space has the filtered homotopy-colimit type by compactness of finite simplicial tests. This elementary valid statement does not authorize replacing a bicategorical pushout by an ordinary one, nor omitting the compatible conjugacies in A.20. □

A.22 (Proved; completion of 2.2 and variance). *Manuscript: LM §2, folio 4, scan 13 (Definition 2.2 and its tower).* The tower requested on scan 13 is constructed and proved in E.1–E.4 below. Its geometric morphisms have direction $\mathcal{E} \to \mathcal{E}_{[i]} \to \mathcal{E}_{[i-1]}$; the associated inverse-image functors have the opposite direction. For $i \geq 2$ its stages must be higher topoi, or formal pro-homotopy types in the absence of the stated local hypotheses. There is no tower of ordinary multi-Galois topoi retaining the higher Postnikov invariants. This is the level-1 limitation proved in the supplied CATHOT C.1–C.4, used here only after checking its groupoid-localization argument.

2 B. The mixed topos and the cohomology of hole filling

We first fix a consequential ambiguity. The assertions about cohomology on LM scans 127–129 concern an ordinary glued topos. Deriving the gluing functor is indispensable. Its underived value does not remember the kernel of a local monodromy homomorphism. This is precisely where the proposed proof on scan 129 can fail.

2.1 The actual construction

B.1 (Proved): the Artin gluing of LM §14. *Manuscript: LM §14, folio 60–62, scans 123–127.* Let G, H_s, and Q_s be discrete groups, or profinite groups with continuous actions on discrete sets. Let

$$r_s : H_s \longrightarrow G, \qquad q_s : H_s \twoheadrightarrow Q_s, \qquad I_s = \ker q_s.$$

We use a finite index set S for the cohomological assertions below; the construction itself permits an arbitrary small index set. There is a Grothendieck topos $\mathcal{B}$ whose objects and morphisms are

$$(M, (N_s), (a_s)), \qquad a_s : q_s^* N_s \longrightarrow r_s^* M,$$

and componentwise equivariant maps commuting with the a_s. Equivalently, a_s is a Q_s-equivariant map $N_s \to (r_s^* M)^{I_s}$. The functor of gluing is $L = (q_{s*} r_s^*)_s$. Its open part is βG, its complementary closed part is $\coprod_s \beta Q_s$, and it is the topos denoted $B_{X,U}$, $B_{U,X}$, and subsequently $B_{X;S}$ in LM §14, folios 60–62, scans 123–127, and §16, scans 163–166.

Proof. The functor L is accessible and preserves finite limits: restriction does so, and fixed points are a right adjoint. The Artin gluing theorem (Artin et al., 1972a, IV, Théorème 9.5.4) applies. Here is also the concrete identification needed for the manuscript. In the displayed adjoint description, finite limits and arbitrary colimits are componentwise, since r_s^* and q_s^* preserve both. The subterminal $u = (1, (\varnothing), a)$ has slice $\mathcal{B}/u \simeq \beta G$: an object mapping to u has all N_s empty. The complementary closed embedding sends (N_s) to $(1, (N_s), !)$. The right adjoint to restriction to the open part is $M \mapsto (M, L(M), \mathrm{id})$, so restriction of this right adjoint to the closed part is exactly L, as asserted on scan 127. This also proves that the closed embedding is not generally an open induction morphism. Evaluation on the open and closed fibres is a conservative family of points. Every point belongs to one of these two parts, because its inverse image sends u to either 0 or 1. □

For a smooth connected curve X/k, a finite reduced closed subset S, and $U = X - S$, take $G = \pi_1(U)$, $H_s = \mathrm{Gal}(K_s^{\mathrm{sep}}/K_s)$ for the fraction field of the henselization at s, and $Q_s = \mathrm{Gal}(k(s)^{\mathrm{sep}}/k(s))$. The maps require compatible geometric points; replacing these choices conjugates the maps and gives canonically equivalent gluing categories. Artin recollement on $X_{\text{ét}}$ gives a fully faithful inverse-image functor

$$f^* : \mathcal{B} \longrightarrow X_{\text{ét}}, \qquad f : X_{\text{ét}} \longrightarrow \mathcal{B}.$$

Its essential image consists of the sheaves whose restriction to U is a continuous G-set. Infinite such objects are allowed; "continuous" is not being replaced by "one finite quotient acts on the entire set". The henselian local topos itself is the gluing of βH_s and βQ_s by q_{s*}: both its generic and closed strata are spectra of fields. Thus the local equivalence asserted at the top of scan 129 is correct. The subsequent derived base-change inference is a different assertion.

B.2 (Proved): filling holes at level one. *Manuscript: LM §14, folio 60, scans 123.* In the preceding geometric setting the continuous surjection

$$\pi_1(U) \longrightarrow \pi_1(X)$$

has kernel the *closed* normal subgroup generated by the images $r_s(I_s)$. For a discrete surface the same formula holds with ordinary normal closure. This proves the formula of LM §14, folio 60, scan 123.

Proof. A finite G-set extends to a finite locally constant sheaf on X exactly when every a_s can be an isomorphism. Equivalently I_s acts trivially on its restriction. Indeed the finite cover of U has a unique normalization over the regular curve X, and at each missing discrete valuation its normalization is ètale precisely when inertia acts trivially. The associated Q_s-set is then

forced. Thus restriction identifies finite covers of X with the full subcategory of finite G-sets annihilated by the asserted normal subgroup. This is exactly the finite representation category of the quotient; the fibre functor identifies its profinite automorphism group with $\pi_1(X)$. The proof does not select a connected component of a cover and hence applies componentwise to groupoids without adding base points to the conclusion. □

B.3 (Proved): the effective local group and the derived comparison. *Manuscript: LM §14, folio 62–63, scans 127–129.* Write

$$K_s = \mathrm{im}(H_s \xrightarrow{(r_s,q_s)} G \times Q_s).$$

In the profinite case this image is closed. For a torsion abelian object $F = (M, N_s, a_s)$ of $\mathcal{B}$ there is a natural fibre sequence

$$R\Gamma(\mathcal{B}, F) \longrightarrow R\Gamma(G, M) \oplus \prod_s R\Gamma(Q_s, N_s) \longrightarrow \prod_s R\Gamma(K_s, M), \tag{1}$$

where K_s acts on M through its projection to G, and the second map is the difference of restriction and the map induced by a_s. For the geometric curve, the corresponding formula is

$$R\Gamma(X, f^*F) \longrightarrow R\Gamma(U, M) \oplus \prod_s R\Gamma(Q_s, N_s) \longrightarrow \prod_s R\Gamma(H_s, M). \tag{2}$$

Consequently, whenever U has the $K(\pi, 1)$ comparison for M,

$$\mathrm{Cone}\big(R\Gamma(\mathcal{B}, F) \to R\Gamma(X, f^*F)\big) \simeq \prod_s \mathrm{Cone}\big(R\Gamma(K_s, M) \to R\Gamma(H_s, M)\big)[-1]. \tag{3}$$

This is an exact coefficientwise necessary and sufficient criterion, not merely a sufficient injectivity hypothesis. In particular an injective (r_s, q_s) for each s suffices.

Proof. For discrete groups form the category with one object u with group G, one object s with group Q_s, no arrows from u to s or between distinct closed objects, and crossing biset $(G \times Q_s)/H_s$. We use the inverse on the second factor to make this a left G, right Q_s biset. Its representations are exactly the triples of B.1. The action groupoid of its crossing arrows has stabilizer K_s, not H_s: the kernel of (r_s, q_s) acts trivially on every crossing arrow. Thus this groupoid is equivalent to BK_s.

For clarity, the cochain assertion follows directly from the bar resolution of this category. A nondegenerate chain lies entirely in one stratum, or contains exactly one crossing arrow. The subcomplexes of the first kind give the two stratum cochain complexes. The chains of the second kind have automorphism strings on the two sides of the crossing arrow; their two-sided bar complex is the cochain complex of the crossing action groupoid. Its two augmentations are restriction from G and the specialization map from Q_s. The extra crossing arrow shifts the latter complex by one. Taking the total complex is therefore the homotopy fibre displayed in (1). This argument includes nonconstant coefficients: their value at a crossing arrow is the value M at its target, and its source value maps there by a_s.

For profinite groups the finite continuous triples form a coherent generating pretopos: they are closed under finite limits, finite coproducts and quotients, and every continuous triple is a filtered union of finite subtriples. Every such finite triple and every finite diagram of them descends to a common finite quotient of G and the Q_s and the corresponding finite images of the H_s. Thus our topos is the coherent inverse limit of these finite-category topoi. Their transition inverse images preserve finite objects and are coherent. The cohomology continuity theorem (Artin et al., 1972b, VI, Corollary 8.7.7, with 8.7.1 and Remark 8.7.8) therefore identifies its derived cohomology with the filtered colimit of the finite-level computations. This verifies

that the following continuous bar calculation really computes topos cohomology, rather than introducing a different cohomology theory. Each continuous map from the compact space of a fixed finite string of arrows to a discrete torsion coefficient module has finite image and factors through a finite quotient. Finite collections of such maps admit a common quotient. Hence the continuous bar complex is the filtered colimit of the finite bar complexes, with coefficients first restricted to their finite submodules. Filtered colimits of abelian groups are exact. The discrete argument therefore proves the same formula for continuous cohomology.

For (2), use the localization triangle for $U \subset X$ and excision at the finitely many points of S. The stalk of Rj_*M at a geometric s is $R\Gamma(I_s, M)$, with its residue Galois action. The henselian local topos has derived sections $R\Gamma(Q_s, N_s)$; after strict henselization this is the exact closed-stalk functor, and descent gives the displayed expression. Hochschild–Serre identifies its local punctured term with $R\Gamma(H_s, M)$. This gives (2). The comparison of the two fibre sequences is identity on the middle term when U has the stipulated comparison. The elementary cone identity in a stable category gives (3). □

Check of the supplied three-puncture calculation. We have checked LM3b H.6–H.8 in its supplied `sections/Higher.md`. For H.7, let $G = F(a, b)$, $I_1 = \langle a \rangle$, $I_2 = \langle b \rangle$, and $I_3 = \langle (ab)^{-1} \rangle$. For each finite-index normal N, its strict quotient category has open group $Q_N = G/N$ and crossing stabilizers $I_{i,N}$, the images of I_i. Its nerve is the homotopy pushout obtained by attaching three cones on $BI_{i,N}$ to BQ_N, by the same crossing-bar decomposition used in B.3. For a finite constant abelian group A, the filtered colimit over N of these cochain triangles computes continuous cohomology of $\widehat{G}$ and the three completed I_i: every continuous cochain factors through a finite quotient, and filtered colimits of abelian groups are exact. Each I_i has its full induced profinite topology, since its generator is primitive in $F(a, b)$ and admits a retraction to $\mathbb{Z}$. The free and procyclic degree-one resolutions identify these continuous cohomology groups with the discrete ones. The two remaining maps are exactly

$$\begin{aligned} A \oplus A^3 &\longrightarrow A^3, \quad (c, d_1, d_2, d_3) \longmapsto (c - d_1, c - d_2, c - d_3), \\ A^2 &\longrightarrow A^3, \qquad\qquad (s, t) \longmapsto (s, t, -s - t). \end{aligned}$$

The first has diagonal kernel A and is surjective. The second is injective with cokernel A, identified by the sum of the three coordinates. Hence the filtered nerve cohomology is A in degrees zero and two and zero elsewhere; the latter cokernel is precisely the oriented peripheral relation and hence the sphere class. This verifies H.7 in its finite-coefficient, filtered-colimit form, without a stagewise cohomology-vanishing assertion. In particular no one finite stage works uniformly for all finite coefficients: for a prime $\ell \nmid |Q_N|$, its positive-degree $\mathbb{F}_\ell$-cohomology vanishes, including degree two. H.8's upgrade from these constant coefficients to all finite homotopy values is separately checked in E.12 and F.43.

The bounded contractible-object comparison. The supplied CATHOT D.1 represents $[NC, Y]$, for categories C of a specified size bound, by a category of bounded contractible simplicial sets over Y, with all maps over Y retained. That bounded resolution is a useful comparison of constructions, but the pieces in B.1 are not individually contractible. Already a geometric punctured disc has group $\mathbb{Z}$, or $\widehat{\mathbb{Z}}(1)$ for finite coverings, and its classifying type has nonzero degree-one cohomology. The mixed topos instead retains the underived crossing biset, whose effective stabilizer is K_s. B.3 proves exactly what is lost on replacing the full local term BH_s by BK_s. Thus the bounded representability result does not imply the mixed-topos comparison and cannot remove its local-inertia condition.

B.4 (Refuted): the proposition of scan 129 without a local hypothesis. *Manuscript: LM §14, folio 63, scans 129.* Let k be algebraically closed of characteristic zero, $X = \mathbf{P}^1_k$, $S = \{\infty\}$, and $U = \mathbf{A}^1_k$. Then $G = Q_s = 1$, $H_s = I_s = \widehat{\mathbb{Z}}(1)$, and $\mathcal{B} \simeq \mathrm{Fun}([1], \mathrm{Set})$. For $A = \mathbb{Z}/n$, $n > 1$,

$$H^2(\mathcal{B}, A) = 0, \qquad H^2(X_{\text{ét}}, A) = A(-1) \neq 0.$$

Nevertheless U is a $K(\pi, 1)$. Thus the proposition on LM §14, folio 63, scan 129, and the prompt's inferred "all nonempty S" conclusion are false.

Proof. The triple is just a map $N \to M$. Its global sections are N, because a compatible pair is uniquely specified by its source value. This functor is exact on abelian objects, so all higher cohomology vanishes. The target computation is the Kummer calculation of (The Stacks Project Authors, 2026i, Tag 03RQ, Lemma 59.69.1); it can also be read as the cokernel of multiplication by n on $\mathrm{Pic}(\mathbf{P}^1) = \mathbb{Z}$. In B.3, $K_s = 1$, whereas $H_s = \widehat{\mathbb{Z}}(1)$. The missing cone is $A(-1)[-2]$. In particular the error on scan 128 is the assertion that local-support derived functors can be computed after restricting an injective global mixed object to the local mixed topos. That restriction need not preserve injectives. Equivalence of the local topoi does not prove this base-change assertion. The program `code/root_checks.py` independently computes the Betti numbers of the nerve [1] and a triangulated S^2 modulo 2, 3, and 5; the degree-two discrepancy is recorded in its output. □

2.2 The complete curve comparison

B.5 (Known, with reduction proved): the curve $K(\pi, 1)$ theorem. *Manuscript: LM §14, folio 63, scans 129.* For a smooth connected curve over any field, an affine curve and a proper curve of geometric genus at least one are $K(\pi, 1)$ for all torsion locally constant abelian coefficients. The same holds for their nonempty open subschemes. A proper curve of geometric genus zero has the comparison for p-primary coefficients in characteristic p, but not for every prime-to-p coefficient.

Proof. The precise all-characteristic theorem is Schmidt and Stix (2016, Lemma 2.7(a)); their definition and the cohomological criterion immediately preceding that lemma are the ones needed here. We spell out the reduction to the present version. Over an algebraic closure, an affine smooth curve has torsion cohomological dimension at most one. For a finite local system, first trivialize its monodromy on a finite ètale cover. Every degree-one class is a torsor under its finite abelian coefficient group and is killed by that finite cover; higher classes vanish. The effaceability criterion for the universal covering pro-object therefore identifies its cohomology with continuous group cohomology.

For a proper curve of positive genus, after the same trivialization only degree two needs attention. For $\ell \neq p$ the trace identification is $H^2(Y, \mathbb{Z}/\ell^a(1)) = \mathbb{Z}/\ell^a$; pullback under a finite ètale map is multiplication by its degree. Such a curve has cyclic ètale covers of degree ℓ^a, by its Jacobian or the prime-to-p surface-group presentation. Its degree-two class is therefore effaced. For p-primary coefficients in positive characteristic, degree two already vanishes; this follows from Artin–Schreier and the surjectivity of $F - 1$ on $H^1(Y, \mathcal{O}_Y)$, and extension through finite filtrations. These are the vanishing statements in (The Stacks Project Authors, 2026b,e, Tags 0A3J and 03SB). The same argument proves the p-primary assertion for genus zero.

For an arbitrary ground field, first replace it, if necessary, by its finite separable algebraic closure in the curve's function field. The curve is geometrically connected over this constant field; neither its etale topos nor its fundamental group changes in this rephrasing. Apply the compatible Hochschild–Serre spectral sequences for the geometric curve and its fundamental group extension by $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. The geometric comparisons just proved identify their second pages and all differentials, hence their abutments. Purely inseparable extensions do not change the ètale site. Infinite discrete torsion modules are filtered unions of finite invariant submodules:

the orbit of each element is finite by continuity, and the subgroup generated by finitely many torsion elements of bounded order is finite. Both cohomology theories commute with these filtered colimits for the quasi-compact curves in question. This proves the assertion in exactly the coefficient generality used in B.3. □

B.6 (Proved): globalization of local covering germs on curves. *Manuscript: LM §14, folio 63, scans 129.* Let U be a smooth connected affine curve over an algebraically closed field k, let $\overline{U}$ be its smooth proper completion, and let $s \in \overline{U} - U$. The full local inertia homomorphism

$$\mathrm{Gal}(k((t))^{\mathrm{sep}}/k((t))) \longrightarrow \pi_1(U)$$

is injective except when $\mathrm{char}\, k = 0$ and $U \simeq \mathbf{A}^1$. In that exception it is the zero map from $\widehat{\mathbb{Z}}(1)$ to 1. The analogous tame local homomorphism is injective, in every characteristic, precisely when $U \not\simeq \mathbf{A}^1$.

Proof. We prove the full assertion by dominating every finite local covering germ by the germ of a global finite ètale covering of U. For a continuous map $h : H \to G$ of profinite groups this criterion implies injectivity: for each open normal $N \lhd H$, dominance gives an open subgroup of G whose inverse image in H is contained in N; the intersection of all such N is 1. Conversely, a closed embedding $H \subset G$ has the criterion. Indeed, for an open subgroup $V \subset H$ choose open normal $N \subset G$ with $N \cap H \subset V$. Then VN is an open subgroup of G and $VN \cap H = V$, so the H-orbit of its distinguished coset in G/VN is H/V. This also specifies the meaning of "sub-germ".

We next globalize cyclic local steps. For n prime to the characteristic, the local cyclic extension of degree n is unique: Hensel's lemma and divisibility of k^* make every local unit an nth power, so its Kummer class is determined by valuation modulo n. If there are two boundary points s, t, divisibility of $\mathrm{Pic}^0(\overline{U})$ supplies a divisor E with $n[E] = [t - s]$. Thus $s - t + nE$ is principal, and its Kummer covering is ètale on U and has ramification index n at s.

If there is one boundary point and the genus is positive, use (Grothendieck and Raynaud, 2003, XIII, Corollaire 2.12). In its prime-to-p presentation, the peripheral generator is the inverse of the product of commutators of $2g$ handle generators. Map one handle pair to

$$x = (1, 0, 0), \quad y = (0, 1, 0) \quad \text{in } \mathrm{UT}_3(\mathbb{Z}/n),$$

where multiplication on triples is $(a, b, c)(d, e, f) = (a + d, b + e, c + f + ae)$. Map all other handle generators to 1. The commutator is $(0, 0, 1)$ of order n. The cited existence clause gives a tame cover with local ramification index n at the unique cusp. Its global group need not be cyclic, which is essential here. The orders for $n = 2, 3, 4, 5, 9$ are checked by the attached program; the displayed multiplication proves the result for every n.

In characteristic p, every cyclic local extension of degree p is Artin–Schreier. It can be globalized without adding a branch point inside U. Put $P = \bigoplus_{x \in \overline{U} - U} k((t_x))/k[[t_x]]$. Principal parts give an exact sequence

$$\Gamma(U, \mathcal{O}_U) \longrightarrow P \xrightarrow{\delta} H^1(\overline{U}, \mathcal{O}_{\overline{U}}) \longrightarrow 0.$$

The map is Frobenius compatible. The map $F - 1$ on the last, finite-dimensional, k-vector space is surjective (The Stacks Project Authors, 2026b, Tag 0A3J, Lemma 59.63.2). Given a desired local principal part a at s, with zero parts at the other boundary points, choose $b \in P$ with $(F - 1)\delta(b) = \delta(a)$. Then $a - (F - 1)b$ is the principal part of a function f regular on U. The covering $z^p - z = f$ has the required local class. The integral remainder is killed by Artin–Schreier on the henselian local ring, since its residue field is algebraically closed. Thus the principal-part calculation really determines the finite local covering germ.

There is just one obstruction to starting this procedure: the one-cusp genus-zero curve has no nontrivial tame cover. In characteristic p first replace it by the finite ètale cover $y^p - y = x^m$, where $p \nmid m$ and $(p-1)(m-1)/2 > 0$. Its completion has that positive genus and one point above infinity. The genus follows from Riemann–Hurwitz: the pole of order m has different exponent $(p-1)(m+1)$, hence $2g-2 = -2p + (p-1)(m+1)$. The different formula follows directly from the single ramification break m of this Artin–Schreier extension. We use $(p,m) = (2,3), (3,2)$ for the small characteristics.

Every finite local Galois group over $k((t))$ is an extension of a p-group by a cyclic group of order prime to p, hence is solvable. Choose a tower with cyclic prime-degree steps. Starting with the preceding positive-genus cover when necessary, suppose a global completion contains the jth local field in the tower. The compositum with the next field is cyclic of that prime degree or trivial. The cyclic construction just given dominates it by a new global cover. Induction on the fixed finite tower terminates. If the initial auxiliary cover was not contained in the prescribed extension, take composita throughout; cyclicity survives this base change. Subsequent finite separable covers preserve positive genus by Riemann–Hurwitz, or preserve the existence of at least two boundary points. The construction therefore never loses its applicability. This proves injectivity in positive characteristic, and also in characteristic zero outside $\mathbf{A}^1$.

For tame inertia it suffices to detect each finite cyclic quotient directly using the Kummer or unitriangular covers above; their orders are prime to p. Conversely a tame finite cover of $\mathbf{A}^1$ would extend to a tame cover of $\mathbf{P}^1$ branched at only one point, contradicting Riemann–Hurwitz. Finally, the equivalence between finite extensions of the excellent henselian local fraction field and of its completion follows by approximating the coefficients of a separable defining polynomial and Hensel's lemma. Thus the same conclusion holds for the local groups used in B.1, not only for completed fields. □

B.7 (Proved): complete comparison and its exact exceptions. *Manuscript: LM §14, folio 63, scans 129.* Let X/k be a smooth geometrically connected separated curve of finite type, $S \subset X$ finite and reduced, and $F = (M, N_s, a_s)$ an arbitrary torsion abelian object of the full ètale mixed topos. The comparison $R\Gamma(\mathcal{B}, F) \to R\Gamma(X, f^*F)$ is an equivalence for every F, except for the following geometric types:

(i) X is proper of genus zero and $S = \varnothing$;

(ii) $\operatorname{char} k = 0$, $X \simeq \mathbf{P}^1_k$, and S consists of one geometric point (necessarily a k-point).

In (i), set $M^{(p')}$ equal to the prime-to-p part of M, and to M in characteristic zero. The comparison cone is

$$R\Gamma(k, M^{(p')}(-1))[-2]. \tag{4}$$

In (ii) the cone is $R\Gamma(k, M(-1))[-2]$, independently of the boundary module N_s and its specialization map. Thus these formulas decide the comparison for *each* coefficient object, including arithmetic coefficients with acyclic Galois cohomology. Over an algebraically closed field the first failure for every nonzero indicated M is exactly degree two. For p-primary coefficients in characteristic p there is no exception.

Proof. When S is nonempty, U is affine. B.5 gives its comparison and B.6 gives injectivity of the geometric inertia maps except in (ii). Over an arbitrary field, purely inseparable invariance first allows passage to the perfect closure, taking the reduced boundary after base change. This leaves the etale topoi and local finite-cover categories unchanged. The residue fields are then finite separable extensions of the ground field. The kernel of (r_s, q_s) is geometric; the residue map is injective into the ground-field Galois group after fixing a residue-field embedding. Thus the geometric result makes (r_s, q_s) injective, and B.3 applies. Alternatively, the geometric

comparisons descend by the compatible Hochschild–Serre spectral sequences. When S is empty, B.5 applies except in (i).

In (i), $\pi_1(X) = G_k$ and f is the structural morphism. On a geometric genus-zero curve, $R^0 f_* f^* M = M$, $R^1 f_* f^* M = 0$, $R^2 f_* f^* M = M^{(p')}(-1)$, and the remaining terms vanish. These are the Kummer and Artin–Schreier calculations in B.5. The cone of the unit $M \to R f_* f^* M$ is therefore $M^{(p')}(-1)[-2]$; apply $R\Gamma(k, -)$. This conclusion does not require the genus-zero curve to split, nor does it assert a splitting of its derived direct image.

In (ii), $\mathcal{B}$ is the arrow topos over βG_k. Geometric calculation B.4 gives its relative comparison cone $M(-1)[-2]$ with the natural Galois action. Descent gives the stated cone. In every other case the cone is zero. The fieldwise formulation avoids a false uniform assertion about the first arithmetic degree: if X is a nonsplit real conic and $M = \mathbb{Z}/2$, its arithmetic cohomology is that of $\mathbf{RP}^2$, whereas $H^*(G_{\mathbb{R}}, \mathbb{Z}/2) = \mathbb{F}_2[t]$. The comparison first fails in degree three. For a split genus-zero curve a rational point splits the unit and the degree-two orientation class already witnesses failure. More generally (4) and its long exact sequence specify the failure in every arithmetic degree without a hidden splitting assumption. For the real example, the complex points of the anisotropic conic identify with S^2, with complex conjugation the free antipodal involution (in a projective coordinate it is $z \mapsto -1/\overline{z}$). Etale–Betti comparison over $\mathbb{C}$ and descent along $X_{\mathbb{C}} \to X$ identify the arithmetic complex with the equivariant cohomology of that sphere. Since the action is free, this is the cohomology of $\mathbf{RP}^2$. The unit is the classifying map of its double cover; it is an isomorphism in degrees zero, one, and two, while degree three of the target curve vanishes and degree three of BC_2 is nonzero. □

B.8 (Refuted, with replacement proved): tame groups for p-torsion. *Manuscript: LM §12 and 14, folio 53 and 63, scans 109 and 129.* The prime-to-p version of B.7 is obtained using tame inertia and tame local systems, with the additional exceptional geometric pair $(X_{\overline{k}}, (S_{\overline{k}})_{\mathrm{red}}) \simeq (\mathbf{P}^1_{\overline{k}}, \{\infty\})$ in *every* characteristic. Over an imperfect field this includes a nonsplit genus-zero curve with one purely inseparable boundary point; the perfection reduction in B.7 covers it. It cannot be extended to p-torsion by retaining only tame groups. For $U = \mathbf{A}^1_k$, with k algebraically closed of characteristic p, its tame group is trivial but

$$H^1_{\text{ét}}(U, \mathbb{F}_p) = k[x]/\{h^p - h : h \in k[x]\} \neq 0.$$

The full-group statement B.7 remains valid for p-torsion.

Proof. For a tame finite local system of order prime to p, a degree-one torsor trivializing a class is again tame: wild inertia acts trivially on the coefficients and has no nontrivial homomorphism to their prime-to-p additive group. Together with the curve cohomological-dimension argument, this proves the tame $K(\pi, 1)$ comparison. B.6 supplies tame local injectivity outside the one-cusp rational case, and B.3 completes the proof. For the counterexample, Artin–Schreier gives the displayed cokernel. The class of x is nonzero: a nonconstant polynomial h has $\deg(h^p - h) = p \deg h$, while a constant gives a constant. This also proves that one cannot erase wild monodromy on the grounds that the underlying variety is a curve. In particular, the demand for an example of failure of the *full* mixed comparison with p-torsion on these smooth curves has a false premise; the failure belongs to its tame replacement. □

2.3 Comparison with logarithmic topoi

B.9 (Proved): the logarithmic comparison, including coefficients. *Manuscript: LM §14, folio 60–62, scans 123–127.* Give the smooth curve X the divisorial log structure of S. There are canonical geometric morphisms

$$X_{\mathrm{ket}} \xrightarrow{\epsilon} X_{\text{ét}} \xrightarrow{f} \mathcal{B}_{X;S}.$$

This does not identify any two of these topoi. On the closed fibre over s, the logarithmic topos is βH_s^t, where $H_s^t = H_s/P_s$ and P_s is wild inertia. The mixed topos has closed fibre βQ_s. Its comparison sends a Q_s-set to its inflation to H_s^t. In characteristic zero $P_s = 1$.

More explicitly, the glued covering model of logarithmic sheaves which are essentially locally constant on U consists of

$$(M, V_s, b_s), \qquad V_s \in \beta H_s^t, \qquad b_s : (H_s \to H_s^t)^* V_s \longrightarrow r_s^* M.$$

Here "covering model" means the associated glued covering topos with its comparison to the full logarithmic topos, not that every full logarithmic sheaf is such an object. The inverse image of (M, N_s, a_s) is $(M, (H_s^t \to Q_s)^* N_s, a_s)$. Thus its boundary monodromy is trivial on tame inertia. A locally constant logarithmic sheaf can have nontrivial boundary inertia, so is in general outside this image.

For $\Lambda = \mathbb{Z}/n$ with n invertible on X, and every torsion Λ-module object F of $\mathcal{B}_{X;S}$, put $\mathcal{F} = f^*F$ and $N_s = i_s^*\mathcal{F}$. There is the exact triangle

$$R\Gamma(X, \mathcal{F}) \longrightarrow R\Gamma(X_{\mathrm{ket}}, \epsilon^*\mathcal{F}) \longrightarrow \bigoplus_{s \in S} R\Gamma(k(s), N_s(-1))[-1] \longrightarrow R\Gamma(X, \mathcal{F})[1]. \tag{B.9.1}$$

The boundary is the logarithmic residue/Gysin boundary; the triangle need not split. For a finite locally constant logarithmic Λ-sheaf L one instead has

$$R\Gamma(X_{\mathrm{ket}}, L) \simeq R\Gamma(U, L|_U). \tag{B.9.2}$$

These assertions do not extend to p-torsion by keeping the same Tate-twist formula.

Proof. The morphism ϵ is induced by viewing an ordinary etale morphism as a strict Kummer-etale morphism. A strict local trait acquires, through all Kummer roots of a uniformizer, the tame inertia group $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$. Its closed logarithmic point therefore has group H_s^t. A section of a nearby H_s-set which survives this system is precisely a P_s-invariant section. Hence its gluing functor is $(r_s^* M)^{P_s}$, now with its H_s^t-action. Adjunction gives the displayed triples. This also proves the inflation formula for the comparison, without conflating wild and tame monodromy. A nontrivial Kummer character of tame inertia gives the promised object outside its image.

Every point of the mixed topos lies in its open part or one of its closed parts: pull its open subterminal back to Set, where it is either 0 or 1. Over an algebraically closed field its closed point has no automorphism, whereas the fibre functor on the logarithmic point has automorphism group $\widehat{\mathbb{Z}}^{(p')}(1)$. In the mixed topos all points on the connected open stratum are identified at the level of its covering theory. Full $X_{\text{ét}}$ and X_{ket} still distinguish different underlying locations by their open subterminals. For instance the mixed topos has only $1 + 2^{|S|}$ subterminals in this connected geometric situation, whereas the full topoi retain every Zariski open of X. These are object-level and point-level distinctions independent of cohomology.

For the cohomological assertion, on a strict local trait the continuous cohomology of tame inertia with constant prime-to-p coefficients is Λ in degree zero, $\Lambda(-1)$ in degree one, and zero above. The Kummer sequence makes this identification canonical. Off S the log structure is trivial. Consequently

$$\Lambda \longrightarrow R\epsilon_*\Lambda \longrightarrow i_*\Lambda(-1)[-1] \longrightarrow \Lambda[1],$$

with $i : S \hookrightarrow X$. The projection formula gives (5) below; it can also be checked on these strict local stalks, whose two cohomology modules are free over Λ:

$$R\epsilon_*\epsilon^*\mathcal{F} \simeq \mathcal{F} \otimes^{\mathbf{L}}_{\Lambda} R\epsilon_*\Lambda. \tag{5}$$

The local calculation and projection formula are the curve case of Illusie (2002, Theorem 5.2 and §5.4). Tensoring the triangle and taking sections proves (B.9.1). For constructible coefficients this is the usual derived projection formula; arbitrary torsion coefficients follow by filtered unions of finite subobjects and continuity.

For (B.9.2), a logarithmically locally constant sheaf is trivialized by a Kummer-etale covering. On such a covering the assertion $L \simeq Rj_{\mathrm{ket}*}L|_U$ reduces to $\Lambda \simeq Rj_{\mathrm{ket}*}\Lambda$. The latter is logarithmic purity, Illusie (2002, Theorem 7.4); in this curve case its proof is the same Kummer calculation of $R^1\epsilon_*\Lambda$ and $R^1j_*\Lambda$, with higher stalks zero. Descent and global sections give (B.9.2). Over $\mathbb{C}$, Theorem 5.9 of that source identifies the same computation with the real oriented blow-up: its boundary circles retain the local loops. Finally B.8's Artin–Schreier class proves the asserted characteristic-p limitation. For example on $(\mathbb{P}^1, \infty)$ in characteristic zero, constant coefficients have mixed $H^2 = 0$, ordinary etale $H^2 = \Lambda(-1)$, and logarithmic $H^2 = 0$. Agreement of two numbers in this example does not give an equivalence of topoi. □

2.4 Orientation, support, and trace

B.10 (Refuted, with replacement): the exact orientation formulas. *Manuscript: LM §14, folio 63–64, scans 130–131.* In this paragraph the groups are geometric, over an algebraically closed field. In characteristic zero put $\widehat{\mathbb{Z}}^\dagger = \widehat{\mathbb{Z}}$ and $T = \widehat{\mathbb{Z}}(1)$; in characteristic $p > 0$ omit the p-factor and write $T = \widehat{\mathbb{Z}}^{(p')}(1)$. Let $\overline{G} = \pi_1(X)$ for a smooth proper connected curve X, and $G = \pi_1(U)$. For $g(X) \geq 1$ there are canonical isomorphisms

$$H^2(\overline{G}, T) \simeq \widehat{\mathbb{Z}}^\dagger, \qquad H^2(\overline{G}, \widehat{\mathbb{Z}}^\dagger) \simeq T^\vee. \tag{B.10.1}$$

If a boundary inertia is cohomologically effective as in B.3, its support group is

$$H^2_s(\mathcal{B}_{X;S}, \widehat{\mathbb{Z}}^\dagger) \simeq L_s^\vee, \qquad L_s = \widehat{\mathbb{Z}}^\dagger(1). \tag{B.10.2}$$

For $g \geq 1$ the local-to-global map identifies all the L_s canonically with the dual of the second group in (B.10.1). If $S \neq \varnothing$, the unfilled group instead has $H^2(G, \widehat{\mathbb{Z}}^\dagger) = 0$. Thus scan 131 must use the *filled* group for its genus-positive H^2 formula. For genus zero and no puncture, the displayed group-cohomology formula is false.

For every nonempty S, including a single puncture on $\mathbb{P}^1$, define the relative support complex using the *full* peripheral homomorphisms by

$$R\Gamma_c(G, \{H_s\}, \Lambda) := \mathrm{fib}\left(R\Gamma(G, \Lambda) \longrightarrow \bigoplus_s R\Gamma(H_s, \Lambda) \right). \tag{B.10.3}$$

Then

$$H^2_c(G, \{H_s\}, \widehat{\mathbb{Z}}^\dagger) \simeq T^\vee. \tag{B.10.4}$$

This equals $H^2(\mathcal{B}, j_!\widehat{\mathbb{Z}}^\dagger)$ when the local maps are effective. For $(\mathbb{P}^1, \infty)$ in characteristic zero the latter group is zero, so the unqualified marginal claim for the ordinary mixed topos is refuted.

Proof. First work with $\Lambda = \mathbb{Z}/n$, n prime to the characteristic. The Kummer sequence and the degree map give $H^2(X, \Lambda(1)) \simeq \Lambda$; for the precise curve calculation see The Stacks Project Authors (2026i, Lemma 59.69.1). The $K(\pi, 1)$ result B.5 gives (B.10.1) in positive genus. For a strict local trait, the cone of $\Lambda \to R\Gamma(I_s, \Lambda)$ is $\mathrm{Hom}(L_s, \Lambda)[-1]$. The support fibre is its shift by -1, proving (B.10.2) when B.3 identifies the effective and full local groups. The boundary class of a uniformizer is the Kummer residue, and its Gysin image is the degree-one point class. Its trace is 1. Therefore the identifications for different s agree after the local-to-global support map. This proves the claimed independence without choosing generators of inertia.

Affineness and B.5 imply the vanishing of $H^2(G, \Lambda)$. Localization on the proper curve gives $R\Gamma(X, j_!\Lambda)$ as the fibre in (B.10.3), because the geometric local punctures are exactly the H_s

and U is a $K(\pi,1)$. Its top compact-support trace is $\Lambda(-1)$, proving (B.10.4). In the exceptional affine line case the fibre is explicitly

$$\mathrm{fib}(\Lambda \to R\Gamma(\widehat{\mathbb{Z}}(1),\Lambda)) \simeq \Lambda(-1)[-2],$$

whereas B.4 proves the mixed $j_!$ group is zero. A proper geometric genus-zero curve has trivial fundamental group, so its positive group cohomology vanishes. Finally take the derived inverse limit in n. The finite groups in these formulas have the surjective transition maps of the stated Tate module, so no unwanted $\varprojlim^1$ term occurs in these degrees. This explains the meaning of the completed coefficients. In characteristic p, $H^2(X,\mathbb{F}_p) = 0$ by Artin–Schreier, hence no all-primes rank-one orientation formula can replace the stated prime-to-p formula. □

B.11 (Proved): the cohomological trace. *Manuscript: LM §14, folio 64, scans 131–132.* Let $G' \to G$ be a continuous homomorphism of geometric curve groups with the peripheral data of scans 131–132: the loops in S_0' map to specified target loops with the specified degrees, and the loops in S_1' map trivially. Include the stated identification $T' \simeq T$ of orientation modules, denoted $\phi : T' \xrightarrow{\sim} T$, and let $\Lambda = \mathbb{Z}/n$ with n invertible in the field. In every case where the group and peripheral complexes compute the curve, there is a functorial trace

$$\mathrm{Tr}_f : R\Gamma_c(U',\Lambda) \longrightarrow R\Gamma_c(U,\Lambda) \tag{B.11.1}$$

which on H_c^2 is $(\phi^{-1})^\vee : T'^\vee \to T^\vee$ modulo n, the contragredient orientation isomorphism. The isomorphism ϕ is explicitly part of the datum on LM §14, folio 64, scan 131; scan 132 specifies its contragredient. Here the cohomology is defined by (B.10.3) for punctured curves, and by group cohomology for proper curves of positive genus. If the ordinary mixed topoi compute these complexes, this is also their support cohomology trace. Without that qualification the requested mixed-topos trace assertion is false, by B.10.

Proof. We give a construction that works for an abstract homomorphism and does not assume it comes from an algebraic map. For a smooth geometric curve the cup product and the trace of its orientation class give Poincare duality

$$R\Gamma_c(U,\Lambda(1)) \simeq R\,\mathrm{Hom}_\Lambda(R\Gamma(U,\Lambda),\Lambda)[-2]. \tag{B.11.2}$$

For the etale statement and its reduction by specialization, see Deligne (1977, VI, Theorem 2.2 and its proof). In the present setting it can be proved by a finite cell complex for an oriented surface with boundary, followed by the prime-to-characteristic comparison, or by the Kummer local calculation and curve duality. The finite complex has one dual cell for each cell, and intersection of a cell with its dual is 1; its boundary matrices are transposes with the usual signs. This proves the perfect pairing, including the boundary case. The etale version has the same trace normalization. All the constant cohomology modules here are finite free over Λ, so these are perfect complexes.

Using B.5, the homomorphism induces restriction $R\Gamma(G,\Lambda) \to R\Gamma(G',\Lambda)$. Apply $R\,\mathrm{Hom}_\Lambda(-,\Lambda)[-2]$ and the two isomorphisms (B.11.2), and then remove the twist by the given $T' \simeq T$. This is (B.11.1). Composition is functorial because restriction and dualization are. In degree two the map is dual to $H^0(U,\Lambda) \to H^0(U',\Lambda)$, which is the identity on Λ for connected curves. Hence it is precisely the orientation isomorphism, rather than multiplication by a global covering degree. For a finite covering the usual relation $\mathrm{Tr}_f\, f^* = \deg(f)$ still holds: pullback, not this top trace, multiplies the fundamental cohomology class by the degree. The construction is equivariant under operators preserving the supplied orientation data, and therefore descends through Hochschild–Serre in arithmetic situations. No general six-functor formalism for arbitrary topoi is asserted. □

2.5 Mixed maps and operators

B.12 (Proved): the mixed morphism, with its commutation data. *Manuscript: LM §14, folio 64–67, scans 132–137.* Let $S' = S'_0 \amalg S'_1$. To construct $g : \mathcal{B}' \to \mathcal{B}$, give the following coherent data: a geometric morphism $\beta G' \to \beta G$; for $t \in S'_0$ a target $\tau(t)$ and maps of local and residue groupoids with invertible commutation two-cells; and, for $t \in S'_1$, a geometric morphism $p_t : \beta Q'_t \to \beta G$ together with an invertible two-cell

$$(\beta H'_t \to \beta G' \to \beta G) \simeq (\beta H'_t \to \beta Q'_t \xrightarrow{p_t} \beta G). \tag{B.12.1}$$

The inverse image on triples is

$$M' = f^*M, \qquad N'_t = \begin{cases} h_t^* N_{\tau(t)}, & t \in S'_0, \\ p_t^* M, & t \in S'_1. \end{cases} \tag{B.12.2}$$

For the first case use the transported $a_{\tau(t)}$; for the second use (B.12.1) itself. These formulas construct the desired geometric morphism and its open and filled comparisons. In geometric discrete notation p_t is the right G-torsor P_t of scan 134. For an infinite profinite G the word torsor here means the compatible system of finite torsors, or the equivalent profinite/localic torsor, not a discrete regular G-set.

Proof. An object of the target is the diagram in B.1. Applying the specified inverse images to this diagram and using the commutation isomorphisms gives exactly (B.12.2), with the required equivariance. The same prescription on componentwise maps is a functor. In the original triple description finite limits and all colimits are computed on M and the N_s: the functors q_s^* and r_s^* preserve both. Each functor in (B.12.2) is an inverse image and has the same preservation properties. Consequently the constructed functor is left exact and cocontinuous, and, by accessibility, has a right adjoint. It is the inverse image of a geometric morphism. Composition of the formulas gives the composition of the maps, with exactly the composition of their two-cells. Restriction to the open part gives f; the filled comparison follows by quotienting the peripheral normal subgroups as in B.2.

For $Q'_t = 1$ the functor p_t^* is the fibre functor $M \mapsto P_t \times^G M$. Equation (B.12.1) says that the local inertia acts trivially on that fibre functor, with a specified identification with the restriction from the open stratum. Merely listing an isomorphism class of torsors and omitting this datum would not specify a mixed map. This is exactly the isocommutative triangle and its collapsed-loop counterpart on scan 134. For profinite groups, replacing P_t by its finite quotients proves the same formula on every continuous set. A free discrete G-torsor would be discontinuous when G is infinite, which explains the qualification. □

B.13 (Proved): operators, splittings, and the exact rigidity claim. *Manuscript: LM §14, folio 67, scans 136–137.* Suppose a group Γ acts coherently on the preceding data. On an orbit of S'_1 choose t with stabilizer Γ_t. Let

$$1 \longrightarrow G \longrightarrow E \longrightarrow \Gamma \longrightarrow 1$$

represent the action on the connected covering groupoid. The datum of an equivariant point over that orbit is a right G-torsor P together with a splitting $q : \Gamma_t \to E^P$ of the twisted extension, over Γ_t. Its automorphism group is

$$C_{G^P}(q(\Gamma_t)) = (G^P)^{\Gamma_t}. \tag{B.13.1}$$

If this group is trivial, every two isomorphic such objects have exactly one isomorphism. This proves the rigidity sentence of scan 136. It neither asserts that a splitting exists nor that different splitting classes are isomorphic.

Proof. An equivariant structure on the fibre torsor is a compatible lift of every element of Γ_t to an automorphism over its prescribed action. Such lifts form the extension E^P; the compatibility for products is exactly the homomorphism condition for a section. Equivalently, after choosing a point of P, it is a lift into E, and changing the point conjugates the section by G. This proves both the torsor-and-section description and its independence of the choice. An automorphism a of P preserves the equivariant structure precisely when $aq(\gamma)a^{-1} = q(\gamma)$ for every γ. This is (B.13.1). The set of isomorphisms between two isomorphic objects is a torsor under the automorphism group of either, proving unique isomorphism when that group is trivial. In the profinite version require all actions and splittings continuous and interpret the torsor as in B.12; the identical equations hold at every finite quotient and pass to the inverse system. Extending over all orbits and using B.12 constructs the equivariant mixed morphism. The subsequent rational-point conjecture on scan 137 is not a consequence of this elementary rigidity calculation. □

3 C. Hole filling, quotient actions, and exterior groups

We retain the notation $r_s : H_s \to G$, $q_s : H_s \twoheadrightarrow Q_s$ of Part B. The word "pushout" below means a bicategorical pushout: cones include invertible comparison two-cells, and the universal property is an equivalence of Hom categories. The mixed-topos triangle has a different, noninvertible comparison cell. This distinction is explicit on LM scans 164–166.

3.1 The two squares and the two kinds of gluing

C.1 (Proved): the squares of (**) and (****). *Manuscript: LM §16, folio 81–83, scans 163–168.* Let $\mathcal{B}_D$ be the gluing of $\coprod_s \beta H_s$ and $\coprod_s \beta Q_s$ along $\prod_s q_{s*}$, and let $R : \mathcal{B}_D \to \mathcal{B}_{X;S}$ be induced by the r_s. Both squares in

$$\begin{array}{ccccc} \coprod_s \beta H_s & \longrightarrow & \mathcal{B}_D & \longleftarrow & \coprod_s \beta Q_s \\ \downarrow & & \downarrow{\scriptstyle R} & & \| \\ \beta G & \longrightarrow & \mathcal{B}_{X;S} & \longleftarrow & \coprod_s \beta Q_s \end{array}$$

are two-cartesian in the two-category of all Grothendieck topoi. The same is true for the open and closed squares in the geometric diagram (****). For a henselian trait its entire etale topos is the indicated $\mathcal{B}_D$. The upper triangle through q_s has only its canonical *lax* cell; it is generally not isocommutative.

Proof. In triples, $R^*(M, N, a) = (r^*M, N, a)$. The open subterminal $(1, 0)$ pulls back to $(1, 0)$. A geometric morphism to a topos factors through its open subtopos precisely when that subterminal pulls back to 1, and through the complementary closed subtopos precisely when it pulls back to 0. These two universal properties identify both pullbacks and their comparison two-cells. They apply equally to the inverse image of the open U and closed S in D.

Every sheaf on a trait is determined by its restriction to its generic field, its restriction to its closed field, and the specialization map from the latter to the inertia invariants of the former. This follows by the ordinary open–closed sheaf gluing construction: on an etale object take pairs of sections whose germs agree on the generic parts of its local branches. The sheaf condition makes these pairs glue uniquely, and restriction recovers the original pair. The generic and closed field topoi are βH_s and βQ_s, and the gluing functor is q_{s*}. Thus it gives the claimed identification. On a triple, the comparison cell is exactly $a_s : q_s^* N_s \to r_s^* M$. It need not be bijective: $N_s = \varnothing$, $M = 1$ already disproves invertibility. This verifies the warning of scan 164 and the two-square assertion of scan 166 separately. □

C.2 (Proved): the exact hole-filling pushout. *Manuscript: LM §16, folio 82–84, scans 166–169.* For discrete groups put

$$\overline{G} = G/\langle\langle r_s(\ker q_s) : s \in S\rangle\rangle. \tag{C.2.1}$$

For profinite groups take the *closed* normal closure. Then $\beta\overline{G}$ is the bicategorical pushout of

$$\beta G \longleftarrow \coprod_s \beta H_s \longrightarrow \coprod_s \beta Q_s \tag{C.2.2}$$

in all ordinary Grothendieck topoi. Hence it is also the pushout among locally connected topoi and among the corresponding multi-Galois topoi. In the profinite case the surjectivity of each q_s is material to this strong statement. Formula (C.2.1) is scan 169's formula, and B.2 identifies it with the geometric filled group.

Proof. For a discrete groupoid C and any topos $\mathcal{T}$,

$$\mathrm{Hom}_{\mathrm{Top}}(\beta C, \mathcal{T}) \simeq \mathrm{Fun}(C, \mathrm{Pt}(\mathcal{T})). \tag{C.2.3}$$

Indeed the components of an inverse image $\mathcal{T} \to \mathrm{Fun}(C, \mathrm{Set})$ are point functors; the C-actions supply their natural isomorphisms. Conversely such a functor assembles a left-exact cocontinuous inverse image. This describes also all two-cells, including noninvertible natural transformations between points. The pushout groupoid of (C.2.2) is obtained by freely adjoining the comparison isomorphisms and imposing the group relations. Its functors into any category, and their natural transformations, are precisely the isocommutative cones. Since each q_s is surjective, one can eliminate all the Q_s generators. The remaining relations are $r_s(\ker q_s) = 1$. The result is the one-object groupoid $\overline{G}$. Equation (C.2.3) proves the full topos universal property, not merely its restriction to Galois targets.

Here is a direct proof which also treats profinite groups. A cone into $\mathcal{T}$ consists of inverse image functors into continuous G-sets and continuous Q_s-sets, identified as H_s-set-valued functors. After using that identification on the underlying point functors, the G-action on every value kills $r_s(\ker q_s)$. The kernel of an action on a discrete set is closed, being the intersection of its open point stabilizers. It therefore kills their closed normal closure. The action factors continuously through $\overline{G}$. Conversely an action of $\overline{G}$ restricts along H_s through Q_s; surjectivity and the quotient topology of q_s give precisely the required continuous Q_s-action. The same argument on each natural transformation proves full faithfulness of the universal equivalence of Hom categories. Left exactness and preservation of colimits are unchanged by the factorization. This proves (C.2.2) in the stated largest two-category. □

C.3 (Refuted, with replacement): arbitrary profinite pushouts and higher shapes. *Manuscript: LM §16, folio 82–84, scans 166–169.* Arbitrary pushouts of profinite groupoids do not become pushouts in all ordinary topoi under $G \mapsto \beta_{\mathrm{cont}}G$. Even locally connected target topoi detect the failure. The correct general statement is the pushout in the two-category of profinite Galois topoi and continuous morphisms. For the surjective right legs of the manuscript, C.2 supplies the stronger result. None of these ordinary two-pushouts is in general the homotopy pushout of the full shapes.

Proof. Take the span $C_2 \leftarrow 1 \to C_2$. Its profinite pushout is $P = C_2 \hat{*} C_2$. Its discrete pushout is the infinite dihedral group $D_\infty = C_2 * C_2 = \langle a, b \mid a^2 = b^2 = 1\rangle$. The two inclusions give a cone of topoi into βD_∞. If $\beta_{\mathrm{cont}}P$ had the asserted ordinary universal property, this cone would induce a geometric morphism to βD_∞. Pull the regular D_∞-torsor back and choose a point of its underlying torsor. The resulting action is a continuous homomorphism $P \to D_\infty$ with discrete codomain. Its image is finite by compactness, but the cone requires its image to contain a and

b, which generate the infinite group D_∞. Contradiction. The target βD_∞ is locally connected. The normal form $(ab)^n$ has length $2n$ for every positive n; the supplied program checks the first ten cases, while this reduced-word argument proves infinitude. The example does not have a surjective right leg and therefore does not contradict C.2.

Within profinite Galois topoi, continuous functors of their fundamental profinite groupoids give exactly the corresponding universal property; the free profinite product is characterized by continuous homomorphisms to profinite groups. Finally, in discrete topoi $\mathrm{Set}\amalg_{\beta\mathbb{Z}}\mathrm{Set} \simeq \mathrm{Set}$ by C.2, whereas the homotopy pushout of their classifying spaces is $* \amalg^h_{S^1} * \simeq S^2$. Its second cohomology is nonzero. Thus a two-pushout of covering topoi cannot be substituted for a homotopy pushout of full spaces or shapes. The finite simplicial calculation of the latter cohomology is included in the program. □

C.4 (Proved): transitivity of partial filling. *Manuscript: LM §16, folio 84, scans 169–170.* If $S = S_0 \amalg S_1$, fill first S_0 and then the images of the loops at S_1. The resulting group and topos are canonically those obtained by filling S at once. The notation on scan 169 must read $U \cup S_0 = X \setminus S_1$.

Proof. Let N_i be the normal closure of the loops in S_i, closed in the profinite case. The iterated quotient is

$$(G/N_0)/\overline{\langle\langle \mathrm{im}(N_1)\rangle\rangle} \simeq G/\overline{\langle\langle N_0, N_1\rangle\rangle}.$$

Both quotients represent exactly the homomorphisms from G killing both families. The same universal description includes conjugators and hence gives the canonical bicategorical equivalence, with its commutation cells, for arbitrary groupoids. In geometry S_0 has been added to U, so the remaining punctures are S_1. This proves both transitivity on scan 170 and the indicated correction of its neighboring subscript. □

3.2 Acyclicity and equivariant quotients

C.5 (Proved): exact acyclicity statements. *Manuscript: LM §16, folio 83, scans 167.* The map $\mathcal{B}_D \to \coprod_s \beta Q_s$ is cohomologically acyclic for all abelian coefficients pulled back from the target; in fact $R\Gamma(\mathcal{B}_D, F) = \prod_s R\Gamma(Q_s, N_s)$ for an arbitrary abelian triple $F = (M_s, N_s, a_s)$. The map $\mathcal{B}_{X;S} \to \beta\pi_1(X)$ need not be acyclic. For a smooth geometrically connected curve and torsion locally constant coefficients from the target, it is acyclic if X is affine or has positive proper genus. For proper geometric genus zero the complete list, with $r = |S_{\bar{k}}|$, is:

case	acyclicity for every torsion local system
$r = 0$	yes, in every characteristic
$\mathrm{char}\, k = 0,\ r = 1$	yes
$\mathrm{char}\, k = 0,\ r \geq 2$	no; first geometric failure H^2
$\mathrm{char}\, k = p > 0,\ r \geq 1$	yes for p-primary; no for prime-to-p.

(C.5.1)

In the failing cases the exact arithmetic obstruction for a coefficient module M is

$$R\Gamma(k, M^{(p')}(-1))[-2],$$

with $M^{(p')} = M$ in characteristic zero. Thus vanishing of that complex is the exact coefficientwise arithmetic criterion. Acyclicity of the mixed-to-filled map does not imply that the curve-to-mixed map is acyclic.

Proof. For a local triple the global section is determined by its Q_s-invariant closed section: its generic section must be its image under a_s. The closed evaluation has the exact left adjoint $N \mapsto (q_s^* N, N, \mathrm{id})$ on abelian objects. It therefore preserves injectives. Deriving the global section

formula proves the first assertion. Equivalently the first and third terms of B.3's fibre diagram are identified by the identity of H_s and cancel.

When X is affine or proper of positive genus, B.5 computes its local-system cohomology by $\pi_1(X)$, while B.6–B.7 compare it with the mixed topos. This proves the positive assertion. If $r = 0$ the mixed and filled topoi are identical by definition. If $r = 1$ in characteristic zero, the mixed geometric topos is the arrow topos of B.4; its constant cohomology is that of a point. If $r \geq 2$ in characteristic zero, or $r \geq 1$ in positive characteristic, B.6 makes the mixed comparison effective, so its cohomology is that of $\mathbb{P}^1$. Its prime-to-p degree-two group is $M^{(p')}(-1)$ and its p-primary positive cohomology is zero. This gives the table. The two-stage geometric cohomology complex and Hochschild–Serre give exactly the stated arithmetic cone, including possible nontrivial transgression. The one-puncture characteristic-zero case now exhibits the final distinction: its mixed-to-filled map is acyclic, while its curve-to-mixed map misses H^2. □

C.6 (Proved): actions, Galois coverings, and the quotient square. *Manuscript: LM §16, folio 85–86, scans 172–174.* For a discrete group Γ, coherent actions on a Grothendieck topos $\mathcal{E}$ are equivalent, as a two-category of data, to a topos $\mathcal{X}$ with a Γ-torsor V and an equivalence $\mathcal{E} \simeq \mathcal{X}_{/V}$. One may take $\mathcal{X} = \mathcal{E}^\Gamma$, the topos of equivariant objects. There is a structure map $\mathcal{E}^\Gamma \to \beta\Gamma$, and an equivariant morphism $\mathcal{E} \to \mathcal{F}$ is equivalently a morphism of these quotients over $\beta\Gamma$. Its square with the torsor covers is two-cartesian. If $\mathcal{E} = \beta C$ is multi-Galois, the quotient is $\beta(C \rtimes \Gamma)$ for the Grothendieck construction of the coherent action. It is connected exactly when Γ is transitive on the nonempty set $\pi_0 C$.

Proof. The underlying-object functor $\mathcal{E}^\Gamma \to \mathcal{E}$ preserves finite limits and all colimits. Equivariant objects form a Grothendieck topos, for example by equivariant sheaves on the disjoint union of the translates of a small generating site. The image of the regular torsor of $\beta\Gamma$ is $V = \coprod_{\gamma \in \Gamma} 1$ with its translation action. An equivariant object over V is recovered from its fibre over the identity and the translated copies of that fibre. This gives $\mathcal{E}^\Gamma_{/V} \simeq \mathcal{E}$. Conversely the effective descent of objects along the epimorphism $V \to 1$ identifies $\mathcal{X}$ with the equivariant objects of $\mathcal{X}_{/V}$; the torsor equation $V \times V \simeq \Gamma \times V$ identifies the descent cocycle with the action law. The same descent applies to inverse image functors and their two-cells. Pulling back the same universal torsor gives the two-cartesian square of scan 173.

For a groupoid action, an equivariant presheaf on C is exactly a presheaf on its Grothendieck construction: arrows are a group element followed by an arrow in C, with the coherence of the action in the composition formula. All these arrows are invertible. Its connected components are the Γ-orbits of $\pi_0 C$, proving the final assertions. □

C.7 (Refuted literally, corrected for profinite operators). *Manuscript: LM §16, folio 85, scans 172.* For infinite profinite Γ, the regular torsor of C.6 is not an object of $\beta_{\mathrm{cont}}\Gamma$. The precise arithmetic analogue uses continuous actions and a *protorsor*, or a torsor for the localic group Γ. Precisely, in the connected profinite Galois setting a continuous coherent action is encoded by an exact sequence of profinite groups

$$1 \longrightarrow G \longrightarrow E \longrightarrow \Gamma \longrightarrow 1.$$

Its quotient is $\beta_{\mathrm{cont}}E \to \beta_{\mathrm{cont}}\Gamma$; base change along the canonical point $\mathrm{Set} \to \beta_{\mathrm{cont}}\Gamma$ is $\beta_{\mathrm{cont}}G$. Changing base points replaces this sequence by the equivalent profinite groupoid extension. This definition, componentwise with a continuous Γ-set of components, is the continuous action meant in the arithmetic applications. It has finite-quotient descent, but is not an ordinary slice by a discrete regular Γ-set. No assertion about an undefined abstract action of the underlying discrete group on an arbitrary topos is implicit here.

Proof. The stabilizer of a point in the regular action is 1, which is not open in an infinite profinite group. Thus the regular set is not continuous. More decisively, the canonical point $\mathrm{Set} \to \beta_{\mathrm{cont}}\Gamma$ cannot be a slice by an object giving the regular cover: every nonempty transitive object has open stabilizer, whereas this point has trivial isotropy. For the corrected statement, an E-set restricts to a G-set together with transport over Γ, and the multiplication in E is exactly its coherent action law. Conversely that transport and its coherence recover the continuous E-action. This proves the quotient assertion by descent. For a precise base-change calculation, write $p : E \to \Gamma$. For every open normal subgroup N of E there is an exact sequence of finite groups

$$1 \to G/(G \cap N) \to E/N \to \Gamma/p(N) \to 1.$$

The finite torsor calculation in C.6 identifies $\beta(E/N) \times_{\beta(\Gamma/p(N))} \mathrm{Set}$ with $\beta(G/(G \cap N))$. The subgroups $p(N)$ are cofinal among the open normal subgroups of Γ: for V open normal, choose $N \subset p^{-1}(V)$. Similarly $G \cap N$ is cofinal among open normal subgroups of G. Take coherent inverse limits of these finite action topoi, using (Artin et al., 1972b, VI, §8). Bicategorical limits commute with this finite pseudopullback by their universal property, giving $\beta_{\mathrm{cont}}E \times_{\beta_{\mathrm{cont}}\Gamma} \mathrm{Set} \simeq \beta_{\mathrm{cont}}G$. The pro-cover is represented by the compatible torsors $\Gamma/p(N)$, not by the regular set Γ. For components, each orbit of a continuous action of the compact group Γ on a discrete set is finite; apply the same calculation to its finite groupoid and then take the disjoint sum over the orbits. Equivalently a localic torsor has the regular compact group as fibre. This is the qualification needed for the parenthesis "profini" on scan 172. □

C.8 (Proved): the component factor and quotient compatibility. *Manuscript: LM §16, folio 87–88, scans 175–178.* For $I = \pi_0 C$ there is the canonical factorization

$$\beta(C \rtimes \Gamma) \longrightarrow \beta(I \rtimes \Gamma) \longrightarrow \beta\Gamma, \qquad \beta(I \rtimes \Gamma) \simeq (\beta\Gamma)_{/I}. \tag{C.8.1}$$

The first functor kills the fibre isotropy; the second is faithful on groupoid arrows. Over an orbit represented by i, the intermediate topos is $\beta\Gamma_i$. Equivariant quotient commutes with both the mixed Artin gluing and the hole-filling pushout, with their respective lax and invertible commutation cells retained. For profinite operators use continuous I and the interpretation of C.7.

Proof. The action groupoid $I \rtimes \Gamma$ has arrows $\gamma : i \to \gamma i$. An equivariant set over I is precisely its presheaf, giving the slice equivalence. Its arrows map injectively to Γ. Its isotropy at i is Γ_i, and its components are the orbits. The quotient functor $C \rtimes \Gamma \to I \rtimes \Gamma$ is identity on the operator and forgets the internal arrow in the fibre. This is exactly the factorization of scans 175–177. In the profinite case continuity of I makes each stabilizer open, as required for the slice assertion.

An equivariant mixed object is an equivariant M, an equivariant family N_s, and an equivariant map a_s. These are exactly the objects obtained by gluing the quotient open and closed topoi with the induced gluing functor. The equivalence on maps is the same componentwise description; it preserves finite limits and all colimits. For the filled diagram, a functor from the quotient pushout to any topos is a compatible cone from all the translates, with its operator cocycle. Imposing the pushout relations and imposing that cocycle commute: both describe precisely the same objects and natural transformations. Equivalently, two bicategorical colimits can be taken in either order by their universal property. This proves the assertion in the slice over $\beta\Gamma$ and, for discrete operators, in ordinary topoi. Continuous descent gives the profinite Galois version. For the surjective hole-filling maps C.2's closed-normal-closure proof also applies directly to the quotient extension groups. This proves the claim of scan 178 with its two-category and continuity hypotheses explicit. □

3.3 Exterior homomorphisms and their transporter groupoids

C.9 (Refuted, with replacement): the precise domain of exterior functoriality. *Manuscript: LM §12, folio 50–51, scans 103–106.* For discrete groups, the Hom groupoid of geometric morphisms $\beta G' \to \beta G$ has objects homomorphisms $u : G' \to G$ and arrows $g : u \to v$ satisfying $v = c_g u$. Its automorphism group at u is $C_G(u(G'))$. The same formula holds for continuous homomorphisms of profinite groups. Hence an exterior homomorphism with trivial centralizer specifies a geometric morphism up to unique isomorphism, and inner changes of a centre-free group model have unique coherent identifications. However such homomorphisms do not, without a further restriction, form a category closed under composition.

Proof. Pulling the regular torsor back in the discrete case gives a G-torsor with commuting G'-action. Choosing one element identifies it with G and identifies that action with u. Changing the element conjugates u; equivariant maps of torsors give exactly the displayed arrows. The finite torsors give the same proof in the profinite case. The stabilizer formula follows immediately. An orbit groupoid with trivial stabilizers is contractible, so the exterior class can be used without a distinguished representative and all its comparison cells are forced. This proves the valid assertion on scans 103–105.

For the compositional qualification take $N = S_3^3$, $H = N \rtimes S_3$ with S_3 permuting the three factors, and $K = S_3$. The centralizer of N in H is trivial: a nontrivial permutation changes a factor, and an element of N centralizing N lies in its trivial centre. The projection $H \to K$ has centralizer $Z(K) = 1$. But the composite $N \to H \to K$ is trivial and has centralizer K. All three groups are centre-free. The exhaustive finite calculation in the program verifies orders 216, 1296, and 6, and the three centralizer sizes $1, 1, 6$.

The strongest categorical replacement is therefore either the full bicategory of homomorphisms and conjugators, which always composes, or an explicitly composition-closed subcategory in which every composite still has trivial centralizer. Surjections between centre-free groups give one such subcategory. On it the contractible representative groupoids descend the classifying-topos pseudofunctor to exterior data with unique coherence. A graph of admissible arrows alone does not justify that conclusion for arbitrary composites. □

C.10 (Proved): the transporter construction and the square on scan 107. *Manuscript: LM §12, folio 51–52, scans 106–108.* Let Λ be the conjugacy class of an injective $\kappa : T \to G$ satisfying $C_G(\kappa(T)) = \kappa(T)$. Define Π_Λ to have the homomorphisms in Λ as objects, and

$$\mathrm{Hom}(\kappa_1, \kappa_2) = \{g \in G : \kappa_2 = c_g \kappa_1\}. \tag{C.10.1}$$

It is connected, its automorphism groups are canonically the indicated copies of T, and its forgetful functor $\Pi_\Lambda \to BG$ sends an arrow to g. This constructs $B_\Lambda \to \beta G$ without choosing a base loop.

For the convention actually written on scan 107,

$$\Lambda = u\Lambda' \circ d, \tag{C.10.2}$$

and the self-centralizing hypotheses there, the functor is

$$\kappa' \longmapsto u\kappa' d, \qquad g' \longmapsto u(g'). \tag{C.10.3}$$

It gives the square of classifying topoi together with its commutation datum. The reversed arrow in the sentence on scan 108 and the unprimed κ in the preceding formula are type errors; the drawn square has the direction in (C.10.3).

Proof. Composition in (C.10.1) is group multiplication, so associativity and inverses are immediate. The stabilizer is the stipulated centralizer. Since an abelian group is its own centralizer inside itself, the hypothesis also implies T is abelian. The forgetful functor is faithful,

although in the profinite case it is a slice induction only when the relevant subgroup is open, as Part A explains.

If $\kappa'_2 = c_{g'}\kappa'_1$, then $u\kappa'_2 d = c_{u(g')}u\kappa'_1 d$, which proves (C.10.3) and its strict commutation with the one-object groupoid maps for a fixed u. If $v = c_h u$, the components h give a natural isomorphism between these diagrams. The condition $C_G(u(G')) = 1$ makes h unique. For three representatives these isomorphisms compose, since their conjugators compose and are unique. Thus descent over the exterior class gives the stated square and its two-cell. Taking disjoint unions over the labels i' gives the family version, with the correctly typed formula $\Lambda_{\tau(i')} = u\Lambda'_{i'}d_{i'}$. This proves the construction rather than treating an exterior homomorphism as a chosen actual homomorphism. □

C.11 (Refuted, with replacement): degrees and uniqueness in scans 107–109. *Manuscript: LM §12, folio 52–53, scans 107–109.* For any abelian group T, the injectivity and self-centralizing hypotheses in (C.10.2) force the integer scalar $d : T \to T$ to be an automorphism. For $T = \mathbb{Z}$ or $\widehat{\mathbb{Z}}$ this forces $d = \pm 1$. The induced map on the identified inertia groups is d^{-1}, not d. In the prime-to-p version an integer d may instead be a unit of $\widehat{\mathbb{Z}}^{(p')}$. The usual geometric convention for a possibly ramified degree is $u\Lambda' = \Lambda_{\tau(i')}d_{i'}$; then the local maps and their commutation cells are included as data, as in B.12.

The two uniqueness assertions in the margin are false without a condition on normalizers and on the indexing family. The exact replacement is that, for each given exterior composite, the set of pairs (i, d) satisfying the displayed exterior equation must have one element. A sufficient intrinsic condition is that the indexing family's allowed scalar orbits are disjoint and each scalar orbit map is injective. Fixing an orientation excludes inversion only when the normalizer respects that orientation.

Proof. Write $a = u\kappa'$ and $\kappa = ad$. Every element of $a(T)$ commutes with $\kappa(T)$, so lies in $\kappa(T)$. Thus $a = \kappa e$ for an endomorphism $e : T \to T$. Since κ is injective, $\kappa = \kappa ed$ gives $ed = 1$. Integer scalars commute with every endomorphism of an abelian group, so also $de = 1$; this proves the assertion for every abelian T (continuously as well when κ is a topological embedding). For the stated procyclic groups their endomorphism rings are $\mathbb{Z}$ or the corresponding completion. Therefore d is a unit and $e = d^{-1}$. Formula (C.10.3) on automorphisms is exactly e. This shows why a nonunit ramification degree cannot be inserted into (C.10.2) under its written hypotheses.

For degree nonuniqueness let $G = D_\infty = \langle r, s \mid s^2 = 1,\ srs^{-1} = r^{-1}\rangle$, $T = \mathbb{Z}$, and $\kappa(1) = r$. The centre of G is trivial and $C_G(\kappa(T)) = \langle r\rangle$. With $u = \mathrm{id}$, the exterior class of κ equals that of $\kappa \circ (-1)$, so $d = 1$ and $d = -1$ both satisfy the same equation. The action of integers on T is faithful, as demanded on scan 109. For label nonuniqueness take $G = F(a, b)$ and the two distinct conjugacy classes $1 \mapsto a$ and $1 \mapsto a^{-1}$. They are self-centralizing and not conjugate, by cyclically reduced word normal form. The same source loop and $u = \mathrm{id}$ satisfy the equation with the first label and $d = 1$, or the second label and $d = -1$. Thus an injective label map alone does not determine τ. Finally the asserted exact replacement is simply the fibre, over $[u\Lambda']$, of the map $(i, d) \mapsto \Lambda_i d^{-1}$ for (C.10.2), or of $(i, d) \mapsto \Lambda_i d$ for the geometric convention: disjointness and injectivity are precisely what make that fibre a singleton. This proves the replacement and both counterexamples completely. □

C.12 (Proved): the operator construction declined on scan 109. *Manuscript: LM §12, folio 51–53, scans 105–109.* Let Σ act externally on a centre-free group G, on the loop labels I, and on T by $\chi : \Sigma \to \mathrm{Aut}(T)$, with $\sigma\Lambda_i = \Lambda_{\sigma i}\chi_\sigma$ as exterior data. Define

$$E = \mathrm{Aut}(G) \times_{\mathrm{Out}(G)} \Sigma. \tag{C.12.1}$$

Then $1 \to G \to E \to \Sigma \to 1$ is the canonical extension. There is a groupoid over $B\Sigma$ whose objects are (i, κ_i) and whose arrows are $e \in E$ over σ satisfying

$$j = \sigma i, \qquad \kappa_j = e\kappa_i \chi_\sigma^{-1}. \tag{C.12.2}$$

Here e acts through its automorphism of G. Its fibre is $\coprod_i \Pi_{\Lambda_i}$; forgetting the loop gives a functor to BE over $B\Sigma$. Its classifying topoi are the equivariant square required in §12. Compatible exterior maps with trivial centralizer, and the loop and degree data of C.10, induce maps of these squares with all coherence cells. Composition is understood in C.9's precise sense.

Proof. Since G has trivial centre, $\mathrm{Inn}(G) = G$, so the kernel of (C.12.1) is exactly G and surjectivity follows from the definition of $\mathrm{Out}(G)$. The formula (C.12.2) is stable under composition: for e over σ and e' over τ, the resulting formula uses $e'e$ and $\chi_{\tau\sigma} = \chi_\tau \chi_\sigma$. It has identities and inverses. External compatibility of the loops supplies an arrow over each allowed σ, after adjusting its lift by an inner automorphism. The fibre over the identity is exactly the transporter groupoid. The operator action reconstructed from this fibration is coherent by its associative composition; no arbitrary lift of each operator is needed.

For clarity, we verify also functoriality of the extension. Let $u : G' \to G$ have trivial centralizer and commute with the external Σ-actions. For $e' \in E'$ over σ, there is an $e \in E$ over σ such that

$$e \circ u = u \circ e'.$$

External compatibility gives existence after an inner adjustment. Two choices differ by an element centralizing $u(G')$, so coincide. Uniqueness then proves that $e' \mapsto e$ preserves multiplication and restricts to u on the kernels. Send (i', κ') to $(\tau(i'), u\kappa' d_{i'})$ and an arrow e' to e. Equivariance of τ, invariance of the degree function, and commutation with the T-action give (C.12.2) for the images. All comparison cells are the unique conjugators already constructed in C.10. This proves the operator version that the manuscript postpones. For profinite groups take continuous automorphisms, continuous extensions and finite-quotient descent; an external action is required to be continuous in precisely that extension sense. An abstract homomorphism into an untopologized outer automorphism group would not suffice. □

4 D. Spheres, their two-type, and the arithmetic quotient

We use $A = \mathrm{Homeo}(S^2)$ with the compact–open topology, A^+ for its identity component, and $M = \mathrm{PGL}_2(\mathbb{C}) \rtimes C_2$ for the conformal and anticonformal subgroup. These are the manuscript's A_0, A_0° and $\widetilde{\mathrm{GP}}(1, \mathbb{C})$, respectively (§20, fols.116v–125v, scans233–250). We write $P_n Y = \tau_{\le n} Y$ and use homotopy fibres and derived mapping spaces throughout. A sphere bundle means a numerable locally trivial bundle with fibre S^2 and structure group A. On a CW complex every locally trivial sphere bundle is numerable. All cohomology in the topological part is singular cohomology. In the arithmetic part all stacks are on the fppf site; they may be restricted to the etale site when 2 is invertible.

4.1 The covering extension and the two-isotopic category

D.1 (Known). The topological input, with reduction. *Manuscript: 20; 117,118; 234,236.*

Manuscript problem locus: §20, fols.117 and 118, scans234 and 236, (64)–(66). The inclusions $O(3) \longrightarrow M \longrightarrow A$ are weak homotopy equivalences. Consequently $BA \simeq BO(3)$, $\pi_0 A = C_2$, $\pi_1 A^+ = C_2$, $\pi_2 A^+ = 0$, $\pi_3 A^+ = \mathbb{Z}$, and $\pi_4 A^+ = C_2$.

Proof. Dobbins's Theorem 1.1 gives a strong deformation retraction of A onto $O(3)$ (Dobbins, 2021). Polar decomposition retracts $\mathrm{SL}_2(\mathbb{C})$ onto $SU(2)$, equivariantly for complex conjugation and the central subgroup $\{\pm 1\}$. It therefore retracts M onto $PU(2) \rtimes C_2 = O(3)$, compatibly with the displayed inclusions. This proves the two asserted weak equivalences by two-out-of-three.

Classifying spaces of these well-pointed topological groups preserve weak equivalences: the bar construction is levelwise a product of the given equivalence, and its fat realization preserves these equivalences. Finally $SU(2) = S^3 \longrightarrow SO(3)$ is a two-sheeted universal cover. Thus the higher homotopy groups of A^+ are those of S^3, and the asserted values follow from $\pi_2 S^3 = 0$, $\pi_3 S^3 = \mathbb{Z}$, and $\pi_4 S^3 = C_2$ (Hatcher, Corollary 4J.4 (Hatcher, 2002)). This is precisely the reduction needed for scans234 and236; no statement about the discrete group A is being made. The local contractibility needed for its covering-space theory follows from Edwards–Kirby, Corollary 1.1 (Edwards and Kirby, 1971). □

D.2 (Proved). The extension of (62), including its exact uniqueness. *Manuscript: 20; 116v-117; 233-234.*

Manuscript problem locus: §20, fols.116v–117, scans233–234, (62)–(64). There is a central covering extension

$$1 \longrightarrow \mu_2 \longrightarrow \widetilde{A} \longrightarrow A \longrightarrow 1$$

whose restriction to M is the specified extension $\mathrm{SL}_2(\mathbb{C}) \rtimes C_2 \longrightarrow \mathrm{PGL}_2(\mathbb{C}) \rtimes C_2$, with the generator of C_2 acting by complex conjugation. The category of such extensions equipped with this identification on M is contractible. Without that identification, "unique up to unique isomorphism" is false: even the chosen extension has a nonidentity automorphism over A, and there are two isomorphism classes of central double covers restricting to the universal cover on A^+.

Proof. Fix the reflection r given by complex conjugation on the Riemann sphere. The orientation character splits $A = A^+ \rtimes \langle r \rangle$. Form the universal covering group $\widetilde{A}^+$ from paths starting at the identity, modulo endpoint-preserving homotopy; multiply paths pointwise. Conjugation by r lifts uniquely to an automorphism $\widetilde{r}$ of this group fixing its identity, and $\widetilde{r}^2 = 1$ by uniqueness. Put $\widetilde{A} = \widetilde{A}^+ \rtimes C_2$. Its kernel is $\pi_1 A^+ = \mu_2$ and is central, since $\mathrm{Aut}(C_2) = 1$. Restriction to the connected subgroup $\mathrm{PGL}_2(\mathbb{C})$ is its universal covering group $\mathrm{SL}_2(\mathbb{C})$ by D.1, and the lifted reflection acts by conjugation on matrix entries. This is exactly (63), not an arbitrary central cover.

For uniqueness with the restriction fixed, an isomorphism on the identity components is forced by universal covering theory. Its value on a lift of r is forced by the prescribed identification on M, and these two requirements determine it on the whole group. They also verify its multiplicativity. For the unrigidified assertion, every automorphism over A has the form $x \mapsto \epsilon(\bar{x})x$ with a continuous character $\epsilon : A \to \mu_2$; the nontrivial orientation character provides a nonidentity automorphism. More generally, after fixing the universal cover of A^+, the square of a chosen lift of r is either $+1$ or -1. Both choices give groups, with multiplication obtained by adding the central cocycle $\epsilon e f$ on C_2. These two groups are not isomorphic over A, since that square cannot change when the lift is multiplied by -1. Their restrictions to $O(3)$ are the two Pin covers. The manuscript's semilinear restriction chooses the reflection-square-$+1$ cover, namely $\mathrm{Pin}^+(3)$. The finite cocycle calculation is reproduced in `code/spheres/spheres_checks.py`. □

D.3 (Proved). The pair $(\omega, \mathcal{R})$ and the missing relation. *Manuscript: 20; 117v-119v; 235-238.*

Manuscript problem locus: §20, fols.117v–119v, scans235–238; the explicit doubt is on fol.119, scan237. The two-groupoid with spheres as objects, homeomorphisms as one-arrows, and endpoint-preserving homotopy classes of isotopies as two-arrows is equivalent to the two-groupoid of pairs consisting of a C_2-torsor ω and a banded μ_2-gerbe $\mathcal{R}$. No additional relation between the two entries is required. The equivalence uses the extension fixed in D.2. On bundles its two characteristic classes are (w_1, w_2); changing to the other covering extension replaces the second class by $w_2 + w_1^2$.

Proof. For a sphere X, let $P_X = \mathrm{Homeo}(S^2, X)$, an A-torsor. Its associated C_2-torsor is the set of its orientations. Let $\mathcal{R}_X$ be the groupoid of $\widetilde{A}$-torsors lifting P_X, with the identification

after extension of structure group included as data. This groupoid is nonempty and connected and has automorphism group μ_2 at every object. Homeomorphisms transport these two objects. An isotopy lifts through the covering in D.2 and gives a natural transformation of lift functors; endpoint-preserving homotopies give the same transformation. Thus we have the stated two-functor.

Its map on classifying two-types is

$$P_2BA \longrightarrow BC_2 \times B^2\mu_2.$$

It induces the orientation isomorphism on π_1 and the covering-boundary isomorphism on π_2; the latter is also the boundary map $\pi_2 BA \to \pi_1 B\mu_2$ of the lifting fibration. Both sides are connected two-types, so it is an equivalence. This proves both essential surjectivity and equivalence on every Hom groupoid, not just equality of object counts. Equivalently, the action of C_2 on μ_2 is trivial and the Postnikov–Sinh class in $H^3(C_2, \mu_2)$ is zero. The vanishing follows also from the splitting by the reflection. The restriction of the selected cover to $O(3)$ is Pin^+, whose obstruction is w_2; the second cover differs by the pullback of the nonzero class in $H^2(BC_2; \mu_2)$, namely w_1^2. This last assertion can be checked on the reflection subgroup and on $SO(3)$, which together determine the two extension classes. It answers the explicit worry on scan237. □

4.2 Bundles through dimension five

D.4 (Proved). Classification through dimension three, by both routes. *Manuscript: 20; 120-122; 239-243.*

Manuscript problem locus: §20, fols.120–122, scans239–243, (67)–(71). For every CW complex S of dimension at most three, the map in (70)–(71) is a bijection

$$[S, BA] \xrightarrow{(w_1, w_2)} H^1(S; C_2) \times H^2(S; \mu_2).$$

There is no orientability hypothesis.

Proof by the Gr-stack. D.3 identifies the classifying space of the fundamental categorical group of A with $Q = BC_2 \times B^2\mu_2$. The fibre of $q : BA \to Q$ is three-connected: q is an isomorphism on π_i for $i \le 3$, and $\pi_4 Q = 0$. To lift a map $S \to Q$, one extends a section of its pulled-back fibration over the cells of S. The obstruction on an n-cell lies in π_{n-1} of the fibre, which is zero for $n \le 4$. Uniqueness up to homotopy uses the relative CW complex $(S \times I, S \times \partial I)$, of dimension at most four, and the same vanishing. Thus the lifting map on components is bijective. This supplies the surjectivity argument left unfinished on scans240–241, and also completes the twisting argument: it works over every chosen ω, since the fibre remains three-connected after pullback.

Direct proof. The group isomorphism $O(3) \cong C_2 \times SO(3)$ sends an orthogonal matrix g to $(\det g, (\det g)g)$. Thus a rank-three real bundle E is determined by its determinant line L and the oriented bundle $V = E \otimes L$. The classifying space $BSO(3)$ has, through degree three, the homotopy groups of $K(C_2, 2)$. Obstruction theory therefore classifies V over S by $w_2(V)$. The splitting principle gives

$$w_1(E) = a = w_1(L), \qquad w_2(E) = w_2(V) + a^2.$$

Hence every desired pair (a, b) is obtained by choosing L with class a and V with class $b + a^2$, and it determines that pair uniquely. D.1 carries this classification back to topological sphere bundles. □

D.5 (Proved). The first failure is dimension four. *Manuscript: 20; 118,122; 236,243.*

Manuscript problem locus: §20, fols.118 and 122, scans236 and 243, (66) and (71). The pair $(\omega, \mathcal{R})$ first fails to distinguish bundle isomorphism classes on four-dimensional CW complexes. Over S^4, the trivial bundle and the sphere bundle associated to the adjoint of the quaternionic

Hopf $SU(2)$-bundle have identical $(w_1, w_2) = (0, 0)$ and are not isomorphic. If c_2 of the Hopf bundle is 1, their first Pontryagin classes are respectively 0 and -4.

Proof. Both low classes vanish because $H^1(S^4; C_2) = H^2(S^4; C_2) = 0$. The clutching map of the Hopf bundle is the identity of $SU(2) = S^3$; its image in $SO(3)$ generates $\pi_3 SO(3) = \mathbb{Z}$, since a covering induces an isomorphism on π_3. Hence its associated oriented three-plane bundle is nontrivial. With formal Chern roots $x, -x$, the adjoint complexification has roots $2x, 0, -2x$, giving $p_1 = -4c_2$. The identity is verified symbolically in the attached program. D.4 excludes any lower-dimensional counterexample. □

D.6 (Proved). The fourth stage, with its necessary homotopy datum. *Manuscript: 20; 118; 236.*

Manuscript problem locus: §20, fol.118, scan236, (66); this statement develops its higher-homotopy question. Let $\mathfrak{P} : H^2(-; C_2) \to H^4(-; \mathbb{Z}/4)$ be the Pontryagin square and β_4 the Bockstein for $0 \to \mathbb{Z} \xrightarrow{4} \mathbb{Z} \to \mathbb{Z}/4 \to 0$. There is an equivalence

$$P_4 BSO(3) \simeq P := K(C_2, 2) \underset{K(\mathbb{Z}/4,4)}{\overset{h}{\times}} K(\mathbb{Z}, 4),$$

where the two maps are $\mathfrak{P}$ and reduction modulo four. In the convention $p_1 = -c_2(E \otimes \mathbb{C})$, the next integral Postnikov invariant is $\beta_4\mathfrak{P} \in H^5(K(C_2, 2); \mathbb{Z})$, up to the sign chosen for the generator of π_4. The equivalent characteristic relation is $\rho_4 p_1 = \mathfrak{P}(w_2)$.

Proof. The universal relation holds for an oriented three-plane bundle. One verification uses the universal oriented two-plane subbundle after pullback to $BSO(2)$: there $p_1 = c_1^2$ and $w_2 = \rho_2 c_1$, so the formula is the defining property of the Pontryagin square. In degree four restriction $H^4(BSO(3); \mathbb{Z}/4) \to H^4(BSO(2); \mathbb{Z}/4)$ is injective; this follows from the Gysin sequence of the universal sphere bundle $S^2 \to BSO(2) \to BSO(3)$, since its possible kernel is the image of $H^1(BSO(3); \mathbb{Z}/4) = 0$. Choose a nullhomotopy of that universal relation and obtain $BSO(3) \to P$. The long exact sequence of the defining homotopy pullback shows that P has $\pi_2 = C_2$, $\pi_4 = 4\mathbb{Z} \cong \mathbb{Z}$, and no other positive homotopy groups. The map is the identity on π_2. Its map on π_4 is an isomorphism by the calculation $p_1 = -4c_2$ in D.5. Hence it is precisely the fourth Postnikov projection. Replacing the pullback by the equivalent fibre of $\beta_4\mathfrak{P} : K(C_2, 2) \to K(\mathbb{Z}, 5)$ gives the invariant. This is the homotopy-fibre calculation used by Antieau–Williams (Antieau and Williams, 2013), with the sign of the integral coordinate fixed by the displayed Chern-class convention.

A map into P consists of a map $x : S \to K(C_2, 2)$, a map $p : S \to K(\mathbb{Z}, 4)$, and a *chosen homotopy* $\mathfrak{P}(x) \simeq \rho_4 p$, all modulo coherent homotopy. The existence of the homotopy is equivalent to equality in cohomology, but forgetting it can lose bundle classes. In particular, it is not legitimate to replace $[S, P]$ by the set of pairs satisfying the cohomological relation on an arbitrary CW complex. □

D.7 (Refuted). Characteristic-pair classification, with a complete replacement. *Manuscript: 20; 118; 236.*

Manuscript problem locus: §20, fol.118, scan236, (66); the refuted uniqueness assertion is the later one in Antieau–Williams, Definition 2. Let $S = M(\mathbb{Z}/4, 3) = S^3 \cup_4 e^4$. There are four oriented three-plane bundles on S, all with $w_2 = p_1 = 0$. The correct invariant is the homotopy-pullback datum of D.6. This refutes the unqualified uniqueness assertion for a map $S \to P_4 BSO(3)$ from a characteristic pair, including the assertion in Definition 2 of the opened arXiv version of Antieau–Williams.

Proof. Cellular cohomology gives $H^1(S; C_2) = H^2(S; C_2) = 0$ and $H^4(S; \mathbb{Z}) = \mathbb{Z}/4$. Consequently every $SO(3)$-bundle has a spin lift, and its spin lift is unique up to isomorphism: the usual ambiguity is the action of $H^1(S; C_2)$, which is zero. Since $BSU(2)$ is three-connected with $\pi_4 = \mathbb{Z}$, its bundles on a four-complex are classified by $c_2 \in H^4(S; \mathbb{Z})$. Thus there are

four. Their Pontryagin classes are all $-4c_2 = 0$, and their second Stiefel–Whitney classes vanish by construction. The cellular differential is the one-by-one matrix (4); its cokernel and the multiplication-by-four calculation are reproduced in the attached program. Dold–Whitney explicitly retain the additional ambiguity in their general four-complex theorem (Dold and Whitney, 1959); the frequently quoted characteristic-class-only version requires a torsion hypothesis. □

D.8 (Proved). The fifth stage and its degree-six invariant. *Manuscript: 20; 118; 236.*

Manuscript problem locus: §20, fol.118, scan236, (66); this is the next Postnikov calculation requested by that question. Put $x \in H^2(P; C_2)$ and $y = \mathrm{Sq}^1 x$. The fifth Postnikov stage is

$$P_5 BSO(3) \simeq \mathrm{hofib}(u : P \longrightarrow K(C_2, 6)),$$

where $u \in H^6(P; C_2)$ is specified completely by the following normalization. It restricts on the fibre $K(\mathbb{Z}, 4)$ to $\mathrm{Sq}^2 \rho_2 \iota_4$; it restricts to zero along the maps induced by the representations

$$SO(2) \hookrightarrow SO(3), \qquad C_2 \times C_2 \hookrightarrow SO(3),$$

the latter being the diagonal representation $L_a \oplus L_b \oplus L_{a+b}$. These requirements determine exactly one class. In particular, this stage is not the product $P \times K(C_2, 5)$.

Proof. We give the low-degree calculation so that the invariant is more than an unnamed obstruction. In the mod-two Serre spectral sequence of $K(\mathbb{Z}, 4) \to P \to K(C_2, 2)$, write $v = \rho_2 \iota_4$ and $z = \mathrm{Sq}^2 \mathrm{Sq}^1 x$. Serre's Theorems 2 and 3 in §2 compute the two Eilenberg–MacLane cohomology algebras used below (Serre, 1953). The integral transgression of ι_4 is $\beta_4 \mathfrak{P}(x)$, whose mod-two reduction is $z + xy$. Thus $d_5 v = z + xy$. Up to degree six the base has polynomial generators x, y, z of degrees $2, 3, 5$, while the fibre contributes v in degree four and $\mathrm{Sq}^2 v$ in degree six. The potential class xv is killed by $d_5(xv) = x(z + xy)$. The compatibility of transgression with Steenrod squares (Serre, §2, proof of Theorem 2) and the Adem relation $\mathrm{Sq}^2 \mathrm{Sq}^2 = \mathrm{Sq}^3 \mathrm{Sq}^1$ give

$$\mathrm{Sq}^2(z + xy) = x(z + xy),$$

which is already a d_5 boundary. Therefore $\mathrm{Sq}^2 v$ survives. There are exactly three independent classes in total degree six: x^3, y^2, and a class restricting to $\mathrm{Sq}^2 v$. This also proves that the kernel of restriction to the fibre has basis x^3, y^2. The same three-dimensional calculation appears as Corollary 3 of Antieau–Williams; we have included the differentials needed here.

The abstract fifth-stage invariant must pull back to zero in $H^6(BSO(3); C_2)$, while x^3, y^2 pull back to the two independent classes w_2^3, w_3^2. Its restriction to the fibre cannot vanish. Indeed, pulling the tower back along $BSU(2) \to BSO(3)$ gives the fifth stage of $BSU(2)$ over $K(\mathbb{Z}, 4)$. If this stage split, its degree-six cohomology would contain the nonzero class $\mathrm{Sq}^2 \rho_2 \iota_4$. But $BSU(2) \to P_5 BSU(2)$ is six-connected, so its map on degree-six cohomology is injective, whereas $H^6(BSU(2); C_2) = 0$ by $H^*(BSU(2); C_2) = C_2[c_2]$. Hence the restriction is the unique nonzero fibre class.

Both displayed representation maps factor through $BSO(3)$, so the actual invariant restricts to zero there. Conversely, two classes satisfying the fibre normalization differ by $\lambda x^3 + \mu y^2$. Restriction to $BSO(2) = \mathbb{C}P^\infty$ forces $\lambda = 0$, since x becomes the reduction of c_1 and y becomes zero. On $B(C_2 \times C_2)$ the formulas are

$$x = a^2 + ab + b^2, \qquad y = a^2 b + ab^2.$$

The polynomial $y^2 = a^4 b^2 + a^2 b^4$ is nonzero, so this restriction forces $\mu = 0$. Thus the normalization specifies the invariant uniquely without an arbitrary choice of a complement to $\langle x^3, y^2 \rangle$. The attached program checks the polynomial identities. The long exact sequence now gives $\pi_5 = C_2$ and proves the asserted fifth-stage model. □

D.9 (Proved). A complete classification on all CW complexes of dimension at most five. *Manuscript: 20; 118,122; 236,243.*

Manuscript problem locus: §20, fols.118 and 122, scans236 and 243, (66) and (71); this extends the dimension bound under discussion. For such an S, sphere bundles are classified by

$$H^1(S;C_2)\times\pi_0\operatorname{Map}\bigl(S,\operatorname{hofib}(u)\bigr).$$

This is an explicit obstruction-theoretic classification: choose $a\in H^1(S;C_2)$, a homotopy-pullback datum $f:S\to P$ as in D.6, and a nullhomotopy of $u\circ f$. Coherent homotopies of these data are the equivalence relation. A nullhomotopy always exists in dimension at most five. Over a fixed component $[f]\in[S,P]$, the possible classes form a torsor under

$$H^5(S;C_2)/\operatorname{im}(\delta_f),\qquad \delta_f:\pi_1\operatorname{Map}(S,P;f)\longrightarrow H^5(S;C_2),$$

where δ_f sends a loop $F:S\times S^1\to P$ to the slant product $F^*u/[S^1]$. The manuscript's second invariant is $b=x+a^2$.

Proof. D.1 and the actual group splitting of $O(3)$ give $BA\simeq BC_2\times BSO(3)$. The map $BSO(3)\to P_5BSO(3)$ has a five-connected fibre. The cell-by-cell lifting argument of D.4, now on S and $S\times I$, gives a bijection on maps from every five-complex. D.8 identifies the target. Since $H^6(S;C_2)=0$, the obstruction to a nullhomotopy vanishes. For a fixed actual map f, nullhomotopies form a torsor under $[S,\Omega K(C_2,6)]=H^5(S;C_2)$. Allowing f itself to move around a loop acts by the stated slant product, by concatenation of the pulled-back homotopy on $S\times I$. This proves the quotient formula, including its ambiguity, rather than asserting an unjustified direct product with H^5. The formula for b is D.4. The model distinguishes the two bundles over S^5, where $[S^5,BSO(3)]=\pi_5BSO(3)=C_2$ although all degree-two and degree-four characteristic classes vanish. □

4.3 Fidelity bounds and the two incompatible calculations

D.10 (Proved). Exact translation of the manuscript's four notions. *Manuscript: 20; 123v-124v; 246-248.*

Manuscript problem locus: §20, fols.123v–124v, scans246–248; the four definitions are on scan247. Let $T_S=\operatorname{Map}(S,BA)$ and $Q_S=\operatorname{Map}(S,Q)$, and interpret their fundamental two-groupoids as the two-isotopic categories of bundles and of pairs. The definitions on scans247–248 translate as follows, at every base point and for every relevant pair of components:

0-faithful	$\pi_2(T_S)\to\pi_2(Q_S)$ injective,
1-faithful	π_2 bijective and π_1 injective,
2-faithful	π_2,π_1 bijective and π_0 injective,
3-faithful	π_2,π_1,π_0 bijective.

The maximal uniform dimension bound guaranteeing any one of these four properties is $\dim S\le 1$. All four hold under that bound; all four can fail for $S=S^2$. In contrast, bijectivity only on bundle isomorphism classes holds through dimension three, as in D.4.

Proof. Two-arrows between fixed one-arrows form torsors under π_2 when nonempty. Faithfulness on these sets is injectivity on π_2. Fullness additionally asks that two image one-arrows which are isomorphic were already isomorphic, which is injectivity on π_1, as well as surjectivity on the corresponding π_2 torsors. Essential surjectivity of each Hom functor adds surjectivity on π_1 and injectivity on π_0; essential surjectivity on objects adds surjectivity on π_0. This proves the dictionary with exactly the cumulative conditions stated on scan247.

The map $q:BA\to Q$ has three-connected fibre. Pulling it back over $S\times D^j$ and extending sections relative to $S\times\partial D^j$ proves that $\operatorname{Map}(S,q)$ is $(4-d)$-connected for $d=\dim S$. Indeed all relative cells have dimension at most $d+j$, and the extension obstructions vanish through dimension four. Hence for $d\le 1$ it is an isomorphism on π_0,π_1,π_2. For sharpness take the

trivial bundle on S^2. Evaluation of mapping spaces has a section by constant maps and fibre $\mathrm{Map}_*(S^2, BA)$. The resulting split long exact sequence in degree two contains

$$\pi_4 BA = \mathbb{Z} \ \subset \ \pi_2 \,\mathrm{Map}(S^2, BA),$$

whereas the corresponding group $\pi_4 Q$ is zero. Thus π_2 is not injective, and even 0-faithfulness fails. This also disproves the proposed bounds $6, 5, 4, 3$ on scans246–247. The later mapping-space calculation on scan248 uses the correct direction of the dimension shift but is still off by one for 0-faithfulness: it needs $\dim(S \times S^2) \leq 3$, not ≤ 4. Also $\pi_i \,\mathrm{Map}(S, B)$ consists of maps relative to $S \times \{*\}$, not arbitrary unbased classes on $S \times S^i$.

For clarity the cellular argument gives the separate sufficient ranges

	π_0	π_1	π_2
bijection	$d \leq 3$	$d \leq 2$	$d \leq 1$
surjection	$d \leq 4$	$d \leq 3$	$d \leq 2$.

These separate assertions must not be conflated with the cumulative four fidelity notions. □

4.4 Arithmetic: forms, gerbes, and the information forgotten

D.11 (Proved). The proposed arithmetic two-category exists. *Manuscript: 20; 125; 249.*

Manuscript problem locus: §20, fol.125, scan249, the proposed arithmetic construction. For a scheme S, let $\mathcal{C}(S)$ have as objects PGL_2-torsors, equivalently forms of $\mathbb{P}^1_S$ or degree-two Azumaya algebras. One-arrows are their ordinary isomorphisms. If P corresponds to A_P, put $G_P = \mathrm{PGL}_1(A_P)$ and $\widetilde{G}_P = \mathrm{SL}_1(A_P)$. A two-arrow $f \Rightarrow g$ is a section t of $\widetilde{G}_P$ satisfying $\partial(t) = f^{-1}g$, where

$$1 \longrightarrow \mu_2 \longrightarrow \mathrm{SL}_1(A_P) \xrightarrow{\partial} \mathrm{PGL}_1(A_P) \longrightarrow 1$$

is exact as fppf sheaves. This defines a two-groupoid. Its lifting-gerbe functor is a two-functor to banded μ_2-gerbes. In characteristic two, fppf sheaves and the group scheme μ_2 are essential; replacing μ_2 by a constant two-element group is incorrect.

Proof. The standard action of PGL_2 on $\mathbb{P}^1$ and on M_2 identifies their automorphism sheaves with PGL_2; descent gives the stated equivalences of groupoids. For the field case this is Stacks Project, Lemma 59.61.5; the local matrix argument is identical over a scheme (The Stacks Project Authors, 2026d). Conjugation by G_P on $\widetilde{G}_P$ is well-defined, since scalar representatives act trivially. Together with ∂ it satisfies the two crossed-module identities

$$\partial({}^g t) = g\partial(t)g^{-1}, \qquad {}^{\partial(t)}s = tst^{-1}.$$

Vertical composition uses multiplication, and horizontal composition uses this conjugation action; the second identity is precisely the interchange law. Every arrow is invertible. Locally splitting A_P identifies the sequence with $\mu_2 \to SL_2 \to PGL_2$; an invertible matrix may be rescaled to determinant one after the fppf cover adjoining a square root of its determinant. Thus it is an epimorphism in that topology, including in characteristic two. The lifting gerbe has objects SL_2-torsors with an identification of their induced PGL_2-torsor with P. Transport gives its functor on one-arrows and sections of the central cover give its natural transformations. These formulas glue, proving the two-functor assertion. The cover here is the simply connected cover of a semisimple group scheme, not the universal etale covering space of its underlying scheme. □

D.12 (Refuted). Bare-gerbe classification, with a complete replacement. *Manuscript: 20; 125-125v; 249-250.*

Manuscript problem locus: §20, fols.125–125v, scans249–250, arithmetic forms and Azumaya algebras. Over a fixed base S there is no independently variable orientation torsor: the algebraic automorphism group PGL_2 is connected. Even after removing that entry, the lifting gerbe alone does not classify the objects of $\mathcal{C}(S)$ in general. On $S = \mathbb{P}^1_k$, the forms

$$\mathbb{P}(\mathcal{O} \oplus \mathcal{O}), \qquad \mathbb{P}(\mathcal{O}(1) \oplus \mathcal{O}(-1))$$

have neutral μ_2-lifting gerbes but are not isomorphic over S. The correct object-level replacement is a banded μ_2-gerbe together with a weight-one locally free sheaf of rank two and a trivialization of its determinant, modulo band-preserving equivalence carrying this sheaf and its determinant trivialization to the other such data. The associated two-stack after forgetting that sheaf and imposing two-descent is $B^2\mu_2$; it is not $BPGL_2$.

Proof. Both displayed rank-two bundles have trivial determinant, so their frame torsors reduce to SL_2 and the lifting gerbes have global objects. Suppose their projectivizations were isomorphic. The exact sequence $\mathbb{G}_m \to GL_2 \to PGL_2$ and descent imply that the two vector bundles differ by tensoring with a line bundle (equivalently, apply Stacks Project, Lemma 59.62.1 (The Stacks Project Authors, 2026c)). Every line bundle on $\mathbb{P}^1$ is $\mathcal{O}(n)$, so this would give $\mathcal{O}(1)\oplus\mathcal{O}(-1) \cong \mathcal{O}(n)^{\oplus 2}$. Determinants force $n = 0$, and twisting by $\mathcal{O}(-1)$ gives a contradiction: the left side has one global section and the right side has none. Thus the two objects are distinct even though their gerbes agree. Adding two-arrows to their ordinary isomorphisms cannot make them equivalent, because an equivalence of objects still requires a one-arrow.

For the replacement, the lifting gerbe of a PGL_2-torsor carries its tautological rank-two bundle, obtained from the standard representation of the tautological SL_2-torsor. Its central inertia acts with weight one, and its determinant is canonically trivial. Conversely, for such $(\mathcal{R}, V, \det V \simeq \mathcal{O})$, inertia acts trivially on $\mathrm{End}(V)$, so that algebra descends to a degree-two Azumaya algebra on S. Its projective frame torsor recovers the form. There is a canonical functor from $\mathcal{R}$ to the lifting gerbe of that torsor. It preserves the band and is locally an equivalence, hence is an equivalence of gerbes. These constructions are inverse on objects and isomorphisms of the stated data. This proves the corrected classification.

Finally, the crossed module $[SL_2 \to PGL_2]$ has sheaf kernel μ_2 and sheaf cokernel zero. Locally it has a single one-arrow up to two-isomorphism, and the two-automorphism sheaf of that arrow is μ_2. Its classifying two-stack is therefore $B^2\mu_2$. Equivalently, the lifting-gerbe two-functor is locally essentially surjective on objects and locally an equivalence on Hom stacks, and hence becomes an equivalence after two-stackification. This operation adds descent data which need not be represented by a degree-two Azumaya algebra. For fixed-base algebraic forms the Tate twist in etale homotopy is fixed by the base; it does not provide the freely selectable C_2-torsor of the topological problem. □

D.13 (Proved). Exact classification and exact loss over a field. *Manuscript: 20; 125v; 250.*

Manuscript problem locus: §20, fol.125v, scan250, the field case and quotient on isomorphisms. For a field K, isomorphism classes of forms of $\mathbb{P}^1$ identify under the lifting-gerbe map with those classes in $H^2_{\mathrm{fppf}}(K, \mu_2) = \mathrm{Br}(K)[2]$ whose index divides two. For a form with Azumaya algebra A, the Hom groupoid from that form to itself has

$$\pi_0 = \mathrm{Nrd}(A^\times)/K^{\times 2}, \qquad \pi_1 = \mu_2(K).$$

Its map to the automorphism groupoid of its gerbe is the inclusion $\mathrm{Nrd}(A^\times)/K^{\times 2} \hookrightarrow K^\times/K^{\times 2}$ on components and the identity on $\mu_2(K)$. In the split case the ordinary automorphism map loses exactly

$$\mathrm{PSL}_2(K) = \mathrm{im}(SL_2(K) \to PGL_2(K)),$$

and its surviving components are $K^\times/K^{\times 2}$.

Proof. The fppf Kummer sequence and $\mathrm{Pic}(K) = 0$ identify $H^2(K, \mu_2)$ with the two-torsion Brauer group. The obstruction gerbe maps to the usual Brauer class of A. A central simple algebra of fixed degree is determined, up to isomorphism, by its Brauer class: writing it as a matrix algebra over its division algebra, the degree fixes the matrix size. Thus the map on degree-two forms is injective, and its image is exactly the index-one and index-two classes.

Skolem–Noether identifies $G_A(K) = A^\times/K^\times$. The obstruction to rescaling a representative $v \in A^\times$ to reduced norm one is $\mathrm{Nrd}(v)$ modulo squares. Its kernel is exactly the image of $SL_1(A)(K)$, its image is the reduced-norm subgroup modulo squares, and its two-automorphisms are the kernel $\mu_2(K)$. Equivalences of a banded gerbe with itself are classified by μ_2-torsors, giving the claimed target and the same two-automorphism group. For $A = M_2(K)$, determinants take every value in $K^\times$, proving the final assertion. In particular the unipotent matrix $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ gives a nonidentity projective automorphism identified with the identity for every field. Over $\mathbb{R}$ and the quaternion division algebra, reduced norms are positive, so the source component group is trivial while the gerbe has two automorphism components. Thus even over a field the gerbe functor need not be full. The finite split cases over $\mathbb{F}_2$ and $\mathbb{F}_3$ are enumerated in the attached program. □

D.14 (Proved). The realizability restriction cannot be removed. *Manuscript: 20; 125v; 250.*

Manuscript problem locus: §20, fol.125v, scan250, realizability by an Azumaya algebra of fixed rank. There exist finitely generated characteristic-zero fields with two-torsion Brauer classes of index four. Such a class is a μ_2-gerbe which is not the lifting gerbe of a form of $\mathbb{P}^1$.

Proof. Take $K = \mathbb{Q}(t_1, t_2, t_3, t_4)$ and

$$D = (t_1, t_2)_K \otimes_K (t_3, t_4)_K.$$

The tensor product has degree four and Brauer exponent dividing two. Embed K in the iterated Laurent series field $L = \mathbb{Q}((t_1))((t_2))((t_3))((t_4))$, with its rank-four valuation, ordered lexicographically in the appropriate Laurent order. In D_L, write the standard basis as $i_1^{e_1} j_1^{e_2} i_2^{e_3} j_2^{e_4}$, $e_i \in \{0, 1\}$, with $i_1^2 = t_1$, $j_1^2 = t_2$, $i_2^2 = t_3$, $j_2^2 = t_4$. Give such a monomial value $\frac{1}{2}(e_1, e_2, e_3, e_4)$ modulo the value group of L. The sixteen values lie in distinct cosets. Every nonzero linear combination therefore has a unique term of least value. The least term of a product is the product of the two least terms and has nonzero coefficient, because commutation changes only a sign. Hence D_L has no zero divisors and is a division algebra, being finite-dimensional over a field. Therefore D is itself division: a zero divisor over K would remain one after extension to L. It has index four and nonzero exponent two. D.13 excludes its gerbe from the image of degree-two forms. The distinct-coset calculation and the signed monomial multiplication are checked in the attached program. □

D.15 (Refuted). The literal 0-faithfulness assertion, with its corrected interpretation. *Manuscript: 20; 124-125v; 247-250.*

Manuscript problem locus: §20, fols.124–125v, scans247–250; compare the definition on scan247 with the claim on scan250. The ordinary functor which sends an algebraic form to the *isomorphism class* of its gerbe-induced automorphism is not faithful: D.13 gives its exact kernel. If the four definitions of scans247–248 are instead imposed literally on the map from the ordinary groupoid, viewed as a two-groupoid with only identity two-arrows, the map is 0-faithful. Thus the wording of scan250 cannot simultaneously be read in that technical sense.

Proof. Ordinary faithfulness fails because the identity and the nonidentity unipotent in D.13 have the same image after passage to two-isomorphism classes. On the other hand the classifying space of a locally discrete groupoid has $\pi_2 = 0$. Its map to any two-groupoid is automatically injective on π_2, which is exactly 0-faithfulness in D.10. The first failure in that hierarchy is

generally at 1-faithfulness (the target can have nonzero $\mu_2(K)$), while failure on the automorphism component group is measured by injectivity on π_1. This is a terminological correction supported by the two explicit manuscript definitions, not a change to the valid observation that all projective automorphisms coming from $SL_2(K)$ are identified. □

D.16 (Refuted). Lead4, with the strongest categorical replacement. *Manuscript: 20; 116v-125v; 233-250.*

Manuscript problem locus: §20, fols.116v–125v, scans233–250; Lead4 of the supplied research prompt. The topological part of Lead4 is correct in the precise sense of D.1–D.10. The assertion that the arithmetic two-category of degree-two Azumaya algebras is simply a gerbe description of $BPGL_2$ is false without the module data of D.12. $BPGL_2$ classifies the original torsors; the imposed crossed-module homotopies produce a different two-category, and its two-stackification is $B^2\mu_2$. None of these constructions determines the full etale or motivic homotopy type of a conic from the gerbe alone.

Proof. The first assertion is D.1 and the explicit equivalences and Postnikov stages in D.3, D.6 and D.8. The scheme counterexample D.12 shows that a bare lifting gerbe does not determine the torsor. The index-four example D.14 shows that not every gerbe is represented by such a torsor. D.13 computes the morphisms lost, and D.12 proves the two-stackification assertion. Together these give the complete corrected statement and refute the proposed identification in all three relevant ways: objects, essential image and Hom groupoids. □

5 Shape, the Postnikov tower, and the precise change of level

Write $\mathcal{S}$ for the infinity-category of spaces, $\mathcal{S}_{\leq n}$ for its n-truncated objects, and $\beta_\infty K = \mathcal{S}/K \simeq \mathrm{Fun}(K, \mathcal{S})$. The suffix on β_∞ distinguishes it from the ordinary set-valued classifying topos. For an ordinary topos $\mathcal{E}$, use its associated 1-localic infinity-topos $\mathcal{X}$ of sheaves of spaces. When hypercompletion is used it is displayed as $\mathcal{X}^\wedge$. All finite shape truncations are independent of this choice. In particular, no assertion below silently identifies ordinary sheaves and hypersheaves.

5.1 The construction and the announced tower

E.1 (Proved; Part E(a), with no point or connectivity hypothesis). *Manuscript: LM §2, folio 4, scans 12–13; §3, folio 6, scan 16.* Every Grothendieck infinity-topos $\mathcal{X}$ has a functorial shape

$$P = \Pi_\infty \mathcal{X} \in \mathrm{Pro}(\mathcal{S}), \qquad \mathrm{Map}_{\mathrm{Pro}(\mathcal{S})}(P, K) \simeq \mathrm{Map}_{\mathcal{X}}(1, \Delta K). \tag{E.1.1}$$

Its fundamental pro-n-groupoid is $P_n = \tau_{\leq n} P$, with truncation applied stagewise in a pro-presentation. There are canonical maps

$$P \longrightarrow \cdots \longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_1 \longrightarrow P_0. \tag{E.1.2}$$

For n-truncated K, (E.1.1) factors uniquely through P_n. Hypercompletion induces $\Pi_n \mathcal{X}^\wedge \simeq \Pi_n \mathcal{X}$ for every finite n. These are statements about pro-objects, with all coherences retained, rather than pro-objects of an ordinary homotopy category.

Proof. The constant-object inverse image $\Delta : \mathcal{S} \to \mathcal{X}$ is accessible and left exact. For each $U \in \mathcal{X}$ the functor $K \mapsto \mathrm{Map}_{\mathcal{X}}(U, \Delta K)$ is accessible and preserves finite limits. It is therefore pro-representable: explicitly take its category of elements, consisting of pairs $(K, U \to \Delta K)$. Finite-limit preservation makes that category cofiltered, and accessibility supplies a small cofinal subcategory. The resulting diagram of its K's is a pro-left adjoint $\pi_! U$ to Δ. Set $P = \pi_! 1$. This construction gives (E.1.1) and is invariant under a change of site. Pullback of torsors along a geometric inverse image gives functoriality.

For a presentation (P_i), the pro-mapping formula and the ordinary truncation adjunction give $\operatorname{colim}_i \operatorname{Map}(P_i, K) = \operatorname{colim}_i \operatorname{Map}(\tau_{\leq n} P_i, K)$ for $K \in \mathcal{S}_{\leq n}$. This proves the universal property of (E.1.2). Hypercompletion fixes every truncated object. The constant object associated to an n-type is already n-truncated, and its mapping space from 1 is unchanged. Formula (E.1.1), restricted to n-types, proves the asserted invariance. This is the construction of Hoyois (2018, Definition 2.3 and §2), with its pro-representability and variance made explicit. It applies in particular to both pointless examples A.2 and A.11. □

E.2 (Proved; the exact universal property of the stages). *Manuscript: LM §2, folio 4, scan 13 (the announced stages).* If P_n is represented by an actual n-type K_n, define

$$\mathcal{X}_{[n]} = \beta_\infty K_n. \tag{E.2.1}$$

There is a canonical geometric morphism $q_n : \mathcal{X} \to \mathcal{X}_{[n]}$, characterized by

$$\operatorname{Map}_{\operatorname{Top}_\infty}(\mathcal{X}_{[n]}, \beta_\infty L) \xrightarrow{\simeq} \operatorname{Map}_{\operatorname{Top}_\infty}(\mathcal{X}, \beta_\infty L) \tag{E.2.2}$$

for every n-type L. Constancy of P_n is necessary and sufficient for a stage of this form with this universal property. If all finite P_n are constant, (E.2.1) gives the entire tower of scan 13. A sufficient intrinsic hypothesis is that $\Delta : \mathcal{S} \to \mathcal{X}$ preserve all small limits, equivalently admit an actual left adjoint $\pi_! : \mathcal{X} \to \mathcal{S}$. This is local constant shape, also called local infinity-connectedness. Under that hypothesis $H = \pi_! 1$ is an actual space and $K_n = \tau_{\leq n} H$.

Proof. The infinity-topos $\beta_\infty L$ classifies L-torsors: its geometric morphisms from $\mathcal{X}$ are sections of ΔL. Explicitly a section $a : 1 \to \Delta L$ sends an object $Z \to L$ to the pullback $1 \times_{\Delta L} \Delta Z$. This functor is left exact and cocontinuous by descent, and conversely applying an inverse image to the universal L-torsor recovers a. Thus (E.1.1) identifies maps to $\beta_\infty L$ with maps from P to L. Apply the same calculation to $\mathcal{S}/K_n$: its shape is K_n, since sections of a constant object are $\operatorname{Map}(K_n, L)$. This proves (E.2.2), including all higher homotopies. Conversely (E.2.2) says exactly that K_n represents P_n, by pro-Yoneda.

If $\pi_!$ is actual, its adjunction identifies P with the constant pro-object H. The equivalence between existence of this left adjoint and local constant shape is Lurie (2017, Proposition A.1.8); it also follows immediately by applying representability to $\operatorname{Map}(U, \Delta -)$ for every U. Truncation maps $K_n \to K_{n-1}$ give geometric morphisms in precisely the manuscript's direction. Their inverse images are restriction functors in the reverse direction. Their universal properties make all the compositions coherent.

Without constancy, P_n itself is the universally defined pro-stage. One may also form $\beta_\infty P_n = \lim_i \beta_\infty(\tau_{\leq n} P_i)$ and its canonical unit, but should not assert that this limit topos retains every arbitrary pro-object: the extension $\beta_\infty : \operatorname{Pro}(\mathcal{S}) \to \operatorname{Top}_\infty$ is not fully faithful in general (Hoyois, 2018, Remark 2.8). A.15–A.18 give the reconstruction of a Galois topos from its *intrinsic* fundamental pro-groupoid; A.21 shows why even a strict presentation of a limit topos need not be that intrinsic pro-groupoid. This qualification is precisely the distinction between a pro-object and a limit demanded in Part A(e). □

E.3 (Proved; Part E(b), the level-one identification). *Manuscript: LM §2, folio 4, scans 12–13 (2.1–2.2).* Under the hypotheses of A.9, if $\operatorname{LC}(\mathcal{E}) = \beta C$, then

$$\Pi_1 \mathcal{X} \simeq C \quad \text{as a constant pro-groupoid}, \qquad \mathcal{E}_{[1]} \simeq (\beta_\infty BC)_{\leq 0}. \tag{E.3.1}$$

The geometric morphism of Definition 2.2 is the ordinary part of the canonical first stage in E.2. The analogous zeroth stage is $\mathbf{Set}^{\pi_0 \mathcal{E}}$. Thus "the 1-truncation" means the classifying topos of the 1-truncated *shape*; it does not mean the subcategory of 1-truncated objects of the original topos.

Proof. A torsor under a discrete groupoid is locally constant. The inclusion $\beta C \hookrightarrow \mathcal{E}$ preserves finite limits and all colimits, so it preserves and reflects its principal-action equations and covering condition. Therefore for every groupoid D,

$$\mathrm{Tors}_{\mathcal{E}}(D) \simeq \mathrm{Tors}_{\beta C}(D) \simeq \mathrm{Fun}(C, D).$$

The last equivalence is A.8. This is the representing universal property for the constant pro-groupoid C, proving the first assertion. Its action on set-valued representations is exactly the full inclusion of A.9, so the inverse images agree, proving the assertion about geometric morphisms. Zeroth truncation discards all automorphisms and retains components. None of this uses actual points of $\mathcal{E}$; A.11 is a direct test of that fact. For merely locally connected $\mathcal{E}$, replace the constant C by A.16's pro-groupoid and retain the distinction between locally constant objects and their generated topos. □

E.4 (Proved; locally contractible spaces and the Postnikov tower). *Manuscript: LM §2, folio 4, scan 13 (local contractibility).* Let T be a topological space with a basis of contractible open subsets. For every finite n,

$$\Pi_n \,\mathrm{Shv}(T) \simeq \tau_{\leq n} \,\mathrm{Sing}(T), \qquad \mathcal{X}_{[n]} \simeq \beta_\infty \tau_{\leq n} \,\mathrm{Sing}(T). \tag{E.4.1}$$

Consequently its stages and maps are exactly the Postnikov tower, expressed as classifying infinity-topoi. Its inverse-image functors include $(n-1)$-truncated local systems into those at higher stages. The full shape of the *hypersheaf* infinity-topos is the actual singular homotopy type. For the ordinary sheaf infinity-topos that untruncated conclusion follows, for example, if it is hypercomplete, or if T is locally of singular shape in the sense of Lurie's Definition A.4.15.

Proof. Here is the hypersheaf comparison underlying the assertion. For a space K define a presheaf $R_K(V) = \mathrm{Map}(\mathrm{Sing}\, V, K)$ on the opens of T. An open hypercover has homotopy colimit weakly equivalent to the open it covers (Dugger and Isaksen, 2004, Theorem 1.3; Theorem 1.2 in the arXiv version). Hence R_K satisfies hyperdescent. On a contractible open its value is K, naturally via constant maps. The comparison theorem for hypersheaves on a basis therefore identifies R_K with the hypersheaf constant object. Taking its value at T gives

$$\Gamma \Delta^{\wedge} K \simeq \mathrm{Map}(\mathrm{Sing}\, T, K).$$

The same construction is a right adjoint to the colimit extension of $V \mapsto \mathrm{Sing}\, V$ from representables. Thus this hypersheaf topos is locally of constant shape, not just of constant global shape. This is precisely Carchedi (2021, Proposition 2.2.7); the argument also states which comparison lemma is being used.

For finite n, E.1 permits removal of hypercompletion in the displayed formula when K is an n-type. Pro-Yoneda then gives (E.4.1), and E.2 supplies its stagewise universal property. The transition maps induce isomorphisms on π_i for $i < n$ and kill π_i for $i > n$; their fibres encode the usual Postnikov extension by $K(\pi_n T, n)$. This is the Postnikov tower by its defining truncation universal property, including its k-invariants. In the actual-space case $\mathrm{Sing}\, T \simeq \lim_n \tau_{\leq n} \,\mathrm{Sing}\, T$, so the whole tower determines the singular weak homotopy type. There is no claim that an arbitrary untruncated pro-shape is recovered from its finite truncations.

The last sufficient hypothesis invokes Lurie (2017, Theorem A.4.19), which identifies local systems with functors on $\mathrm{Sing}\, T$ whenever every open has singular shape. Ordinary manifolds and CW spaces satisfy this condition. The use of hypersheaves above covers the requested general locally contractible case without silently importing a finite-dimensional or a paracompactness hypothesis. □

5.2 Which constructions survive a change of coefficients

E.5 (Proved; higher recognition and the correction to 1.1). *Manuscript: LM §1, folio 1, scan 6; §2, folio 4, scan 12.* If $\mathcal{X}$ is locally of constant shape, the canonical inverse image

$$q^* : \beta_\infty(\Pi_\infty \mathcal{X}) \longrightarrow \mathcal{X}$$

is fully faithful with essential image the locally constant objects. Hence all objects are locally constant if and only if $\mathcal{X} \simeq \beta_\infty H$ for an actual space H. At finite height the correct version uses n-localic infinity-topoi, local $(n-1)$-connectedness, generation by locally constant objects, and $(n-1)$-strict pro-n-groupoids. At height one this is A.15; dropping local connectedness is already refuted by A.2.

Proof. Put $H = \pi_! 1$. The unit $1 \to \Delta H$ defines q^* by pullback as in E.2, with left adjoint $q_! U = (\pi_! U \to H)$. The projection formula

$$\pi_!(\Delta A \times_{\Delta B} U) \simeq A \times_B \pi_! U$$

is Lurie (2017, Proposition A.1.9). Applying it to the defining pullback gives $q_! q^* Z \simeq Z$ in $\mathcal{S}/H$, so q^* is fully faithful. Its image is exactly the locally constant objects by Theorem A.1.15: its applicability here follows from the actual left adjoint in E.2, with no point hypothesis. An object in the image is visibly locally constant after pulling back the covering family of points of H; the converse is the theorem's effective-descent assertion, with the projection formula identifying the gluing maps. If every object is locally constant, full faithfulness plus essential surjectivity gives the equivalence; conversely every object of $\mathcal{S}/H$ is split by the representable covers $* \to H$.

For finite height, apply Hoyois (2018, Theorem 3.13 and Corollary 3.20) with the stated indices: set-valued objects correspond to $n = 1$ topoi but to 0-truncated coefficients. The theorem's strictness is exactly the connected-fibre condition of A.15, and the completion from split representations to the classifying topos is exactly A.16. This derives the manuscript's cases while retaining the shifted index and the generation condition. The Boolean example A.2 prevents the unconditional level-one recognition and therefore prevents a purported higher theorem with that false specialization. □

E.6 (Proved; slices, points, universal covers, and induction). *Manuscript: LM §1, folios 1–3, scans 6–11; §4, folios 6–7, scans 16–18.* The slice, point and Hom assertions A.5–A.8 extend to all levels with spaces of morphisms and coherent transformations. For $Z \to H$,

$$(\mathcal{S}/H)/(Z \to H) \simeq \mathcal{S}/Z, \qquad \mathrm{Map}_{\mathrm{Top}_\infty}(\beta_\infty H, \beta_\infty K) \simeq \mathrm{Map}(H, K). \tag{E.6.1}$$

The space of points of $\beta_\infty H$ is H. A pointed universal representable has contractible space of pointed identifications. The faithful-functor criterion of A.13 is specific to *set-valued* slices: a map of groupoids is faithful exactly when its map of classifying spaces has 0-truncated homotopy fibres. With arbitrary space-valued slices every map $Z \to H$ gives a slice morphism, so there is no faithful-functor restriction at that level.

Proof. The first identity is associativity of slice categories. The second is the torsor calculation of E.2, or the full faithfulness of $H \mapsto \mathcal{S}/H$. Setting $H = *$ gives the assertion about points. Under straightening the point h is represented by the path-space functor $\mathrm{Map}_H(-, h)$; enriched Yoneda gives its representing property, and the infinity-categorical uniqueness of a representing pair gives a contractible space of pointed identifications. Sums of components are handled by products of these topoi, exactly as in A.1 and A.5.

For $D \to C$ between ordinary groupoids, the homotopy fibre over c is its comma groupoid of lifts. It is 0-truncated precisely when it is equivalent to a discrete set, equivalently when there is at most one morphism between each two of its objects. This is the faithfulness condition on $D \to C$. The slice then comes from that set-valued local system, recovering A.13 and its thin

Hom-groupoids. If the fibre is a higher space, the corresponding slice object is space-valued; its mapping spaces need not be discrete. Replacing these by sets or by isomorphism classes discards information. For example $\beta_\infty S^2$ has ordinary locally constant set category **Set** but shape S^2, whereas $\beta_\infty *$ has the same set category and shape $*$. Their degree-two cohomology distinguishes them. The supplied CATHOT C.1–C.4 describes exactly this level-one boundary for functors modulo natural transformations; no new proof of that representability theorem is being substituted here. □

E.7 (Proved; delimited statement: higher van Kampen). *Manuscript: LM §16, folios 81–84, scans 163–170.* Shape sends colimit diagrams of infinity-topoi to colimit diagrams of pro-spaces. Consequently any genuine gluing equivalence of infinity-topoi gives a van Kampen theorem at every truncated level, and after profinite completion. Effective descent for an open cover gives such an equivalence. The ordinary 2-pushout of C.2 is the correct level-one statement; it cannot be substituted for that higher gluing hypothesis.

Proof. The torsor calculation in E.2 gives the adjunction $\Pi_\infty : \mathrm{Top}_\infty \rightleftarrows \mathrm{Pro}(\mathcal{S}) : \beta_\infty$. The left adjoint preserves colimits. Truncation to pro-n-types and completion to pro-π-finite types are also left adjoints, so the same conclusion holds after either operation. For an effective epimorphism $U \to 1$ in an infinity-topos, descent identifies the original topos with the colimit of its Čech diagram of slices in Top_∞. An open cover is such a diagram. For a two-member cover this gives the familiar homotopy pushout of the two pieces along their intersection. In an algebraic setting one may use an elementary Nisnevich square: Barwick, Glasman, and Haine (2020, Proposition 11.6.3 and Corollary 11.6.6) prove that it is a pushout of étale infinity-topoi and hence of their shapes. This identifies a precise descent hypothesis, rather than assuming that an ordinary topos pushout suffices.

For the failure of that substitution, consider the span of ordinary groupoids $* \leftarrow B\mathbb{Z} \to *$. Its bicategorical pushout is the terminal groupoid, and its classifying ordinary topoi have pushout **Set** by C.2. The homotopy pushout of the corresponding spaces is

$$* \underset{S^1}{\overset{h}{\sqcup}} * \simeq \Sigma S^1 \simeq S^2. \tag{E.7.1}$$

For example its reduced integral cellular chain complex is $\mathbb{Z}$ in degree two, whereas that of a point is zero. Hence the comparison loses π_2 and degree-two cohomology. In fact the corresponding pushout of classifying infinity-topoi is $\beta_\infty S^2$: a map from $\mathcal{S}/K$ into any infinity-topos is a K-family of its points, since left exactness and colimit preservation are checked pointwise in $\mathrm{Fun}(K, \mathcal{S})$. Thus $K \mapsto \mathcal{S}/K$ preserves colimits of actual spaces as well as being fully faithful. Its extension to arbitrary pro-spaces need not do so (E.2).

The topology on automorphism groups is a separate issue. The generic profinite pushout in C.3 fails in all ordinary topoi, whereas the surjective hole-filling pushout in C.2 is valid there. Neither assertion repairs (E.7.1): even when the ordinary 2-pushout has the right universal property in all ordinary topoi, the higher universal property is different. Transitivity of filling, when formulated by actual homotopy pushouts or topos descent, does extend: both iterated orders are the colimit of the same full diagram. This proves the higher form of C.4 without replacing a homotopy colimit by a quotient of groups. □

E.8 (Refuted; exact higher replacement: acyclicity). *Manuscript: LM §14, folio 63, scan 129; §16, folio 83, scan 167.* The map to the fundamental-groupoid stage preserves locally constant sets and their monodromy. It need not preserve their higher cohomology. Thus the acyclicity of C.5 is not a formal consequence of being a multi-Galois reflection. At all levels its exact replacement is equivalence of the relevant shapes, or the stated coefficient-relative

equivalence when only a coefficient class is required. The curve hypotheses and local inertia condition of B.3–B.8 are substantive conditions for that replacement.

Proof. The map $S^2 \to *$ induces an equivalence on locally constant sets and an isomorphism on π_1, but $H^2(S^2;\mathbb{Z}/m) = \mathbb{Z}/m$ and $H^2(*;\mathbb{Z}/m) = 0$ for $m > 1$. This is the simplest counterexample. For a local system A of abelian groups, cohomology is represented by sections of its fibrewise Eilenberg–Mac Lane objects. A shape equivalence therefore gives cohomology isomorphisms with every such local system, including twisted ones. Conversely, for a map inducing an equivalence on finite covering groupoids, isomorphisms for every finite abelian local system in every degree imply an equivalence on all π-finite coefficient spaces. To see this, use induction on the finite Postnikov tower of a target. The first stage is the finite-covering hypothesis; at the next stage lifts of a map are the sections of a twisted Eilenberg–Mac Lane fibration, their obstruction is the indicated cohomology class, and their homotopy groups are the lower cohomology groups with the same local system. Isomorphism of these groups gives equivalence of each lift space. This proves the claimed profinite-shape criterion, not merely a comparison of individual cohomology groups.

It follows that B.7's full torsion comparison, together with B.2's finite-cover comparison, upgrades to a comparison with all finite homotopy coefficients in precisely its valid cases. Its characteristic-zero exceptions $(g,\nu) = (0,0),(0,1)$ remain exceptions, already in degree two. The prime-to-p variant only gives the correspondingly restricted completion; it cannot imply a statement for p-torsion, for the reason and example of B.8. The kernel criterion in B.3 is the exact local test before making these global deductions.

Finite homotopy coefficients still do not determine an unrestricted shape. For example $K(\mathbb{Q},2)$ has contractible profinite completion but is not contractible. One elementary verification presents it as the mapping telescope of $K(\mathbb{Z},2) = \mathbb{C}P^\infty$ under multiplications $2,3,4,\ldots$. In each positive even degree the induced maps on cohomology with $\mathbb{Z}/m$ multiply by the corresponding power of those integers, so their inverse system is pro-zero; odd positive cohomology is zero. The telescope's Milnor sequence gives zero reduced finite cohomology, including the $\lim^1$ term. The space is simply connected, and the Postnikov induction just given makes every map into a π-finite target equivalent to a constant one. Its nonzero $\pi_2 = \mathbb{Q}$ distinguishes it from a point.

For stronger coefficient theories we use the supplied GALOIS results in their verified scope. GALOIS C.7, C.9–C.10 and F.7 show that equivalence of the *specified restriction functor* on integral derived local systems, or on local spectra, detects a weak equivalence of actual spaces. The proof passes from restriction to the specified map of loop algebras, then to π_1 and integral homology of universal covers, and finally uses the homology Whitehead theorem. That argument has been checked in the attached return. It is stronger than ordinary acyclicity with constant coefficients, and makes no assertion that a bare abstract ordinary category reconstructs its enhancement. The latter assertion is expressly not proved by GALOIS F.8 and is not used here. □

E.9 (Proved; the higher content of the mixed gluing and the Remords). *Manuscript: LM §14, folios 60–63, scans 123–129; cote 147, scans 179–180 (Remords).* The open–closed gluing of B.1 has a higher analogue using the full nearby-cycle functor. Its set-valued gluing functor $q_* r^*$ records only degree-zero invariants. Replacing it by homotopy fixed points, or equivalently retaining the full homotopy link, is necessary for a general comparison of shapes. For one closed stratum, the ordinary mixed category has shape

$$BG \underset{BK}{\overset{h}{\sqcup}} BQ, \qquad K = \operatorname{im}(H \longrightarrow G \times Q), \tag{E.9.1}$$

in the discrete case. The geometric link model, when the pieces and link have the indicated $K(\pi, 1)$ types, has shape

$$BG \underset{BH}{\overset{h}{\sqcup}} BQ. \tag{E.9.2}$$

The map $H \to K$, not just the homotopy types of the individual strata, controls their difference.

Proof. Let $\mathcal{U}, \mathcal{Z}$ be infinity-topoi with an accessible left exact gluing functor $\Phi : \mathcal{U} \to \mathcal{Z}$. Their Artin gluing consists of triples $(M, N, N \to \Phi M)$ with all homotopies retained. Finite limits are computed on the two components; effective descent gives its open and closed embeddings, and reconstructs every object from its two restrictions and its gluing map. Applying this construction to an open–closed decomposition uses $\Phi = i^* j_*$. The right adjoint j_* in sheaves of spaces already computes the homotopy-coherent sections on the punctured neighbourhood, so it retains the higher link data. On abelian coefficients its linear form uses $Ri^* Rj_*$, yielding the fibre sequence for derived sections. This is the recollement construction of Barwick, Glasman, and Haine (2020, §5.1), and the ordinary truncation is precisely B.1. The resulting corrected infinity-topos need not be 1-localic: its category of 0-truncated objects can agree with the ordinary mixed topos while its higher objects differ. Thus the operation is not asserted to be the automatic 1-localic enhancement of that ordinary topos.

For the ordinary mixed category, crossing arrows from the closed to the open object form the biset $G \times_H Q$. The groupoid of crossing arrows and their source and target isomorphisms is the action groupoid for $G \times Q$ on this biset. Its action is transitive and a stabilizer is K (up to the harmless inverse in the right-action convention); it therefore has classifying space BK. A simplex of the mixed category either stays in one stratum or contains exactly one crossing block. Dividing it at that block identifies its realization with the two-sided mapping cylinder of the source and target maps of the crossing groupoid. This proves (E.9.1). The identical argument with the actual homotopy link proves (E.9.2) under the stated local product and descent hypotheses. In the profinite case these are statements about the completed, homotopy-coherent diagrams; finite-stage descent and SGA 4 VI, Corollary 8.7.7, with hypotheses 8.7.1 and Remark 8.7.8, justify the torsion cohomology passage used in B.3.

Applying a stable coefficient theory to the two homotopy pushouts gives the two fibre sequences with common terms $R\Gamma(G, M) \oplus R\Gamma(Q, N)$ and last terms $R\Gamma(K, M)$ or $R\Gamma(H, M)$. The comparison's cofiber is the cofiber of inflation on those last terms, shifted by -1. This rederives B.3's obstruction and locates it exactly in the lost homotopy link. If $H \to G \times Q$ is injective, $BH \to BK$ is an equivalence, so this particular loss is absent. If it is not injective, its homotopy fibre is the classifying space of the kernel, and its effect must be tested on the requested coefficients.

For $(\mathbb{P}^1, \{\infty\})$ over $\mathbb{C}$, the strata have types $* = B1$ and $* = B1$, while the punctured link is $S^1 = B\mathbb{Z}$. Here $K = 1$, so (E.9.1) is a point and (E.9.2) is S^2. The mixed ordinary category is [1], as in the manuscript's Remords, and cannot have the cohomology of the sphere. In the full exit infinity-category the crossing mapping space is S^1; passing to its homotopy category replaces that space by a singleton. Notice the exact lesson: an ordinary category can have a classifying space of arbitrarily high homotopy dimension. What loses the information here is replacing a non-discrete crossing mapping space by its components, not merely writing down an ordinary category. E.12 gives cases where the crossing spaces really are discrete and the ordinary category retains the full higher shape. □

E.10 (Proved; delimited statement: direct images and coefficient-relative descent). *Manuscript: LM §2, folio 4, scans 12–13; GALOIS B.3 comparison requested in Part E.* For a map $g : H \to K$ of spaces and any presentable coefficient infinity-category $\mathcal{C}$, restriction on local systems has left and right Kan extension adjoints. Their pointwise formulas use the homotopy fibres of g. This is the all-level direct-image statement for the Galois envelopes. A locally

connected geometric morphism of ambient topoi need not carry locally constant objects to locally constant objects under its left adjoint, even if its usual base-change maps are isomorphisms.

Proof. For $F : H \to \mathcal{C}$ and $k \in K$, the comma infinity-groupoid of lifts of k is the homotopy fibre H_k. Pointwise Kan extension gives

$$(g_! F)(k) = \operatorname{colim}_{H_k} F, \qquad (g_* F)(k) = \lim_{H_k} F. \tag{E.10.1}$$

Transport in K makes these expressions local systems and supplies all the coherence maps. These are the coefficient-relative formulas proved in GALOIS; no separate derivator theory is required here. They apply directly to every morphism between the classifying infinity-topoi of E.2.

For the ambient warning let $j : (0,1) \hookrightarrow \mathbb{R}$ be the open inclusion. Its geometric morphism is essential and locally connected; its left adjoint is extension by the empty sheaf, and that operation commutes with pullback along maps of spaces. Nevertheless $j_!1$ is the open subterminal $(0,1) \subset 1$ on $\mathbb{R}$, which is not locally constant near either endpoint. Thus neither local connectedness nor this base-change property proves the assertion that would be needed to apply (E.10.1) directly to $j_!$.

Here is a useful non-tautological sufficient hypothesis. Let $f : T \to S$ be a locally trivial fibre bundle of spaces with contractible open bases and locally contractible fibre, and take hypersheaves. Over a contractible trivializing open $V \subset S$, E.4 identifies every locally constant object on $V \times F$ with a local system on F; the equivalence is natural under smaller contractible opens. The relative left adjoint exists there by the local-constant-shape construction of E.2, and is the constant sheaf with value the homotopy colimit over F. These local adjoints glue, since their defining adjunctions and their comparison on overlaps are unique up to contractible choice. Thus $f_!$ preserves locally constant objects and is (E.10.1) on their Galois envelopes. More generally, whenever this preservation has been separately proved, restricting the adjunction identifies the restriction with $g_!$; the corresponding mate is an equivalence. This supplies the exact geometric hypothesis missing from an unrestricted transfer of the fibrewise argument GALOIS B.3. □

E.11 (Proved; delimited statement: operators, exterior maps, transporters, and rigidity). *Manuscript: LM §14, folios 64–67, scans 132–137; §16, folios 85–88, scans 172–178; §12, folios 50–53, scans 103–109.* Actions and quotient compatibility in C.6 and C.8 extend to all levels as homotopy-coherent actions and colimits. The same is true of mixed morphisms once their full gluing transformation is included. Centralizer computations describe the level-one automorphisms of a specified map; higher uniqueness requires the entire space of choices to be contractible. The domain and degree obstructions in C.9 and C.11 remain actual obstructions and are not removed by changing coefficients.

Proof. A coherent action of a discrete group Γ is a functor $B\Gamma \to \mathrm{Top}_\infty$. Its colimit is the quotient infinity-topos. Descent along the universal Γ-torsor identifies such actions with infinity-topoi over $\beta_\infty B\Gamma$ whose pullback along its point is the specified topos. The action, its associativity homotopies and all higher coherences are recovered by the Čech nerve of that torsor. Taking a quotient and taking the colimit of an equivariant gluing diagram commute: both are the colimit over the Grothendieck construction of the same diagram. This proves the all-level versions of actions, the equivariant diagram, and passage to the quotient. Continuous profinite actions require continuous torsor data and pro-objects; C.7's distinction from an internal discrete torsor persists unchanged.

A morphism of glued infinity-topoi is specified by the maps on its pieces and a transformation between the two composites with their gluing functors. Composing transformations supplies the mixed morphism construction of B.12 at all levels. Taking an equivariant torsor and a coherent splitting gives its operator form, as in B.13. In a 1-type the automorphisms of a map $BG \to BH$ are $C_H(\varphi G)$, so triviality of that group makes its component of the choice space contractible.

This is exactly A.8 and B.13's rigidity calculation. For higher types, a zero automorphism group at level one only kills π_1 of the choice space. For example $\mathrm{Map}(*, K(A,2)) = K(A,2)$ has trivial π_1 and nontrivial π_2 when $A \neq 0$. Higher uniqueness therefore means a contractible fibre of the relevant mapping-space comparison, not just trivial centralizer.

Exterior homomorphisms are components of mapping spaces. Descent from a representative to its exterior class is unambiguous exactly when the appropriate component is contractible; this recovers C.9's sufficient centralizer condition at level one, but does not make its proposed class of morphisms closed under composition. The explicit compositional counterexample in C.9 already occurs in discrete groups. Transporters in C.10 are homotopy fibres of conjugacy-orbit maps; replacing their ordinary groupoids by those full fibres gives the coherent all-level construction. The equation of scan 107 still forces the scalar degree to be invertible under its stated self-centralizing hypotheses, by C.11's algebraic proof. Keeping coherent higher data cannot turn that equation into the differently directed ramified equation. Once the corrected local maps and their comparison cells are supplied, the operator construction is the equivariant homotopy-fibre construction and gives C.12's extension. □

E.12 (Proved; the finite-homotopy tests $n = 4$ and $n = 5$). *Manuscript: Part E(c), the LM3b H.8–H.9 tests; cote 147, scan 179.* For the boundary stratifications of $\overline{\mathcal{M}}_{0,4}$ and $\overline{\mathcal{M}}_{0,5}$ over $\mathbb{C}$, the full exit infinity-category has 0-truncated mapping spaces and hence is equivalent to an ordinary EI category. Its profinite completion by finite categorical congruences represents constructible sheaves with values in all π-finite homotopy types. Thus the comparison left open at $n = 5$ in LM3b H.9 is positive in this case. This statement concerns the full category with its noninvertible generization arrows and their compositions; its groupoidification retains only the fundamental groupoid and does not suffice.

Proof. Here is the geometric test and the additional computation at $n = 5$; F.43 supplies the flag construction and the full profinite descent argument. In a conically stratified manifold whose incident strata and homotopy links are $K(\pi, 1)$ spaces, the space of exits from a fixed point of a stratum S to a fixed point of an incident stratum T is a homotopy fibre of the link map

$$L_{S,T} \longrightarrow S \times T. \tag{E.12.1}$$

The long exact homotopy sequence shows that all homotopy groups of this fibre above degree one vanish, and that its fundamental group is $\ker[\pi_1 L_{S,T} \to \pi_1 S \times \pi_1 T]$. The kernel of projection to $\pi_1 S$ is the relevant normal meridian lattice. Hence injectivity of that lattice into $\pi_1 T$ makes the entire mapping space 0-truncated. This includes every component of the mapping space, not just a preferred exit. For $n = 4$ the open stratum is a three-punctured sphere, and each peripheral $\mathbb{Z}$ injects into its free group. Formula (E.12.1) proves the ordinary-category claim and verifies the structural input of LM3b H.8. Its ordinary category still has a sphere as its classifying space, in agreement with its degree-two calculation.

For $n = 5$ use

$$U = \{(x, y) \in (\mathbb{P}^1 \setminus \{0, 1, \infty\})^2 : x \neq y\}.$$

Blowing up $(0,0), (1,1), (\infty,\infty)$ in $\mathbb{P}^1 \times \mathbb{P}^1$ gives the stable compactification. Its ten boundary components are copies of $\overline{\mathcal{M}}_{0,4}$, their open parts are three-punctured spheres, and there are fifteen crossing points. The map forgetting y is a locally trivial configuration-space fibration with base a three-punctured sphere and fibre a four-punctured sphere. Both are aspherical, and its homotopy exact sequence proves that U is aspherical. Its fundamental group is an extension of a free group of rank three by one of rank two. The abelianization is free on the meridians $a = x_0$, $b = x_1$, $c = y_0$, $d = y_1$, $e = (x = y)$; this also follows from the five affine hyperplane equations defining the arrangement complement.

In this basis the boundary meridians are

$$\begin{array}{c|c|c|c} x_0 & a & x_1 & b \\ x_\infty & -a-b-e & y_0 & c \\ y_1 & d & y_\infty & -c-d-e \\ \mathrm{diag} & e & E_0 & a+c+e \\ E_1 & b+d+e & E_\infty & -a-b-c-d-e. \end{array} \tag{E.12.2}$$

The compatible pairs are (x_i, y_j) for $i \neq j$ and, for each i, the three pairs $(E_i, x_i), (E_i, y_i), (E_i, \mathrm{diag})$. Each corresponding two-column integer matrix has a 2×2 minor equal to ± 1. For the finite indices this is read off from the coefficients of a, c, e or b, d, e; for E_∞ use the rows (a, c) for its pairs with x_∞, y_∞ and the rows (a, e) for its pair with the diagonal. The six remaining pairs have a unit minor in their two separate coordinate directions. Therefore every incident normal $\mathbb{Z}^2$ is split injective in $H_1(U, \mathbb{Z})$, and every rank-one meridian is primitive. The attached computation checks all fifteen Smith invariants $(1, 1)$, while this unit-minor argument proves the injections. The maps into $\pi_1 U$ are consequently injective as well. Incidences from a crossing point into a boundary curve use the already verified three-punctured-sphere case. The links are torus bundles over these aspherical strata, hence are aspherical by their homotopy exact sequences. Formula (E.12.1) now proves that every exit mapping space is 0-truncated. Composition along flags is retained; no third-stratum coherence is discarded.

For clarity, two further steps are required before concluding the finite-homotopy comparison. First, these groups are good for finite coefficients. The only nontrivial new stratum group is the extension $F_3 \rtimes F_2$: the extension splits because its quotient is free, and finite-index characteristic subgroups of the finitely generated kernel show that its profinite topology is the full induced one. The completed extension is therefore exact. Free groups are good and have finite cohomology with finite modules; the ordinary and continuous Hochschild–Serre spectral sequences, whose kernel rows are only $0, 1$, identify its finite cohomology. The torus bundles are treated by the same argument, or by the triviality of their line bundles over a punctured sphere and the groups $\mathbb{Z}^r \times F_2$. The split lattice injections in (E.12.2) remain injective after profinite completion. Thus the completed flag pieces are their continuous group classifying types and the profinite version of (E.12.1) again has 0-truncated fibres.

Second, finite set representations alone would not prove the assertion for higher finite values. For a finite-object category with compact zero-dimensional arrow spaces and continuous composition, every finite collection of clopen arrow partitions is refined by a finite categorical congruence. Define $a \sim b$ when every composable context $u(-)v$ gives the same value in each given partition. Joint continuity and compactness of the spaces of u, v imply that this equivalence has open classes; compactness then makes the quotient finite. Inserting contexts proves compatibility with composition. A diagram with finitely many π-finite values has a uniform finite truncation bound; a degreewise finite coskeletal model for the full coefficient category on these values reduces its coherent data to finitely many simplicial degrees. The continuous data in those degrees first give finite clopen partitions of compact spaces of composable arrow tuples. Because there are finitely many objects, these spaces are finite unions of products of compact zero-dimensional Hom spaces. Every finite clopen partition of such a product has a finite coordinatewise clopen refinement: cover each clopen cell by basic clopen rectangles and use compactness, then take the finite Boolean refinements in each coordinate. The preceding context-congruence argument now makes all these data factor through a common finite categorical quotient. It applies also with a finite simplex parameter, so it proves the assertion on mapping spaces and all coherences, not just on isomorphism classes of objects. This is the finite congruence realization of the profinite categorical completion used here.

Finally Barwick, Glasman, and Haine (2020, Theorem 10.1.8) identify π-finite constructible sheaves with continuous representations of the profinite stratified shape. Their stratified Riemann Existence Theorem, Corollary 12.6.6, and its finite-stratification form Corollary 12.6.7, identify

the analytic and étale versions; Proposition 12.6.4 is the constructible-sheaf comparison. Applying these theorems to the just-computed exit category gives the asserted comparison over $\mathbb{C}$. The goodness and congruence steps explain why this application is stronger than simply citing exodromy for finite sets. The counterexample in E.9 fails exactly the meridian-injectivity test and so does not conflict with either case. □

E.13 (Refuted; corrected statement: the Alexandrov and enhancement qualifications). *Manuscript: Part E(c), the STRATA A.1–A.3 qualification; LM §2, folio 4, scan 13.* The level-one equivalence between set-valued sheaves on a poset's Alexandrov space and diagrams on the poset holds for every small poset. Its all-level extension holds for hypersheaves, and without hypercompletion when every principal open is finite. It is false without that qualification for arbitrary posets. Thus a higher model for the tower or the gluing diagram cannot use unrestricted Alexandrov sheaves and presheaves interchangeably.

Proof. Use the upward-open convention. Principal opens have trivial covering topology: any covering family of the principal open at p has a member containing p, hence containing that entire open. Restriction to this basis and right Kan extension give

$$\mathrm{Shv}(P_{\uparrow})^{\wedge} \simeq \mathrm{Fun}(P, \mathcal{S}). \tag{E.13.1}$$

The set-valued comparison is the ordinary basis theorem; for higher values the relevant basis theorem is the hypercomplete one. A finite principal open has finite homotopy dimension and its sheaf topos is hypercomplete. Hypercompleteness can be tested on this open cover, giving the indicated sufficient condition. These are Aoki (2023, Theorem A.6 and Examples A.11–A.12) and Barwick, Glasman, and Haine (2020, Proposition 3.12.11 and Example 3.12.15), and they verify the corresponding statements of STRATA A.1–A.3.

Here is the actual obstruction from Aoki's Example A.13. Let $Q = \mathbb{N} \times \{0, 1\}$ with $(i, a) < (j, b)$ exactly when $i < j$. Every interval is finite, but its principal upper opens are infinite. Every upper open has at most two minimal elements; finite gluing therefore makes filtered colimits of ordinary sheaves sectionwise, and the terminal sheaf is compact. In $\mathrm{Fun}(Q, \mathcal{S})$ let Q_n consist of ranks $0, \dots, n$ and let F_n be the left Kan extension of its terminal diagram. At ranks greater than n its value is

$$BQ_n = (S^0)^{*(n+1)} \simeq S^n,$$

whereas $\mathrm{colim}_n F_n = 1$ pointwise. If 1 were compact, its identity would factor through some F_n, making 1 a retract of F_n. The essential image of this fully faithful left Kan extension is closed under retracts (apply its counit), so 1 itself would be the left Kan extension of its restriction to Q_n. That contradicts the value S^n at a later rank. The two terminal objects have different compactness properties, proving the categories inequivalent. The finite joins and their homology are included among the program's checks.

There is a related distinction even for locally contractible spaces: Carchedi (2021, Example 2.2.4) exhibits a compact Hausdorff locally contractible space with nonzero Betti numbers in arbitrarily high degrees. Compactness of global sections implies that, if its ordinary sheaf shape were an actual space, that space would be a retract of a finite CW complex. Its unbounded Betti numbers prevent agreement with its singular type. Hypercompletion does give its singular type by Proposition 2.2.7, and every finite truncation agrees by E.1. This is why E.4 states the exact finite-stage result and marks the untruncated hypotheses explicitly. □

E.14 (Proved; decided list for every statement of Parts A–C). *Manuscript: LM §§1–4, 12, 14, 16; exact passage references in the indexed statements.* The following index gives the change of level for every preceding statement. "Extends" always means with spaces of coefficients, morphisms, and coherences; a statement already refuted at level one is not reasserted at higher levels. The proofs and counterexamples are the numbered statements just established, together with the specified calculations in Parts B and C.

Statement	Higher conclusion and its proof or obstruction
A.1–A.4; 1.1–1.2	Recognition extends under local constant shape, or under the finite-height local-connectedness and generation hypotheses of E.5. The unconditional implication is already false by A.2. The exactness axioms of A.4 are ordinary axioms; their higher replacement is infinity-topos descent plus the recognition hypothesis, not the same axioms imposed on a homotopy category.
A.5–A.8; 1.3–1.12	Slices, sums, represented points, pointed uniqueness and Hom classification extend by E.6. Connected-projective representability and five equivalent set-theoretic conditions of 1.5 belong to height one; their replacement is enriched representability and contractible fibres, as E.6 proves. Ordinary centres and centralizers are the corresponding first homotopy groups, by E.11.
A.9; 2.1–2.2	The actual locally constant envelope extends under local constant shape, by E.2 and E.5. Local simple connectedness alone only makes the first stage actual: S^2 satisfies it but has a nontrivial second stage (E.4, E.6). The general replacement is pro-shape, E.1.
A.10–A.12; 2.3	The failures of faithful restriction of actual points, and of point-defined universal covers, already occur at height one and persist. Intrinsic shape does not need points (E.1–E.3). Higher pointed uniqueness uses a contractible space, E.6; restriction to ordinary point groupoids is insufficient.
A.13–A.14; §4	Induction extends as the slice identity of E.6. Faithfulness is precisely the additional 0-truncated-fibre condition, hence is special to set-valued slice objects. The open-stabilizer condition remains necessary in the continuous setting, A.14.
A.15–A.19; §3 and schemes	Strict pro-groupoids and finite Galois envelopes extend to their strict pro-n and finite-homotopy versions, E.1–E.2, E.5. Generation and continuity remain necessary. Full étale topos reconstruction from an untopologized point groupoid remains false, A.17–A.19.
A.20–A.22	The finite-test and inverse-limit counterexamples persist under the embeddings into spaces; adding higher coefficients does not make a non-surjective shift conjugate to the identity. The correct pro-Hom retains coherent mapping spaces, A.21 and E.1. The announced tower is E.2–E.4.
B.1–B.2	Open–closed gluing extends via the full nearby-cycle functor, E.9. The quotient by normal inertia computes only π_1; full homotopy filling uses E.7's pushout.
B.3–B.4	The comparison defect extends as the fibre sequence of E.9 for stable coefficients. Inertia kernels obstruct it; $\mathbb{P}^1$ with one marked point disproves an unrestricted all-level comparison as well as the literal scan 129 proposition.
B.5–B.8	Valid curve comparisons extend to the indicated completed finite-homotopy theory by the obstruction-theoretic argument E.8. Their characteristic and coefficient restrictions remain. They do not imply an uncompleted shape equivalence.
B.9	The distinction from the log or Kummer-étale topos persists: retaining the inertia classifying fibre is different from filling the hole. E.9 identifies the lost link data and E.8 specifies which coefficients can see them.
B.10–B.11	Orientation classes and the specified trace are stable-coefficient constructions, so survive via the same cohomological or Verdier-duality operations. They require the dimension, coefficient and degree hypotheses proved there. Degree two is visible at shape level two; it cannot generally be computed from π_1, by E.8–E.9. No universal sphere-orientation formula is asserted for arbitrary dimensions.

B.12–B.13	Mixed morphisms and operator splittings extend with coherent gluing cells. Rigidity becomes contractibility of the choice space; triviality of its level-one centralizer alone is insufficient, E.11.
C.1	Slice pullback and open–closed complement squares extend to infinity-topoi by the same universal properties; E.6 and E.9 prove them. A pullback square alone is not a shape pullback theorem, since shape preserves colimits, not arbitrary limits.
C.2–C.4	The genuine higher descent pushout and transitivity extend, E.7. The ordinary pushout of even valid discrete or surjective profinite data need not preserve the higher shape, counterexample (E.7.1). The generic profinite failure in C.3 is separate.
C.5	Acyclicity is coefficient-relative and conditional, E.8–E.9. Its unrestricted form fails in degree two on the sphere.
C.6–C.8	Coherent actions, torsor descent, equivariant component factoring and quotient compatibility extend by E.11. The distinction between discrete and continuous operators in C.7 persists.
C.9–C.12	Exterior descent and transporters extend as mapping-space components and homotopy fibres, E.11. The compositional and scalar-degree counterexamples remain; higher contractibility replaces the ordinary unique-isomorphism condition.

The finite-homotopy stratified cases $n = 4, 5$ requested in the returns are separately settled by E.12; E.13 records the model-category qualification needed when posets are used. Every row has either a positive universal-property proof with its hypotheses, or a displayed counterexample with the corrected construction. In particular none infers higher reconstruction merely from a category of set-valued local systems.

E.15 (Proved). What the enhanced and marked coefficient categories reconstruct. *Manuscript context: LM §1, folios 1–3, scans 6–11; Part E(c), the explicit STRATA B.6, B.10 and B.12 leads.* For small infinity-categories C, D, put $\mathcal{P}(C) = \mathrm{Fun}(C^{\mathrm{op}}, \mathcal{S})$. Then

$$\mathcal{P}(C) \simeq \mathcal{P}(D) \quad \Longleftrightarrow \quad \mathrm{Idem}(C) \simeq \mathrm{Idem}(D). \tag{E.15.1}$$

For ordinary categories the right side is equivalence of their ordinary Karoubi completions. If every endomorphism of D is an equivalence, an equivalence in (E.15.1) forces every endomorphism of C to be an equivalence as well. In particular an enhanced presheaf category remembers this obstruction to being a layered exit category.

For a small space H, the ordinary category $\mathrm{Ho}\,\mathrm{Fun}(H, \mathcal{S})$ together with its specified constant-diagram functor $\Delta_H : \mathrm{Ho}\,\mathcal{S} \to \mathrm{Ho}\,\mathrm{Fun}(H, \mathcal{S})$ determines H in $\mathrm{Ho}\,\mathcal{S}$. Set-valued local systems together with their evaluation functors and all natural transformations recover $\Pi_1 H$; finite-valued local systems recover the corresponding profinite invariant. For connected pointed H, an enhanced category of R-module local systems together with its specified evaluation functor recovers the E_1-algebra $R[\Omega H]$.

Proof. An object A of $\mathcal{P}(C)$ is *completely compact* when $\mathrm{Map}(A, -)$ preserves every small colimit. Representables have this property because mapping out of one is evaluation, and retracts retain it. Conversely express A as the colimit of its category of representables. Complete compactness gives

$$\mathrm{Map}(A, A) \simeq \operatorname*{colim}_{(y(c) \to A)} \mathrm{Map}(A, y(c)).$$

The component of the identity comes from a point of one term: π_0 preserves colimits of spaces, and a colimit of sets has no elements outside the images of its terms. Thus A is a retract of some $y(c)$. This identifies the intrinsic full subcategory of completely compact objects with $\mathrm{Idem}(C)$ and proves the forward implication. The reverse follows from the universal property of idempotent completion, since $\mathcal{S}$ is idempotent complete. This is exactly the reduction in Lurie

(2009, Definition 5.1.6.2, Lemma 5.1.6.4 and Proposition 5.1.6.8) (Lurie, 2009), and verifies the supplied STRATA B.6 proof. If all endomorphisms are equivalences, every idempotent is an identity in the homotopy category and splits; hence D is already idempotent complete. The fully faithful inclusion of C into its completion then reflects the asserted property of endomorphisms.

For the assertion with specified constants, the intrinsic terminal object satisfies

$$\mathrm{Hom}_{\mathrm{Ho}\,\mathrm{Fun}(H,\mathcal{S})}(1, \Delta_H K) = [H, K] \tag{E.15.2}$$

naturally in K. Ordinary Yoneda in the homotopy category of spaces recovers the represented object H. For sets write $G = \Pi_1 H$. The covariant representable $G(x, -)$ represents evaluation at x on $\mathrm{Fun}(G, \mathbf{Set})$. Yoneda identifies all transformations between evaluations with the arrows of G, with their composition. This is also consistent with the intrinsic point-category reconstruction A.8. For a connected groupoid, automorphisms of evaluation on finite actions form the inverse limit of the finite quotient groups, explaining precisely the profinite qualification; A.17 supplies the analogous topological and pro-groupoid distinctions.

Finally evaluation U at the chosen point of connected H is conservative and preserves colimits in the enhanced R-linear category. Its left adjoint is

$$(LM)(h) = R[\mathrm{Map}_H(x, h)] \otimes_R M.$$

Consequently the monad UL is tensor product with $R[\Omega_x H]$, and its multiplication remembers loop composition. Monadicity applies because U preserves geometric realizations; this is Lurie (2017, Theorem 4.7.3.5) (Lurie, 2017). Thus the specified enhanced evaluation recovers that algebra. These arguments independently check the exact marked assertions of STRATA B.10. □

The scope of STRATA B.12 remains the bare *ordinary homotopy* categories of space-valued or linear coefficient systems, with no enhancement, constants, or evaluation specified. Neither the completely compact test, (E.15.2), nor the monad argument reconstructs that missing structure. This report therefore retains its stated open qualification for those bare categories, without asserting a counterexample or a literature-wide open-problem claim. It does not extend that qualification to the ordinary groupoid-presheaf topoi of A.8, whose intrinsic point groupoids already give the required level-one reconstruction. This distinguishes the actual statements of STRATA B.6, B.10 and B.12 from the shorter summary in the prompt.

Comparison with the variable-shape route. The other construction named in the supplied GALOIS F.5–F.6 and PATHS D.14–D.16 varies the indexing poset and retains the groupoids $\Pi_1 \mathrm{Map}(BJ, X)$. The stated class of tests consists of posets of finite chain dimension, not merely finite posets; in particular it contains infinite discrete sets. This is distinct from increasing the coefficient truncation in E.1–E.4. Even on finite tests it cannot be constructed from $\Pi_1 X$ alone. Take J to be the face poset of the boundary of a tetrahedron, so $BJ \simeq S^2$, and take $X = K(\mathbb{Z}, 2)$. Then

$$\pi_0 \mathrm{Map}(BJ, X) = [S^2, K(\mathbb{Z}, 2)] = H^2(S^2; \mathbb{Z}) = \mathbb{Z}, \qquad \Pi_1 X \simeq *.$$

Thus $\Pi_1 \mathrm{Map}(BJ, X)$ has infinitely many components, whereas $\mathrm{Map}(BJ, \Pi_1 X)$ is contractible. This proves the needed distinction directly. No reconstruction theorem from the preliminary variable-shape chapters is used as an input to the proofs of this return.

6 F. The later Montpellier topoi

We distinguish three constructions throughout this section: a pro-object of full neighbourhood topoi, the classifying topos of covering germs, and a topos of sheaves constructible for a fixed stratification. They have different universal properties. We write BG for the topos of continuous discrete sets with an action of a profinite group G, and $B\mathcal{G}$ for its groupoid version. A morphism over a base topos includes its commutation *isomorphism*; forgetting that isomorphism changes several answers below.

6.1 F(a): the six presentations and their rigidifications

F.1 (Proved). The six presentations, with the information retained. *Manuscript: MPL 149; section1; folios1-1v and8v-9; scans3-4 and18-19.*

Fix a profinite group Γ and its one-object profinite groupoid $B_{\mathrm{gr}}\Gamma$. The following are presentations of the same unpointed object, provided the indicated two-dimensional data are retained.

1. Connected point-admitting profinite Galois topoi with a geometric morphism to $B\Gamma$ whose induced group homomorphism has open image. No point is selected. Morphisms are geometric morphisms together with a specified isomorphism over $B\Gamma$; two-morphisms respect that isomorphism.

2. Connected profinite groupoids over $B_{\mathrm{gr}}\Gamma$ with open image, with functors, commutation isomorphisms, and compatible natural isomorphisms.

3. One-object presentations of (2), modulo inner conjugacy, *together with the conjugating elements as two-morphisms.* If only exterior classes are retained, this is the homotopy category, not the original two-category.

4. Homomorphisms $p : E \to \Gamma$ of profinite groups with open image. A morphism from $p' : E' \to \Gamma$ to $p : E \to \Gamma$ is a pair (u, γ) with $u : E' \to E$ continuous and $pu = \mathrm{Ad}(\gamma)p'$. A two-morphism $e : (u, \gamma) \Rightarrow (v, \delta)$ satisfies $v = \mathrm{Ad}(e)u$ and $\delta = p(e)\gamma$ (the equivalent inverse convention gives the same two-category).

5. If $H = p(E)$ is open, the extension $1 \to \pi \to E \to H \to 1$, together with its embedding $H \hookrightarrow \Gamma$. If $Z(\pi) = 1$, one can instead retain the continuous outer action $H \to \mathrm{Out}(\pi)$, with its extension topology; morphisms require the intertwiners and their coherence.

6. A profinite groupoid with continuous weak Γ-action, whose homotopy quotient is connected and whose object-component Γ-set is finite; equivariant functors and equivariant natural isomorphisms are its morphisms. Strict actions may be used after the usual transport-groupoid strictification, without discarding its coherence.

Without the open-image restriction, replace the finite component set in (6) by an arbitrary profinite homogeneous Γ-space, and omit the word *open* in (5). These descriptions are equivalent for the data arising from a finitely generated field $K/\mathbb{Q}$. In (5), $H = \Gamma_k$, where k is the algebraic closure of $\mathbb{Q}$ in K, embedded in a fixed $\overline{\mathbb{Q}}$; in (6) the groupoid is that of $K \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$. If K is finite over $\mathbb{Q}$, then $\pi = 1$: the subgroup H, or equivalently the finite Γ-set Γ/H, must still be retained. The trivial geometric group alone does not distinguish number fields.

Proof. For a profinite groupoid, evaluation at each object is a fibre functor on continuous actions. Equivariant natural transformations of evaluation are exactly the continuous arrows of the groupoid: finite transitive actions detect each finite quotient, and their inverse limit detects the arrow. Conversely the fibre functors of a pointed connected profinite Galois topos recover the finite Galois category and its continuous-action topos. This proves (1)$\simeq$(2), including all natural isomorphisms, without choosing one object globally. Choosing an object of a connected groupoid gives its one-object skeleton; changing that object is an isomorphism torsor, which is exactly the inner-conjugacy datum in (3)–(4). On one-object groupoids the naturality equation is the displayed conjugacy equation; compatibility with the base is its second displayed equation. Consequently forgetting the conjugating elements loses the centralizer of $u(E')$ intersected with $\ker p$, and is a biequivalence only on Hom-groupoids where that group is trivial.

The geometric fibre of $B_{\mathrm{gr}}E \to B_{\mathrm{gr}}\Gamma$ is the transport groupoid with objects Γ and arrows given by right translation by E through p. Its components are Γ/H, its vertex groups are π, and left translation supplies the Γ-action. Taking the transport homotopy quotient reverses this

construction. This is also a direct verification of effective Galois descent of actions, so it proves (4)≃(6), rather than identifying a quotient group with a quotient groupoid.

For a centreless kernel, conjugation identifies an extension with

$$E \;\simeq\; \mathrm{Aut}(\pi) \times_{\mathrm{Out}(\pi)} H.$$

Indeed its kernel is $Z(\pi)$, and any pair on the right differs from the conjugation action of a lift of its H-coordinate by a unique inner automorphism. This proves both injectivity and surjectivity and also the uniqueness of the extension isomorphism. In the profinite situation all maps are continuous; the extension topology is retained, so the assertion does not silently replace continuous extensions by abstract extensions. For a general kernel the outer action alone has the usual central extension ambiguity: already $H = C_2$, $\pi = C_2$ and trivial outer action admit C_4 and $C_2 \times C_2$. This proves exactly what description (5) would otherwise lose. Finally the components of $K \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ are indexed by embeddings $k \hookrightarrow \overline{\mathbb{Q}}$, and their stabilizer is Γ_k. □

F.2 (Proved). Openness and the centralizer used at scan 20. *Manuscript: MPL 149; section1; folio9v; scan20.*

Let $F \subset F'$ be finitely generated extensions of an algebraically closed characteristic-zero field. Then the image of $G_{F'} \to G_F$ is open. Every open subgroup of G_F has trivial centralizer in G_F when $\mathrm{trdeg}\, F > 0$. For transcendence degree zero, $G_F = 1$ and its centralizer is again the trivial group. Consequently the geometric exterior homomorphism at MPL 149, fol. 9, scan 20 determines a geometric morphism of classifying topoi up to unique isomorphism. This assertion is about the geometric groups; openness alone is not a centralizer theorem for arbitrary profinite groups.

Proof. The relative algebraic closure L of F in F' is finite over F. The regular extension F'/L is linearly disjoint from every finite separable extension of L. Every automorphism of L^{sep}/L therefore extends to the compositum with F', and then to a separable closure of F'. Thus the image is G_L, an open subgroup of G_F.

We give the needed slimness argument. For every finite Galois extension L/F with group Q and every prime ℓ, there exists a finite Galois extension M/F above it such that the action of Q on $N = \mathrm{Gal}(M/L)$ is faithful. Choose a normal model of F and shrink it so that the normalization in L is finite etale. A closed point in its smooth locus has a full Q-orbit above it, since the constant field is algebraically closed. A divisorial valuation centred there (use the exceptional divisor of the blow-up when the dimension exceeds one) has $|Q|$ distinct prolongations to L. Approximation for these finitely many independent discrete valuations gives $a \in L^\times$ with valuation one at one prolongation and zero at all the others. Its conjugates are independent in $L^\times/L^{\times\ell}$, as their valuation vectors are the standard basis. Since $\mu_\ell \subset F$, adjoining all their ℓth roots gives $N \simeq (C_\ell)^Q$, with faithful permutation action. If z is central in G_F, its image in Q acts trivially on N, so faithfulness gives that image equal to one. Since L/F was arbitrary, $z = 1$. The same argument applies to every open subgroup, because its fixed field is again finitely generated over the same algebraically closed constant field. Finally, if z centralizes an open subgroup H, then $J = \langle H, z\rangle$ is open, H is normal in J, and z lies in $Z(J)$. The already proved centrelessness of J gives $z = 1$. Thus every open subgroup has trivial centralizer in G_F. Natural automorphisms of a restriction functor are precisely the centralizer of the image, proving the final assertion. □

F.3 (Proved). Faithfulness of fields in every characteristic. *Manuscript: MPL 149; section1 Theorem2; folios10-10v; scans21-22.*

Let k be any field and K, K' finitely generated extensions. Two k-embeddings $f, g : K \to K'$ inducing isomorphic morphisms of profinite Galois groupoids *over* Π_k are equal. Thus the

characteristic-zero margin to Theorem 2, MPL 149, fol. 10, scan 22, is correct; a positive-characteristic counterexample with the same relative data does not exist.

Proof. For each integer n invertible in k, the coefficient μ_n is pulled back from B_k. The specified relative isomorphism therefore identifies the pullbacks in Kummer cohomology, with the specified identification of coefficients, so

$$f(a)/g(a) \in K'^{\times n} \qquad (a \in K^\times).$$

The same remains true after any algebraic extension of the constants and on each connected component of the geometric base change. Choose a component over an algebraic closure $\bar{k}$ of k and write F' for its function field. For a transcendental over k, both $f(a)$ and $g(a)$ remain nonconstant in F'. On a normal proper integral $\bar{k}$-model of F', every prime-divisor valuation of their ratio is divisible by all integers prime to the characteristic, and hence zero. A rational function with zero divisor on a normal proper integral variety over $\bar{k}$ is a constant: regularity in codimension one and normality make it an everywhere regular unit, and properness makes it an element of $\bar{k}^\times$. Consequently $g(a) = \lambda f(a)$. Applying the same argument to $a - 1$ gives $g(a) - 1 = \mu(f(a) - 1)$. Nonconstancy gives $\lambda = \mu = 1$.

For a algebraic over k, the finite separable part of $k(a)/k$ is an object of the finite etale category of k. The relative commutation datum recovers its embedding in K' by the universal property of the associated slice topos. A purely inseparable extension has at most one embedding extending that separable embedding. Thus $f(a) = g(a)$ also in this case. The map from K' into the chosen geometric component field is injective (it is a map from a field), so equality there implies equality in K'. □

F.4 (Proved). Relative faithfulness of reduced schemes. *Manuscript: MPL 149; section2; folio25v; scan52.*

Let S be locally noetherian and let X, Y be locally of finite presentation over S, with X reduced. If $f, g : X \to Y$ induce isomorphic morphisms $X_{\mathrm{et}} \to Y_{\mathrm{et}}$ in the two-category over S_{et}, then $f = g$. No characteristic-zero hypothesis and no reducedness hypothesis on Y are necessary. In particular the requested positive-characteristic refutation of this relative statement cannot be given.

Proof. First suppose $S = \operatorname{Spec} k$. Isomorphic geometric morphisms induce the same map on subterminal objects and hence the same map on underlying Zariski spaces. At a closed point $x \in X$ their common image $y \in Y$ is closed, and both residue fields are finite over k. The two maps on residue fields have the same effect on the maximal separable subextension of $\kappa(y)/k$: a finite separable extension represents a finite etale object of B_k, and the slice-topos universal property recovers its actual map, not merely a conjugacy class. The remaining purely inseparable extension has a unique embedding. Thus f and g agree on every closed residue point. Locally on a common inverse image of an affine open in Y, their difference on any coordinate function vanishes in every closed residue field. The source is reduced and Jacobson, so the intersection of its maximal ideals is zero. Hence the two ring maps, and therefore the morphisms, agree.

For general S, the square of etale topoi obtained by base change along $\operatorname{Spec} \kappa(s) \to S$ is cartesian. One can verify this by writing the local scheme as the inverse limit of affine open neighbourhoods of s and then taking its closed residue subtopos; this is also (Carlson et al., 2024, Lemma 2.9). Pull the relative isomorphism to this square, and then restrict to $(X_s)_{\mathrm{red}}$. The field case applies to this reduced finite-type source and proves equality on every reduced fibre. The morphism $\coprod_s (X_s)_{\mathrm{red}} \to X$ is schematically dominant for reduced X: a function vanishing on every such fibre vanishes in every residue field and hence is zero. Since the underlying point maps already agree, the same affine-local argument proves $f = g$. All reductions are local on S, X, Y, so no global quasi-compactness is required.

For comparison with (Carlson et al., 2024, Theorem 3.9), the extra reduced-fibre hypothesis in its stated version can be removed here by explicitly restricting to the reduced fibres, as above. This argument asserts faithfulness, not fullness. □

F.5 (Refuted). Unrestricted full faithfulness is false; pinned reconstruction is the correct theorem. *Manuscript: MPL 149; section2; folio25v; scan52.*

For every characteristic-zero field k, morphisms of etale topoi over B_k need not arise from scheme morphisms, even after universal homeomorphisms are inverted. If k is an infinite finitely generated field, the functor becomes fully faithful into the category of *pinned* geometric morphisms, which send closed points to closed points. The same holds for sub-p-adic characteristic-zero fields.

Proof. The t-adic decomposition subgroup gives a section of $G_{k(t)} \to G_k$: explicitly embed $k(t)$ in $k((t))$, use the splitting of $G_{k((t))} \to G_k$ obtained by acting on coefficients and fixing compatible roots of t, and restrict to a separable closure of $k(t)$. The resulting morphism $B_k \to B_{k(t)} \to (\mathbb{P}^1_k)_{\mathrm{et}}$ over B_k has image the generic point. A k-point of $\mathbb{P}^1_k$ has closed image, and universal homeomorphisms do not change that obstruction. This is the explicit obstruction in (Carlson et al., 2024, Example 2.30).

For an infinite field finitely generated over its prime field, (Carlson et al., 2024, Theorem 0.1) gives an equivalence on the Hom-sets of the localization at universal homeomorphisms and the pinned Hom-groupoids. The latter are equivalent to sets by its Proposition 2.22. The source localization is the category of absolute weak normalizations: in characteristic zero this is seminormalization, and in positive characteristic it is perfection. Thus the theorem applies to the manuscript's reduced finite-type schemes with exactly this localization and this target restriction. The assertion for sub-p-adic fields is (Carlson and Stix, 2024, Theorem A, equivalently Theorem 4.1); its identification of the localization with seminormal schemes gives the same reduction. These statements do not classify arbitrary base fields. □

F.6 (Refuted). Even pinned reconstruction fails over every algebraically closed characteristic-zero field. *Manuscript: MPL 149; section2; folio25v; scan52.*

Let k be algebraically closed of characteristic zero. Put $j = i$ and $j' = -i$, and let

$$E_j : \quad y^2 = x^3 - \frac{3j}{j - 1728}x - \frac{2j}{j - 1728}.$$

Then E_j and $E_{j'}$ are nonisomorphic smooth proper k-curves but have equivalent etale topoi over $B_k \simeq \mathbf{Set}$. This remains a counterexample after universal homeomorphisms are inverted.

Proof. The displayed Weierstrass equation has nonzero discriminant and j-invariant j; the symbolic computation is in `code/montpellier/montpellier_checks.py`. Extend the automorphism $i \mapsto -i$ of $\overline{\mathbb{Q}}$ to k, by choosing and fixing a transcendence basis and then extending to an algebraic closure. Applying this automorphism to coefficients gives an isomorphism of the underlying schemes and hence an equivalence of their small etale topoi. It is over **Set** automatically and is pinned because it is a homeomorphism. A k-isomorphism of the curves would preserve the j-invariant, contradicting $i \neq -i$. The curves are normal, hence seminormal, so their objects in the localized characteristic-zero category remain themselves. □

F.7 (Known). The experiment with a proper curve over a number field. *Manuscript: MPL 149; section2; folio25v; end of scan52.*

Let k be a number field, C/k a smooth proper geometrically connected curve of any genus, $K = k(C)$, and $\pi = G_{K\bar{k}}$. The k-automorphisms of C are in bijection with continuous outer automorphisms of G_K over G_k, and equivalently with continuous outer automorphisms of π commuting with the outer G_k-action. Requiring preservation of the conjugacy classes of the inertia groups of closed points does not change this answer. Smoothness and normality are essential if the curve, rather than its normalization, is to be recovered from its function field.

Proof. Apply (Mochizuki, 1999, Theorem B, Theorem 17.1 in the body) to the regular function field K/k on both sides; k is sub-p-adic. An isomorphism of groups is open, so it and its inverse give field embeddings. Their composites are the identity by the theorem's uniqueness, hence are inverse k-automorphisms. A birational map between smooth proper curves extends uniquely at every discrete valuation ring by the valuative criterion of properness, and the inverse extends as well. Thus it is an automorphism of C. Conversely any such automorphism induces the stated outer group automorphism and permutes the local valuations and their inertia. F.2 makes π centreless; the extension pullback in F.1 therefore turns a G_k-equivariant outer automorphism of π into the required outer automorphism of G_K. This verifies the proposed experiment at MPL 149, §2, fol. 25v, scan 52, including genus zero and genus one; it uses the birational group, not just the proper curve's fundamental group. □

6.2 F(c): quasi-Galois gluing and ordinary exit categories

F.8 (Proved). A precise finite quasi-Galois characterization. *Manuscript: MPL 147; section12; folio85; scans179-180; MPL149 folio67 scan135.*

Suppose a Grothendieck topos $\mathcal{T}$ has a finite filtration by open subtopoi, each successive locally closed stratum being the presheaf topos of a groupoid with finitely many connected components. At each step write $F = i^*j_* : \mathcal{T}_{\mathrm{open}} \to \mathcal{T}_{\mathrm{closed}}$ for the accessible left-exact gluing functor. Then $\mathcal{T}$ is, compatibly with the filtration, a presheaf topos on an EI-category with finitely many isomorphism classes if and only if every F preserves arbitrary products. Here an EI-category is a category in which every endomorphism is invertible, and its isomorphism classes are partially ordered by the existence of a morphism. Without the finiteness of the component sets of the strata, the corresponding conclusion uses a small EI-category and need not have finitely many isomorphism classes.

Proof. We prove one induction step. Write the open topos as $\widehat{A} = \mathrm{Fun}(A^{\mathrm{op}}, \mathbf{Set})$ and the closed one as $\widehat{B}$, with B a groupoid. A sheaf in the glued topos is a triple $(P, Q, q : Q \to F(P))$. If F preserves products, left exactness makes it preserve every small limit. Accessibility and the adjoint functor theorem give a left adjoint L to F. For each representable $h_b \in \widehat{B}$ put $M_b = L(h_b) \in \widehat{A}$. Yoneda and adjunction give

$$F(P)(b) = \mathrm{Hom}_{\widehat{B}}(h_b, F(P)) \simeq \mathrm{Hom}_{\widehat{A}}(M_b, P).$$

Define a category C by adjoining to A the objects of B, putting $\mathrm{Hom}_C(a, b) = M_b(a)$, no morphisms from B to A, and using the actions supplied by the presheaves M_b and functoriality in b for composition. A presheaf on C is exactly the triple above. This proves $\mathcal{T} \simeq \widehat{C}$. All new endomorphisms lie in A or B, so the EI property follows by induction. No cycle can cross the finite filtration; after passing to isomorphism classes the resulting preorder is a partial order.

Conversely, for an EI-category with the given filtration, crossing arrows form the bimodule $M_b(a) = \mathrm{Hom}_C(a, b)$. The gluing functor has the displayed representable form and preserves all limits, particularly products. The numbers of isomorphism classes add under the finite construction. A single stratum $\mathbf{Set}^{\mathbb{N}}$ shows why that finite-component condition cannot be omitted: its indecomposable projective generators have infinitely many nonisomorphic types.

This proves the categorical case asserted in STRATA B.1–B.2 rather than assuming that arbitrary left-exact gluing is representable by crossing sets. □

F.9 (Proved). A two-stratum gluing which is not of presheaf EI type. *Manuscript: MPL 147; section12; folio85; scan179; task F(c).*

Let $C = \{0,1\}^{\mathbb{N}}$ be the Cantor space, and define $F(A) = \mathrm{LC}(C, A)$ for a set A with the discrete topology. The Artin gluing of two copies of **Set** along F is a Grothendieck topos with two Galois strata, but it is not the presheaf EI-topos asserted in F.8.

Proof. Every locally constant map from compact C has finite image and factors through a finite clopen partition. Consequently $F = \mathrm{colim}_{\mathcal{P}} (-)^{\mathcal{P}}$, where $\mathcal{P}$ runs through the filtered set of finite clopen partitions. It is accessible and preserves finite limits, so its Artin gluing is a Grothendieck topos. For the countable family $A_n = \{0,1\}$, the coordinate projections $C \to A_n$ are locally constant. They define an element of $\prod_n F(A_n)$ which does not come from $F(\prod_n A_n)$: the only map with these coordinates is the identity of the underlying set of C, which is not continuous from Cantor space to that set with the discrete topology. Thus F fails to preserve this product, and F.8 proves the claim. This example also specifies the missing gluing condition in the undefined term "topos quasi-galoisien" at MPL 147, fol. 85, scan 179. □

F.10 (Proved). Constructible sets and the crossing double cosets. *Manuscript: MPL 147; sections11-12; folios56-64 and85; scans114-130 and179-180; MPL149 scans133-135.*

Let (X, D) be a complex manifold with a locally finite normal-crossings divisor, or an analytic Deligne–Mumford stack with such an etale local presentation. Assume the usual paracompact local-conical hypotheses for constructible exodromy. Write $\mathcal{E} = \mathrm{Exit}(X, D)$, with arrows from closed to more open strata. Then

$$\mathrm{Sh}_{\mathrm{str}}(X; \mathbf{Set}) \simeq \mathrm{Fun}(h\mathcal{E}, \mathbf{Set}) = \widehat{(h\mathcal{E})^{\mathrm{op}}}.$$

This is a quasi-Galois topos in the sense of F.8 (without finite component restriction if necessary). For one connected incidence link L between connected strata S, T, choose endpoints and put $A = \pi_1 S$, $B = \pi_1 T$, $G = \pi_1 L$. The crossing-arrow set, with its A–B actions, is

$$(A \times B) / \operatorname{im}(G \to A \times B),$$

with the harmless opposite convention on one factor dictated by endpoint transport. If several link components occur, take the disjoint union of these orbit sets. Composition is induced by concatenation on the full flag spaces, not by composing their sets of components without the flag maps.

Proof. A constructible sheaf transports germs along paths within each stratum. In a conical chart a generization from a closed stratum is transport along an outgoing radial arc; homotopic exit arcs induce the same map of germs. The sheaf axiom on conical charts and their intersections gives precisely compatibility under concatenation, and produces a functor on $h\mathcal{E}$. Conversely, for such a functor assign on a conical open chart the compatible germs indexed by its exit objects. Restriction is restriction of compatible families. The charts form a basis stable under refinement, and compatibility on a cover is exactly the sheaf condition; the stalk at an exit object is the assigned set. On each stratum the construction is the usual locally constant sheaf associated to a fundamental-groupoid action. These two constructions are inverse on stalks, hence on sheaves, and on natural transformations. For the stack apply this construction on the entire etale atlas nerve and descend: compatible charts include the stabilizer actions. This is the set-valued reduction of the exodromy theorem, with all transition data retained.

The exit mapping space for this incidence is the homotopy fibre of $L \to S \times T$ at the chosen endpoints. The long exact sequence in components of that fibre gives the orbit-set formula. Its residual endpoint actions give the two module actions. Two composable exits are represented by a flag with two steps; the face forgetting the middle step is their composition. Three-step flags verify associativity. Within a stratum every exit is invertible, and a strict exit cannot return to its source stratum, so all endomorphisms are invertible. Finally the bimodule formula for restrictions shows that its gluing functor is represented as in F.8. This verifies the relevant STRATA C.1 and C.6 calculation in the geometric setting; it does not identify the full sheaf topos of X with this constructible subcategory. □

F.11 (Refuted). The Boolean source axiom does not imply thinness. *Manuscript: MPL147; section11; folios56-59; scans114-121.*

The weighted Boolean axiom (174)c at MPL 147, fols. 56–57, scans 115–116, implies (174)a and b. It does *not* imply that an arrow is determined by its source and target. The additional geometric transversality hypothesis later imposed at scans 120–121 gives this property in the intended branch-incidence examples.

Proof. The coslice of a rank-d object has the weighted Boolean lattice of subsets of a d-element set. Its ranks are between zero and d, and it has a unique object of rank d, namely the identity. This proves a and b. For the false extra implication use the finite category with objects x, y, z of ranks $2, 1, 0$, arrows

$$a, b : x \longrightarrow y, \qquad c : y \longrightarrow z, \qquad d : x \longrightarrow z, \qquad ca = cb = d,$$

and identities. Its coslices are the Boolean diamond, the two-element chain, and the one-point poset, respectively. Thus it satisfies the full weighted Boolean coslice condition, but $a \neq b$ have identical source and target. Regard all these finite sets as discrete zero-dimensional analytic spaces to obtain an internal analytic category. The program `montpellier_checks.py` verifies every composition and every coslice relation.

The exact correction to the axiom is injectivity of the target map on every source coslice, or equivalently monicity of $(s, t) : D_{**} \to D_* \times D_*$. For genuine normalized branch germs, an object is a specified set of local branches, so an arrow exists exactly when one specified set contains the other, and its inclusion is unique. Under the later transverse geometric realization, distinct subsets give distinct intersection germs by independent local coordinates, so this extra condition holds. It is the geometric realization, not the abstract Boolean axiom alone, that supplies it. □

6.3 F(d): germs, full sheaves, and descent

F.12 (Proved). The punctured germ is a pro-topos; its covering reflection is a different object. *Manuscript: MPL147; section11; folios70-73; scans143-148; LM8 E.10-E.11.*

For a locally closed analytic embedding $Z \hookrightarrow Y$, let $\mathfrak{V}_{Z,Y}$ be the pro-object of full sheaf topoi $(\mathrm{Sh}(V))_V$ indexed by open neighbourhoods of Z in an ambient open where Z is closed. With a divisor D, put $\mathfrak{V}^*_{Z,Y} = (\mathrm{Sh}(V \setminus D))_V$. These definitions are invariant under cofinal refinement. Their covering-germ categories are the filtered two-colimits of the covering categories at the stages. For a normal-crossing point with r branches this category is $\mathbb{Z}^r$-sets, and its finite part is finite continuous $\widehat{\mathbb{Z}}(1)^r$-sets. In contrast the literal inverse two-limit of the full sheaf topoi of shrinking punctured discs is degenerate.

Proof. Cofinal systems present the same pro-object by the definition of pro-morphisms. Polar coordinates identify a cofinal system of punctured normal polydiscs with a contractible radial factor times $(S^1)^r$; shrinking acts identically on the coordinate winding numbers. Covering

classification therefore gives the asserted filtered category, including its morphisms after common restriction. This verifies all steps of LM8 E.10 independently.

For the last assertion let $D_n^* = \{0 < |z| < 1/n\}$ and suppose a topos $\mathcal{A}$ maps compatibly to every $\mathrm{Sh}(D_n^*)$. Viewed as opens of D_1^*, the annuli $U_k = \{1/k < |z| < 1\}$ cover D_1^*. Each inverse image of U_k is initial because the morphism factors through D_k^*, disjoint from U_k. Inverse images preserve unions, so the terminal object of $\mathcal{A}$ is initial. The degenerate topos has the unique such cone, proving the universal property. This also verifies LM8 E.11. The manuscript's arcs and C^∞ embeddings at scans 143–148 identify these as classical analytic germs; a profinite etale replacement is a separate construction, not a new reading of the word "germe". □

F.13 (Proved). The precise full-topos gluing theorem. *Manuscript: MPL147; sections11-12; folios74 and76; scans150 and155.*

Let X be paracompact and locally compact, D a locally finite normal-crossings divisor, and $\Sigma \subset D$ a closed union of strata. A cofinal system of controlled tubular neighbourhoods of Σ can be covered by tubular neighbourhoods of its normalized branches, with all their intersections. The resulting full Cech diagram, or its full flag subdivision, has two-colimit $\mathfrak{V}^*_{\Sigma,X}$ in pro-Grothendieck topoi. This is the correct full-sheaf meaning of both versions of the gluing theorem at MPL 147, fols. 74 and 76, scans 150 and 155. The same theorem holds for analytic Deligne–Mumford stacks presented by proper etale groupoids with paracompact locally compact orbit space, or for an explicitly controlled atlas whose descent conditions admit the common-stage refinement used below. Arbitrary nonproper atlases, even with finite isotropy, require that extra hypothesis: F.51 refutes passage from objectwise atlas germs to an intrinsic stack neighbourhood without it. For a closed D one also has

$$\mathrm{Sh}(X) \;\simeq\; \mathrm{Sh}(X \setminus D)\, \amalg_{\mathfrak{V}^*_{D,X}} \mathfrak{V}_{D,X} \quad \text{in pro-topoi,}$$

where the left side is constant. This proves the full-topos version of (254) and (266), with their germ objects correctly typed.

Proof. At a fixed neighbourhood stage V, choose open tubular sets V_a whose union is V (and intersect them with $X \setminus D$ for the punctured version). Sheaves on V are exactly sheaves on the V_a with isomorphisms on double intersections satisfying the triple cocycle condition. Here is the topoi-level universal property: for any target $\mathcal{A}$, a geometric morphism $\mathrm{Sh}(V) \to \mathcal{A}$ is a finite-limit- and colimit-preserving functor $\mathcal{A} \to \mathrm{Sh}(V)$. Restricting that functor produces such functors on every V_a with compatible isomorphisms. Conversely objectwise sheaf descent glues these functors and natural transformations; finite limits and colimits are checked after the conservative open restriction family. Thus the gluing is a bicategorical colimit of *topoi*, not merely a statement about covering categories.

For finitely many local branches, subdividing the Cech index category by all nonempty nested intersections does not change descent data: a subdivided datum is recovered by restriction from its one-step data, and the flag triangles are precisely the cocycle relations. For completeness the passage from local germs to a global pro-object requires a uniformization argument. Fix a target topos $\mathcal{A}$ and a compatible germ datum. On a locally finite, relatively compact shrinking of the coordinate cover choose a representative neighbourhood for every chart datum, every overlap isomorphism, and every triple cocycle equality. Near any given point only finitely many of these conditions occur. After shrinking the coordinate sets once more, their allowed radii have a positive lower bound on each compact member of that shrinking. A partition of unity gives a positive continuous function smaller than all these locally finite bounds. Restricting to that variable-radius tube realizes every datum and every compatibility at one stage. The same argument applied to two representatives, and then to their two-morphisms, proves full faithfulness as well as essential surjectivity. Thus finite chart descent passes to germs and global open descent is legitimate; we have not commuted a filtered colimit with an arbitrary infinite

limit. For a pro-target first apply this argument to each fixed target stage, and then take the limit over target stages in the pro-mapping formula. This proves the first assertion and its version for Σ.

For the second assertion, at every stage V the pair $X \setminus D, V$ is an open cover of X, with intersection $V \setminus D$. Apply the just-proved two-open universal property and pass through the same cofinal system. For non-simple normal crossings, choose an etale chart atlas and all its fibre products; the normalized local branches are handled on this atlas, not assumed globally embedded. For a proper etale groupoid the local orbifold charts have finite stabilizers. Choose a locally finite shrinking on the orbit space and take the minimum over the finitely many stabilizer translates of each chart radius. Over compact chart closures, properness makes the relevant transition conditions finite after an etale refinement. The preceding uniformization therefore applies to the overlap maps and their cocycles as well as to the branch data. Tubular constructions and intersections pull back etale locally, and these synchronized descent diagrams may be composed. The result retains stabilizer groups and requires no invariant tubular embedding. For a nonproper groupoid this finite-control step must be supplied separately; F.51 gives a case where it fails. The intrinsic pro-object indexed by actual stack neighbourhoods still makes sense, and its two-open pushout is proved at each actual common stage; it is not thereby identified with the objectwise atlas-germ diagram. □

F.14 (Refuted). Covering topoi cannot replace full elementary topoi in an ordinary two-colimit. *Manuscript: MPL147; section12; folios85-86; scans179-180; LM8 E.11.*

For discrete groupoids $\mathcal{G}_i$, the ordinary bicategorical colimit of the topoi $B\mathcal{G}_i$ is $B(2\text{-}\mathrm{colim}_i\, \mathcal{G}_i)$. It generally has the wrong higher shape for the geometric gluing. In particular

$$\mathbf{Set} \amalg_{B\mathbb{Z}} \mathbf{Set} \simeq \mathbf{Set}, \qquad * \amalg^h_{S^1} * \simeq S^2.$$

Thus neither "aspherical strata" nor "all flags" repairs the replacement of full elementary topoi by covering topoi inside the ordinary two-category. One must instead take the homotopy colimit of classifying *spaces*, or use the corresponding infinity-topoi.

Proof. Present a topos $\mathcal{A}$ by a small site. A geometric morphism $B\mathcal{G} \to \mathcal{A}$ is a flat continuous site functor valued in $\mathcal{G}$-sets. Evaluation at the objects of $\mathcal{G}$ identifies this with a $\mathcal{G}$-diagram of points of $\mathcal{A}$, with natural transformations: flatness and covering epimorphisms are checked objectwise. Hence $\mathrm{Geom}(B\mathcal{G}, \mathcal{A}) \simeq \mathrm{Fun}(\mathcal{G}, \mathrm{Pt}\,\mathcal{A})$. The bicategorical colimit universal property for groupoids now proves the first assertion. The groupoid pushout of $1 \leftarrow \mathbb{Z} \to 1$ is the trivial groupoid, whereas the homotopy pushout of its classifying spaces is the suspension of S^1. The pair $(\mathbb{P}^1, \infty)$ is the geometric instance. This correction is consistent with F.13: its full neighbourhood topoi have not been replaced by their covering reflections. □

6.4 Finite flag truncations and van Kampen

F.15 (Refuted). The two-depth diagram (246). *Manuscript: MPL147; section11; folio75; scan152; (246).*

For the coordinate divisor in $\mathbb{C}^3$, the bicategorical colimit of the depth-one and depth-two groupoids in (246), with the incidence isomorphisms but without the depth-three flag relations, has vertex group $\mathbb{Z}^4$. The punctured normal boundary has fundamental group $\mathbb{Z}^3$. These statements concern discrete groups; their finite-covering analogues have the indicated profinite completions.

Proof. In polar coordinates the boundary of the real oriented blow-up is

$$L = (S^1)^3 \times R, \qquad R = \{(r_1, r_2, r_3) \in \mathbb{R}^3_{\geq 0} : \min r_i = 0\}.$$

The factor R contracts to its vertex. Removing depth three deletes that vertex, and radial normalization gives $R \setminus \{0\} \simeq \partial\Delta^2 \simeq S^1$. Thus $L_{\leq 2} \simeq (S^1)^4$. The index category has three singleton and three two-element subsets, with six incidences: its nerve is a hexagon. Each total-link group is $\mathbb{Z}^3$ and each incidence identifies its three coordinate meridians. A spanning tree eliminates five incidence generators; the remaining generator commutes with the three meridians, giving $\mathbb{Z}^3 \times \mathbb{Z}$. Adding the three-element subset and all flags makes the index nerve a cone on that hexagon. Its six triangular relations eliminate the six non-tree generators in the enlarged presentation. This gives $\mathbb{Z}^3$.

The attached program constructs the subset posets, their edges and triangles, chooses a spanning tree, and performs single-occurrence Tietze eliminations; the outputs are respectively $(6, 6, 0)$ and $(7, 12, 6)$ for vertices, edges, triangles, with one and zero surviving index generators. This checks the actual bicategorical presentation, not just its abelianization. The extra generator is precisely the coherence lost by a strict colimit of one-object groups. □

F.16 (Proved). Sharp uniform depth bound. *Manuscript: MPL147; section11; folio75; scan152; EXIT P5a2.*

Let L be the boundary of the real oriented blow-up of a smooth normal-crossings pair, and let $L_{\leq r}$ be its part over points with at most r local branches. Include every incidence flag among the retained depths. For $r \geq 1$ the map $L_{\leq r} \to L$ is $(r-1)$-connected, with the convention that an m-connected map induces isomorphisms on π_i for $i < m$ and a surjection for $i = m$. Hence depths at most $k + 2$ compute $\tau_{\leq k} L$. This is the smallest uniform depth bound. The same assertion holds on the classifying spaces of analytic Deligne–Mumford stacks. In this last statement L means the actual oriented-blow-up boundary stack; identifying it with an intrinsic punctured-neighbourhood germ still requires the hypotheses of F.19.

Proof. At a depth-a point, the resolved boundary has a angular coordinates and a radial corner with at least one coordinate zero. Its depth-a locus has real codimension $a - 1$ in the boundary. Round corners and triangulate compatibly. The deleted strata for $L_{\leq r}$ therefore have codimension at least r. Relative general position moves a map of a compact polyhedron of dimension less than r off them, fixing any closed subpolyhedron already disjoint. Apply this first to representatives of homotopy groups and then to nullhomotopies. It gives precisely the asserted connectivity. Local finiteness suffices: the image of each such map meets only finitely many charts.

For sharpness take $m = r + 1$ coordinate hyperplanes in $\mathbb{C}^m$. The radial boundary is contractible, whereas its complement of the vertex retracts onto $\partial\Delta^{m-1}$; consequently

$$L \simeq (S^1)^{r+1}, \qquad L_{\leq r} \simeq (S^1)^{r+1} \times S^{r-1}.$$

For $r = 1$ the components differ. For $r > 1$ the factor S^{r-1} changes π_{r-1}. Thus $r \leq k + 1$ cannot determine every k-type. Exact integral boundary-matrix ranks for the simplex boundaries of dimensions zero through four were computed independently; the radial argument proves the formula in all dimensions. For a stack apply the manifold result to every term of an etale atlas nerve. Realization preserves an m-connectivity bound: relative CW models have cells only above degree m, and realization adds nonnegative simplicial degrees. This proves the assertion with isotropy retained. □

F.17 (Proved). Corollary 3 and the topoi that are truncated. *Manuscript: MPL147; section11; folio75; scan152; (247).*

Depths one, two and three, with every flag between them, compute $\Pi_1(L)$. If $L \to X^* = X \setminus D$ is a one-type equivalence, formula (247) has target $\Pi_1(X^*)$. The unstarred $\Pi_1(X)$ visibly written in scan 152 is false under that hypothesis. For the full *sheaf pro-topos* gluing of F.13 all nonempty depths must be retained; no dimension-independent cutoff computes it. The depth bound in F.16 applies to the associated shape, not to that topos.

Proof. The fundamental-groupoid functor is the reflection from spaces to one-types and preserves homotopy colimits, interpreted as bicategorical colimits of groupoids. Apply F.16 with $k = 1$. The counterexample to the unstarred formula is $(\mathbb{C}, \{0\})$: $L \simeq \mathbb{C}^* \simeq S^1$, whereas $\mathbb{C}$ is contractible. This identifies a mathematical correction, not an alternative reading of the ink. For full topoi a deleted deeper stratum is detected by sheaves supported on it; even its locale of opens has changed. Thus arbitrarily high depths cannot be removed in arbitrary dimension. Ordinary sheaf descent uses objects, arrows and the cocycle condition on triple overlaps; higher simplicial degrees are then forced. This fact about *simplicial coherence degree* does not permit omission of high-depth objects: a depth-three object already supplies the missing triangular relations in F.15. □

F.18 (Proved). Formula (254) in every dimension. *Manuscript: MPL147; section12; folio74 as transcription; scan156; (254).*

With the germ conventions of F.12, put $\mathcal{V} = \{\mathrm{Sh}(V)\}_{V \supset D}$ and $\mathcal{V}^* = \{\mathrm{Sh}(V \setminus D)\}_{V \supset D}$. Then in the pro-two-category of full Grothendieck topoi

$$\mathrm{Sh}(X) \simeq \mathrm{Sh}(X^*) \amalg_{\mathcal{V}^*} \mathcal{V}.$$

For locally conical pairs the analogous gluing holds for sheaves of spaces, and its shape is the homotopy pushout of the shapes of the three terms. In particular

$$\Pi_1 X \simeq \Pi_1 X^* \amalg_{\Pi_1 \mathcal{V}^*} \Pi_1 \mathcal{V}.$$

The two germ terms may be computed by the full flag diagrams of their elementary pieces; for the punctured boundary one-type F.17 gives the sharp depth-three replacement. For curves the closed term is the disjoint union of the boundary points and the punctured term the disjoint union of their meridian circles. This is the geometric form of STRATA C.2 and C.5, not merely its presheaf special case.

Proof. For every neighbourhood V, X^* and V form an open cover of X. Their sheaves are glued by identifying their restrictions to $V \setminus D$, with the ordinary cocycle condition; F.13 verifies this universally for geometric morphisms and passes to cofinal germs. For space-valued sheaves use the entire Cech nerve: descent identifies mapping spaces, not just their components. Applying this to a constant space K identifies maps from the shape with the homotopy limit of maps from the three shapes. This is exactly the homotopy-pushout universal property. Passing to one-types gives the displayed groupoid formula. The tube retraction to a point and the punctured-disc retraction to a circle give the curve case directly, without an asphericity assumption on X. The intrinsic two-open argument also applies to actual stack neighbourhoods. Computing those germs through an atlas requires the proper or common-stage controlled class of F.13; F.51 shows why the unrestricted atlas-germ substitution is invalid. □

6.5 Normal directions, tangential points and arithmetic tubes

F.19 (Proved). The canonical functor (243). *Manuscript: MPL147; section11; folios68-73; scans139-148; (231),(234),(239),(243).*

Let Y be a paracompact complex manifold with normal-crossings divisor D, and let $D' \to Y$ be a normalized locally immersed intersection of branches. Put $S = D'^*$, its open stratum, and let $P_S = \prod_{j \in J,S} N_j^{\times}$ be the product of the punctured normal line bundles. There is an equivalence

$$\Pi_1^{\mathrm{arcs}}(P_S) \longrightarrow \Pi_1^{\mathrm{rev}}(\mathcal{V}^*_{D',Y}),$$

canonical as a functor with its specified natural comparison isomorphisms. It is compatible with the generizations (231) and with (234), including the coherence for every longer flag. No

tubular embedding is part of its definition. The intrinsic covering-germ equivalence descends to the proper or common-stage controlled analytic etale stacks of F.13. For an arbitrary etale presentation the sector construction still defines a canonical coherent functor on its chartwise germs, but these need not compute the intrinsic stack-open neighbourhood germ. F.51 gives a parabolic quotient for which even the latter covering equivalence fails. The quotient $P_S/(\mathbb{R}_{>0})^J$ is the open face over S of the real oriented blow-up of (Y, D).

Proof. Define the oriented blow-up intrinsically: a point over y assigns a unit complex argument to each local defining equation of a branch, subject to $\arg(uf) = u(y)/|u(y)| \arg(f)$. A normal vector has the same transformation rule. Its modulus is forgotten by the quotient by positive real scalars; this gives the asserted face identification without coordinates or a metric.

Here is the functor on coverings, which makes canonicity explicit. Given a covering of a sufficiently small deleted neighbourhood and a normal direction $v \in P_S$, choose a simply connected angular sector about its arguments and a sufficiently small base neighbourhood in S. The fibre at v is the set of germs of sections on that sector with all normal moduli positive and sufficiently small. Shrinking sectors and radii does not change this set: in a normal crossings chart these regions are contractible, and each covering is a disjoint union of such regions. These sets form a covering of P_S. A path in P_S is covered by finitely many sectors; overlap identification transports the germ of a section. Subdivision of the path and homotopies give the same transport by uniqueness of lifting. Thus evaluation gives a fibre functor on covering germs at every v, and transport gives exactly (243).

In a chart $\Delta^a \times (\Delta^*)^{|J|}$ this operation identifies both covering categories with $\mathbb{Z}^{|J|}$-sets, with the same coordinate monodromies. It is fully faithful and essentially surjective there. Coverings and their isomorphisms satisfy descent, so these equivalences glue along all chart intersections. If D' meets further branches, apply the same construction with their coordinates also punctured. Pushing their positive radial coordinates slightly into the interior shows that its covering-germ category is unchanged upon replacing D' by S; in each chart both categories have exactly the same punctured coordinates and monodromy generators. This proves (239) in the stated, not merely smooth-divisor, case.

For a nested flag use sectors with all its branch coordinates at once. Successive restriction of a section germ is literal restriction, so both orders agree; this proves (231), (234) and all longer compatibilities before taking fundamental groupoids. On an etale atlas all these constructions pull back and retain the isotropy coherence. To identify the result with coverings of intrinsic stack neighbourhoods, use the synchronized descent of F.13 in its stated proper or controlled range. Without that hypothesis this argument constructs the atlas-germ functor only; it does not silently exchange an arbitrary infinite descent diagram with shrinking neighbourhoods. In dimension one the construction evaluates a cover on a germ of a sector tangent to a nonzero vector. The finite algebraic version is Deligne's specialization to the punctured tangent line (Deligne, 1989, §15.13–15.27). In a parameter t, it selects compatible roots of t and the fibre over the specified tangent vector; changing t by ut multiplies the root coordinate by the corresponding roots of $u(0)$. This is precisely the sector transition rule. Hence the analytic, profinite, and tangential-point functors coincide, with a path torsor rather than a falsely chosen canonical basepoint identification. The specified cote 140-4 pages of *Longue Marche* were subsequently recovered inside the LM8 background archive; F.48 supplies the literal manuscript cross-check. □

F.20 (Refuted). An invariant tubular embedding need not exist. *Manuscript: MPL147; section11; folio72; scan146.*

The matrix in MPL 147, fol. 72, scan 146 must have upper-right entry 1 for the example to work. Let the discrete group $\mathbb{Z}$ act on $\mathbb{C}^2$ by $(x, y) \mapsto (x + y, y)$ and let $D = \{y = 0\}$. There is no $\mathbb{Z}$-equivariant differentiable tubular embedding from the normal line bundle to $\mathbb{C}^2$ which

induces the identity on its normal quotient. This is an example for an analytic etale quotient stack with infinite isotropy, not for a proper finite-stabilizer orbifold.

Proof. The action on D and on its normal line is trivial. Differentiating such an embedding at $(0,0)$ would give an equivariant splitting of

$$0 \longrightarrow \mathbb{C}(1,0) \longrightarrow \mathbb{C}^2 \longrightarrow \mathbb{C} \longrightarrow 0.$$

A fixed vector (a,b) satisfies $(a+b,b) = (a,b)$, hence $b = 0$; none projects to 1. This rules out even a real differentiable splitting. The attached exact matrix calculation verifies this obstruction. Entry 0 would give the identity matrix and a splitting, so it would not be a counterexample. For a finite group, averaging an invariant metric supplies an equivariant smooth tubular neighbourhood; the infinite stabilizer is material. The sector construction in F.19 needs no such splitting and therefore survives. □

F.21 (Proved). The arithmetic normal tube and (37). *Manuscript: MPL149; section4; folios65v-67; scans132-135; (36)-(38).*

Let (X,D) be a smooth normal-crossings pair over a characteristic-zero field K, with X of finite type, and let $S = D_J^*$ be a connected open stratum on the normalization that labels its $r = |J|$ branches. Define its tame etale tube by the finite-covering category of the punctured henselian neighbourhood along S; equivalently use the Kummer etale normal tube. This is independent of the chosen etale charts. Its geometric profinite shape is that of $P_S = \prod_{j\in J,S} N_j^{\times}$. With geometric basepoints and paths specified, its fundamental groups have exact sequence

$$\pi_2^{\mathrm{et}}(S_{\bar K}) \xrightarrow{\partial} T^J = \widehat{\mathbb{Z}}(1)^J \longrightarrow \pi_1^{\mathrm{et}}((\mathcal{V}_S^*)_{\bar K}) \longrightarrow \pi_1^{\mathrm{et}}(S_{\bar K}) \longrightarrow 1.$$

Here π_2^{et} is the profinite homotopy group of the geometric etale shape, not the abstract completion of an unspecified ordinary group. The map ∂ is the tuple of normal-line Chern transgressions. Thus (37) is exact with D_J meaning the open stratum; its initial 1 is valid exactly when $\partial = 0$. Injectivity of $T^J \to \pi_1^{\mathrm{et}}(U_{\bar K})$ implies that vanishing. Arithmetic groups fit into the additional extensions over G_K; Galois acts cyclotomically on each labelled factor of T^J and permutes factors if branch labels have not been split.

Proof. Locally along S choose parameters $t_1,\dots,t_r$ for its branches. The henselian normal comparison sends a tame covering to its normal specialization. A finite covering is trivialized locally by adjoining roots $t_j^{1/n}$: normalization after these adjunctions is unramified, by the valuation form of Abhyankar's lemma. On overlapping charts $t_j' = u_j t_{\sigma(j)}$, the transition is the Kummer torsor of the unit $u_j|_S$. These are exactly the transitions of $N_j^{\times}$. This gives mutually inverse functors on finite covers and on their isomorphisms, with the same descent cocycles; it proves the assertion about the tube and its functor to U.

For the higher exact sequence, the root gerbes of the normal lines provide a precise profinite model. For each n, the product of gerbes of nth roots of the N_j has fibre $B\mu_n^J$ over S and is classified by the Kummer Chern classes $c_1(N_j) \bmod n$. Their compatible inverse system has fibre $B\widehat{\mathbb{Z}}(1)^J$. The preceding local calculation, now for finite locally constant truncated spaces by descent and their finite Postnikov towers, identifies it with the profinite normal-tube shape. Equivalently this is the local normal-bundle instance of the log comparison theorem (Carchedi et al., 2019, Theorem A and Corollary 5.5, preprint numbering). The homotopy exact sequence of this fibre sequence gives the displayed sequence, and the boundary maps of the root gerbes are precisely the Chern transgressions. Over $\mathbb{C}$ their evaluation on a sphere is the integer degree of the pulled-back normal line bundle; reduction modulo every n gives the profinite statement. Descent from a finitely generated field of definition and geometric extension

of constants preserves the finite-covering and finite-coefficient data, so the same argument applies over arbitrary characteristic-zero K.

The composite $T^J \to \pi_1 \mathcal{V}_S^* \to \pi_1 U$ kills $\operatorname{im} \partial$. If this composite is injective then $\partial = 0$, proving the requested direction. It need not vanish: for the zero section in $\operatorname{Tot} \mathcal{O}_{\mathbb{P}^1}(1)$, ∂ is an isomorphism and the punctured bundle is simply connected; for a smooth conic in $\mathbb{P}^2$, its normal bundle has degree four and the tube has fundamental group $\mathbb{Z}/4$, with kernel $4\widehat{\mathbb{Z}}$ in its meridian group. Finally take $X = \mathbb{A}^2$, $D = \{xy = 0\}$ and the branch $D_1 = \{x = 0\}$. The punctured tube has group $\widehat{\mathbb{Z}}^2$, its normal meridian has rank one, and $D_1^* = \mathbb{G}_m$ supplies the other factor. Replacing this open stratum by the closed $\mathbb{A}^1$ would make the last group trivial and break exactness. This also confirms the mathematical force of (36) and the definition of U_J as the trace of U on the localized neighbourhood in scans 132–133. □

F.22 (Refuted). Finite roots do not have a constant inertia kernel. *Manuscript: MPL149; section4; folios66-67; scans133-135; EXIT task5.*

Write $G = \pi_1^{\mathrm{tame}}(U_{\bar{K}})$ and $\phi : T^J \to G$. The Kummer etale site of (X, D) and the infinite root stack have this same group and hence the same kernel $\ker \phi$. For a finite root stack with orders n_i along the boundary components, put $N_{\mathbf{n}}$ equal to the closed normal subgroup of G generated by all n_ith powers of its corresponding boundary inertia groups. Then

$$\pi_1(\sqrt[\mathbf{n}]{(X, D)}) = G/N_{\mathbf{n}}, \qquad \ker(T^J \to \pi_1 \sqrt[\mathbf{n}]{(X, D)}) = \phi^{-1}(N_{\mathbf{n}}).$$

These kernels can be strictly larger than $\ker \phi$. Their intersection over all divisible root orders is $\ker \phi$. A finite Kummer *cover* likewise cannot remove a nonzero kernel, but this is a different assertion from equality for every finite root stack.

Proof. A finite etale cover of the finite root stack is a tame cover of U whose inertia orders divide the specified n_i. The assertion is checked in a root chart $[\operatorname{Spec} A[z_i]/(z_i^{n_i} - t_i)/\prod \mu_{n_i}]$: unramifiedness on the chart is precisely that inertia bound; the charts and their covers descend. Therefore its finite quotient actions are exactly those killing $N_{\mathbf{n}}$, proving the displayed formulas. Every finite quotient of G kills $N_{\mathbf{n}}$ for some common multiple of its finite inertia orders; thus $\bigcap_{\mathbf{n}} N_{\mathbf{n}} = 1$. All tame covers extend on the Kummer site by the same local normalization argument, proving the infinite-level assertion.

For an explicit counterexample take $(\mathbb{A}^1, \{0\})$. Here $G = T = \widehat{\mathbb{Z}}(1)$ and ϕ is the identity, while $\sqrt[n]{(\mathbb{A}^1, 0)} = [\mathbb{A}^1/\mu_n]$ has group μ_n. The kernel has changed from zero to $n\widehat{\mathbb{Z}}(1)$. Finite quotients of this calculation were checked by the attached program; the Kummer classification proves it in the inverse limit.

For a Kummer cover, the local lattice is a finite-index sublattice $T' \hookrightarrow T$, and its global group embeds as an open subgroup of G. Its kernel is the inverse image of $\ker \phi \cap T'$. For any nonzero closed subgroup $A \subset \widehat{\mathbb{Z}}^r$, and any finite-index sublattice T', choose a positive integer m with $mT \subset T'$. Multiplication by m is injective on T, so $0 \neq mA \subset A \cap T'$. Thus the kernel persists. This proves the valid content of EXIT_ONE_TYPE P1e-roots and refutes the stronger finite-root-stack sentence added in the present prompt. □

F.23 (Proved). The stable cusp quotient changes the model. *Manuscript: MPL149; section4; folios66-67; scans133-135; supplemental EXIT P1e-cusp-stack test, not a manuscript assertion.*

Over a characteristic-zero field containing μ_6, put $F = \{Y^2 = X^3 + 1\}$ and

$$\mathcal{X} = [F \times \mathbb{A}^1_s/\mu_6], \qquad \zeta(X, Y, s) = (\zeta^{-2}X, \zeta^{-3}Y, \zeta s).$$

Its boundary is $[F/\mu_6]$ at $s = 0$; its open part is the family $y^2 = x^3 + t$, $t \neq 0$, via $t = s^6, x = s^2X, y = s^3Y$. The normal meridian injects both discretely and profinitely into the

open fundamental group. Here groups are geometric; the discrete calculation uses the complex realization of this model defined over $\mathbb{Q}(\mu_6)$. The same holds for the full rank-two corner when the infinity section in the projective elliptic completion is retained as boundary. The original embedded cusp resolution has the exact nonzero node kernels $\mathbb{Z}(-3,1)$ and $\mathbb{Z}(-2,1)$, and their profinite counterparts, computed below. Thus the stable quotient removes explicit local meridian kernels by changing the model. It is not a finite root stack of that fixed resolved divisor.

Proof. The displayed substitutions verify the equation and the action is free when $s \neq 0$, so the asserted identification of the open part is an isomorphism. A loop in the normal coordinate s has degree six under $t = s^6$. Its composite with the degree homomorphism to $\pi_1(\mathbb{G}_m) = \mathbb{Z}$ is multiplication by six, which is injective; the same calculation on $\widehat{\mathbb{Z}}(1)$ proves profinite injectivity. No residual-finiteness inference from the discrete injection is being used.

The affine curve F is a once-punctured elliptic curve; its completion is elliptic. Both are aspherical, and their finite quotient stacks are aspherical by the fibration over $B\mu_6$. In the affine case the coarse boundary coordinate $w = Y^2 = X^3 + 1$ has cone orders two and three, giving group $C_2 * C_3$; the projective case adds the order-six point at infinity. Thus the quotient retains finite monodromy in boundary isotropy while retaining the infinite normal meridian in the open part. F.22 shows why merely extracting finite roots of the old boundary could not eliminate an existing kernel. Here is the explicit kernel in the old resolved model, so that the comparison with the stable quotient does not rely on the supplied EXIT calculation. In the affine plane put $f = y^2 - x^3$. Three successive blow-up charts give

$$f = x^2(v^2 - x), \qquad f = u^2v^3(v - u), \qquad f = u^6w^3(w - 1). \tag{F23.1}$$

The substitutions are respectively $y = xv$, $x = uv$, and $v = uw$. In the second substitution the preceding coordinate v is retained; in the third the preceding u is retained. After the third blow-up, the strict cusp and the first two exceptional components meet the last exceptional component E_3 at three distinct points. The first two exceptional components E_1, E_2 each have only that one boundary intersection. Their self-intersections are $-3, -2$, since E_1 is blown up twice after its creation and E_2 once. The last component has square -1. The multiplicities of f on E_1, E_2, E_3 are $2, 3, 6$.

Write b, a, h for the positive normal meridians of these three components. The plumbing relation at the two exceptional tails is

$$h = b^3, \qquad h = a^2. \tag{F23.2}$$

To verify the signs and the relation, consider a rational component of square $-d$ meeting the remaining divisor only at q. Its normal line is $\mathcal{O}_{\mathbb{P}^1}(-d)$. With u a coordinate at q and z_∞ its normal coordinate there, a trivialization over $\mathbb{P}^1 \setminus \{q\} \simeq \mathbb{A}^1$ has normal coordinate $z_0 = u^d z_\infty$. Thus the positive loop in u with z_∞ fixed winds d times in z_0. Its base loop contracts in $\mathbb{A}^1$, so in the punctured normal tube it is the dth power of the component's normal meridian. The ordinary smooth tubular-neighbourhood theorem identifies these maps with the local plumbing maps into the divisor complement. Taking $d = 3, 2$ gives (F23.2); no equivariant tubular embedding is used.

The map $f : U \to \mathbb{G}_m$ sends b, a, h to $2, 3, 6$ in $\pi_1(\mathbb{C}^*) = \mathbb{Z}$. Hence a and b have infinite order, and the two full node maps have the exact kernels

$$\begin{aligned} \ker(\mathbb{Z}^2 \longrightarrow \pi_1 U, (r,s) \longmapsto b^r h^s) &= \mathbb{Z}(-3,1), \\ \ker(\mathbb{Z}^2 \longrightarrow \pi_1 U, (r,s) \longmapsto a^r h^s) &= \mathbb{Z}(-2,1). \end{aligned} \tag{F23.3}$$

Indeed (F23.2) gives one inclusion, and composition with f_* gives the reverse one. The same argument is exact after profinite completion: multiplication by 2 or 3 is injective on $\widehat{\mathbb{Z}}$, so the kernels are respectively $\widehat{\mathbb{Z}}(-3,1)$ and $\widehat{\mathbb{Z}}(-2,1)$. Thus the kernel that the stable quotient removes is explicit, even without computing the entire trefoil group. The polynomial identities,

self-intersection updates, and finite congruence certificates for these kernel equalities are executed in `code/montpellier/F23_cusp_certificates.py`, with its output attached.

There are two different uses of the projective completion. If only its vertical divisor is retained, the preceding multiplication-by-six argument proves the asserted rank-one injection. If the infinity section is also retained as boundary, the finite atlas is $(\overline{F} \times \mathbb{A}^1, \overline{F} \times \{0\} \cup \{\infty\} \times \mathbb{A}^1)$, with open part $F \times \mathbb{G}_m$. Its rank-two corner map is

$$\mathbb{Z}^2 \longrightarrow \pi_1(F) \times \mathbb{Z}, (r, s) \longmapsto (\delta^r, s), \tag{F23.4}$$

where $\pi_1(F)$ is free on two generators and δ is their commutator, the meridian of the missing point. A reduced commutator has infinite order in this free group, so (F23.4) is injective. Its profinite cyclic factor is injective as well: for every n send the two free generators to the elementary generators of the finite upper-unitriangular Heisenberg group modulo n; their commutator has exact order n. Consequently the map $\widehat{\mathbb{Z}} \to \widehat{\pi_1(F)}$ generated by δ has zero kernel. The product lattice therefore injects profinitely. The finite atlas is etale over the quotient stack and over its open part, so injection into the open atlas group remains injection into the open quotient group. This proves the full corner assertion for the marked projective model, rather than identifying a pointed completion with its unmarked version.

The construction uses inversion of 2 and 3; nothing here asserts its validity for wild reduction at those primes. □

F.24 (Proved). The tame constructible topos and its descriptive system. *Manuscript: MPL147; section12; folios85-85v; scans179-180; MPL149 section4 folio67 scan135.*

For a finite normal-crossings stratification over a characteristic-zero field, finite etale sheaves constructible on its strata are finite continuous representations of the profinite completion of its ordinary exit category. The Ind-category of these finite representations is the corresponding coherent constructible topos. Its stratum groups are the geometric etale fundamental groups, with arithmetic descent over G_K; its crossing sets and compositions are the profinite versions of the coset and flag constructions in F.10. The groupoids of punctured normal tubes are exactly the extensions in F.21, and the maps obtained by forgetting branch flags are (38). These identify the claimed "1-descriptif" system without suppressing its kernels or its commutation isomorphisms.

The arbitrary-product condition in F.8 is a characterization for *discrete* groupoid presheaf topoi. It is false as a necessary condition for this profinite variant.

Proof. Over $\mathbb{C}$, finite constructible sheaves are recovered by their finite stalks, path transport on each stratum, and the sector specialization maps of F.19. The inverse construction glues constant finite sheaves on the sectors; compatibility on double intersections and on triple flags gives ordinary sheaf descent. Thus it is exactly the set-valued case of F.10 with finite fibres. Every such finite representation factors through a finite quotient of the category: for each pair of the finitely many stratum basepoints, identify arrows with the same map on all the specified finite fibres, and identify endomorphisms with the same permutations. Composition descends because these are a congruence. Taking all finite representations and their finite products gives the cofiltered system of finite quotient categories and proves the continuous-representation assertion. The same construction for a finite family of representations proves that the category is a pretopos, whose Ind-category is the asserted coherent topos.

Analytification identifies the algebraic finite constructible sheaves with these analytic ones, including their gluing functors, by (Barwick et al., 2020, Proposition 12.6.4 and Corollary 12.6.7). More explicitly, comparison holds on each smooth stratum for finite local systems, and on its punctured normal charts by the Kummer calculation of F.21; induction over the finite closed filtration identifies the recollement triples, so it holds on the whole pair. Descent from a field of definition and finite separable extension of constants gives the arithmetic groupoids and their

weak G_K-action as in F.1. The descriptions of crossing sets and compositions follow by applying finite stalk functors to the same sector and flag constructions. In positive characteristic the same assertion concerns tame sheaves and prime-to-characteristic local normal inertia; no wild cover is silently included.

To see why products must be treated differently, let G be a profinite group and $H \subset G$ a closed subgroup which is not open. In continuous discrete G-sets the product of a family is the subset of the set-theoretic product whose points have open stabilizer. Choose open subgroups $G_\alpha \supset H$ with intersection H. Each coset G_α in G/G_α is fixed by H. Their tuple is not in the continuous product, since its stabilizer is H. Therefore

$$\left(\prod_\alpha^{\mathrm{cont}} G/G_\alpha\right)^H \longrightarrow \prod_\alpha (G/G_\alpha)^H$$

is not surjective. This is the actual cusp gluing functor for a hyperbolic curve whenever its inertia subgroup has infinite index, for example the thrice-punctured projective line. Nevertheless its finite continuous exit description has just been proved. Finite representations and coherent descent, rather than an unchanged arbitrary-product test, give the valid profinite statement. □

F.25 (Refuted). Components of the analytic flag category need not be a category. *Manuscript: MPL147; section11; folios56-59; scans114-121; EXIT P4a1-P4a2.*

Even when the analytic incidence spaces satisfy all Segal identities, their sets of connected components need not do so. The correct object is the semisimplicial set of components of *all* strict flags, with repeated ranks adjoined if a simplicial object is desired.

Proof. Let $\tau = (12)(34)$ act freely on $\mathbb{C}^4 \times \mathbb{C}^*$ by permuting the coordinates and sending w to $-w$. Coordinate hyperplanes descend to a normal-crossings divisor in a smooth quotient. Every coordinate intersection times $\mathbb{C}^*$ is connected, so connected flag components are the τ-orbits of branch-set flags. Put $B = \{1, 2\}$. The upper pairs $(123, B)$ and $(124, B)$ are in the same orbit, and the lower pairs $(B, 1)$ and $(B, 2)$ are in the same orbit. However $(123, B, 1)$ and $(123, B, 2)$ are different orbits: applying τ changes 123 to 124. Their composites $(123, 1)$ and $(123, 2)$ are also different orbits. Thus a composable pair of component arrows has two possible composites. The attached orbit enumeration checks these claims. Forgetting entries of a complete flag always gives well-defined face maps and satisfies the semisimplicial identities before and after taking components; that is the stated replacement. Locally its realization is the barycentric subdivision of the simplex of branches, so its global realization is the usual coarse dual complex. The normalized flag spaces themselves, with stabilizers and their full fibre products, remain the correct geometric descent data. □

6.6 Elementary topoi and the two shapes

The intrinsic neighbourhood-germ statements in F.26–F.30 concern analytic spaces and the proper or common-stage controlled stack presentations specified in F.13 and F.19. The abstract exit-category and bar-construction statements are separate from that geometric identification. No assertion here replaces intrinsic germs by an uncontrolled nonproper atlas, as ruled out by F.51.

F.26 (Proved). The four elementary pieces and their products. *Manuscript: MPL147; section12; folios78-81; scans163-169; (257),(264)-(277).*

Over $(\mathbb{C}, \{0\})$ write $P = \mathrm{Sh}(\{0\})$, $A = \mathrm{Sh}(\mathbb{C}^*)$, $D = \{\mathrm{Sh}(\Delta_\epsilon)\}_\epsilon$ and $D^* = \{\mathrm{Sh}(\Delta_\epsilon^*)\}_\epsilon$. The canonical incidence maps *over the ambient line* are $P \to D$, $D^* \to D$, $D^* \to A$, their identities and composites with the ambient inclusions. These generate exactly diagram (265). There is no

incidence $P \to A$ over the line. Structural maps to the terminal topos exist, but they are not these ambient incidences. For I finite the elementary pieces are

$$E_{I_P,I_A,I_D,I_*} = P^{I_P} \times A^{I_A} \times D^{I_D} \times (D^*)^{I_*}, \qquad I = I_P \amalg I_A \amalg I_D \amalg I_*.$$

Their canonical incidence maps are the coordinatewise products of the preceding three maps. For $|I| = 2$ the complete list is

	P	A	D	D^*
P	PP	PA	PD	PD^*
A	AP	AA	AD	AD^*
D	DP	DA	DD	DD^*
D^*	D^*P	D^*A	D^*D	D^*D^*

where juxtaposition denotes the product, not an identification. The shape of such a piece is $(S^1)^{|I_A|+|I_*|}$, although its full sheaf pro-topos is generally not the classifying topos of that group.

Every canonical deleted germ (257), defined as the neighbourhood germ of Σ' in Σ'' followed by deletion of its trace of Σ, is the bicategorical colimit of its contained elementary pieces with their full incidence flags. This assertion uses full sheaf pro-topoi. For shapes it is a homotopy colimit, and not the shape of an arbitrary ordinary two-colimit of covering topoi. For stack presentations this intrinsic-germ assertion has the proper or common-stage controlled scope of F.13 and F.19.

Proof. The list is the independent choice of one of four states for each coordinate: fixed zero, globally nonzero, near zero, and near zero but nonzero. An inclusion over the ambient line can only fix zero inside its germ, or include a punctured germ into that germ or the nonzero line. This proves the generators and their coordinatewise product description; it also explains why an apparent arrow from P to $\mathbb{C}^*$ in the typographical list (264) cannot be an ambient inclusion. Polar retraction gives the displayed shapes.

At each positive radius the cover $\mathbb{C} = \mathbb{C}^* \cup \Delta_\epsilon$ has intersection Δ_ϵ^*. Its products form a finite cubical open cover, and their intersections are exactly the stated punctured and unpunctured coordinate pieces. Restriction to a coordinate closed subspace replaces its zero coordinates by P. Restriction to an admissible union, deletion of another admissible union, and taking neighbourhoods of a third are computed in the same coordinates; finite unions of tubes and their intersections give the corresponding subdiagram. Apply full sheaf descent, including its triple-overlap relations, and pass to the cofinal system of positive radius functions. This is F.13 applied to this explicit cover, and proves the assertion for (257). Applying space-valued descent gives the homotopy-colimit assertion by the mapping-space argument of F.18. On non-simple crossings in a Hausdorff analytic space use normalized branch charts and their full etale nerve; their fibre products supply the flag incidences and descend the same calculation. A stack atlas additionally requires the synchronization hypothesis in F.13; no unrestricted nonproper atlas-germ replacement is used. The order of taking a germ and deleting its support matters: the punctured germ of a nonempty stratum is not the germ of the empty stratum, exactly as scan 158 observes. □

F.27 (Proved). The Scholie and (295). *Manuscript: MPL147; section12; folio84; scans177-178; (295).*

For a general flag d_* and two of its ranks $d_0' \geq d_0''$, the elementary germ $\mathcal{V}^{d_0'}_{d_*,d_0''}$ has the homotopy type of the punctured graded normal bundle over $D^*_{d_*}$ with fibre $(\mathbb{C}^*)^{d_0'-d_0''}$. Its fundamental groupoid is canonically equivalent, with the sector comparison data, to that of this punctured bundle. This is (295). Normal-line twisting and branch monodromy must be retained.

Proof. The two nested normal subbundles give an exact sequence $0 \to N' \to N'' \to Q \to 0$ as in (287). Remove the coordinate hyperplanes from Q and take its inverse image in N''. The resulting map to $Q^\times$ is an affine bundle under the pullback of N'; its fibres are contractible. Local trivializations and a numerable cover show that this map is a homotopy equivalence, without choosing a global splitting. Restrict to $D^*_{d_*}$. The sector construction of F.19 compares its covering category with the elementary germ and is compatible with refinement of the flag. It therefore supplies the canonical equivalence on fundamental groupoids claimed in (295). The homotopy equivalence need not be a specified global analytic map, consistent with the warning at (292). A line bundle of nonzero Chern class already shows why one must keep the torus bundle rather than replace it by a product with the base. For a stack the normal sequence and its punctured quotient are formed on an atlas and descend; no invariant splitting is used. That vector-bundle descent is distinct from identifying the result with an intrinsic neighbourhood germ, for which the proper or common-stage controlled hypotheses of F.19 remain necessary. □

F.28 (Proved). Exactly what the Remarque recovers. *Manuscript: MPL147; section12; folios85-86; scans179-180; Remarque andRemords.*

Let $\mathcal{E} = \mathrm{Exit}(X, D)$ be the full exit infinity-category, with exits directed from deeper to more open strata. The system of fundamental groupoids of the elementary pieces, with every flag and its comparison isomorphisms, always recovers $\Pi_1 X$ and the topos of constructible *sets*, $\mathrm{Fun}(h\mathcal{E}, \mathbf{Set})$. If all open incidence strata are aspherical, the homotopy colimit of the *classifying spaces* of that same coherent system recovers the homotopy type of X and of every canonical germ. Finally $\mathcal{E}$ is equivalent to an ordinary category if and only if all these strata are aspherical and every full local meridian lattice injects into $\pi_1(U)$. Under both hypotheses the shape of the ordinary constructible topos is also the correct shape. Asphericity alone does not imply this last conclusion.

Proof. The first groupoid assertion follows from F.18 and the preservation of homotopy colimits by Π_1. For constructible sets, F.10 obtains crossing Hom-sets as the components of the endpoint homotopy fibres; only the fundamental groups of links and endpoints enter these components. Full flag spaces give composition. This reconstructs $h\mathcal{E}$ and its representation topos.

By F.27 every elementary link is a torus bundle over an incidence cover of an open stratum, with possible additional contractible factors. A finite cover of an aspherical space is aspherical, and the long exact homotopy sequence of a torus bundle shows that its total space is aspherical. Thus the natural maps from all diagram values to their $B\Pi_1$ are equivalences, coherently with every map and homotopy. F.26 now proves the second assertion.

For the criterion, endomorphism spaces in $\mathcal{E}$ are loop spaces of strata; they are homotopy-discrete exactly when the strata are aspherical. For incident S, T, choose its incidence cover $F_{ST} \to S$ and normal link L_{ST} with torus fibre. Normal sectors give

$$\mathrm{Map}_{\mathcal{E}}(s, t) \simeq \mathrm{hofib}_{(s,t)}(L_{ST} \to S \times T).$$

If the strata are aspherical, so is L_{ST}. A component of this homotopy fibre is the classifying space of the kernel of $\pi_1 L_{ST} \to \pi_1 S \times \pi_1 T$. The incidence cover injects on π_1, so this kernel is precisely that of the relative normal lattice in $\pi_1 T$. Hence all mapping spaces are zero-types exactly when all relative meridian maps are injective.

It remains to justify that full injection suffices for every relative map. Partition the local branches as $I = J \amalg K$, with the J branches remaining zero on T. Push the full torus into the normal torus bundle over T. If $v \in \mathbb{Z}^K$ dies in $\pi_1 T$, exactness lifts it to the image of some $w \in \mathbb{Z}^J$ in that normal bundle. Then $v - w$ dies in $\pi_1 U$. Full injection forces $v - w = 0$, and disjoint coordinate supports force $v = w = 0$. This step does not presuppose asphericity or injection of the intermediate normal lattice. Taking $T = U$ proves the converse. It proves the criterion, including all incidence covers and isotropy. □

F.29 (Proved). Shapes of the two constructible topoi and the first obstruction. *Manuscript: MPL147; section12; folios85-86; scans179-180; EXIT P1d3.*

The genuine constructible infinity-topos has shape $B\mathcal{E} \simeq X$; the infinity-topos associated with the ordinary constructible-set topos has shape $B(h\mathcal{E})$. Their comparison $q : X \to B(h\mathcal{E})$ is two-connected. If every mapping-space component is $(r-1)$-connected, $r \geq 1$, it is $(r+1)$-connected. For its homotopy fibre F, its first possible homotopy group is

$$\pi_{r+1}F \cong \operatorname{coker}\left[H_r(\widetilde{V}_2;\mathbb{Z}) \xrightarrow{d_0-d_1+d_2} H_r(\widetilde{V}_1;\mathbb{Z})\right].$$

Here V_p is the bar construction of a well-based enriched model of $\mathcal{E}$ with discrete objects, and tildes denote pullback of the fundamental-groupoid local system of paths on the chosen component of $B(h\mathcal{E})$. The formula is equivariant for that component's fundamental group.

Proof. In a presheaf infinity-topos on a category C, global sections of the constant space K are $\lim_C K = \operatorname{Map}(BC, K)$; this proves the shape assertions. The exit category groupoidifies to X by conical sector descent, or by F.26 with its space-valued coherence. Replace the exit category by a cofibrant enriched model and put

$$V_p = \coprod_{x_0,\dots,x_p} \prod_{i=1}^{p} \operatorname{Map}(x_{i-1}, x_i), \qquad W_p = \pi_0 V_p.$$

The degree-zero map is an isomorphism. In positive simplicial degree, $V_p \to W_p$ is r-connected. A relative cell first occurring there has dimension at least $r+1$, and realization adds the positive simplicial degree; the first relative cells thus have dimension $r+2$. This proves $(r+1)$-connectivity, and $r = 1$ always applies.

Pull back the universal path local system of $|W|$ to both bars; faces transport the lift along the corresponding edge. Its relative homology spectral sequence is

$$E^1_{p,j} = H_j(\widetilde{W}_p, \widetilde{V}_p;\mathbb{Z}) \Longrightarrow H_{p+j}(\widetilde{|W|}, \widetilde{|V|};\mathbb{Z}).$$

Column zero vanishes and all rows $j \leq r$ vanish. In the first possible total degree $r+2$, the only position is $(1, r+1)$, with $H_{r+1}(\widetilde{W}_p, \widetilde{V}_p) = H_r(\widetilde{V}_p)$. Its incoming differential is the three-face map displayed above; higher differentials have zero source or target. Relative Hurewicz and the exact sequence of the homotopy fibre identify this cokernel with $\pi_{r+1}F$. This supplies the topos version of EXIT_ONE_TYPE P1d3 rather than merely referring to it. □

F.30 (Proved). The gluing question and loss of homotopy. *Manuscript: MPL147; section12; folio85; scans179-180; closing question.*

For a two-stratum conical space with closed stratum S, open stratum T and link L, the genuine constructible shape is $S \amalg^h_L T$. When S, T, L are aspherical with groups A, B, G, the ordinary constructible shape replaces G in this pushout by $\operatorname{im}(G \to A \times B)$. Therefore the two shapes can differ. For $(\mathbb{P}^1, \infty)$ they are respectively S^2 and a point. For $(\mathbb{P}^n, \mathbb{P}^{n-1})$ the comparison fibre is $\mathbb{P}^n$; for the square of $(\mathbb{P}^1, \infty)$ it is $S^2 \times S^2$. The first nonzero comparison-fibre group can occur in arbitrarily high degrees, even for smooth algebraic pairs.

For a presheaf topos on C with a full sieve A and complementary full subcategory B, let E be the category of crossing arrows $a \to b$, with commuting squares as morphisms. Then its shape is

$$BC \simeq BA \amalg^h_{BE} BB.$$

This is the unconditional categorical answer to the question at scan 180. For full geometric topoi the link is an actual topos if a specified collared link exists; otherwise the deleted-neighbourhood pro-topos is the precise object, as in F.12–F.18.

Proof. The conical mapping cylinder of $L \to S$ is a neighbourhood of S; open descent with T gives the first pushout. In the aspherical case the crossing space is the homotopy fibre of $BG \to B(A \times B)$. Its components are cosets of the image, and their homotopy quotient by $A \times B$ is the classifying space of the image stabilizer. Passing to the ordinary exit category keeps only these components, which proves the replacement rule. For the projective-line pair both endpoints are contractible and the link is S^1. The ordinary category is [1] and has contractible nerve. For the hyperplane pair the open and closed strata have trivial fundamental groups, and the Hopf-link endpoint fibre is connected; again $h\mathcal{E} = [1]$. Products of exit categories are computed coordinatewise, giving $[1] \times [1]$ for the square. These prove the three fibre assertions.

For arbitrary high degree take the smooth affine quadric $Q_m = \{\sum_{i=0}^{m} z_i^2 = 1\}$, $m \geq 2$, and the pair $(Q_m \times \mathbb{C}, Q_m \times \{0\})$. Writing $z = x + iy$ identifies Q_m with TS^m up to the harmless positive rescaling $|x|^2 = 1 + |y|^2$, $x \cdot y = 0$; scaling y to zero retracts it onto S^m. Its ordinary fundamental group is trivial. The second factor's ordinary exit category has contractible classifying space, so $B(h\mathcal{E})$ is contractible while the genuine shape is S^m. The first discrepancy is therefore degree m, as asserted.

For the categorical statement, a nondegenerate string of arrows in C crosses from the sieve A to B at most once. Divide the nerve into strings lying in either side and strings with their unique crossing marked. The latter form the simplicial prism resolution of NE; the two prism ends are its source and target maps. Realizing this decomposition is the double mapping cylinder, which proves the pushout, including disconnected and empty cases. This checks STRATA C.3 and iterates to C.5. Its space-valued coefficient form is the lax pullback of restriction functors: objects are $(F_A, F_B, F_B \to i^* j_* F_A)$, and mapping spaces are homotopy pullbacks. Restriction and the units of the two adjunctions construct inverse functors, objectwise, proving STRATA C.6. For locally constant coefficients the gluing arrow is an equivalence; functors out of the just-proved homotopy pushout give the ordinary infinity-categorical pullback of STRATA C.8. One cannot first take homotopy categories: for the arrows $S^0 \to *$ and $* \to S^1$, the mapping-space components are $\operatorname{coker}(\mathbb{Z} \xrightarrow{(1,1)} \mathbb{Z}^2) = \mathbb{Z}$, whereas the ordinary comma category of homotopy categories has a single morphism. This directly checks STRATA C.7 and locates the lost higher coherence. □

6.7 F(h): local stratifications, flags, and controlled links

The topology used in this subsection is essential. A stratum has its subspace topology in a global chart; a global leaf of a local stratification has the topology obtained by gluing its local plaques. A neighbourhood germ remains a pro-object of *full* sheaf topoi. Neither taking connected components nor taking covering categories is part of that definition.

F.31 (Proved). With a counterexample to the sheaf assertion: the presheaf is separated, and its sheafification is the atlas of local stratifications. *Manuscript: MPL 147; section 13, folios 86-87; scans 181-184.*

Use conditions a)–d) of MPL 147, fol. 86, scans 181–182: a locally finite partition into locally closed, connected, locally connected strata; the frontier condition; and, for every stratum S and $x \in \overline{S}$, a fundamental system of neighbourhoods N of x for which $N \cap S$ is nonempty and connected. Restrictions split strata into connected components. These restrictions define a separated presheaf $\mathrm{Strat}_{\mathrm{glob}}$. It is not a sheaf, even if all the charts satisfy e). A section of its associated sheaf is exactly an open atlas of such partitions agreeing, after component splitting, on overlaps. Its leaves and all its local flag spaces are intrinsic to that section, rather than to a chosen atlas.

Proof. We first verify the restriction assertion, which is needed before asking whether the presheaf is separated. Let $U \subset X$ be open. A component C of $S \cap U$ is open in $S \cap U$ because S is locally connected, and is therefore locally closed in U. For $x \in U$, only finitely many original strata

meet a sufficiently small neighbourhood. For each such stratum whose closure contains x, choose $N_S \subset U$ with $N_S \cap S$ connected. This set lies in just one component of $S \cap U$. Intersect the finitely many N_S, and avoid the closures of the remaining strata. The resulting neighbourhood meets at most one component of each original stratum, proving local finiteness. If $x \in \overline{C}^{,U}$, every sufficiently small connected intersection $N \cap S$, with $N \subset U$, meets C and is contained in it. This proves condition d) for C.

Let $C \subset S \cap U$ and $E \subset R \cap U$ be components and suppose $x \in E \cap \overline{C}^{,U}$. The original frontier condition gives $R \subset \overline{S}$. Choose $N \subset U$ about x with $N \cap S$ connected, hence contained in C. For every $y \in E \cap N$, every neighbourhood of y inside N meets S, hence C. Thus $E \cap \overline{C}^{,U}$ is open in E; it is also closed and nonempty. Connectedness of E gives $E \subset \overline{C}^{,U}$. This is the frontier condition for the restricted partition. Successive component splitting agrees with direct restriction. Condition e), when imposed as a condition on local product germs, is preserved by the same restriction.

Suppose global partitions P, Q have the same restriction to each member of an open cover. Fix a stratum S of P. Its intersections with the cover split into connected plaques, open in S. Each plaque is also a plaque for Q. Two overlapping plaques belong to the same stratum of Q. The graph whose vertices are these plaques and whose edges are nonempty intersections is connected: a partition into two graph components would partition S into two nonempty disjoint open subsets. Consequently S lies in one Q-stratum. Interchanging P, Q proves equality of the partitions. This proves separatedness, answering the margin of scan 182 affirmatively. It also explains why the deleted word "separated" cannot be restored in the preceding sentence.

For failure of gluing, choose distinct $p, q \in S^1$ and put $U = S^1 \setminus \{q\}$, $V = S^1 \setminus \{p\}$. Stratify U by p and the two components of $U \setminus \{p\}$, and stratify V by its single stratum. Both restrictions to $U \cap V$ are its two connected components. Both charts satisfy a)–e). A global partition with these restrictions must have strata $\{p\}$ and $S^1 \setminus \{p\}$: the second follows from its connectedness and the preceding plaque argument. The germ of $S^1 \setminus \{p\}$ at p has two components, violating d). This is also the boundary phenomenon in the marked-disc picture of scan 181.

Because the presheaf is separated, its sheafification is its usual "plus" construction: a compatible atlas modulo common refinement. To construct the leaves, generate an equivalence relation on X by membership in a connected local plaque. A leaf is given the final topology for all its plaques. Plaques agree on open subplaques, so they are an atlas of locally closed embeddings into X. The leaf inclusion is a local immersion, and the disjoint union of leaves is the bijective local immersion $D_0^* \to X$ of (296)–(297). This construction commutes with open restriction. If X is compact Hausdorff, a finite collection of charts, each reduced to finitely many plaques, covers X; every leaf contains one of these plaques. There are therefore finitely many leaves, and they are locally compact. The Hausdorff condition is included here to justify the local compactness assertion of scan 184. □

F.32 (Proved). The germ flags form the nerve of an internal topological category in every degree. *Manuscript: MPL 147; section 13, folios 88-90; scans 185-189.*

For a chart V with strata S_i, order them by $S_i \le S_j \iff \overline{S}_i^{,V} \subset \overline{S}_j^{,V}$. The flag spaces of (299)–(303), with the literal topology specified there, are

$$D_r|_V = \coprod_{i_0 \le \cdots \le i_r} \overline{S}_{i_0}^{,V}, \qquad D_r^*|_V = \coprod_{i_0 \le \cdots \le i_r} S_{i_0}. \tag{F.h.1}$$

These formulas glue canonically for every local stratification of F.31. Every projection $D_r \to X$ is a local immersion; $D_r^* \subset D_r$ is open. Deleting or repeating entries gives continuous face and degeneracy maps. For every $r \ge 2$ the Segal map is a homeomorphism

$$D_r \xrightarrow{\sim} \underbrace{D_1 \times_{D_0} \cdots \times_{D_0} D_1}_{r \text{ factors}}. \tag{F.h.2}$$

Thus $D_\bullet$ is exactly the nerve of an internal category in topological spaces. Its fibre over x is the nerve of the finite poset of incident stratum germs at x. The first-vertex map $D_r \to D_0$ is a finite covering locally on D_0; in particular it has the asserted property over D_0^*. Under the literal model (F.h.1), this property over all of D_0 is not special to normal crossings.

Proof. Let $W \subset V$ be open and $x \in \overline{S}_i \cap W$. Condition d) selects a unique component C_i of $S_i \cap W$ whose closure contains x: a small connected $N \cap S_i$ lies in such a component, and any other component accumulating at x would meet that same N. The choice is locally constant as x moves in $\overline{S}_i \cap W$. Indeed $N \cap S_i \subset C_i$ implies $N \cap \overline{S}_i \subset \overline{C}_i$. For a chain $S_{i_0} \leq \cdots \leq S_{i_r}$ these uniquely selected components form a chain: near x, the lower component is contained in the closure of the higher one, and the restricted frontier condition proved in F.31 extends this inclusion over the entire connected lower component. This gives a canonical homeomorphism between the restriction of (F.h.1) and the formula computed after component splitting on W. The identifications compose exactly, because they select the same germs. They therefore glue on an arbitrary atlas and prove independence of that atlas. This also gives the precise sense in which the associated sheaf is described by local flags: an atlas defines compatible germ flags, and the degree-zero strict pieces recover its local plaques. One must retain the topology and the maps to X; bare incidence posets do not determine a local stratification.

Deleting the first entry of a chain uses the inclusion $\overline{S}_{i_0} \hookrightarrow \overline{S}_{i_1}$; deleting any other entry uses the identity on the first closure. Repetition also uses the identity. All are continuous and commute with chart changes. The simplicial identities are the identities for deleting and repeating entries, before any topological quotient is taken.

In the iterated fibre product in (F.h.2), composable arrows have the same origin x and the same middle *germ*. Their concatenation is a unique chain. On the component indexed by this chain, the fibre product is

$$\overline{S}_{i_0} \times_{\overline{S}_{i_1}} \overline{S}_{i_1} \times_{\overline{S}_{i_2}} \cdots \times_{\overline{S}_{i_{r-1}}} \overline{S}_{i_{r-1}} \cong \overline{S}_{i_0}$$

with its subspace topology. This proves a homeomorphism, not just a bijection, and works for every r. Composition is concatenation followed by forgetting the intermediate entry; associativity and units follow from the same identities. The strict spaces alone do not form this category, since deleting the initial stratum need not preserve strictness.

Fix a chart stratum S_i. Every stratum above it has closure containing any selected point of S_i, so local finiteness implies that there are only finitely many such strata. Over the copy of $\overline{S}_i$ in D_0, the first-vertex map is a disjoint union of a finite number of identity maps, one for every chain beginning with i. This proves the covering assertion, including at boundary points of that copy. Thus the aside in scan 189 claiming that extension from D_0^* to all D_0 is special to normal crossings is not a consequence of the definitions as written. The manifolds and the Boolean incident posets of normal crossings *are* additional properties. Likewise the phrase about degeneracies giving isomorphisms at scan 187 means isomorphisms onto the equal-adjacent-entry summands, not $D_r \cong D_{r+1}$: the two-element chain already has two vertices and three arrows. Passing to π_0 before the fibre products would invalidate this proof; F.25 addresses that separate error. □

F.33 (Refuted). Product germs in e) do not imply a stationary link or a conical tube. *Manuscript: MPL 147; section 13, folios 86 and 90; scans 181-182 and 190.*

There is a compact metrizable space with two strata satisfying a)–e) whose punctured neighbourhood pro-homology at the closed stratum is not constant. Hence e) alone cannot justify replacing its elementary neighbourhood pro-topoi by fixed link models. This does not obstruct the full pro-topos descent theorem below.

Proof. Construct an orientable surface M by starting with a disc and attaching successively compact genus-one cobordisms with two boundary circles. Choose a compact exhaustion K_n by connected genus-n subsurfaces with one boundary circle and $K_n \subset \operatorname{int} K_{n+1}$. Then M has one end and infinite genus. Let $X = M \cup \{\infty\}$ be its one-point compactification. It is compact metrizable and is stratified by M and $\{\infty\}$. Both strata are connected and locally connected; the partition is finite and satisfies the frontier condition. The sets $T_n = M \setminus K_n$ are a connected cofinal system of punctured neighbourhoods of ∞, proving d). Along M, e) is the product with a point. Along the point ∞, e) is the tautological product $\{\infty\} \times (X, \infty)$. Its transverse stratum has connected germ. Thus even the asserted transverse connectedness in scan 182 holds.

Choose the two handle generators a_j, b_j in the jth cobordism and let c_n be its inner boundary class. Cutting off after the mth handle gives a genus-$(m-n)$ surface with two boundary components, whose first homology has basis $c_n, a_{n+1}, b_{n+1}, \dots, a_m, b_m$. Passing to the union yields

$$H_1(T_n; \mathbb{Z}) = \mathbb{Z}c_n \oplus \bigoplus_{j>n} (\mathbb{Z}a_j \oplus \mathbb{Z}b_j).$$

Inclusion $T_m \hookrightarrow T_n$, $m > n$, sends c_m to c_n and preserves the surviving handle generators. Its image is

$$\mathbb{Z}c_n \oplus \bigoplus_{j>m} (\mathbb{Z}a_j \oplus \mathbb{Z}b_j), \tag{F.h.3}$$

which strictly decreases with every m.

For completeness, an inverse system isomorphic in pro-abelian groups to a constant group has eventually stationary images at each fixed stage. Write such inverse isomorphisms as $f : A_n \to B$ and compatible $g_m : B \to A_m$ after choosing a representative of f. The equation $gf = 1$ says that for sufficiently large k the transition $A_k \to A_m$ factors through g_m; the compatibility $g_m = (A_k \to A_m)g_k$ gives the reverse inclusion of images. Hence the eventual image equals $g_m(B)$. Formula (F.h.3) contradicts this conclusion. A conical punctured tube $L \times (0, \epsilon)$ with compatible shrinkings would have constant pro-homology $H_1(L)$, so no such link model exists here. A finite truncation of the image calculation is reproduced in the accompanying program; the argument above proves strict decrease at every stage. □

F.34 (Proved). Elementary pro-topoi, their gluing, and the precise additional link hypotheses. *Manuscript: MPL 147; section 13, folio 90; scan 190; gluing scans 150 and 155.*

There are two different assertions. First, full sheaf descent holds for neighbourhood *pro-topoi* without a conical or asphericity assumption. In an ambient open Y where a locally closed set Z is closed, define

$$\mathfrak{T}_{Z,Y} = (\operatorname{Sh}(N))_{N \supset Z}, \qquad \mathfrak{L}_{Z,Y;W} = (\operatorname{Sh}(N \cap W))_{N \supset Z}, \tag{F.h.4}$$

where W is a specified open part of Y; in particular $W = Y \setminus Z$ gives the punctured tube. Restriction to local stratification charts and all their overlaps defines the same objects for local plaques. For an incident flag $\sigma = (S_0 < \cdots < S_r)$, its elementary pro-topoi are the iterated neighbourhood objects

$$\mathfrak{E}_{\sigma,W} = \big(\operatorname{Sh}(W \cap N_0 \cap \cdots \cap N_r)\big)_{(N_0,\dots,N_r)}, \tag{F.h.5}$$

where N_j ranges over neighbourhoods of the indicated plaque S_j in the chart, and W is an independently specified locally closed union of strata, with its *subspace topology*. It need not be open in the ambient chart. The empty flag gives the constant topos $\operatorname{Sh}(W)$. One may take $W = S_r$, in which case the last neighbourhood condition is redundant, or take a stratum closure, or the specified punctured part of such a closure. Ambient open restrictions make every closed embedding in this expression legitimate. In a Cech or flag diagram W is held fixed. Refining neighbourhoods and deleting a neighbourhood condition give the comparison maps; a specified

inclusion $W \subset W'$ gives a further comparison. There is no inclusion from one disjoint stratum into another when a terminal vertex is deleted. Conical retractions, when available, are separate geometric maps and retain their specified control data. Common refinement makes the restriction identities strict. These definitions do not assert that $\mathfrak{E}_{\sigma,W}$ is constant.

Second, to model these objects by locally trivial link bundles, it is sufficient to assume a locally compact paracompact Hausdorff space of finite depth with compatible conical control data: local manifold plaques S, tubes T_S, retractions π_S, and radius functions ρ_S with $\rho_S^{-1}(0) = S$, locally identified over S with

$$S \times C(L_S), \qquad T_S \setminus S \cong L_S^{\mathrm{bun}} \times_S (S \times (0, \epsilon_S)). \tag{F.h.6}$$

Here L_S is compact and recursively locally contractible and conically stratified; $L_S^{\mathrm{bun}} \to S$ is a locally trivial bundle, so monodromy is retained. Changes of charts preserve the strata, cone radius, and these control data. For incident plaques $S < T$ require $\pi_S \pi_T = \pi_S$ and $\rho_S \pi_T = \rho_S$ wherever defined; tubes of incomparable plaques are disjoint after shrinking. The data may be given on a locally finite atlas with its full overlap nerve, which allows distinct local branches of one global leaf. This is an explicit sufficient hypothesis, not a conclusion from e). Whitney stratifications in the usual locally compact smooth setting provide such controlled tube germs by the control-data and isotopy results of Mather (1970, §7 and Corollaries 10.2–10.6)(Mather, 1970).

With these data, the full Cech diagram of tubes covering a closed union of strata, or its subdivision by all flags, has the required two-colimit in pro-Grothendieck topoi. In particular, for a closed union $Z \subset Y$,

$$\mathrm{Sh}(Y) \;\simeq\; \mathrm{Sh}(Y \setminus Z) \amalg_{\mathfrak{L}_{Z,Y;Y\setminus Z}} \mathfrak{T}_{Z,Y} \quad \text{in pro-Grothendieck topoi.} \tag{F.h.7}$$

At the level of homotopy types these are homotopy colimits of the *full* link and iterated-link bundles. No $K(\pi, 1)$ assumption is required for this gluing. To replace every elementary homotopy type by its fundamental groupoid, the exact condition is that every such type be a 1-type. A sufficient geometric condition is that the base strata and all the relevant iterated-link fibres be 1-types; for connected components this means $K(\pi, 1)$ spaces. The reconstruction then uses the homotopy colimit of their classifying *spaces*, not an ordinary two-colimit of classifying topoi or groupoids.

Proof. At a fixed open-neighbourhood stage, an open cover $N = \bigcup N_a$ gives the usual sheaf descent equivalence, including morphisms: a sheaf on N is a family of sheaves on N_a with isomorphisms on $N_a \cap N_b$ satisfying the triple cocycle. For any target topos $\mathcal{A}$, apply this objectwise to inverse-image functors $\mathcal{A} \to \mathrm{Sh}(N_a)$. Their finite-limit and colimit preservation can be checked on the conservative open restriction family; hence the glued functor is again the inverse image of a geometric morphism. Natural transformations glue as well. This proves the bicategorical universal property for full topoi, not only for local systems.

The two-open case makes passage to pro-objects completely explicit. For a fixed target $\mathcal{A}$, filtered two-colimits of mapping *categories* commute with finite pseudolimits in Cat. This follows by representing the finitely many objects, arrows, invertible comparison arrows, and their coherence equalities at one common stage; the same argument handles arbitrary morphisms and equality of morphisms. Apply this to

$$\mathrm{Geom}(\mathrm{Sh}(Y), \mathcal{A}) \simeq \mathrm{Geom}(\mathrm{Sh}(Y \setminus Z), \mathcal{A}) \times^{\mathrm{ps}}_{\mathrm{Geom}(\mathrm{Sh}(N\setminus Z),\mathcal{A})} \mathrm{Geom}(\mathrm{Sh}(N), \mathcal{A}).$$

The left side and first factor are constant in N. Taking the filtered two-colimit gives precisely the mapping-out universal property of (F.h.7). For a pro-target $(\mathcal{A}_j)$, take the additional limit over j; pseudolimits commute with the displayed finite pseudolimits. This proves (F.h.7) for arbitrary Z as stated, without appealing to a putative inverse limit of punctured full sheaf topoi. The latter can be degenerate, as F.12 proves.

We next explain the geometric content of the extra hypothesis. On a conical chart, compactness of L_S implies that every neighbourhood of the cone vertex contains a full radial ball of some positive radius. Over each relatively small open subset of S, the same holds with a positive local bound. A locally finite partition of unity produces a positive continuous minorant of these bounds. Consequently tubes with positive continuous radii form a cofinal system of all tube germs. The control identities make these choices compatible on incident plaques. Choose the radii successively in the finite depth, reducing them to satisfy all earlier conditions. Only finitely many conditions occur near a given point. Incomparable tubes are disjoint, so a nonempty intersection of distinct tubes is indexed by an incident flag. It is exactly a stage of (F.h.5). Recursing inside L_S shows that its angular part is the corresponding iterated-link bundle; the remaining coordinates are positive radial intervals. This is proved inductively on depth from (F.h.6): an incident higher plaque is S times a stratum in L_S times the radial interval, and the next tube condition is its corresponding link-tube condition. There is no assertion that the angular bundle is a product of its fibres.

At every simultaneous radius stage, the tubes covering the given closed union give an actual open cover of their union, and their full intersections give its Cech nerve. Thus the preceding full-topos descent applies. Passing from ordered intersections to all incident flags is subdivision of the same descent datum: its one-step maps are restrictions and its triangles are the cocycle relations. It does not discard higher flags. For an infinite locally finite atlas, one must also justify synchronization of pro-stages. A datum and its isomorphisms are represented on individual charts, double overlaps, and triple overlaps. Choose a common refinement on each such finite family; at any point only finitely many bounds are imposed. A positive continuous minorant gives one global radius assignment satisfying all of them. The same argument applies to transformations and equality of transformations. This proves essential surjectivity and full faithfulness on the mapping categories, and justifies the pro-colimit without commuting a filtered colimit with an arbitrary infinite product. The same argument applies to the Cech groupoid of the stated locally finite Hausdorff chart cover when plaques have branch identifications: on compact chart pieces only finitely many overlap conditions occur. This is a chart presentation of the given space, not a claim about an arbitrary nonproper etale quotient atlas. These constructions establish the extension of both versions of F.13 to the stated controlled local stratifications.

For shapes, apply the open-cover homotopy-colimit statement at every stage. It can be proved directly by the thick Cech realization: a partition of unity subordinate to a numerable cover provides a section up to homotopy, and the simplex of possible partition weights over each point is contractible. Refining charts and radii preserves this comparison. A conical radial interval is contractible and its shrinkings are homotopy equivalences, so the iterated-link bundles give constant models for the resulting local *pro-homotopy types*. They do not give equivalences of full sheaf topoi: even $\mathrm{Sh}((0,\epsilon))$ is not Set. The homotopy long exact sequence of each locally trivial link bundle shows that an aspherical fibre over an aspherical base has aspherical total space, proving the sufficient 1-type criterion. Conversely a space with nonzero homotopy above degree one cannot be replaced by the classifying space of its fundamental groupoid. This is precisely the information discarded by a level-one replacement.

The categorical cases are STRATA C.3, C.5, and C.6. The relevant calculation in C.6 is independently checkable: for a full sieve $A \subset C$ and complementary cosieve B, restriction identifies a presheaf with $(F_A, F_B, F_B \to i^*j_*F_A)$, where $(i^*j_*F_A)(b) = \lim_{(A/b)^{\mathrm{op}}} F_A$. The inverse sends this triple to $j_*F_A \times_{i_*i^*j_*F_A} i_*F_B$, as is verified on A and B separately. With spaces one must take the coherent limit and pullback. Repeating this argument along a finite filtration gives the actual crossing-flag data of C.5; its shape is the homotopy pushout of C.3. This verifies the categorical pattern but is not substituted for the full-topos geometric descent just proved.

Finally consider the two requested tests. For $(S^2, \{p\})$, take N a small open disc and $U = S^2 \setminus \{p\} \cong \mathbb{R}^2$. The punctured tube is $S^1 \times (0,\epsilon)$, so (F.h.7) has shape

$$\mathrm{hocolim}(* \leftarrow S^1 \longrightarrow *) \simeq S^2.$$

Both open and closed strata are simply connected. The topos of sheaves constructible for these two strata is the arrow topos: the local-system gluing functor sends a set to that same set, since the link is connected and the local system on U is constant. Its classifying shape is contractible. Similarly the ordinary groupoid pushout of the displayed diagram is trivial. The full diagram with its link retains the missing $H_2(S^2;\mathbb{Z}) = \mathbb{Z}$. Thus even aspherical elementary pieces do not license an ordinary level-one colimit.

The drawing at scan 190 is three half-lines in the *plane*: there are seven strata, namely the origin, three open rays, and three open sectors. The origin's link is a circle divided into three points and three open arcs. The incident poset has seven elements, twelve strict comparable pairs, and six strict three-element chains. Removing the origin gives the six-vertex, six-edge incidence cycle, with first Betti number one. Adjoining the origin cones off that cycle, as the disc tube does. The three ray links are two-point spaces and the sector links are empty. All these elementary angular pieces are 1-types, but the seven-element incident poset is not Boolean, since seven is not a power of two. Treating the picture as only a tripod would remove the three sectors and give a three-point link, contradicting both the written count and the topology. The program checks the flag counts, the Segal bijections through degree six, and the circle and sphere homology calculations; the proof of (F.h.2) establishes all degrees. □

6.8 Additional arithmetic tests and coherent limits

F.35 (Proved). Inertia injection and extension as a sub-germ. *Manuscript: Additional F(h)(1); MPL149 section4 folios66-67 scans133-135.*

Let X/K be smooth in characteristic zero, D a normal-crossings divisor, $U = X \setminus D$, and $\bar{x}$ a geometric point of D. Choose a geometric component of U and a path from its base point to the strict local punctured neighbourhood at $\bar{x}$. Write

$$H = \widehat{\mathbb{Z}}(1)^{I(\bar{x})}, \qquad G = \pi_1(U_{\overline{K}}), \qquad \iota : H \longrightarrow G.$$

Then ι is injective if and only if every finite tame covering germ at $\bar{x}$ is an open-and-closed sub-germ of the restriction of some finite etale covering of $U_{\overline{K}}$. Equivalently, that covering may be taken over U after allowing a finite extension of the constants and then composing its covering with $U_{K'} \to U$. The statement does not assert that the entire restriction of the global covering is the prescribed germ. This proves the first additional test F(h).

Proof. We prove the group-theoretic assertion for any continuous homomorphism $\iota : H \to G$ of profinite groups. If it is injective, identify the compact H with a closed subgroup of G. A transitive finite H-set is H/K for an open subgroup $K \subset H$. The compact subset $H \setminus K$ does not contain 1, so some open normal $N \lhd G$ satisfies $N \cap H \subset K$. Put $J = KN$. Since N is normal, J is a subgroup; since it contains N, it is open. Also

$$J \cap H = K : \quad kn \in H \Longrightarrow n \in H \cap N \subset K.$$

Consequently $hK \mapsto hJ$ embeds H/K into the restriction of the finite G-set G/J. Apply this to each of the finitely many orbits of an arbitrary finite H-set. Conversely, if $1 \neq h \in \ker \iota$, some finite continuous quotient of H detects h. Its regular H-set cannot embed in a restricted G-set, on which h acts trivially. This proves the equivalence.

Finite tame germs are finite continuous H-sets by the strict local Kummer calculation of F.21. Finite etale coverings of the geometric component of U are finite G-sets. An invariant subset of a finite covering is precisely an open-and-closed subcover, so the group assertion has exactly the stated geometric interpretation. The global covering and its finite presentation descend to a finite extension K'/K; in characteristic zero this extension is separable. Composition with the finite etale map $U_{K'} \to U$ gives the stated version over K. The comparison remains geometric: for a nonclosed point one must not substitute the whole arithmetic decomposition group for H. □

F.36 (Proved). The coherent etale limit is nondegenerate. *Manuscript: Supplemental LM8 coherent-etale-limit question; context MPL147 sections11-12 folios70-81 scans143-148 and157-169; MPL149 section4 folios66-67 scans133-135.*

Let x be a closed point on a smooth curve over an algebraically closed field k of characteristic zero. Let $R = \mathcal{O}^{\mathrm{sh}}_{X,x}$, $F = \operatorname{Frac} R$, and let $(V_i, \bar{x}_i)$ run through pointed affine etale neighbourhoods of $\bar{x}$, shrunk so that a local parameter t is defined and its only zero in V_i lies over x. Put $V_i^* = V_i[1/t]$. Then

$$\varprojlim_i (V_i^*)_{\mathrm{et}} \simeq (\operatorname{Spec} F)_{\mathrm{et}} \simeq B\widehat{\mathbb{Z}}(1)$$

as a bicategorical limit of Grothendieck topoi. The hypotheses of the coherent-limit assertion in the LM8 intake are therefore sufficient. The exact general references are SGA 4, VI, Theorems 8.2.3 and 8.3.13, and VII, Lemma 5.6; the last statement is also the Stacks Project, Lemma 59.51.2 (Artin et al., 1972b; The Stacks Project Authors, 2026f).

Proof. The schemes V_i^* are affine, the transition maps are affine, and $\varinjlim_i \mathcal{O}(V_i) = R$. Localization commutes with that colimit, so their scheme limit is $\operatorname{Spec} R[1/t] = \operatorname{Spec} F$. An affine etale algebra of finite presentation over this limit is given by finitely many coefficients and finitely many equations; it descends to some V_i^*. The same finite-presentation argument descends morphisms and equalities. A covering can be replaced by a finite affine covering of each affine object, and surjectivity of that finite family descends after increasing i. Thus the restricted etale site of the limit is the filtered colimit of the restricted sites, including the covering families. Passing to sheaves gives exactly the inverse-topos universal property of VI 8.2.3.

Each stage is a coherent topos and every transition morphism is coherent. One can see the extra assertion of VI 8.3.13 directly here: a coherent object is the quotient of a finite diagram of coherent generators by a finitely presented equivalence relation. Generators, relation, and its finite reflexivity, symmetry and transitivity identities all descend to one stage. Morphisms descend by the same finite presentation. Hence coherent objects in the limit are the filtered two-colimit of coherent objects at the stages; no analytic infinite annulus covering is present.

The ring R is a strictly henselian discrete valuation ring. Its residue field has characteristic zero, so all finite extensions of F are tame. Every unit of R has an nth root by Hensel's lemma. The tame extension classification therefore gives the separable closure $\bigcup_n F(t^{1/n})$, with Galois group $\varprojlim_n \mu_n = \widehat{\mathbb{Z}}(1)$; the classification follows by removing the ramification index with $t^{1/n}$, after which the extension is unramified and hence trivial over a strictly henselian ring. This also follows from the tame valuation calculation in the Stacks Project, Lemma 15.116.2 (The Stacks Project Authors, 2026h). The topos is not degenerate: its terminal finite set is not its empty set. In residue characteristic p, the tame quotient is instead $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$ and the full group has wild inertia.

There is no conflict with F.12 or LM8 E.11. The classical shrinking-disc system has the annular covering that kills every cone. The etale system has finite coherent covering descent and the nonempty scheme limit $\operatorname{Spec} F$. MPL 147, scans 143–148 and 157–169, explicitly works with analytic neighbourhoods, arcs, and analytic multiplicities; its D, D^* are the classical pro-objects. The phrase "tubulures etales" of MPL 149, scans 133–135, proposes this different arithmetic construction. □

F.37 (Refuted). Pinned fullness over a finite field. *Manuscript: MPL149; section2; folio25v; scan52; taskF(b).*

Even after localization at universal homeomorphisms, the etale-topos functor on finite-type $\mathbb{F}_q$-schemes is not full onto pinned morphisms over $B\mathbb{F}_q$. This does not contradict F.4, which proves faithfulness. It also explains why the infinite-field hypothesis in (Carlson et al., 2024, Theorem 0.1) must not be silently dropped.

Proof. Let $X = \mathbb{A}^1_{\mathbb{F}_q}$, let $p : X_{\mathrm{et}} \to B\mathbb{F}_q$ be the structure morphism, and identify $\Gamma_{\mathbb{F}_q}$ with $\widehat{\mathbb{Z}}$. Since $\pi_1(X) \to \Gamma_{\mathbb{F}_q}$ is onto and the latter is abelian, every $a \in \widehat{\mathbb{Z}}$ gives a natural automorphism of p^*: on a finite constant-field covering it acts by a. Thus

$$(\mathrm{id}_{X_{\mathrm{et}}}, a) : X_{\mathrm{et}} \longrightarrow X_{\mathrm{et}} \quad \text{over } B\mathbb{F}_q$$

is a relative geometric morphism. Its underlying map of spaces is the identity, so it is pinned.

We claim that the identity geometric morphism of X_{et} has no nontrivial natural automorphism. Such an automorphism restricts to every nonempty open of X, and then to its generic point. Every finite separable extension of $\mathbb{F}_q(t)$ spreads to a finite etale cover of some such open, so restriction gives an element of $Z(G_{\mathbb{F}_q(t)})$. This restriction is injective. Indeed affine etale schemes are generators; an automorphism of such a representable which is generically the identity is the identity, because its source is reduced and its affine target is separated. Affine generators are essential here: one need not make a generic-detection assertion for arbitrary nonseparated etale objects.

To prove $Z(G_{\mathbb{F}_q(t)}) = 1$, let a central element z project to $\sigma \in \Gamma_{\mathbb{F}_q}$. Write $N = G_{\overline{\mathbb{F}}_q(t)}$. For any prime $\ell \neq p$, the Kummer classes of $t - b$, $b \in \overline{\mathbb{F}}_q$, are linearly independent in $\overline{\mathbb{F}}_q(t)^*/(\overline{\mathbb{F}}_q(t)^*)^\ell$, as their valuations at the distinct points b show. The conjugation outer action of σ on N permutes these classes by $b \mapsto \sigma(b)$, with the scalar factor prescribed by the action on μ_ℓ. If σ moves any b, this cannot be the identity on $\mathrm{Hom}(N, \mu_\ell)$: distinct valuation basis vectors cannot become one another by a nonzero scalar. Centrality therefore forces $\sigma = 1$. Hence $z \in Z(N)$, which is trivial by the valuation and Kummer proof of F.2, unchanged in characteristic p when its auxiliary prime is chosen different from p. We have proved the claim.

An isomorphism between the two relative maps indexed by a, b would now have to be the identity automorphism of the underlying identity map; compatibility with their base isomorphisms forces $a = b$. There are therefore uncountably many isomorphism classes of pinned relative maps of this form. In contrast, the category of finite-type schemes over a finite field has a countable skeleton and countable Hom-sets: all its affine presentations and finite gluing data use finite strings over a countable set. Its localization at universal homeomorphisms still has countable Hom-sets, since morphisms are represented by finite zigzags. It cannot surject onto this uncountable set of pinned morphisms. This is a complete cardinality counterexample; it does not require choosing an unspecified nonintegral a and asserting that particular one is nonalgebraic. Powers of Frobenius account for some, but not all, of the relative twists. □

F.38 (Known). The verified reconstruction range over larger bases. *Manuscript: MPL149; section2; folio25v; scan52.*

The following are the precise positive ranges used for MPL 149, scan 52. Over an infinite finitely generated field, use pinned morphisms and invert universal homeomorphisms (F.5). The same holds over every sub-p-adic field, meaning a subfield of a finitely generated extension of $\mathbb{Q}_p$. More generally, let S be noetherian, Nagata and geometrically unibranch, with sub-p-adic function field on every irreducible component. On finite-type S-schemes which are *componentwise universally open*, the localization at universal homeomorphisms is fully faithful into topoi over S_{et} with *chivalrous* morphisms: the induced continuous map sends constructible subsets to constructible subsets. In particular normal flat finite-type schemes over $\operatorname{Spec}\mathbb{Z}$ are reconstructed, with all their morphisms. These are Theorem 4.1, Proposition 6.11 and Theorem 6.12 of the opened September 20, 2026 version of Carlson–Stix (Carlson and Stix, 2026); the relative hypothesis is part of the theorem and is not a theorem about all schemes over S.

Proof. We spell out the reduction from the quoted statements to the formulation at scan 52. A universal homeomorphism induces an equivalence of small etale topoi, so the functor factors through the stated localization. Proposition 6.11 identifies that localization, in the componentwise

universally open category, with absolute weak normalizations. Theorem 6.12 identifies the Hom-set from X^{awn} to Y with the groupoid of chivalrous relative topoi morphisms; its rigidity assertion says that this groupoid is equivalent to that set, with no extra automorphisms. For a normal X in this category, $X = X^{\mathrm{awn}}$ by their Lemma 6.10. Thus the asserted Hom-equivalence follows, and applying it to a topos equivalence and its inverse gives inverse scheme morphisms by faithfulness. Flat finite-type schemes over $\mathbb{Z}$ satisfy the componentwise openness condition, giving the stated example.

Over a field all these schemes are componentwise universally open and finite-type field morphisms send closed points to closed points. Their Theorem 4.1, together with the seminormalization description of the localization in characteristic zero, gives the pinned field assertion. In the general-base theorem, pinned is the wrong substitute for chivalrous: for a discrete valuation ring R and nonunit $a \neq 0$, the closed point $(aT - 1)$ of $\mathbb{A}^1_R$ maps to the generic point of $\operatorname{Spec} R$. This genuine scheme morphism is chivalrous by constructibility of images, but not pinned. The proof of the relative theorem reconstructs over the generic fields and spreads out using normal quasi-sections; this is exactly where geometric unibranchness, Nagata finiteness and componentwise openness enter. Thus its application here does not discard any of those hypotheses.

Combining F.5–F.7 and F.37 gives the necessary qualifications to the provisional question: unrestricted fullness is false even over number fields; pinned fullness holds in the preceding ranges, fails over finite fields, and fails over algebraically closed characteristic-zero fields. Faithfulness of genuine scheme maps remains the all-characteristic statement F.4. The remaining general-field question is recorded precisely in F.44, not identified with the false finite-field assertion. □

F.39 (Proved). Stable towers, full inertia and cyclotomic normalization. *Manuscript: MPL149 section4 folios66-67 scans133-135; EXIT task6 andP3b rows.*

Let $U \to B$ be an elementary hyperbolic fibration over a field K finitely generated over $\mathbb{Q}$, with B a connected smooth finite-type K-variety: a smooth proper curve family with a nonempty finite etale marked divisor removed, of type (g, n) with $2g - 2 + n > 0$. Then each horizontal geometric puncture group injects into $\pi_1(U_{\bar{K}})$. For a recursively compatible tower of these fibrations, equip its base with a normal-crossings completion whose strata are aspherical and whose full meridian lattices inject both discretely and profinitely. Suppose the fibration extends to a stable pointed family there, after finite root or quotient changes, with a smooth toroidal total model and fibre incidence type constant on each base stratum. Then every total full meridian group injects:

$$I_{\bar{x}} = \widehat{\mathbb{Z}}(1)^{I(\bar{x})} \hookrightarrow \pi_1(U_{\bar{K}}).$$

These assertions are natural for etale changes and compatible monomial maps, with path conjugacy retained. On a splitting field of the branches, conjugation on each normal factor is the cyclotomic character; without splitting, it is the product of that action with the branch permutation action. A monomial map $t'_j = u_j \prod_i t_i^{m_{ji}}$ induces the integer contact matrix (m_{ji}) on inertia. This supplies a common geometric injection and normalization theorem for the arithmetic issue raised in the sixth EXIT task. F.48 audits the recovered `CYCLOTOMIC_KERNEL` rows; the remaining literal comparison with the unlocated `FLAGGED_149` rows is recorded in F.45.

Proof. We give the completion argument, since discrete injection alone is insufficient. Every stage of the tower has this smooth finite-type base geometry. After choosing $\bar{K} \hookrightarrow \mathbb{C}$, the proper smooth family, with its finite etale markings, is locally a trivial differentiable family of marked compact surfaces: a horizontal distribution tangent to the markings is obtained by a partition of unity and its local flows give the trivializations. Removing the markings therefore gives a locally trivial analytic topological bundle. Its fibre is a punctured surface with group a free group $N = F_r$, $r \geq 2$. For a bundle over any connected base, the image of $\pi_2(B) \to \pi_1(F)$ is central: pull back over a sphere, write its clutching loop as $h_t \in \mathrm{Homeo}(F)$, and use the square $h_t(\gamma(s))$

to commute the evaluation loop $h_t(x)$ with every $\gamma \in \pi_1(F, x)$. The free group's centre is trivial, so the fibre injects. A puncture loop is nontrivial: if there are several punctures it survives in homology; with one puncture it is the reduced nonempty product of handle commutators. Thus each such loop has infinite order.

We prove that for *every* extension $1 \to F_r \to G \to Q \to 1$ the completed sequence is exact on the left. First $Z(\widehat{F}_r) = 1$. Given a finite quotient $F_r \to A$, with kernel L, choose a prime $\ell \nmid |A|$. The cellular chain complex of the finite regular covering of the r-rose gives

$$0 \longrightarrow H_1(L, \mathbb{F}_\ell) \longrightarrow \mathbb{F}_\ell[A]^r \longrightarrow \mathbb{F}_\ell[A] \longrightarrow \mathbb{F}_\ell \longrightarrow 0.$$

Averaging over A splits these representations, so $H_1(L, \mathbb{F}_\ell) \simeq \mathbb{F}_\ell \oplus \mathbb{F}_\ell[A]^{r-1}$ is a faithful A-module. The finite quotient $F_r/[L, L]L^\ell$ consequently detects the image in A of any purported central element of $\widehat{F}_r$: centrality forces its action on this module to be trivial, and hence its image in A is 1. This for every A proves centrelessness.

The automorphism group of a finitely generated profinite group is profinite: the intersections of its finitely many open subgroups of index at most m form a characteristic open basis, and automorphisms are the compatible invertible maps on the resulting finite quotients. The conjugation map $G \to \mathrm{Aut}(\widehat{F}_r)$ therefore extends to $\widehat{G}$. Its composite with $\widehat{F}_r \to \widehat{G}$ is the inner-automorphism map, which is injective by centrelessness. Thus $\widehat{F}_r \to \widehat{G}$ is injective. Its image is the closure of F_r, and the quotient of $\widehat{G}$ by that closure is $\widehat{Q}$, since both classify the finite quotients killing F_r.

A nontrivial cyclic subgroup $\langle w \rangle \subset F_r$ has its full profinite topology. Indeed represent a cyclically reduced conjugate of w by an immersed labelled circle in a rose. For every label, pair the missing incoming and outgoing edges to complete this finite graph to a finite covering of the rose, without identifying its vertices. The original circle is an embedded cycle in that graph, so its integral homology class is primitive. The corresponding finite-index subgroup $H \subset F_r$ has a homomorphism $H \to \mathbb{Z}$ sending w to 1. For any m the core in F_r of the kernel of $H \to \mathbb{Z}/m$ gives a finite quotient in which w has order divisible by m. This proves $\widehat{\langle w \rangle} \simeq \widehat{\mathbb{Z}} \hookrightarrow \widehat{F}_r$. Together with the extension argument it proves horizontal profinite injection over an arbitrary base. By induction a tower is aspherical, and its discrete group is residually finite: separate an element either in the completed base or in the embedded completed free fibre.

For the vertical assertion let A be a total local normal lattice, A_B its base lattice, and $M : A \to A_B$ the integral monomial map. At a smooth vertical point $\ker M = 0$; at a marking it is generated by the fibre puncture; at a node $xy = t$ it is generated by $(1, -1)$ in the map $(u, v) \mapsto u + v$. In the latter case its image in the smooth fibre is the vanishing cycle. Root changes and smooth toroidal charts replace these lattices by finite-index or primitive sublattices; hence in general $\ker M$ has rank at most one and maps to a nonzero power of the corresponding cycle. The cycle is essential. If a neck bounded an unpunctured disc, the side it cut off would be an unmarked genus-zero tree attached at one node, impossible for a stable pointed curve: the sum of the special-point counts on such a tree is less than three times its number of components. A neck around one marked point remains a nontrivial puncture loop, and the extra two-punctured bridges of a toroidal model also have essential meridians.

The diagram

$$\begin{array}{ccccc} \ker M & \longrightarrow & A & \longrightarrow & A_B \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(F) & \longrightarrow & \pi_1(U) & \longrightarrow & \pi_1(B) \end{array}$$

has the needed exactness in its rows. An element of A killed in $\pi_1(U)$ maps to zero in the injective base lattice, then belongs to $\ker M$, where the essential-cycle map is injective. It is zero. Complete the lower sequence by the free-kernel result. Smith normal form proves $\ker \widehat{M} = \widehat{\ker M}$: the image of M is free abelian, the sequence onto it splits, and its inclusion into A_B remains injective after completion. The same diagram chase, using completed base injection and cyclic fibre injection, proves the full profinite assertion. The strata are bundles of punctured stable

components or tori over finite incidence covers of base strata; their homotopy sequences prove their asphericity. These statements give the induction for every recursively compatible tower in the theorem.

The Riemann existence comparison is SGA 1, XII, Theorem 5.1 and Corollary 5.2 (Grothendieck and Raynaud, 2003). It identifies these completed groups and Kummer meridians with the geometric etale groups, so it transports the proved injections. In local parameters the automorphisms of $t_i^{1/n}$ are multiplication by μ_n; Galois conjugation acts on these roots of unity by the cyclotomic character. Taking all n gives the asserted Tate modules. Pulling back t'_j gives the contact matrix, while the roots of the units u_j alter the tangential splittings by their Kummer cocycles. They do not alter the homomorphism on geometric meridians. Thus neither a preferred generator of $\widehat{\mathbb{Z}}(1)$ nor a preferred tangential path has been asserted.

Finally, if a local arithmetic extension $1 \to I_x \to E_x \to G_{\kappa(x)} \to 1$ maps to that of U/K and the chosen map $G_{\kappa(x)} \to G_K$ is injective, then

$$\ker(E_x \to \pi_1(U)) = \ker(I_x \to \pi_1(U_{\bar{K}})).$$

Project a kernel element to $G_{\kappa(x)}$ to prove this equality. The condition holds at a closed point after its finite residue-field extension, but is not automatic at a transcendental point.

There are two precise limits on this theorem. First, cyclotomic action on a subgroup is a necessary normalization of genuine inertia; we have not asserted that it characterizes inertia among all subgroups. Second, stable extension alone is insufficient for an arbitrarily chosen base completion. Blow up a smooth boundary point of the product completion of $(\mathbb{P}^1 \setminus \{0, 1, \infty\})^2$. At the new double point the exceptional and strict-transform meridians both map to the old meridian, so $(1, -1)$ is killed. A constant stable curve family over that base retains this kernel, discretely and profinitely. The compatibility and full base-lattice assumptions above are therefore necessary to this induction. This also verifies, rather than merely quotes, the distinction between EXIT P3b-vertical and its counterexample. □

6.9 F(h), added item (2): the punctured completion and the end

F.40 (Proved). Nonarchimedean invariance, with the missing hypothesis. *Manuscript: MPL 149; section4; folio66; scan133; additional F(h)(2) tests.*

Let k be any field, equipped with the trivial absolute value, let X be a separated k-scheme of finite type, and let $U \subset X$ be a dense open subscheme with complement D. Denote by $Y^{\beth}$ the generic fibre of the scheme Y regarded as a formal k-scheme with the discrete topology. The analytic punctured tube is

$$W(X, D) = X^{\beth} \setminus (U^{\beth} \cup D^{\beth}).$$

It is the punctured generic fibre associated with $\widehat{X}_D$. For every proper morphism $X' \to X$ which is the identity on U, the induced morphism $W(X', X' \setminus U) \to W(X, D)$ is an isomorphism of k-analytic spaces. If X is proper over k, then

$$W(X, D) = U^{\mathrm{an}} \setminus U^{\beth} =: U^{\mathrm{na}}_{\infty}.$$

Consequently it is independent of the proper compactification, without smoothness, normal-crossings, or characteristic-zero assumptions. The phrase "with no hypothesis" is false if it also removes properness.

Proof. A point of $Y^{\beth}$ is a multiplicative seminorm whose valuation ring has a centre on Y; separatedness makes the centre unique. A point of $W(X, D)$ has its generic point in U and its centre in D. The valuative criterion for a proper $X' \to X$ gives a unique lifted centre, necessarily in $X' \setminus U$. Thus both tubes are the same analytic domain of U^{an}, with the same induced analytic structure. If X is proper, every analytic point has a centre, so $X^{\beth} = X^{\mathrm{an}}$ and

$D^{\beth} = D^{\mathrm{an}}$, giving the formula. This is also the precise statement of Ben-Bassat–Temkin, §3.3.2 and Proposition 4.5.1 (Ben-Bassat and Temkin, 2012).

For necessity, take $U = \mathbb{G}_m$ over an algebraically closed field. Its embedding in $\mathbb{A}^1$ has boundary $\{0\}$ and tube $0 < |t| < 1$, a single open interval. Its proper compactification $\mathbb{P}^1$ has two boundary points and tube $\{0 < |t| < 1\} \amalg \{1 < |t| < \infty\}$, two open intervals. These are not even equally connected. The first ambient space is a partial compactification, so it is not a counterexample to the properly stated theorem.

The notation $\widehat{X}_D \setminus D$ on MPL 149, fol. 66, scan 133 must be given this punctured-generic-fibre meaning: literally deleting D from the underlying topological space of the formal completion gives the empty space. In particular, the invariant does not assert that the formal schemes $\widehat{X}_D$ themselves are isomorphic. □

F.41 (Refuted). Stronger replacement: the archimedean link is invariant too. *Manuscript: MPL 149; section4; folio66; scan133; additional F(h)(2) tests.*

The proposed failure of compactification invariance for the archimedean punctured neighbourhood when X is compact is false. For every compact Hausdorff compactification $U \subset X$, with $D = X \setminus U$, its punctured-neighbourhood pro-space is canonically

$$\{N \setminus D\}_{N \supset D} = \{U \setminus K\}_{K \subset U \text{ compact}}.$$

Here N ranges over open neighbourhoods, and the equality is an isomorphism of indexing systems, not just equality of cohomology. For a smooth proper complex variety with a simple normal-crossings boundary, this pro-homotopy type is constant, represented by the boundary L of its real oriented blow-up, after rounding corners. Thus the homotopy type of L is independent of that compactification. In general the natural map from this end to U is not a homotopy equivalence.

Proof. If N is open and contains D, then $K = X \setminus N$ is a compact subset of U and $N \setminus D = U \setminus K$. Conversely, every compact $K \subset U$ is closed in the Hausdorff space X, so $N = X \setminus K$ is such a neighbourhood. These assignments are inverse and preserve the refinement maps. The same proof applies to punctured analytic neighbourhoods, since these are open subsets of the common complex manifold U.

In a normal-crossings coordinate chart $z_1 \cdots z_r = 0$, write $z_j = \rho_j e^{i\theta_j}$. The real oriented blow-up replaces the vanishing coordinates by $\rho_j \geq 0$ and their angles. Its local model is $[0, \infty)^r \times (S^1)^r \times \mathbb{R}^{2n-2r}$, so it is a compact manifold with corners with interior U. Rounding corners and taking a boundary collar gives a cofinal family of end neighbourhoods $L \times (0, \epsilon)$. All refinement inclusions are homotopy equivalences, which proves the constant-model assertion. The constant embedding of the homotopy category into its pro-category is fully faithful; hence two such models have equivalent homotopy types. Weber's §2 identifies the corresponding cohomology directly with $\varinjlim_K H^*(U \setminus K)$ (Weber, 2012); the argument here retains the whole pro-homotopy type. The explicit failures of $L \to U$ are proved in F.42. □

F.42 (Proved). Both requested examples and the three different invariants. *Manuscript: MPL 149; section4; folio66; scan133; additional F(h)(2) tests.*

For $(X, D) = (\mathbb{P}^1, \{0, 1, \infty\})$ over $\mathbb{C}$, one has

$$L \simeq S^1 \amalg S^1 \amalg S^1, \qquad U(\mathbb{C}) \simeq S^1 \vee S^1,$$

while over trivially valued $\mathbb{C}$ the underlying topological space of U_∞^{na} is three disjoint open intervals. For $(X, D) = (\overline{M}_{0,5}, \partial\overline{M}_{0,5})$ the invariants are

	b_0	b_1	b_2	b_3
$U(\mathbb{C})$	1	5	6	0
L	1	11	11	1
U_∞^{na} (underlying topology)	1	6	0	0.

All displayed integral homology groups are free. Both end constructions are independent of a proper compactification in the precise senses of F.40–F.41; neither the archimedean link nor the underlying topological space of the nonarchimedean tube is the homotopy type of $U(\mathbb{C})$. The last row concerns Berkovich topology, not its etale topos.

Proof. For the three-punctured sphere, small punctured disks retract to three circles and a pair of pants retracts to a wedge of two circles. A multiplicative seminorm in the formal punctured disk satisfies $|\sum_{n\geq m} a_n t^n| = r^m$, where $a_m \neq 0$ and $0 < r < 1$; a unit and its inverse are bounded, so they have norm one. Thus r parametrizes an open interval. The three boundary points give the asserted disjoint union. Its normalized quotient by rescaling valuations is three points.

Normalize three marked points of a smooth five-pointed curve to $0, 1, \infty$. Then

$$U(\mathbb{C}) = \{(x, y) \in (\mathbb{C} \setminus \{0, 1\})^2 : x \neq y\}.$$

Projection to x is a locally trivial bundle, with base the plane minus two points and fibre the plane minus three points. A local trivialization is obtained by an isotopy moving x in a small disk disjoint from $0, 1$. Both base and fibre retract to graphs, so the homotopy exact sequence proves that U is a $K(\pi, 1)$. Monodromy preserves each oriented puncture-meridian in first homology. The integral Serre spectral sequence therefore has only its 0 and 1 rows and columns, all free, with ranks given by $(1 + 2t)(1 + 3t)$. It has no possible differential, yielding the first row of the table.

The boundary divisors are indexed by two-element subsets of $\{1, \ldots, 5\}$. Two distinct divisors meet exactly when these subsets are disjoint, as follows by drawing the unique stable three-component tree with the two prescribed nodes. They intersect transversely once; there are no triple intersections. Thus the incidence graph is the Petersen graph, with 10 vertices and 15 edges, and its first Betti number is $15 - 10 + 1 = 6$. Every component is a projective line of self-intersection -1. These geometric descriptions also follow from the identification with the blow-up of $\mathbb{P}^2$ in four points, where the boundary consists of four exceptional curves and the six transforms of their joining lines; see Dolgachev–Farb–Looijenga, §§2.1–2.2 (Dolgachev et al., 2018).

For the nonarchimedean row, the normal-crossings skeleton theorem gives a deformation retraction of the tube onto $|\Delta(D)| \times \mathbb{R}_{>0}$: in a chart its coordinates are the nonnegative numbers $-\log|z_j|$, not all zero, and normalization divides by their sum. The coordinate retractions glue under etale changes of normal-crossings charts. This is Thuillier's Theorem 3.25 and Proposition 4.5 in the opened 2006 preprint (Thuillier, 2006). Consequently the tube has the homotopy type of the Petersen graph.

For the archimedean row, let N be a closed regular neighbourhood of D in X. It is obtained by plumbing the disk normal bundles of the ten components, and retracts to D. It follows that $H_1(N; \mathbb{Z}) = \mathbb{Z}^6$, $H_2(N; \mathbb{Z}) = \mathbb{Z}^{10}$, and $H_3(N; \mathbb{Z}) = 0$. The boundary $L = \partial N$ is a connected closed oriented three-manifold. Let Q be the intersection matrix of the ten components. Its diagonal is -1 and its off-diagonal entries are the adjacency matrix of the Petersen graph. Poincare–Lefschetz duality and the homology exact sequence of (N, L) give

$$0 \longrightarrow \operatorname{coker} Q \longrightarrow H_1(L; \mathbb{Z}) \longrightarrow \mathbb{Z}^6 \longrightarrow 0.$$

To compute the cokernel integrally, use the basis $H, E_1, \dots, E_4$ of $H_2(X;\mathbb{Z})$ with form $\operatorname{diag}(1,-1,-1,-1,-1)$. The boundary classes are E_i and $H - E_i - E_j$. They generate this unimodular lattice, so their column map $V : \mathbb{Z}^{10} \to \mathbb{Z}^5$ is onto. Since $Q = V^t \operatorname{diag}(1,-1,-1,-1,-1)V$, its image is the primitive rank-five sublattice $\operatorname{im} V^t$. Hence $\operatorname{coker} Q = \mathbb{Z}^5$ and $H_1(L) = \mathbb{Z}^{11}$. Duality now gives $H_2(L) = \mathbb{Z}^{11}$ and $H_3(L) = \mathbb{Z}$. The attached program independently computes the graph incidence rank and Smith normal form of Q.

Already connectedness rules out $L \simeq U$ in the curve example; in the surface example the nonzero fundamental class in $H_3(L)$ rules it out. Neither fact contradicts compactification invariance. The manuscript's cautious assertion on scan 133 is therefore supported when "punctured completion" denotes the nonarchimedean tube or the archimedean punctured germ; its identification with the open variety is not asserted by that sentence and is false in these examples. □

F.43 (Proved). Finite-homotopy exodromy for $\overline{M}_{0,5}$. *Manuscript: Supplemental PartE(c) n=5 question; LM3 F.18 and LM3b H.9; manuscript context MPL147 section12 folio85 scan179.*

Let $X = \overline{M}_{0,5,\mathbb{C}}$ with its normal-crossings boundary stratification, and let $C = h\,\mathrm{Exit}(X^{\mathrm{an}})$. The continuous functors from the strict profinite completion of C to π-finite spaces classify the etale sheaves of π-finite spaces on X constructible for these strata. Stalks and all generization maps are preserved. Thus the $n = 5$ question of LM3b H.9 and Part E(c) is decided; one must retain the category and its flags, not just its groupoidification.

Proof. Use

$$U = M_{0,5} = \{(x,y) : x, y \in \mathbb{P}^1 \setminus \{0,1,\infty\},\ x \neq y\}.$$

The compactification is the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ at $(0,0),(1,1),(\infty,\infty)$. Forgetting y is a locally trivial bundle with base the thrice-punctured sphere and fibre the four-times-punctured sphere: small disjoint discs around the moving puncture give local trivializations by an isotopy supported in those discs. The homotopy exact sequence proves that U is aspherical and that $1 \to F_3 \to \pi_1(U) \to F_2 \to 1$ is exact. All ten one-dimensional boundary strata are thrice-punctured spheres and the fifteen deepest strata are points, so every stratum is aspherical.

Let a, b, c, d, e denote the homology classes of meridians of $x = 0, x = 1, y = 0, y = 1, x = y$. They are a basis of $H_1(U,\mathbb{Z})$: project the five complex affine line complements to their defining linear functions to see independence, and attach the normal disc boundaries to the simply connected affine plane to see generation. The ten boundary vectors are

x_0	a	y_0	c
x_1	b	y_1	d
x_∞	$-a-b-e$	y_∞	$-c-d-e$
Δ	e	E_0	$a+c+e$
E_1	$b+d+e$	E_∞	$-a-b-c-d-e.$

The infinity formulas are the product-one meridian relations on a projective line; an exceptional meridian is the sum of the three meridians through the blown-up point. The intersecting pairs are (x_i, y_j) for $i \neq j$, and $(E_i, x_i), (E_i, y_i), (E_i, \Delta)$. For each of these fifteen pairs the 5×2 matrix has a 2×2 minor 1 or -1 after an integral change of rows; equivalently the gcd of its maximal minors is 1. The attached exact computation lists all fifteen gcds and obtains Smith invariants $(1,1)$ in each case. Consequently every local meridian lattice $\mathbb{Z}^r$, $r \leq 2$, embeds as a direct summand of $H_1(U,\mathbb{Z})$, hence injects into $\pi_1(U)$ and, after completion, into $\widehat{\pi_1(U)}$.

For any incident strata S, T, let L be the normal link with its two maps. The exit mapping space is the homotopy fibre of $L \to S \times T$. Its higher homotopy groups vanish because these three spaces are aspherical. Its fundamental group is $\ker(\pi_1 L \to \pi_1 S \times \pi_1 T)$; an element in that kernel is a normal meridian and is zero by the injection just established (with the relative-face

argument of F.28). Thus every component of every exit mapping space is contractible. The exit infinity-category is therefore equivalent to its ordinary category C, with composition specified by the actual three-stratum flags. This is the extra step not furnished by asphericity of strata alone.

We next justify completion; it is not a formal assertion that profinite completion preserves all homotopy pullbacks. A finitely generated free group is good: the cellular two-term resolution of a rose computes its cohomology with every finite module, and the same free resolution in continuous modules computes the cohomology of the free profinite group. The full profinite topology on F_3 inside $\pi_1(U)$ is induced by conjugation into $\mathrm{Aut}(\widehat{F}_3)$, as proved in F.39. Thus the profinite completion of the displayed free-by-free sequence is exact. Comparing its discrete and continuous Hochschild–Serre spectral sequences proves goodness of $\pi_1(U)$: the cohomology groups of the fibre with finite coefficients are finite, and only rows $0, 1$ occur, so goodness of the free quotient applies to every term. Links over a one-dimensional stratum have group $\mathbb{Z}^r \rtimes F_2$ (the extension splits since F_2 is free), and links over a point have group $\mathbb{Z}^r$. Their completions are exact by embedding the normal lattice into $\widehat{\mathbb{Z}}^r \rtimes \widehat{F}_2$; the finite Koszul resolution of $\mathbb{Z}^r$ and the same spectral-sequence comparison prove goodness. Hence all the stratum and link profinite homotopy types are the classifying types of their profinite groups. The completed meridian injection now proves, by the same homotopy-fibre calculation, that every profinite exit mapping object is a profinite set. Descent on the finitely many full flag diagrams reconstructs a profinite ordinary EI category with these Hom spaces and compositions.

Here is the finite-congruence point explicitly. A compact totally disconnected category with finitely many objects is the inverse limit of its finite categorical quotients. Indeed, choose finite clopen partitions of its Hom spaces, and put $a \sim b$ when every defined left and right context $u(-)v$ puts uav and ubv in the same partition piece. Joint continuity and compactness of the context spaces imply that this equivalence is open: finitely many neighbourhoods of contexts suffice to find one neighbourhood of a on which all their partition values are fixed. Compactness then makes the number of classes finite. Inserting further contexts shows that the equivalence is compatible with composition. Thus finite congruences are cofinal among the finite clopen data. In our case their restrictions to the dense discrete category C are precisely the strict finite congruences used in LM3 F.2–F.4. Here one also uses the topology assertion in F.39 and the completed link descriptions above: finite quotient data on the discrete stratum groups and incident cosets extend continuously to their completions, and the finitely many composition constraints extend by density. Density alone would not imply this extension property. Thus the inverse system is the strict profinite completion.

The same statement works with finite homotopy values, not only finite sets. Given finitely many π-finite values, take a degreewise finite simplicial model of the full infinity-category they span; its mapping spaces have one common finite truncation bound. Its coherent diagram data are determined by finitely many simplicial degrees, using a coskeletal model above that bound. Continuous maps in those degrees factor through finite clopen partitions of finite products of Hom spaces. Every such partition admits a finite coordinatewise clopen refinement: compactness gives a finite cover by clopen rectangles, and the finitely many possible object sequences cause no additional infinite choice. Refine the resulting individual Hom partitions by the context congruence of the preceding paragraph. Thus one finite categorical congruence suffices for all the diagram data. The same argument with an additional finite simplex Δ^m proves the assertion on mapping spaces, for every m. Consequently continuous representations in π-finite spaces are the infinity-categorical filtered colimit of their finite-stage representations; this is stronger than a statement about finite sets.

Finally, the algebraic-to-analytic comparison on these strata and their normal links identifies their finite covers and their finite-coefficient higher descent data. In the notation of Barwick–Glasman–Haine, this is Proposition 12.6.4 and Corollary 12.6.7, followed by the exodromy Theorem 10.1.8 (Barwick et al., 2020). The preceding paragraphs verify in this particular geometry that the resulting profinite exit object is the strict completion of the ordinary category,

rather than a profinite infinity-category with hidden higher mapping spaces. Applying the comparison and exodromy therefore gives precisely the asserted stalk-preserving equivalence. The classifying space of this category still has higher homotopy; for example $H^2(X,\mathbb{Z}) = \mathbb{Z}^5$ and $H^4(X,\mathbb{Z}) = \mathbb{Z}$. Ordinary exit mapping spaces being discrete does not mean that this classifying space is a 1-type. □

Two explicit completion checks for the preceding argument. The flag description can be made completely concrete here. The only strict flags with three strata are $p < S < U$, where p is a boundary crossing and S is a thrice-punctured boundary component. The normal circle bundle over S is topologically trivial, since $H^2(S,\mathbb{Z}) = 0$. Consequently its link group is $G_S \times \mathbb{Z}$ and its map to G_S is the projection. The other adjacent link has group $\mathbb{Z}$, mapping to G_S by the relevant puncture loop. The three-stratum flag space is their homotopy pullback, hence has group

$$\mathbb{Z} \times_{G_S} (G_S \times \mathbb{Z}) \simeq \mathbb{Z}^2.$$

After profinite completion the same identity holds, with hats: the projection is split and all these products carry the full induced profinite topology. Thus the finitely many nontrivial Segal gluing identities survive completion in this geometry. This supplies the hypothesis needed to apply the spatial and profinite decollage equivalences of (Barwick et al., 2020, Theorem 2.7.4 and equivalence (2.8.9)); no preservation of arbitrary homotopy pullbacks by completion is asserted.

It also verifies universality for *strict* finite categorical quotients, which does not follow from density alone. Choose one point in each stratum. For each incident pair $S < T$, the link is connected, so its Hom set is a transitive (G_T, G_S)-biset. Upon choosing one arrow its stabilizer is the image of its link group $H_{S,T}$ in $G_T \times G_S$ (using the inverse right action). The injectivity already proved gives the completed Hom space

$$(\widehat{G}_T \times \widehat{G}_S)/\operatorname{im}(\widehat{H}_{S,T}).$$

The denominator is compact and therefore closed. A functor from the discrete exit category to a finite category restricts to homomorphisms from each G_S to finite automorphism groups; these extend to $\widehat{G}_S$. Its map on an incident Hom set is determined by the image of one arrow and the finite group actions. The stabilizer condition extends to the closure, so this map extends uniquely and continuously to the displayed profinite Hom space. Composable pairs of discrete arrows are dense in the corresponding product of Hom spaces. Hence the extended maps preserve composition, since the two resulting continuous maps to a finite discrete set agree on that dense subset. This proves that every finite categorical quotient extends, and hence that the compact category constructed from the completed flags has precisely the strict finite quotients of C. The context-congruence argument then identifies it with their inverse limit.

Finally, the finite bound used for coherent representations has an exact model-theoretic justification. The full coefficient category on finitely many π-finite spaces has finitely many object classes, finite homotopy groups in every mapping space, and a common finite truncation bound on those mapping spaces. A minimal quasicategory model is degreewise finite by Blom–Moerdijk, Lemma A.10, and has a finite coskeletality bound by Lurie, Proposition 2.3.4.18 and Remark 2.3.4.4; this is exactly the conclusion recorded in Blom–Moerdijk, Remark A.11 (Blom and Moerdijk, 2023; Lurie, 2009). One takes this equivalent minimal model before using finite coherent data; no strict coskeletality assertion about an arbitrary homotopy coherent nerve is necessary. Thus a continuous coherent diagram is determined by finitely many clopen data on products of Hom spaces. The bound is taken separately for each finite-object diagram and for each finite simplex of its mapping space, rather than for the whole class of all π-finite spaces.

F.44 (Open). Delimitation beyond the proved reconstruction ranges. *Manuscript: MPL149; section2; folio25v; scan52; intermediate-base delimitation.*

The requested base-field delimitation now includes the following proved positive ranges: infinite finitely generated fields by F.5, sub-p-adic fields and the specified general bases by F.38, and every algebraic subfield of $F\mathbb{Q}^{\mathrm{ab}}$, for a number field F, by F.47, and, more generally, every subfield of a finitely generated extension of $\mathbb{Q}^{\mathrm{ab}}$ by F.50. These assertions include arbitrary cyclotomic extensions of number fields and fields finitely generated over them. In particular, the test over $\mathbb{Q}(\mu_{2^\infty})$ considered during this work is settled, including full site realization. F.46 records its initial point-function proof and F.47 supplies the additional argument needed for the full geometric morphism. The proposed unrestricted fullness is false by F.5. F.49 strengthens the counterexamples F.6 and F.37: pinned fullness fails over every algebraic extension of a finite field, every algebraically closed field, and every infinite perfect field with nontrivial centre of its absolute Galois group, including every real closed field. Faithfulness of actual reduced-scheme maps remains the all-characteristic theorem F.4.

The exact general criterion is Carlson–Haine–Wolf, Theorem 4.18 (Carlson et al., 2024): a field has pinned reconstruction if and only if every pinned morphism from a connected regular affine scheme with a rational point to $(\mathbb{G}_m)_{\mathrm{et}}$ has the geometric-point values of a regular unit (on the absolute weak normalization in positive characteristic). Its implication to full site realization is proved through all sliced etale objects, as explained in F.47. Their Theorem 6.24 further supplies the positive range of amenable fields admitting a nontrivial discrete valuation. Amenability means that rational points of every torus have no nonidentity prime-to-characteristic divisible element and that the prime-to-characteristic Tate module of the Picard group of every regular finite-type scheme vanishes (Definition 5.15). This is an exact verified sufficient condition, not a claim that every field outside it is a failure or a literature-wide open case.

A concrete test not decided by this return is

$$k_{\mathrm{solv}} = \mathbb{Q}^{\mathrm{solv}}, \qquad \mathrm{Hom}_{k_{\mathrm{solv}}}(\mathbb{G}_m, \mathbb{G}_m) \longrightarrow \pi_0 \,\mathrm{Hom}^{\mathrm{pin}}_{BG_{k_{\mathrm{solv}}}}((\mathbb{G}_m)_{\mathrm{et}}, (\mathbb{G}_m)_{\mathrm{et}}), \tag{F44}$$

where $\mathbb{Q}^{\mathrm{solv}}$ is the maximal solvable algebraic extension in a fixed algebraic closure. The map is injective by F.4; its surjectivity is not asserted. Finite-type dimension-zero schemes, after universal-homeomorphism localization, are finite etale and their relative Galois topoi already determine their maps. Thus (F44) is a first positive-dimensional test of the remaining criterion, rather than a claim of a smallest base field or of literature-wide minimality. Constant morphisms and the general nonconstant, possibly noninvertible, locus are part of the question; the unresolved realization issue concerns the latter locus.

Here are the precise failed extensions of the proof, rather than an appeal to the absence of a cited theorem. The field k_{solv} is radically closed. Indeed, an element lies in a finite solvable Galois extension $E/\mathbb{Q}$. Adjoin the needed roots of unity and then nth roots of all its conjugates; the latter extension is abelian over the enlarged E, so its normal closure is still solvable over $\mathbb{Q}$. Consequently

$$D(k^*_{\mathrm{solv}}) = k^*_{\mathrm{solv}}.$$

The height argument used in F.47 cannot reduce these elements to roots of unity: this field contains $2^{1/n}$ for every n, of positive heights tending to zero. The same observation shows that it cannot be contained in $F\mathbb{Q}^{\mathrm{ab}}$ for any fixed number field F, by the verified Amoroso–Zannier bound. It is also outside the broader range F.50: the algebraic constants in a finitely generated extension of $\mathbb{Q}^{\mathrm{ab}}$ form a finite extension of that field, to which the same height bound applies. It is not amenable. It is not sub-p-adic either, since it contains all roots of unity whereas the algebraic constants in a finitely generated extension of $\mathbb{Q}_p$ form a finite extension and have only finitely many roots of unity.

The Picard input of F.47 also fails here. For an elliptic curve $E/\mathbb{Q}$, every 2^r-division field is solvable: its Galois group is a subgroup of $\mathrm{GL}_2(\mathbb{Z}/2^r)$, an extension of $\mathrm{GL}_2(\mathbb{F}_2) \simeq S_3$ by a 2-group.

Hence $E[2^\infty] \subset E(k_{\mathrm{solv}})$, providing nonzero infinitely divisible points and a nonzero Tate module. Neither the integral-residue argument followed by Picard divisibility nor the translated-function proof can therefore be copied with its hypotheses unchanged. Kummer classes at the k_{solv}-points already forget all nonzero constants.

The relative birational isomorphism results checked in Tsujimura's Theorem 4.7 and Corollary 4.10 (Tsujimura, 2022) do not supply this noninvertible realization step; the requisite AVKF hypotheses also fail in this example. Descent of a scheme to a number field does not imply descent of an arbitrary geometric morphism on the whole etale site. No exotic pinned morphism in (F44) has been constructed here, and no argument using the nonabelian finite covers has yet eliminated one. The next step would have to exploit those nonsolvable-cover compatibilities, rather than the Kummer and Picard invariants just shown to vanish or to have the wrong divisibility. This is the exact residual test and obstruction left by this return; no broader assertion about the status of all other fields is made.

A further literature check does not settle (F44). Mochizuki's *Global Solvably Closed Anabelian Geometry* (Mochizuki, 2006), Definitions 1.1 and 2.3 and Theorem 2.5, concerns quotient groups $Q_F = \mathrm{Gal}(\widetilde{F}/F)$ for a solvably closed extension of a number field, together with the required Σ-compatible quotients of hyperbolic-curve fundamental groups. Its conclusion reconstructs compatible isomorphisms. The absolute group $G_{\mathbb{Q}^{\mathrm{solv}}}$ in (F44) is instead a kernel in $G_{\mathbb{Q}}$, and (F44) asks about all pinned morphisms, including noninvertible ones. Neither changing the quotient to this kernel nor replacing the Isom conclusion by Hom is justified by that theorem.

Finite extensions do not repair these particular inputs. One possible approach would pass to a proper finite extension L/k_{solv}, which is Hilbertian by the Weissauer theorem used explicitly in the proof of (Jarden, 2011, Proposition 2.5). Descent of a reconstruction theorem would indeed be valid here. More generally, in characteristic zero pinned reconstruction descends along a finite extension L/k. To prove this, apply the criterion of (Carlson et al., 2024, Theorem 4.18) to a connected regular affine X/k with a rational point and a pinned morphism to $(\mathbb{G}_m)_{\mathrm{et}}$. The curve-free criterion applies after finite etale base change to L, where it gives a unit f_L with the prescribed geometric-point values. On $X_{L\otimes_k L}$ the two pullbacks of f_L have the same values at every geometric point, by the descent data of the original morphism. This scheme is reduced, so the units agree. Faithfully flat descent gives $f \in \Gamma(X, \mathcal{O}_X)^*$ with the required values over k; the same criterion proves reconstruction. The source remains connected after base change: regularity in characteristic zero makes it normal and smooth, and its rational point forces its finite algebraic constant field to be k.

The obstacle is therefore the arithmetic input over L, not this descent argument. For *every algebraic* extension L/k_{solv}, including each proper finite one, one has

$$k_{\mathrm{solv}}^* \subset D(L^*), \qquad E[2^\infty] \subset E(L).$$

Thus L^*/μ_∞ contains the nonzero divisible subgroup $k_{\mathrm{solv}}^*/\mu_\infty$ and cannot be free abelian; the same elliptic curve still violates the AVKF and Tate-module conditions. Nor can such an L admit a nontrivial discrete valuation. A homomorphism $v : L^* \to \mathbb{Z}$ arising from a valuation vanishes on the divisible subgroup k_{solv}^*. For any element algebraic over this trivially valued field, its minimal polynomial has all nonzero coefficients of value zero. A positive or negative value of the element would make the minimum value among its nonzero polynomial terms occur exactly once, which is impossible. Hence v vanishes on L^*. Hilbertianity of a finite extension therefore does not restore the multiplicative, Picard, or discrete-valuation hypotheses used in the reconstruction proofs verified here. This eliminates that specific route without producing an exotic pinned morphism or deciding (F44).

Additional audit of the residual test $k = \mathbb{Q}^{\mathrm{solv}}$. Neither of the elementary nongeometric constructions in F.49 settles this test. In fact, with $G = G_{\mathbb{Q}}$ and $H = G_{\mathbb{Q}^{\mathrm{solv}}}$, one has

$$C_G(H) = 1. \tag{F44.C1}$$

We give the proof, verifying the precise scope of the recovered `CYCLOTOMIC_KERNEL` row B2 instead of importing a stronger unqualified assertion about arbitrary closed subgroups.

The following argument works for every Hilbertian field F and every nontrivial closed normal subgroup $N \triangleleft G_F$. Two primary results suffice. Jarden's Proposition 2.5 says that if E/F is algebraic Galois and E is not separably closed, then G_E is almost locally free; in particular it contains a free profinite subgroup of rank two (Jarden, 2011). Haran–Jarden, Corollary 2.5, says that F^{sep} cannot be the compositum of two proper subfields which are Galois over F (Haran and Jarden, 1991). Both statements allow infinite Galois extensions.

There is no nontrivial closed abelian normal subgroup A of G_F: its fixed field is Galois over F and is not separably closed, so the first result would place a nonabelian free profinite rank-two group inside A. Put $C = C_{G_F}(N)$. It is closed and normal, and $N \cap C = Z(N)$ is a closed abelian normal subgroup of G_F; hence $N \cap C = 1$. If $C \neq 1$, the two proper fixed fields of N and C would be Galois over F, and their compositum would have absolute Galois group $N \cap C = 1$. This contradicts the second result. Thus $C_{G_F}(N) = 1$.

To apply it to $F = \mathbb{Q}$, we check that $\mathbb{Q}^{\mathrm{solv}} \neq \overline{\mathbb{Q}}$. The polynomial $P = X^5 - X - 1$ is irreducible modulo 3: the attached exact program divides it by all twelve monic polynomials of degree one or two and finds no divisor. A reducible quintic must have a factor of one of these degrees. Modulo 2 its squarefree factorization is

$$(X^2 + X + 1)(X^3 + X^2 + 1),$$

with both factors irreducible. Its rational Galois group is therefore transitive of degree five and contains a permutation of cycle type $(2)(3)$, whose cube is a transposition. A transitive group of prime degree is primitive. The graph whose edges are the conjugates of that transposition has components forming blocks, and is therefore connected. Its edge transpositions generate S_5, so the Galois group is S_5. The program also computes its derived-subgroup orders $120, 60, 60$, certifying nonsolvability. Its splitting field cannot lie in the maximal solvable extension. Thus $H \neq 1$, and Hilbert irreducibility for $\mathbb{Q}$ gives (F44.C1). In particular $Z(H) = 1$, so F.49 supplies no nonidentity central comparison twist over this field.

The semilinear field-automorphism construction also does not apply. Since $k/\mathbb{Q}$ is Galois, $\mathrm{Aut}_{\mathbb{Q}}(k) = G/H$. For an automorphism σ choose a lift $g \in G$. Its outer action on H is conjugation by g. If this action were inner, say equal to conjugation by $h \in H$, then

$$h^{-1}g \in C_G(H) = 1.$$

Hence $g = h$ and $\sigma = 1$. A nonidentity coefficient-semilinear automorphism of $\mathbb{A}^1_k$ consequently cannot be supplied with the relative comparison used in F.49: such a comparison, restricted along the rational point 0, would make its base automorphism of BH two-isomorphic to the identity and thus its action on H inner. This excludes that construction; it makes no assertion that all pinned topoi morphisms over k have been classified.

The finite certificate is `code/montpellier/F44_solvable_certificates.py`; its exact output is included with the deliverable.

F.45 (Open). The remaining literal FLAGGED_149 audit. *Manuscript: Supplemental EXIT task6 source audit; manuscript context MPL149 section4 folios65v-67 scans132-135.*

The required literal comparison with the `FLAGGED_149` register rows F11–F16 and F25 remains incomplete. No exact matching return was found in the supplied files, the targeted title and content searches, or the complete 749-item inventory of accessible ZIP filenames. That inventory does not exclude an unindexed copy nested inside another archive. The actual register and its proofs are therefore still needed to verify the intake's row numbers and claims.

Two earlier source gaps were closed during this work. The `LM8_LIE_APPROXIMATIONS_background.zip` archive contains the actual `CYCLOTOMIC_KERNEL` return and the §53 scans and transcriptions. F.48 audits its exact rows B5 and C8 and proves the literal sector-functor

comparison with §53. Earlier claims that these particular sources had not been located are superseded by that recovery. The initial unsuccessful searches are retained only as a record of the work, not as current descriptions of availability.

The EXIT rows P3b-discrete, P3b-completion, P3b-horizontal-profinite, P3b-vertical, P3b-vertical-counterexample and P3b-profinite were inspected with their proofs; F.39 gives the relevant stable-tower theorem and its precise contact-matrix and cyclotomic qualifications. F.22 refutes the separate assertion that finite root stacks have one fixed uncompleted inertia kernel. These independent mathematical results do not authenticate the unlocated FLAGGED register. Only that literal cross-return comparison remains open in this source-audit row; F.48 closes the previously recorded CYCLOTOMIC and §53 gaps.

F.46 (Proved). The point function of every cyclotomic test morphism. *Manuscript: MPL149; section2; folio25v; scan52; prompt Part F remaining-field test.*

Let $k = \mathbb{Q}(\mu_{2^\infty})$, fix $K = \overline{\mathbb{Q}}$, and let

$$\phi : (\mathbb{G}_{m,k})_{\mathrm{et}} \longrightarrow (\mathbb{G}_{m,k})_{\mathrm{et}}$$

be a pinned geometric morphism over BG_k, with the relative identification included in the data. Its induced map on geometric points is

$$f : K^* \longrightarrow K^*, \qquad f(a) = ca^m \quad (c \in k^*,\ m \in \mathbb{Z}). \tag{F46}$$

Thus the point-function obstruction in the cyclotomic test over $\mathbb{Q}(\mu_{2^\infty})$ vanishes. This assertion by itself does not identify the full geometric morphism: the hypothesis of Carlson–Haine–Wolf, Proposition 4.14, concerns every quasicompact separated etale target object and its pullback, not only the terminal object.

Problem locus. Montpellier cote149, §2, fol.25v, scan52, and the supplied prompt's Part F intermediate-field reconstruction question. This develops the arithmetic reconstruction question there. It is a new consequence of the proved elementary arguments below and the explicitly cited arithmetic input; it is not attributed to a sentence of the manuscript.

Proof: arithmetic facts. For a field L put $D(L^*) = \bigcap_{n\geq 1}(L^*)^n$. For every finite extension L/k one has

$$D(L^*) = \mu_{2^\infty}. \tag{F46.1}$$

Here is the precise arithmetic input and its application. Amoroso–Zannier, Theorem 1.1, gives a positive lower bound for the height of a non-torsion algebraic number in terms of its degree over an abelian extension of a fixed number field (Amoroso and Zannier, 2000). Apply it with the abelian extension $\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}$. The degrees of elements of $L\mathbb{Q}^{\mathrm{ab}}$ over $\mathbb{Q}^{\mathrm{ab}}$ are bounded by $[L : k]$. Consequently the heights of non-torsion elements of L^* have a uniform positive lower bound. If $a \in D(L^*)$ were not a root of unity, choosing $b_n \in L$ with $b_n^n = a$ would give $h(b_n) = h(a)/n > 0$, a contradiction. Thus a is torsion.

The field k is unramified at each odd prime ℓ: every finite subextension is contained in a 2-power cyclotomic extension, whose defining roots of unity are etale over $\mathbb{Z}_\ell$. The value group of any extension of the ℓ-adic valuation to L therefore has index at most $[L : k]$ over $\mathbb{Z}$. In particular, the orders of the ℓ-primary roots of unity in L are bounded, since

$$v_\ell(\zeta_{\ell^r} - 1) = \frac{1}{\ell^{r-1}(\ell - 1)}.$$

If a torsion element of $D(L^*)$ had nontrivial ℓ-primary part, its ℓ^n-th roots in L would contradict this bound. Only its 2-primary part remains. Conversely μ_{2^∞} is divisible by every positive integer and is contained in k, proving (F46.1).

We also use that K/k has infinite degree, and that every finite extension of k admits a discrete valuation. Choose a valuation over 3 with $v(3) = 1$. Its value group on k is $\mathbb{Z}$; on a finite

extension it has finite index over $\mathbb{Z}$ and can be renormalized to $\mathbb{Z}$. Finally $T^n - 3$ is Eisenstein for this valuation, so $[K : k]$ is unbounded. Equivalently, a root has valuation $1/n$, which forces the degree of its extension of k to be at least n.

Proof: a rational-source Kummer lemma. Let L/k be finite, and let

$$U = \mathbb{P}^1_L \setminus \{a_1, \dots, a_r, \infty\}, \qquad a_i \in L \text{ distinct.}$$

Suppose a compatible system of Kummer classes $\alpha \in \varprojlim_n H^1(U, \mu_n)$ has the following property: at every $x \in U(L')$, for every finite L'/L, its evaluation is the Kummer class of a specified element $F(x) \in (L')^*$. Then there is a regular unit $g \in \mathcal{O}(U)^*$ such that

$$\alpha = [g], \qquad F(x)/g(x) \in D((L')^*) \quad (x \in U(L')). \tag{F46.2}$$

Indeed, $\mathrm{Pic}(U) = 0$, and unique factorization gives

$$\alpha = \widehat{c} \prod_{i=1}^{r} (t - a_i)^{\lambda_i}, \qquad \widehat{c} \in \widehat{L^*}, \quad \lambda_i \in \widehat{\mathbb{Z}}.$$

Choose a discrete valuation $v : L^* \to \mathbb{Z}$ and an element π of value one. For fixed j, put $x_N = a_j + \pi^N$. For all sufficiently large integers N, these are points of $U(L)$, $v(x_N - a_j) = N$, and $v(x_N - a_i) = v(a_j - a_i)$ for $i \neq j$. Evaluating each Kummer quotient and subtracting the equations for $N + 1$ and N yields

$$\lambda_j = v(F(x_{N+1})) - v(F(x_N)) \quad \text{in } \widehat{\mathbb{Z}}.$$

Its right side is an integer, so write $\lambda_j = m_j \in \mathbb{Z}$. Choose $x_0 \in U(L)$ and set

$$c = F(x_0) \Big/ \prod_i (x_0 - a_i)^{m_i}.$$

Evaluation at x_0 identifies $\widehat{c}$ with the completion of this actual $c \in L^*$. Thus $g = c \prod_i (t - a_i)^{m_i}$ proves (F46.2), including its assertion after every finite extension. No finite-presentation descent for the topos morphism is used here.

Proof: apply the lemma to every translate. A relative pinned morphism sends L-rational points to L-rational points, for every finite L/k. One way to check this is to compose the point's section $G_L \to \pi_1$ with the given map. If its image is a closed point with residue field E, the resulting map into G_E lies over the inclusion $G_L \subset G_k$; hence $G_L \subset G_E$ and the specified embedding of E lies in L. The induced function f is therefore G_k-equivariant and has the stated finite-extension compatibility. In particular $f(1) \in k^*$; divide by this constant and assume from now on that $f(1) = 1$.

If the generic point maps to a closed point, continuity and pinning make f constant, necessarily equal to 1, and (F46) follows with $m = 0$. Otherwise its image is generic. For each $b \in K$, after a finite extension containing b, pull back the open $\mathbb{G}_m \setminus \{b\}$ of the target. This is a nonempty open U_b of the source, with finitely many missing points. Its geometric points are exactly the x with $f(x) \neq b$. After a further finite extension L/k all the missing points are rational. Pullback of the compatible Kummer classes of $t - b$ to U_b satisfies the preceding lemma, with $F(x) = f(x) - b$. Thus there is $g_b \in L(t)^*$, a unit on U_b, such that

$$f(x) - b = d_b(x) g_b(x), \qquad d_b(x) \in \mu_{2^\infty} \subset k^*, \quad x \in U_b(K). \tag{F46.3}$$

Each x is contained in a finite extension of L, so (F46.1) applies. For $b = 0$ the domain is all of $\mathbb{G}_m$. The lemma and the normalization at 1 give

$$f(x) = d_0(x) x^m, \qquad m \in \mathbb{Z}, \quad d_0(x) \in k^*. \tag{F46.4}$$

The stronger assertion $d_0(x) \in \mu_{2^\infty}$ will not be needed until the end.

Proof: the case $m = 0$. In this case all values of f lie in k. Choose $b \in K \setminus k$. Then $U_b = \mathbb{G}_m$, and (F46.3) gives $f(x) - b = d_b(x)c_b x^n$ for an integer n and $c_b \in K^*$. If $n \neq 0$, the values of $c_b x^n$ fill K^*; the equality would place all these values in the k-vector space $k + kb$, of dimension at most two. This contradicts $[K : k] = \infty$. Therefore $n = 0$. If f took two distinct values, subtracting their two equations would give $c_b \in k$ and then $b \in k$, again a contradiction. So f is constant, and $f(1) = 1$ gives $f = 1$. This treats the zero exponent without applying a nonconstant quasi-rationality criterion to it.

Proof: a finite-exception quasi-rationality lemma. We give the argument needed to use (F46.3) safely at its missing points. It is the finite-exception form of the mechanism in Tsujimura, Lemmas 3.3–3.5 and Proposition 3.6 (Tsujimura, 2022). Suppose K is algebraically closed of characteristic zero, algebraic over a subfield M, and $[K : M] = \infty$. Let C/K be a smooth proper connected curve, let q be a function on a cofinite subset of $C(K)$, and suppose that for each $b \in K$ one has

$$q(x) - b = e_b(x)r_b(x), \qquad e_b(x) \in M^*, \quad r_b \in K(C)^*, \tag{F46.5}$$

outside a finite set depending on b. Then q agrees with one rational function on a cofinite subset. If r_0 is constant, this rational function is constant as well.

First suppose r_0 is nonconstant and apply (F46.5) to q and $1 - q$. Write their rational factors as r, s. Choose a finite extension N/M in K over which C, r, s and a finite nonconstant map $C \to \mathbb{P}^1$ are defined. For fixed $A, B \in M^*$ the degree of the pole divisor of $Ar + Bs - 1$, if it is nonzero, is bounded by one integer d, independent of A, B. Thus a zero of this nonzero rational function has residue degree at most d over N. Choose x outside the exceptional sets whose residue field over N has degree greater than d. Such points exist: choose a value of the fixed finite map of degree greater than d over N, and any point above it. The values, and hence the points, can avoid any given finite set because K/N has infinite degree. Substituting $A = e_0(x)$ and the corresponding coefficient B of $1 - q$ shows that $Ar + Bs - 1$ vanishes at x. The degree bound forces the rational-function identity $Ar + Bs = 1$. Rescale r, s by these constants so that $r + s = 1$. The pointwise equation then gives

$$(e_0(x) - e_1(x))r(x) = 1 - e_1(x).$$

Consequently $q(x) = r(x)$ wherever $r(x) \notin M$, outside the finite exceptional set.

Choose $a \in K \setminus M$ and repeat this argument for $q + a$ and $1 - q - a$. Its rational factor r_a is nonconstant: if it were a constant c, the equation $q + a = e_a c$, together with $q = e_0 r_0$, would place all cofinite values of the nonconstant rational function r_0 in the finite-dimensional M-vector space $Mc + Ma$. A nonconstant rational function has cofinite image, so this is impossible. After rescaling we obtain $q(x) = r_a(x) - a$ wherever $r_a(x) \notin M$. The two sets $r(x) \notin M$ and $r_a(x) \notin M$ cover the cofinite domain under consideration: if both rational values were in M, (F46.5) would put both $q(x)$ and $q(x) + a$ in M. Their intersection is infinite. To check this last point without a cardinality assumption, enlarge N by their finitely many coefficients. Points of sufficiently large degree over N have neither rational value in M, since a fibre of either fixed nonconstant rational function has bounded degree. Infinitely many such points exist. On this intersection $r = r_a - a$; equality of rational functions follows. Their common value therefore agrees with q outside a finite set.

If $r_0 = c$ is a nonzero constant, divide q by c; the hypothesis for all translates is preserved, and $q(x) \in M$ outside a finite set. Choose $b \in K \setminus M$. Its rational factor r_b cannot be nonconstant, because its cofinite image would be contained in $M + Mb$ by (F46.5). Hence r_b is constant. If q took two distinct values on the common cofinite domain, subtracting the equations $q(x) - b = e_b(x)r_b$ would put r_b and then b in M. Thus q is constant there. This proves the lemma in both cases, for an arbitrary smooth proper curve, as needed also in F.47.

Proof: remove the exceptional values and finish. For $m \neq 0$, apply the lemma with $M = k$, $q = f$, and $r_0 = t^m$. Let $R \in K(t)$ be the resulting nonconstant rational function, equal to f

outside a finite set. For each b, equation (F46.3) implies that

$$(R-b)/g_b$$

takes values in k^* on a cofinite set. A nonconstant rational function has cofinite image in K, whereas k is a proper subfield; hence this quotient is a constant $c_b \in k^*$. Thus $R - b = c_b g_b$ as rational functions.

Fix any $x \in K^*$. Taking $b \neq f(x)$ ensures $x \in U_b$, where g_b is a regular unit. The rational identity just proved first shows that R has no pole at x. For every $b \notin \{f(x), R(x)\}$ it then gives

$$\frac{f(x)-b}{R(x)-b} \in k^*.$$

If $f(x) \neq R(x)$, the displayed fractional linear expression, as b varies, takes every value in $K \setminus \{0,1\}$. This contradicts $k \neq K$. Therefore $f(x) = R(x)$ for every $x \in K^*$, including the initially exceptional points. In particular R is a regular unit on $\mathbb{G}_{m,K}$, so $R = ct^j$. Alternatively, (F46.4) shows directly that R/t^m is a rational function whose cofinite values lie in k; it must be constant. Normalization at 1 makes that constant one. Undoing the initial normalization gives exactly (F46). □

F.47 (Proved). Pinned reconstruction over cyclotomic algebraic bases. *Manuscript: MPL149; section2; folio25v; scan52; intermediate-base reconstruction.*

Let $k \subset \overline{\mathbb{Q}}$ be any subfield contained in $F\mathbb{Q}^{\mathrm{ab}}$ for some number field F. For all finite-type k-schemes X, Y, the natural map

$$\mathrm{Hom}_k(X^{\mathrm{awn}}, Y) \longrightarrow \pi_0 \,\mathrm{Hom}^{\mathrm{pin}}_{BG_k}(X_{\mathrm{et}}, Y_{\mathrm{et}})$$

is a bijection, where a relative geometric morphism includes its commutation isomorphism over BG_k. Equivalently, the etale-topos functor is fully faithful on finite-type schemes after inverting universal homeomorphisms, with pinned geometric morphisms as the target morphisms. In characteristic zero the absolute weak normalization is seminormalization. In particular this settles the test over $\mathbb{Q}(\mu_{2^\infty})$ in full, including constants and noninvertible morphisms, and applies to $\mathbb{Q}^{\mathrm{ab}}$ and all its finite extensions. We give the derivation from the verified arithmetic theorems and the reconstruction criterion; no priority assertion is intended.

Proof. **An integrality lemma for cyclotomic algebraic bases**

Let $k \subset \overline{\mathbb{Q}}$ be contained in $F\mathbb{Q}^{\mathrm{ab}}$ for a number field F, and let L/k be finite. Let $b_1, \dots, b_r \in L$ be distinct and put $V = \mathbb{A}^1_L \setminus \{b_1, \dots, b_r\}$. Suppose

$$\xi \in \varprojlim_n H^1_{\mathrm{et}}(V, \mu_n)$$

has the following evaluation property: for every $a \in V(L)$, $a^*\xi$ is the Kummer class of an element of L^*. Then ξ is the Kummer class of an actual unit of V. This is an integrality assertion for a cohomology class; the individual elements realizing its evaluations need not initially form a rational function.

Since $\mathrm{Pic}(V) = 0$ and $\mathcal{O}(V)^* = L^* \times \prod_i (t-b_i)^{\mathbb{Z}}$, Kummer theory gives

$$\xi = \gamma + \sum_{i=1}^{r} \lambda_i [t - b_i], \qquad \gamma \in \widehat{L^*}, \quad \lambda_i \in \widehat{\mathbb{Z}}. \tag{F47.R1}$$

All completions in this argument are $\varprojlim_n A/nA$ (in multiplicative notation, $\varprojlim_n A/A^n$). We first prove $\lambda_i \in \mathbb{Z}$ without assuming that L has a discrete valuation.

Choose a number field $K_0 \subset L$ containing all the b_i. For each i, choose a different prime ideal $\mathfrak{p}_i$ of $\mathcal{O}_{K_0}$ such that the b_j are integral there, their pairwise differences are units there, and the residue field has more than r elements. Only finitely many primes are excluded. The Chinese remainder theorem supplies $x, y \in K_0$ such that

$$v_{\mathfrak{p}_i}(x - b_i) = 1, \qquad v_{\mathfrak{p}_i}(x - b_j) = 0 \ (j \neq i), \qquad v_{\mathfrak{p}_i}(y - b_j) = 0 \ (\text{all } j). \tag{F47.R2}$$

For the first condition specify x modulo $\mathfrak{p}_i^2$; for the last choose a residue different from each b_j. Thus $x, y \in V(L)$. The elements $r_j = (x - b_j)/(y - b_j)$ are multiplicatively independent modulo roots of unity: applying the valuations in (F47.R2) to a relation $\prod_j r_j^{e_j} \in \mu_\infty$ gives $e_i = 0$ for every i. This independence persists in every algebraic overfield, since the relation itself is an equality of the same algebraic numbers.

There is a number field F_0 for which $L \subset M = F_0\mathbb{Q}^{\mathrm{ab}}$: take F_0 to contain F and finitely many algebraic generators of L/k. Theorem 1.1 of Amoroso–Zannier (Amoroso and Zannier, 2000) implies

$$\bigcap_{n \geq 1} (M^*)^n = \mu_\infty. \tag{F47.R3}$$

Indeed $[\mathbb{Q}^{\mathrm{ab}}(a) : \mathbb{Q}^{\mathrm{ab}}] \leq [M : \mathbb{Q}^{\mathrm{ab}}]$ for $a \in M$, so their height bound is uniformly positive on the non-torsion elements of M. If $z_n^n = z \neq 0$ for every n, then $h(z_n) = h(z)/n$, forcing z to be a root of unity. Conversely every root of unity is divisible in M, which contains all roots of unity. Tsujimura's Proposition 4.9 (Tsujimura, 2022), applied with its parameter $f = 1$ and number field F_0, now says that

$$A = M^*/\mu_\infty$$

is a free abelian group. This is the exact specialization of that theorem used here.

The independent elements $[r_1], \ldots, [r_r]$ of A have finite support in a basis of the free abelian group. Their coordinate matrix has rank r, so we may choose homomorphisms $\chi_1, \ldots, \chi_r : A \to \mathbb{Z}$ such that the integer matrix $N = (\chi_i([r_j]))_{i,j}$ has nonzero determinant. Choose actual elements $u_x, u_y \in L^*$ realizing the two evaluations of ξ. Subtracting the two evaluations of (F47.R1), then mapping to $\widehat{A}$ and applying the completed characters, gives

$$N(\lambda_1, \ldots, \lambda_r)^{\mathrm{t}} = \big(\chi_i([u_x/u_y])\big)_i \in \mathbb{Z}^r. \tag{F47.R4}$$

The adjugate matrix shows that $d\lambda_i \in \mathbb{Z}$, where $d = \det N \neq 0$. For each prime dividing d, membership in $\mathbb{Z}_p$ forces the corresponding divisibility of that integer by the prime power in d. Consequently $\lambda_i \in \mathbb{Z}$. Equivalently, $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$ inside the finite adeles. Put $m_i = \lambda_i$.

Finally choose any $a \in V(L)$ and an element $u_a \in L^*$ realizing $a^*\xi$. The constant class in (F47.R1) is then the Kummer class of $c = u_a/\prod_i (a - b_i)^{m_i}$. Thus $\xi = [c \prod_i (t - b_i)^{m_i}]$, as required. When $r = 0$ this last evaluation argument proves the result directly. The actual unit is unique modulo $\bigcap_n (L^*)^n$, a subgroup of the roots of unity; the assertion does not identify different pointwise choices before the translated-function argument is applied.

The smooth-curve step in cyclotomic reconstruction

Fix an algebraic subfield $k \subset \overline{\mathbb{Q}}$ for which $k \subset F\mathbb{Q}^{\mathrm{ab}}$ for some number field F. Every finite extension of k has the same property, after enlarging F. Write $K = \overline{\mathbb{Q}}$. We use the rational-source integrality lemma proved above, with all multiplicative completions taken as $\varprojlim_n A/A^n$.

Arithmetic input on abelian varieties. For each finite extension L/k and each abelian variety A/L,

$$\bigcap_{n \geq 1} nA(L) = 0. \tag{F47.1}$$

Indeed, the finite presentation of A and a finite set of generators of L/k allow a number field F_0 such that $L \subset F_0\mathbb{Q}^{\mathrm{ab}}$ and A is obtained by base change from an abelian variety over F_0.

Tsujimura's Theorem 3.1 (Tsujimura, 2020) says exactly that the Kummer map on $A(F_0\mathbb{Q}^{\mathrm{ab}})$ is injective. Its kernel is the intersection of the multiplication-by-n images, so the injection $A(L) \hookrightarrow A(F_0\mathbb{Q}^{\mathrm{ab}})$ proves (F47.1). The compatible total Tate module of $A(L)$ is also zero: each coordinate of a compatible torsion system is divisible by every integer, and (F47.1) kills it. Separately, the height argument in the rational-source lemma gives

$$D(L^*) := \bigcap_n (L^*)^n \subset \mu_\infty. \tag{F47.2}$$

Neither assertion requires that L have a discrete valuation.

A smooth-curve Kummer specialization lemma. Let C/L be a smooth geometrically connected affine curve and let

$$\xi \in \varprojlim_n H^1_{\mathrm{et}}(C, \mu_n).$$

Assume that for every finite L'/L and every $z \in C(L')$ the specialization $z^*\xi$ is the Kummer class of an actual element of $(L')^*$. After a finite extension of L there is an actual regular unit g on C such that $\xi = [g]$. This extension is only an extension of constants used in the proof; no descent of the given topos morphism is assumed.

To prove the lemma, let $\overline{C}$ be the smooth proper completion and $S = \overline{C} \setminus C$. Enlarge L so that the points of S and one point $z_0 \in C$ are rational. Purity for a smooth curve and the Kummer exact sequence give compatible residue maps

$$H^1(C, \mu_n) \longrightarrow \bigoplus_{s \in S} \mathbb{Z}/n, \qquad \mathrm{res}(\xi) = (\lambda_s)_{s \in S}, \quad \lambda_s \in \widehat{\mathbb{Z}}. \tag{F47.3}$$

We first prove that every λ_s is an integer.

Fix $s \in S$. By Riemann–Roch and interpolation of values and a first derivative at finitely many points, there is a nonconstant rational function

$$\pi : \overline{C} \longrightarrow \mathbb{P}^1_L$$

which has a simple zero at s and takes nonzero values at every other point of S. For example allow a sufficiently large pole divisor at a point different from S, prescribe the first-order jet of a uniformizer at s, and prescribe the value 1 at the other boundary points. The map is finite. Remove from $\mathbb{P}^1$ the values $0, \infty$, its branch values, and $\pi(S)$, and enlarge L once more to split this finite set if necessary. This gives

$$W = \pi^{-1}(V) \subset C, \qquad W \longrightarrow V \quad \text{finite etale},$$

where V is a punctured affine line. Apply the compatible norm maps to obtain

$$\eta = \mathrm{Cor}_{W/V}(\xi|_W) \in \varprojlim_n H^1(V, \mu_n).$$

The specialization property is preserved. If a fibre above $a \in V(L')$ is the disjoint union of $\operatorname{Spec} L_i$, etale base change for the norm and the usual Kummer norm identity give the Kummer class of

$$\prod_i N_{L_i/L'}(u_i) \in (L')^*,$$

where u_i realizes the evaluation of ξ at the corresponding point of W. Thus the rational-source integrality lemma applies to η.

The residue at 0 of a corestriction is $\sum_{z|0} [\kappa(z) : L] \operatorname{res}_z(\xi)$. This is the divisor-of-norm formula, and can also be read directly from the norm of a local parameter in the Kummer boundary map. Points of C contribute zero because ξ extends over them; among the points of S only the rational point s lies over 0. Consequently this residue is exactly λ_s, with coefficient one. The lemma on

V makes it integral. The choice of a simple zero also makes explicit that no ramification factor has been silently inserted.

Put $D = \sum_s \lambda_s[s]$, an integral divisor. The next arrow in the localization sequence sends the residue vector to the Kummer Chern class of D. Therefore

$$c_1(D) = 0 \quad \text{in } H^2(\overline{C}, \mu_n) \quad \text{for every } n,$$

and the Kummer injection $\operatorname{Pic}(\overline{C})/n \hookrightarrow H^2(\overline{C}, \mu_n)$ implies $[D] \in \bigcap_n n \operatorname{Pic}(\overline{C})$. Its degree is zero. Any divisor class E with $nE = [D]$ likewise has degree zero. The rational point z_0 identifies degree-zero line bundles with $J(L)$ for the Jacobian J and removes the Brauer obstruction to representing a Picard-scheme point by a line bundle. Equation (F47.1) therefore gives $[D] = 0$. Choose $g_0 \in L(C)^*$ with $\operatorname{div}(g_0) = D$; it is a unit on C.

All residues of $\xi - [g_0]$ now vanish. For every n the map $H^1(\overline{C}, \mu_n) \to H^1(C, \mu_n)$ is injective: purity identifies its preceding support group with a negative-degree cohomology group of S. Thus its residue-free lifts are unique, and hence compatible as n varies. The Kummer sequences give

$$0 \longrightarrow \widehat{L^*} \longrightarrow \varprojlim_n H^1(\overline{C}, \mu_n) \longrightarrow T \operatorname{Pic}(\overline{C}) \longrightarrow 0.$$

Here the left inverse system has surjective transition maps, so there is no $\varprojlim^1$ obstruction. The total Tate module on the right is zero by (F47.1). Thus $\xi - [g_0]$ is a constant completed Kummer class. Evaluate at z_0: the hypothesis identifies it with the Kummer class of an actual constant $c \in L^*$. Then $g = cg_0$ proves the lemma.

Algebraicity of the point function for a smooth curve. Let $\phi : C_{\mathrm{et}} \to (\mathbb{G}_m)_{\mathrm{et}}$ be pinned and relative to BG_L, and write $f : C(K) \to K^*$ for its point function. It is Galois equivariant and sends L'-points to L'-points for every finite extension, as checked in F.46. Its pulled-back Kummer class satisfies the specialization lemma. After a finite extension choose $g_0 \in \mathcal{O}(C)^*$ representing it. Equation (F47.2) gives

$$f(x)/g_0(x) \in \mu_\infty \qquad (x \in C(K)). \tag{F47.4}$$

If the generic image is a closed point, the inverse-image of its open complement is empty by continuity and the Jacobson property. The morphism factors through that closed subtopos. After splitting its residue field it has a constant point function. We may therefore assume the generic image is generic.

For each $b \in K$, base change to contain b and pull back the open $\mathbb{G}_m \setminus \{b\}$. It is a dense open $C_b \subset C$ whose omitted geometric points are finite, and $x \in C_b(K)$ exactly when $f(x) \neq b$. Applying the curve lemma to the Kummer classes of $t - b$ gives a rational function g_b, a unit on C_b, with

$$f(x) - b = d_b(x) g_b(x), \qquad d_b(x) \in \mu_\infty, \quad x \in C_b(K). \tag{F47.5}$$

Use the fixed proper field $M = F\mathbb{Q}^{\mathrm{ab}} \subset K$. In any one argument, a finite extension N/M contains the finitely many coefficients needed there; the field M itself stays fixed. All the coefficients $d_b(x)$ lie in M^*. The extension K/M has infinite degree, as does K/N for every finite extension N/M in K. For clarity, the latter assertion can also be seen from the height bound: if $K = N$ were finite over M, that bound would be uniformly positive on its nontorsion elements, whereas K contains all roots of the rational number 2.

The finite-exception quasi-rationality argument in F.46 applies to an arbitrary smooth curve, with the following explicit replacement for its polynomial-degree step. Descend the curve and the finitely many rational functions under consideration to a finite extension N/M. The poles of any fixed linear combination of two such functions lie in one fixed divisor. If the combination is nonzero, each zero has residue degree over N bounded by the degree of that divisor. Points of arbitrarily large residue degree exist: choose a finite map of the proper curve to $\mathbb{P}^1_N$, choose values of arbitrarily large degree over N, and take points above them. They avoid any prescribed finite

exceptional set. Thus a vanishing at such a point forces the same rational-function identity as in F.46. The two shifted outside-M sets cover the cofinite domain, and have infinite intersection by this same degree bound.

If g_0 is nonconstant, this proves that f equals a rational function $R \in K(C)$ on a cofinite subset. If g_0 is constant, divide f by one of its values and use (F47.4) to put its values in $\mu_\infty \subset M$. Choose $b \notin M$. If g_b were nonconstant, (F47.5) would put its cofinite image in the vector space $M + Mb$, which cannot contain a cofinite subset of K. Hence g_b is constant. Two different values of f would, by subtracting their equations, force this constant into M, and then force $b \in M$. Thus f is constant in this case.

We remove the finite exceptional set in the nonconstant case. For every b, the rational function $(R-b)/g_b$ takes its values in μ_∞ on a cofinite set. A nonconstant rational function on a smooth curve has cofinite image in K, so this quotient is a constant $c_b \in \mu_\infty$. Fix $x \in C(K)$ and choose $b \neq f(x)$. Since g_b is a unit at x, the identity $R - b = c_b g_b$ first excludes a pole of R at x. For every $b \notin \{f(x), R(x)\}$, it then gives

$$\frac{f(x)-b}{R(x)-b} \in \mu_\infty.$$

If $f(x) \neq R(x)$ this fractional linear expression takes all values of $K \setminus \{0,1\}$ as b varies, a contradiction. Therefore $f = R$ on every geometric point. It follows that R is a regular unit on C_K. Galois equivariance and equality on the Zariski-dense geometric points imply that R descends to a regular unit on the original curve over L. This proves the full point-function assertion for every smooth affine curve and every finite extension of the base field.

It also gives the equality of compatible Kummer classes that will be used below. Over the finite extension L'/L employed in the specialization lemma, the rational function R/g_0 has values in μ_∞ at every geometric point and is therefore a constant $\zeta \in (L')^*$. Evaluation at a rational point of the curve gives $\zeta \in D((L')^*)$, so $[R] = [g_0] = \xi$ over L'. If $C(L)$ is nonempty, equality already holds over L. Indeed, for every n the residual class in $H^1(C, \mu_n)$ is geometrically zero, and hence is a constant class by the low-degree Hochschild–Serre sequence. Its evaluation at an L-point is zero, because R is the actual point function there. The same sequence, or the rational-point section, shows that this constant class is zero. Without an L-point, equality after a finite extension is all that is asserted or needed for the geometric residue tests.

From the curve calculation to arbitrary dimension

We give the higher-dimensional step independently, so that the passage from point values to a geometric morphism does not rest on a fundamental-group calculation. Let $k \subset \overline{\mathbb{Q}}$ be contained in $F\mathbb{Q}^{\mathrm{ab}}$ for some number field F. We use the preceding curve calculation in precisely the following form: after every finite extension L/k, a pinned relative geometric morphism $C_{\mathrm{et}} \to (\mathbb{G}_{m,L})_{\mathrm{et}}$, with C a smooth geometrically connected affine curve having an L-point, has the geometric-point values of a unit of C. Its compatible Kummer class is the class of that unit. The conclusion below is conditional only on that already proved curve calculation and the arithmetic inputs explicitly recorded here.

It is enough that the curve calculation initially represents its Kummer class after a further finite extension. Indeed, suppose the pointwise unit is $f_C \in \mathcal{O}(C)^*$ and, over a finite extension L'/L, the Kummer class is $[g_0]$. At every algebraic point the value of f_C/g_0 is divisible in the multiplicative group of a finite extension of L' and hence is a root of unity, by the height input used in the curve calculation. A nonconstant rational function on a geometrically integral curve has cofinite image in $\mathbb{G}_m(\overline{L})$, so this ratio is constant. Evaluation at an L-point shows that its constant value is itself divisible in L'^*; hence its Kummer class is zero. Thus the two classes agree after L'. At each finite Kummer level, their difference on C lies in the image of $H^1(L, \mu_n)$ by the Hochschild–Serre exact sequence, since it is zero geometrically. Evaluation at the L-point is a retraction onto this image and is zero on the difference. Consequently the classes already agree

over L. This proves exactly the curve input used below, without requiring a chosen function to descend merely because its Kummer class descends.

The arithmetic Picard lemma. For every finite extension L/k and every abelian variety A/L,

$$\bigcap_{n\geq 1} nA(L) = 0. \tag{F47.H1}$$

Indeed, choose a number field F_0 over which A is defined and large enough that $L \subset M = F_0\mathbb{Q}^{\mathrm{ab}}$. Theorem 3.1 of Tsujimura (2020) states that the Kummer map on $A(M)$ is injective (Tsujimura, 2020). Its proof explicitly factors this map as

$$A(M) \longrightarrow \varprojlim_n A(M)/nA(M) \hookrightarrow H^1(G_M, TA),$$

so its kernel is exactly $\bigcap_n nA(M)$. Thus (F47.H1) follows by inclusion. Equivalently, the finite nonempty division fibres of a point in $\bigcap_n nA(M)$ have a compatible inverse-limit point: use the factorial levels and the finite intersection property. There is no distinction here between unrelated divisions and coherent divisions.

If Y/L is smooth, projective, and geometrically integral, put $P = \mathrm{Pic}(Y)$ and let P^0 be the subgroup whose geometric classes are algebraically trivial. Then

$$\bigcap_{n\geq 1} nP = 0, \qquad TP = \varprojlim_n P[n] = 0. \tag{F47.H2}$$

Here P^0 injects into the group of L-points of the abelian variety $\mathrm{Pic}^0_{Y/L}$, and $N = P/P^0$ is a subgroup of the finitely generated geometric Neron–Severi group; for the latter finiteness statement see (Cheng et al., 2021, Theorem 7.3 and Lemma 7.2). These assertions do not require the Picard scheme to represent every rational Picard class: the possible Brauer obstruction only changes the image of P^0 inside $\mathrm{Pic}^0_{Y/L}(L)$. Injectivity of the passage from actual line bundles follows from Hilbert 90.

For completeness, let $x \in \bigcap_n nP$. Its image in N vanishes, since N is finitely generated. Let e annihilate the torsion subgroup of N. For every $m \geq 1$, choose $y \in P$ with $emy = x$. Then the image of y in N is torsion, so $z = ey$ lies in P^0 and $mz = x$. Thus x is infinitely divisible in $\mathrm{Pic}^0_{Y/L}(L)$, and (F47.H1) gives $x = 0$. Finally, each coordinate of a compatible torsion system in TP is divisible by every positive integer, so the first equality in (F47.H2) implies the second. This proves the lemma, including the torsion in N.

Curves with prescribed points and transverse directions. We will use the following elementary consequence of Bertini. In a smooth geometrically integral projective variety over an infinite characteristic-zero field, a finite set of rational points lies on a smooth geometrically integral complete-intersection curve. At any specified point one may require its tangent line to be transverse to a given smooth divisor. The construction is over the ground field; one first makes the chosen points rational by a finite extension when necessary. To check the imposed conditions, take sufficiently high powers of an ample bundle. Serre vanishing makes evaluation on the finite first-order neighbourhoods surjective. The linear systems vanishing at the prescribed points embed the complement of those points, and a general member is smooth there and geometrically irreducible. Smoothness and the required independent differentials at the finitely many prescribed points are nonempty open conditions on the first jets. Intersect these opens and repeat until dimension one. This is the usual prescribed-point version of Bertini; the needed linear-system and irreducibility statements are Stacks, Lemmas 33.47.1, 33.47.3, and 37.32.3 (The Stacks Project Authors, 2026j).

Integral boundary residues and the global unit. Let X/k be a connected regular affine scheme of finite type with $x_0 \in X(k)$, and let

$$\phi : X_{\mathrm{et}} \longrightarrow (\mathbb{G}_{m,k})_{\mathrm{et}}$$

be pinned over B_k. Write $\phi(a)$ for its value at a geometric point. Over a finite extension L/k, a point $a \in X(L)$ has $\phi(a) \in L^*$: the relative morphism of closed-point Galois topoi forces the residue field of the image to embed in L. Base change of ϕ to L is obtained by slicing by the finite etale constant-field object. These observations justify every evaluation used below.

The scheme X is smooth and geometrically integral. In fact, in characteristic zero regularity implies smoothness, and the algebraic closure of k in $k(X)$ is a finite field extension contained in all normal local rings of X; evaluation at x_0 embeds it in k. Choose a smooth projective compactification Y of X whose boundary is a divisor with normal crossings, by characteristic-zero embedded resolution. Concretely, take a projective closure, resolve it while preserving the smooth open X, and principalize its boundary ideal; the blowups are projective and supported outside X. The strong resolution and monomialization theorem in (Encinas and Hauser, 2002, "The result" and "Idea of proof", pp. 1–3) supplies these operations. Write the boundary prime components over k as $D_1, \dots, D_s$. Only smooth points of individual boundary components will be used.

Pulling back the Kummer torsors of the coordinate t gives

$$h = (h_n)_n \in \varprojlim_n H^1_{\mathrm{et}}(X, \mu_n). \tag{F47.H3}$$

Let $K = k(Y)$. The map $H^1(X, \mu_n) \to K^*/K^{*n}$ is injective: a generically trivial finite etale torsor on a normal integral scheme has a section, by normalization. For every prime divisor D_i, the valuation of the generic Kummer representative of h_n defines a residue $r_{i,n} \in \mathbb{Z}/n$. Compatibility gives $r_i \in \widehat{\mathbb{Z}}$.

We claim $r_i \in \mathbb{Z}$. After a finite extension L/k, choose an L-rational point q on a geometric component above D_i, away from the other boundary components, and a smooth complete-intersection curve $\overline{C} \subset Y_L$ passing through q transversely to that component and through x_0. Its open part $C = \overline{C} \cap X_L$ is an affine smooth geometrically integral curve with an L-point. The curve calculation makes $h|_C$ the Kummer class of an actual unit; its residue at q is therefore an integer. That residue is exactly r_i.

One can check this equality without an inverse-limit purity argument. At q, the regular local ring R is factorial and the only boundary has one prime equation z. Hence $\mathrm{Pic}(R[1/z]) = 0$ and $R[1/z]^* = R^* z^{\mathbb{Z}}$. For each n, the restricted Kummer class has a representative uz^j, with $u \in R^*$, and residue j mod n. On the transverse curve z is a local parameter and u remains a unit, giving the same residue for every n. The base change from D_i to a geometric component has multiplicity one in characteristic zero. This proves the claim.

Set $D = \sum_i r_i D_i$, now an integral Cartier divisor on Y. Its class lies in $\bigcap_n n\,\mathrm{Pic}(Y)$. Indeed, choose a rational function $a_n \in K^*$ representing the generic class of h_n. Its valuations at divisors inside X are multiples of n, by the Kummer description over a discrete valuation ring; at D_i its valuation is r_i mod n. Consequently

$$\mathrm{div}(a_n) = D + nE_n \tag{F47.H4}$$

for an integral divisor E_n on the regular scheme Y. Thus $[D] = -n[E_n]$. By (F47.H2), $[D] = 0$, and there is a rational function $g \in k(Y)^*$ with $\mathrm{div}(g) = D$. In particular, $g|_X$ is an actual regular unit.

Subtracting $[g]$ from h leaves a compatible class unramified on Y. Explicitly, the divisor of a_n/g is divisible by n, so its associated line bundle with an nth-power trivialization defines a μ_n-torsor on Y extending $h_n - [g]$. Its uniqueness follows from injectivity into K^*/K^{*n}, so these extensions are compatible as n varies. The Kummer exact sequences on Y and (F47.H2) give

$$\varprojlim_n H^1(Y, \mu_n) = \widehat{k^*}. \tag{F47.H5}$$

To see that no derived-limit term is being suppressed, the maps $k^*/k^{*mn} \to k^*/k^{*n}$ are surjective; alternatively, since the compatible Picard-torsion class is zero, each class is represented uniquely

in k^*/k^{*n}, and these representatives are automatically compatible. Here $\Gamma(Y,\mathcal{O}_Y)^* = k^*$ by geometric integrality and properness.

Evaluation at x_0 identifies the constant class in (F47.H5) with the actual element $c = \phi(x_0)/g(x_0) \in k^*$. Therefore

$$f = cg|_X \in \Gamma(X,\mathcal{O}_X)^*, \qquad h = [f], \qquad f(x_0) = \phi(x_0). \tag{F47.H6}$$

Pointwise equality in every dimension. Let $a \in X(\overline{k})$. Choose a finite extension L/k making a rational and a smooth geometrically integral curve $\overline{C} \subset Y_L$ through both x_0 and a; put $C = \overline{C} \cap X_L$. In dimension one take $C = X_L$ itself. The curve theorem gives a unit f_C whose values equal those of $\phi|_C$. Since both f_C and $f|_C$ have Kummer class $h|_C$, the unit $u = f_C/(f|_C)$ has zero Kummer class for every n. The injection

$$\Gamma(C,\mathcal{O}_C)^*/\Gamma(C,\mathcal{O}_C)^{*n} \hookrightarrow H^1(C,\mu_n)$$

shows that u has an nth root among regular units for every n. Its divisor on $\overline{C}$ is therefore divisible by every positive integer and is zero. Thus $u \in L^*$ is constant. Evaluation at x_0 gives $u = 1$, and consequently

$$\phi(a) = f(a) \quad \text{for all } a \in X(\overline{k}). \tag{F47.H7}$$

In dimension zero $X = \operatorname{Spec} k$ and the assertion is immediate. Notice that this final argument does not infer equality from a finite collection of Kummer quotients: it uses the integral curve realization and equality in every Kummer degree. It also does not require the residual roots of unity to form a chosen compatible system.

Recovery of the entire geometric morphism. The assertion just proved holds for every connected regular affine finite-type source with a rational point, and after every finite extension of the ground field. It is exactly condition (4.18.5) of Carlson–Haine–Wolf, Theorem 4.18 (Carlson et al., 2024). Their equivalence with (4.18.1) therefore gives the full pinned reconstruction statement for this field class, with absolute weak normalization as in their definition. This application addresses all etale objects, not just the point map of one morphism.

Here is the categorical part of that implication. Two copies of $\mathbb{G}_m$ cover $\mathbb{A}^1$, so the pointwise units just constructed give pointwise algebraic maps to $\mathbb{A}^1$. Affine coordinates and equality on geometric points of a reduced source then reconstruct the addition and multiplication relations and hence a map to every affine target. Apply this construction also to each sliced pinned morphism over a quasicompact separated etale target object V. Its inverse image is again represented by a quasicompact separated etale source scheme, by CHW Corollary A.9; one works on its affine components and, when needed, after a finite extension supplying a rational point. Uniqueness on geometric points glues these maps and makes them compatible with all morphisms of etale objects. Proposition 4.14 then identifies the original inverse image functor with the functor of the reconstructed scheme map. Finally, the h-descent part of Theorem 4.18 handles arbitrary finite-type sources. Thus no openness assertion for an unconstructed birational Galois homomorphism, and no inference from finite set coefficients alone, is needed.

□

F.48 (Proved). Recovered Section 53 comparison and the exact cyclotomic rows. *Manuscript: Longue Marche, §53.5A–C, folios 768–779, cote 140-4 scans 242–262, especially (47), (56), (81)–(94); Montpellier cote 149, §1, folios 6–7, scans 13–15, and §4, folios 63–65, scans 128–131; cote 147, §11, scans 143–148.* The one-dimensional specialization of F.19 on a smooth surface is the sector-filter functor actually constructed in §53, not just an abstract equivalence between two groupoids having fundamental group $\mathbb{Z}$. Its finite completion gives the complex comparison of the tangential fibre functor. The recovered `CYCLOTOMIC_KERNEL` register has exact row identifiers `B5` and `C8`. B5 is a slimness statement; C8 is diagonal boundary nonvanishing

together with a counterexample to unrestricted local injectivity. Their mathematical contents agree with the corrected statements below, and do not assert a universal converse recognizing cuspidal sections.

Proof. The germ and its coverings. Let x be a point of a smooth real surface X and write D_x^* for the germ of punctured neighbourhoods of x. Let $S_x = (T_xX \setminus 0)/\mathbb{R}_{>0}$ be its circle of positive half-rays. A covering of a germ means the filtered colimit of the categories of coverings of its representatives, exactly (47) on scan 242. Small punctured discs form a cofinal system; restriction between them induces an equivalence on all covering spaces. Therefore

$$\mathrm{Rev}(D_x^*) \simeq \mathbb{Z}\text{-}\mathbf{Set}, \qquad \mathrm{Rev}_{\mathrm{uni}}(D_x^*) \simeq B\mathbb{Z}. \tag{F48.1}$$

Here the latter is the groupoid of connected simply connected covering germs and their isomorphisms. Choosing an orientation identifies its automorphism group with $\mathbb{Z}$; intrinsically it is the orientation module denoted $\overline{T}$ in (56), scan 248.

The poset $\mathcal{A}_x$ in (82), scan 256, consists of nonempty *closed proper connected arcs* of S_x, including single points and excluding the whole circle. Its groupoid completion is $\Pi_1(S_x)$. Indeed, choose a point in every arc. An inclusion of arcs determines a unique homotopy class of paths in the larger arc. Conversely, subdividing any path into finitely many arc neighbourhoods expresses it by these generators. A homotopy of paths has a finite subdivision into triangles lying in such neighbourhoods, so the only relations required are compositions of inclusions and their inverses. These constructions are inverse up to the evident changes of the chosen points. This proves (83)–(85), scans 257–258, without assuming that the arc poset is simply connected.

Strict thickening and coordinate independence. For $A \in \mathcal{A}_x$, let C_A be its cone in T_xX. Choose a local differentiable chart from the tangent plane to X inducing the identity on T_xX. The germ associated to A is represented by the filter generated by

$$\{v : 0 < \|v\| < \epsilon,\ [v] \in A'\}, \qquad A \subset \mathrm{int}(A'), \quad A' \subsetneq S_x, \tag{F48.2}$$

where A' runs through strict angular enlargements and $\epsilon > 0$. One may take A' open or take interiors of closed strict enlargements; the generated filters are cofinal. Every such representative is contractible, by polar coordinates on an angular interval. No convexity assertion is needed when the interval is wider than a semicircle.

If h changes the chosen chart and $dh_0 = 1$, then $h(v) = v + o(\|v\|)$ uniformly in direction. Given a strict enlargement A' choose an intermediate enlargement A'' whose closure is contained in $\mathrm{int}(A')$. For sufficiently small δ, h sends the sector for (A'', δ) into the sector for (A', ϵ). Apply the same observation to h^{-1}. Thus both charts define the same filter germ. This is precisely the strict-enlargement construction (89)–(90) on scans 259–260. Using just a fixed ray would fail: in coordinates (s, t) the diffeomorphism $(s, t) \mapsto (s, t + s^2)$ has identity derivative at zero but does not preserve the germ of the positive s-axis.

Pulling a covering back to this simply connected sector germ gives a set of sheets and hence a fibre functor F_A on $\mathrm{Rev}(D_x^*)$. Inclusions of arcs give canonical isomorphisms between these fibre functors, by restriction to the smaller sector germ; their compatibility is the compatibility of restriction. The associated universal covering germs therefore define

$$\Theta : \Pi_1(S_x) \longrightarrow \mathrm{Rev}_{\mathrm{uni}}(D_x^*), \tag{F48.3}$$

the functor (81), scan 256. Rotating the direction once lifts, in angular coordinates, to a translation by one full turn on the universal covering of a punctured disc. Thus Θ sends the generator of $\pi_1(S_x)$ to a generator of the automorphism group in (F48.1), with the sign fixed by the chosen fibre-functor and representing-cover convention. Both groupoids are connected and nonempty, so Θ is fully faithful and essentially surjective. This proves the equivalence by the winding criterion in (86bis), scan 258.

Arcs and tangential points. An outgoing differentiable arc with nonzero derivative and tangent half-ray Δ eventually lies in every strict sector enlargement of Δ. It consequently gives a map of germs into (F48.2). On coverings this identifies its fibre functor with $F_{\{\Delta\}}$, canonically and compatibly with morphisms of universal covers. These are (91)–(94), scans 261–262. The normal-direction construction in F.19 uses these same simply connected sectors and restriction maps; its rank-one specialization is consequently (F48.3). This verifies agreement of the actual functors, including their path morphisms.

For a complex algebraic curve, apply the finite covering comparison and complete the automorphism group to $\widehat{\mathbb{Z}}(1)$. In a local parameter z, the finite connected covers are cofinally the Kummer covers $z^{1/n}$; continuation along the sector picks their compatible sheets. This is the fibre functor in Deligne's §15.13–15.27, as identified in F.19. Replacing z by uz changes the compatible lifts by choices of nth roots of the unit u; their ratios are exactly the Kummer transition torsor. Thus the comparison retains the parameter-change identifications. In particular it does not invent a canonical map of spaces $D_x^* \to S_x$, whose absence is explicitly discussed on scan 262.

There is a separate correction on scans 242–243: the speculative identification with the inverse limit of the *full small topoi* of all shrinking punctured neighbourhoods is false. Their localic intersection is empty, so that inverse limit is the degenerate topos, whereas (F48.1) is not degenerate. The correct objects are the pro-topos of neighbourhoods, or the category of covering germs. This is the exact LM8 E.10/E.11 distinction proved in F.12, now checked against the original wording, including the inserted word *petits* on scan 243.

The recovered register B5. Write $\mathsf{Slim}(G)$ for the condition $C_G(H) = 1$ for every open $H \subseteq G$. It is preserved by extensions: the image of a centralizer element is trivial in a slim quotient, and its remaining element in the kernel centralizes the open intersection there. It is also preserved by inverse limits of slim profinite groups with surjective transition maps. A centralizer element has trivial coordinates, since projections are open and send the given open subgroup to open subgroups. The nonabelian finite-rank free profinite case is the case established in the free-group argument of F.39. More explicitly, an open normal subgroup has free rank $1 + [F : H](\operatorname{rank} F - 1) > 1$ and trivial centre. Its centralizer injects into the finite group F/H and is trivial by torsion-freeness. Rank one is excluded because $F = \widehat{\mathbb{Z}}$ is its own centralizer. This proves exactly the three assertions of the recovered B5, including its correction to scan 15. The identifier does not designate an inertia-recognition or cyclotomic-normalization theorem.

Here torsion-freeness can be checked without a separate group-theoretic black box. If a nontrivial finite-order element $z \in F$ survives in a finite quotient F/N, put $V = N\overline{\langle z \rangle}$. The open group V is free profinite, so its abelianization is a finite product of copies of $\widehat{\mathbb{Z}}$, a torsion-free group. But the nonzero cyclic quotient V/N generated by zN factors through that abelianization, while the finite-order element z maps to zero there. This contradiction proves the assertion used above.

The recovered register C8. Let U/k be a tower of relative hyperbolic curves in characteristic zero, $U \subset X$ a smooth proper normal-crossings compactification, and x a rational boundary point with geometric branch set I. The local geometric group is T^I, $T = \widehat{\mathbb{Z}}(1)$, with action given by cyclotomic scaling and branch permutation (Carlson and Corwin, 2026, Proposition 3.4). For its map $j_x : T^I \to \pi_1(U_{\bar{k}})$, the diagonal $T \to T^I$ has nonzero image. Here is the additional argument, beyond local normalization. Hoshi's valuative criterion for a relative hyperbolic curve says that a map from the fraction field of a strictly henselian trait extends over the trait precisely when its fundamental-group image is trivial (Hoshi, 2014, Lemma 2.8, pp. 175–176). Induction up the tower gives that criterion for U: extend the map to the previous stage first, and then apply the relative-curve criterion at the last stage.

In formal branch coordinates at a geometric point over x, take the arc $z_i = t$ for every boundary parameter, and take the remaining parameters to be zero. It has generic point in U and closed point x. Pullback of $z_i^{1/n}$ to $t^{1/n}$ shows that its fundamental group map is exactly

the diagonal followed by j_x. If this were zero the criterion would extend the generic-point map to U. Separatedness of X would identify this extension with the original arc, contradicting its boundary closed point. This proves nonvanishing. The cyclotomic kernel fixes the diagonal regardless of branch permutation, and fixes all of T^I when the branches are rational. The stated cyclotomic centralizer conclusion follows.

Full injectivity fails in the blown-up product already computed in F.39: at the new crossing the map is $T^2 \to T$, $(a, b) \mapsto a + b$, with kernel generated by $(1, -1)$. The diagonal maps by multiplication by two and is nonzero. Thus the stable-model hypotheses in F.39 and the counterexample in C8 are compatible. A converse boundary-local factorization of all sections with nonzero cyclotomic centralizer would imply the point-or-cusp section assertion for curves: apply it to the product of an arbitrary curve section and a rational cusp section of the triply punctured line, and project back to the curve. Neither B5 nor C8 supplies this converse. Carlson and Corwin's Theorem 3.7 and Proposition 4.12 concern closures of the corresponding section sets; dropping those closure bars would change their statements. □

Audit evidence. The recovered register, the B5 and C8 proof files, the two §53 transcriptions, and their file hashes are retained in the source-audit appendix of the package. The decisive original scans 242, 243, 256, 259–262 were inspected visually. Earlier unsuccessful-search notes are preserved as dated history and superseded here. The separate `FLAGGED_149` register remains the source obligation in F.45.

F.49 (Proved). Further field ranges where pinned reconstruction fails. *Manuscript: MPL149; section2; folio25v; scan52; prompt Part F field-range classification.*

The functor from finite-type k-schemes with universal homeomorphisms inverted to pinned etale topoi over BG_k is not full in each of the following cases:

1. k is any algebraic extension of a finite field, including $\overline{\mathbb{F}}_p$;
2. k is infinite and perfect and $Z(G_k) \neq 1$; in particular, k is any real closed field;
3. k is any algebraically closed field.

In every case an endomorphism of $(\mathbb{A}^1_k)_{\mathrm{et}}$ already gives a counterexample. The relative comparison with the base topos is part of the morphism. Forgetting that comparison would erase the counterexamples constructed in the second case.

Manuscript problem locus. Montpellier cote149, §2, fol.25v, scan52, including the margin asking about characteristic zero; these statements sharpen the field-range delimitation in F.44.

Proof: the central comparison twist and its point function. Let k be perfect, choose a separable closure K, put $H = G_k$, $X = \mathbb{A}^1_k$, and let $p : X_{\mathrm{et}} \to BH$ be the structure morphism. A central element $z \in H$ acts naturally and H-equivariantly on every discrete continuous H-set. Pulling back this natural automorphism to X_{et} gives an invertible two-cell of p. Define

$$T_z = (\mathrm{id}_{X_{\mathrm{et}}}, \alpha_z) \quad \text{in the category of topoi over } BH, \qquad \alpha_z : p \circ \mathrm{id} \Longrightarrow p. \tag{F49.1}$$

Choose the orientation of this comparison so that on the fibre of a constant finite etale scheme E/k it is

$$\mathrm{Hom}_k(E, K) \longrightarrow \mathrm{Hom}_k(E, K), \qquad u \longmapsto z \circ u. \tag{F49.2}$$

The underlying geometric morphism is the identity, hence is pinned.

Its induced function on framed geometric points is

$$a \longmapsto z(a), \qquad a \in K. \tag{F49.3}$$

Here is an explicit check of the relative data. A standard geometric point at a has the standard identification of its restriction to constant etale schemes with the fixed fibre functor of BH.

Composition with (F49.1) changes that identification to (F49.2). Applying z to every lift of the point through an etale neighbourhood sends a lift above a to a lift above $z(a)$. On constant covers this is exactly (F49.2), so it identifies the twisted framed point with the standard framed point at $z(a)$.

Distinct elements of K give nonisomorphic relative closed geometric points. For completeness, over a closed point with residue field $E \subset K$ their groupoid is the homotopy fibre of $BG_E \to BG_k$ at the fixed fibre functor. Since $G_E \to G_k$ is injective, this groupoid has trivial automorphism groups, and its isomorphism classes are the embeddings $E \hookrightarrow K$. Different closed supports have no isomorphisms between them. This is also the case $L = K$ of Carlson–Haine–Wolf, Proposition 3.6 (Carlson et al., 2024). Consequently (F49.3) distinguishes the relative morphisms T_z without any assertion about the centre of the generic Galois group of X.

Proof: infinite perfect fields with nontrivial Galois centre. Take $1 \neq z \in Z(G_k)$. Formula (F49.3) fixes every $a \in k$ and moves some $a \in K$. No endomorphism from the scheme localization can have this point function. We make this assertion explicit, including the inverted universal homeomorphisms. By the absolute-weak-normalization description of that localization (F.4; Carlson–Haine–Wolf, Proposition 2.28), its endomorphisms from $\mathbb{A}^1_k$ to itself are represented by

$$\mathrm{Hom}_k((\mathbb{A}^1_k)^{\mathrm{awn}}, \mathbb{A}^1_k). \tag{F49.4}$$

In characteristic zero $\mathbb{A}^1_k$ is already absolutely weakly normal, so a member is given by $P(t) \in k[t]$. If $P(a) = a$ for every $a \in k$, the infinitude of k gives $P = t$. In characteristic p, (F49.4) is the set

$$k[t^{1/p^\infty}] = \bigcup_{n \geq 0} k[t^{1/p^n}].$$

For an element h of this ring choose n so that $h^{p^n} = P(t) \in k[t]$. If $h(a) = a$ for every $a \in k$, then $P(a) = a^{p^n}$ on an infinite set; hence $P = t^{p^n}$. The ring is reduced, so Frobenius is injective and $h = t$. In either characteristic the only possible scheme morphism is the identity. Its geometric-point function fixes all of K, whereas (F49.3) does not. Thus T_z is an explicit morphism outside the image. A real closed field has algebraic closure $k(i)$ and $G_k = C_2$; complex conjugation supplies the stated example.

Proof: all algebraic extensions of finite fields. Let $k \subset \overline{\mathbb{F}}_p$ be algebraic over $\mathbb{F}_p$ and write

$$H = \mathrm{Gal}(\overline{\mathbb{F}}_p/k) \subset \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) = \widehat{\mathbb{Z}}.$$

If $H \neq 1$, it is abelian and has cardinality $2^{\aleph_0}$. Indeed a nonzero element has a nonzero projection to some $\mathbb{Z}_\ell$; the closure of its integer multiples projects onto a nonzero subgroup of $\mathbb{Z}_\ell$, already of that cardinality. Every $z \in H$ gives (F49.1), and (F49.3) makes their isomorphism classes distinct. On the scheme side there are only countably many localized endomorphisms: k is countable and the ring $k[t^{1/p^\infty}]$ in (F49.4) is countable. Thus the functor cannot be full. This handles finite fields as well as all proper infinite algebraic subfields. For the infinite subfields the preceding argument also gives the stronger explicit obstruction for every nonidentity $z \in H$.

If $H = 1$, then $k = \overline{\mathbb{F}}_p$ and $BH = \mathrm{Set}$. For each $\sigma \in \widehat{\mathbb{Z}}$ the ring automorphism

$$k[t] \longrightarrow k[t], \qquad \sum_i a_i t^i \longmapsto \sum_i \sigma^{-1}(a_i) t^i \tag{F49.5}$$

gives a semilinear automorphism of the underlying scheme. Its map on closed points is $a \mapsto \sigma(a)$. The resulting topos automorphism is automatically relative to $BH = \mathrm{Set}$, and it is pinned. Distinct σ induce distinct point maps, so they give $2^{\aleph_0}$ distinct morphisms again, compared with the same countable ring (F49.4). This proves the first assertion, including the algebraically closed endpoint.

Proof: arbitrary algebraically closed fields. Let k now be any algebraically closed field. It suffices to find a nonidentity field automorphism σ fixing an infinite subfield $k_0 \subset k$. The

semilinear construction (F49.5), relative to Set, has point function $a \mapsto \sigma(a)$. It fixes all k_0-points and moves another point. Exactly the polynomial or finite-p-denominator calculation used above shows that a morphism in (F49.4) fixing this infinite subset must be the identity. Therefore (F49.5) is outside the image.

In characteristic zero, choose a copy of $\overline{\mathbb{Q}}$ in k, take a nonidentity automorphism of it fixing $\mathbb{Q}$, and extend it to k. Such extension is obtained by fixing a transcendence basis over $\overline{\mathbb{Q}}$ and extending the resulting automorphism of the rational-function field to its algebraic closure. The fixed subfield contains $\mathbb{Q}$. In characteristic p, choose $\delta \in \widehat{\mathbb{Z}} = \prod_\ell \mathbb{Z}_\ell$ whose 2-adic coordinate is zero and whose odd-primary coordinates are one. The automorphism Frob_p^δ of $\overline{\mathbb{F}}_p$ is nonidentity and fixes the infinite field

$$\bigcup_{n \geq 0} \mathbb{F}_{p^{2^n}}.$$

Extend it to k by the same transcendence-basis argument. This supplies σ and k_0 in every characteristic, and completes all three assertions. The attached program `code/montpellier/F49_checks.py` checks the framed-point distinction over $\mathbb{F}_9$ and 2726 finite compatibilities of the displayed profinite exponent; the infinitary claims follow from the proofs above. □

F.50 (Proved). Reconstruction over subfields of finitely generated cyclotomic extensions. *Manuscript: MPL149; section2; folio25v; scan52; intermediate-base reconstruction.*

Let k be any subfield of a field finitely generated over $\mathbb{Q}^{\mathrm{ab}}$. Then k has pinned etale reconstruction: for all finite-type k-schemes X, Y,

$$\mathrm{Hom}_k(X^{\mathrm{awn}}, Y) \xrightarrow{\sim} \pi_0 \,\mathrm{Hom}^{\mathrm{pin}}_{BG_k}(X_{\mathrm{et}}, Y_{\mathrm{et}}). \tag{F50}$$

The relative commutation isomorphism is part of the morphism data. This includes F.47 and also every subfield of a finitely generated extension of an arbitrary cyclotomic extension of a number field. We prove the additional arithmetic and quasi-rational steps that extend F.47; its geometric and categorical deductions then apply with all their hypotheses verified.

Proof. Choose $k \subset M$, where $M/\mathbb{Q}^{\mathrm{ab}}$ is finitely generated, and choose an algebraic closure Ω of M. Let $K \subset \Omega$ be the algebraic closure of k. In particular $\overline{\mathbb{Q}} \subset K$. For every finite extension L/k inside K, the compositum $M_L = ML$ is finite over M, hence again finitely generated over $\mathbb{Q}^{\mathrm{ab}}$. Thus the field class in the statement is stable under every finite extension needed for points, boundary components, or coefficient descent.

1. Divisibility and the free multiplicative quotient. Let $E/\mathbb{Q}^{\mathrm{ab}}$ be finitely generated, and let E_0 be the algebraic closure of $\mathbb{Q}^{\mathrm{ab}}$ in E. This is a finite extension (Stacks, Lemma 9.26.11 (The Stacks Project Authors, 2026a)); hence $E_0 = F_0\mathbb{Q}^{\mathrm{ab}}$ for a number field F_0. The extension E/E_0 is regular in characteristic zero. Choose a proper normal geometrically integral E_0-model Z of E. Its divisor map gives an injection

$$E^*/E_0^* \hookrightarrow \mathrm{Div}(Z), \tag{F50.1}$$

because a rational function with zero divisor on a proper normal variety is a global unit, and its constant field is E_0. The divisor group is free abelian, so its subgroup E^*/E_0^* is free abelian. By Tsujimura's Proposition 4.9 with parameter $f = 1$, E_0^*/μ_∞ is free abelian; the identification of its divisible subgroup with μ_∞ is the Amoroso–Zannier height argument in F.47. Consequently the exact sequence

$$0 \longrightarrow E_0^*/\mu_\infty \longrightarrow E^*/\mu_\infty \longrightarrow E^*/E_0^* \longrightarrow 0$$

splits as a sequence of abelian groups, because its quotient is free. Therefore E^*/μ_∞ is free abelian.

Also $D(E^*) = \bigcap_n (E^*)^n = \mu_\infty$. If u is infinitely divisible, its divisor is divisible by every integer, hence zero. Every chosen nth root of u has zero divisor as well, so u and all these roots

lie in E_0. The height theorem applies there and forces u to be a root of unity. The converse holds since E_0 contains all roots of unity. In particular, for every finite L/k,

$$D(L^*) \subset \mu_\infty. \tag{F50.2}$$

This follows by embedding L and all its division witnesses into M_L; no assertion that L itself contains all roots of unity is needed.

2. Abelian-variety Kummer faithfulness. For each finite L/k and every abelian variety A/L,

$$\bigcap_n nA(L) = 0. \tag{F50.3}$$

This is also the stability statement in Tsujimura, Proposition 1.2(ii) (Tsujimura, 2022), combined with his Theorem 3.1 over finite extensions of $\mathbb{Q}^{\mathrm{ab}}$ (Tsujimura, 2020). We give the specialization argument, including the uniformity needed for all divisions.

For a finitely generated field $E/\mathbb{Q}^{\mathrm{ab}}$, an abelian variety A/E, and a point $P \in \bigcap_n nA(E)$, spread the abelian variety and this one point to an abelian scheme $\mathcal{A} \to U$ and a section P_U, where U is a normal integral finite-type scheme over $\mathbb{Q}^{\mathrm{ab}}$ with function field E. For every n, choose $Q_n \in A(E)$ with $nQ_n = P$. The finite etale scheme $[n]^{-1}(P_U) \to U$ has the generic section Q_n. Normality of U extends it to a section on the same U: the generic section selects a finite birational component, which is isomorphic to the normal base. Thus all the divisions extend without an infinite sequence of shrinkings. At every closed point $x \in U$, the residue field is finite over $\mathbb{Q}^{\mathrm{ab}}$, and $P_U(x)$ is divisible by all n. Theorem 3.1 gives $P_U(x) = 0$. Closed points are dense and the abelian scheme is separated, so $P_U = 0$ and $P = 0$. Apply this to $E = M_L$ and the base change of A; the injection $A(L) \hookrightarrow A(M_L)$ proves (F50.3). The Picard and total Tate-module conclusions of F.47 follow exactly as there, including the treatment of Neron–Severi torsion.

3. Rational-source integrality without algebraic coefficients. The rational-source lemma of F.47 must now allow distinct punctures $b_1, \dots, b_r \in L$ which can be transcendental over $\mathbb{Q}$. Put $E = \mathbb{Q}(b_1, \dots, b_r)$, a finitely generated field. There is an integral domain $A \subset E$, smooth and of finite type over $\mathbb{Z}[1/N]$, with fraction field E, containing all b_i and all $(b_i - b_j)^{-1}$ for $i \neq j$. To construct it, take a smooth affine model over $\mathbb{Q}$, shrink it to contain these finitely many functions as regular functions and units, and spread the finitely presented smooth algebra over $\mathbb{Z}[1/N]$; Stacks, Lemma 10.138.14 (The Stacks Project Authors, 2026g) supplies this last step. Flatness makes the model inject into its generic fibre, so it remains a domain. Its smooth image in $\operatorname{Spec}\mathbb{Z}[1/N]$ is an open set containing the generic point, hence contains all but finitely many primes.

Choose different primes $p_i > r$ with nonempty fibres, and choose a generic point $\mathfrak{p}_i$ of a component of the fibre over p_i. The local ring $A_{\mathfrak{p}_i}$ is a discrete valuation ring and p_i is a uniformizer: the model is regular and flat and the fibre is smooth, hence generically reduced. Let v_i be this valuation of E. Since the pairwise differences are units in A, their residues at $\mathfrak{p}_i$ are nonzero. Among the p_i elements of $\mathbb{F}_{p_i}$ choose c_i different from all the at most r residues of b_j that lie in that prime field. The Chinese remainder theorem in A now supplies $x, y \in A$ such that

$$x \equiv b_i + p_i \pmod{p_i^2 A}, \qquad y \equiv c_i \pmod{p_i A} \quad (1 \le i \le r).$$

The ideals for different rational primes are comaximal. Therefore

$$v_i(x - b_j) = \delta_{ij}, \qquad v_i(y - b_j) = 0. \tag{F50.4}$$

In particular x, y lie in the punctured affine line over L, and the ratios $(x - b_j)/(y - b_j)$ are multiplicatively independent modulo roots of unity. Independence persists in the overfield M_L, since it is a statement about equalities of these same elements.

Part 1 makes M_L^*/μ_∞ free abelian. The character-matrix argument (F47.R4) therefore applies without alteration: finitely many integer characters give a nonsingular integer matrix N on these

independent ratios; two actual Kummer evaluations give $N\lambda \in \mathbb{Z}^r$. The adjugate matrix and $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$ force $\lambda \in \mathbb{Z}^r$. A single further evaluation makes the constant completed class an actual element of L^*. Thus every compatible Kummer class on a punctured affine line with actual specializations is an actual regular-unit class, in the same generality as required for the norm calculation in F.47. No valuation on the possibly much larger field L was asserted to be discrete.

4. The quasi-rational degree argument in the larger algebraic closure. Use the fixed subfield $B = \mathbb{Q}^{\mathrm{ab}} \subset K$. All pointwise ambiguities furnished by (F50.2) lie in $\mu_\infty \subset B^*$. The proof of the finite-exception lemma in F.46 uses high-degree closed points after adjoining finitely many coefficients. Here those coefficients generate a finitely generated field N/B inside K, which need not be algebraic. We verify the required high-degree property directly.

Let $N_0 = N \cap \overline{\mathbb{Q}}$. It is finite over B by the constants lemma, and N/N_0 is regular. Hence N and $\overline{\mathbb{Q}}$ are linearly disjoint over N_0; this is the geometric irreducibility assertion in Stacks, Lemma 10.47.13 (The Stacks Project Authors, 2026a), using characteristic zero for geometric reducedness. Thus, for algebraic numbers a,

$$[N(a) : N] = [N_0(a) : N_0]. \tag{F50.5}$$

The right side is unbounded. If $\overline{\mathbb{Q}}/N_0$ had finite degree, its elements would have uniformly bounded degree over B, and the Amoroso–Zannier height bound would contradict the elements $2^{1/n}$. Finite extensions of the infinite algebraic extension therefore have unbounded degrees; characteristic zero supplies primitive elements.

Descend the smooth curve and the finitely many rational functions in a given translate argument to such an N. A finite map to $\mathbb{P}^1_N$ and values a as in (F50.5) give closed points of arbitrarily large residue degree. A nonzero linear combination of the fixed rational functions has a uniformly bounded zero divisor, so cannot vanish at those points. The same choice avoids any finite exceptional set: its points algebraic over N have bounded degrees, and its points transcendental over N cannot coincide with the chosen closed points. This is exactly the degree argument used to obtain the rational identities, the two covering outside-B sets, and their infinite intersection in F.46. Also K/B is infinite dimensional, since it contains $\overline{\mathbb{Q}}/B$, so the constant-factor case of that proof applies. Consequently the lemma holds here without assuming that K/B is algebraic. We have verified the proof's actual hypothesis, rather than using the algebraic-extension version of Tsujimura's criterion beyond its stated range.

5. Completion of the reconstruction argument. The curve argument of F.47 now applies with its three inputs verified for every finite L/k: the norm class on the punctured line has integral residues by part 3; the integral divisor class and the total Picard Tate module vanish by (F50.3); and the pointwise quotient is a root of unity by (F50.2). Part 4 gives a rational point function from all translates, including every exceptional value. Galois equivariance descends that function, and the rational-point Hochschild–Serre argument identifies its entire compatible Kummer class.

The higher-dimensional argument of F.47 uses only characteristic zero, these curve assertions after finite extensions, and (F50.3). Smooth compactification and transverse curves make the boundary residues integral; the Picard/Neron–Severi argument makes their divisor principal; the remaining class is constant; and a curve through the fixed rational point and an arbitrary geometric point proves pointwise equality with one global unit. These hypotheses all hold here. Finally Carlson–Haine–Wolf, Theorem 4.18, applies to every regular affine source and all its sliced etale objects, giving (F50). Thus the conclusion concerns the full relative geometric morphisms after localization, including constants and noninvertible maps, not only their point functions or their fundamental-group homomorphisms. □

F.51 (Refuted). Independent atlas germs need not be intrinsic stack neighbourhoods. *Manuscript context: MPL147, §§11–12, folios72,74,76, scans146,150,155; the unrestricted stack extensions of the gluing and tangential assertions.* The following two constructions must be distinguished. For a closed substack $Z \subset \mathcal{X}$, the *intrinsic neighbourhood pro-topos* is

$$\mathfrak{V}_{Z,\mathcal{X}} = \big(\mathrm{Sh}(\mathcal{U})\big)_{Z\subset\mathcal{U}\subset\mathcal{X}\ \text{open}}. \tag{F51.1}$$

Given an etale presentation $X_\bullet \to \mathcal{X}$ with $Z_q = Z \times_{\mathcal{X}} X_q$, one can instead first take the full neighbourhood germ of Z_q in each X_q, independently, and then form its simplicial diagram in pro-topoi. The canonical cocone from this diagram to (F51.1) is *not* always a two-colimit. The failure occurs even for an analytic etale groupoid with trivial isotropy. Moreover, for a nonproper presentation the intrinsic punctured neighbourhood can have a different covering groupoid from the punctured normal stack.

The strongest unconditional replacement retains the entire atlas diagram over one actual open-substack stage before passing to the pro-object. Ordinary intrinsic two-open gluing remains valid for every such stack. Independent chart-germ descent and F.13's controlled flag construction agree with the intrinsic object for a proper etale groupoid with paracompact locally compact orbit space, or under the explicit common-stage synchronization hypothesis of F.13. Finite isotropy alone does not replace properness.

Proof. 1. A full-topos counterexample. Let $\Gamma = \mu_\infty$ be the countable group of all complex roots of unity, given the discrete topology, acting on the unit disc Δ by rotation. Put $\mathcal{X} = [\Delta/\Gamma]$ and $Z = [\{0\}/\Gamma]$. This is an analytic etale quotient stack; its groupoid is not proper. Its atlas nerve is $X_q = \Gamma^q \times \Delta$, with $Z_q = \Gamma^q \times \{0\}$. A neighbourhood of Z_q permits an independent disc radius on each of the discrete components Γ^q. By contrast an intrinsic neighbourhood is $[U/\Gamma]$ for a single invariant open $U \subset \Delta$ containing 0.

For $n \geq 1$, put

$$A_n = \{2^{-n-1} < |z| < 2^{-n}\}, \qquad b_n(r) = 2^{-n}\max\{0, 1 - 16|2^n r - 3/4|\}.$$

The support of b_n is strictly inside the radial interval of A_n. Define a continuous real function by

$$f(0) = 0, \qquad f(re^{i\theta}) = b_n(r)\cos(n!\theta) \quad (r \in A_n), \tag{F51.2}$$

and set it to zero elsewhere. Continuity at the annular boundaries comes from the support condition; continuity at zero follows from $|f| \leq 2^{-n}$ on A_n. If ζ has order m, then for every $n \geq m$ the integer $n!$ is divisible by m. Hence

$$f(\zeta z) = f(z) \quad \text{for } |z| < 2^{-m}. \tag{F51.3}$$

For every finite tuple of roots, choose n at least their maximum order; the same equality holds for all products in the subgroup they generate. Consequently the germ of f gives compatible maps from every term of the atlas-germ diagram to $\mathrm{Sh}(\mathbb{R})$. All face compatibilities and their cocycles hold: each fixed tuple has a representative radius on which the required function equalities are literal. The target is localic and Hausdorff, so the corresponding mapping categories here are discrete; no unrecorded two-cell choice is required.

This cone does not come from (F51.1). A geometric morphism $\mathrm{Sh}([U/\Gamma]) \to \mathrm{Sh}(\mathbb{R})$ pulls back along U to a continuous Γ-invariant real function. This follows either from fixed-stage etale descent or from the inverse-image frame homomorphism into the invariant open subsets of U. Since Γ is dense in the circle group, continuity makes the function constant on each circle contained in U. Every neighbourhood of zero contains a disc. But at $r_n = 3 \cdot 2^{-n-2}$, the two angles 0 and $\pi/n!$ give values 2^{-n} and -2^{-n} in (F51.2). Thus f is nonradial on every disc. It cannot agree as a germ with any such invariant representative. The canonical mapping-out

functor to this fixed target is not essentially surjective, proving the failure of the pro-topos colimit universal property.

2. The failure is not repaired by finite stabilizers. Let Γ instead act on $Y = \mathbb{C}^* \times \Delta$ by $\zeta(z, w) = (\zeta z, w)$, and put $Z_0 = \mathbb{C}^* \times \{0\}$. This action is free. Define

$$F(z, w) = b_n(|w|) \cos(n! \arg z) \quad (|w| \in A_n), \qquad F(z, 0) = 0, \tag{F51.4}$$

with zero values in the remaining radial intervals. The expression is single-valued because $n!$ is integral. Equations (F51.3) hold uniformly in z after shrinking the w-radius for each fixed group element, so again this is a full coherent atlas-germ cone.

If it came from an invariant neighbourhood representative, equality of the two germs would hold on an ordinary neighbourhood of Z_0. Compactness of $\{|z| = 1\} \times \{0\}$ supplies a single positive w-radius on which that equality holds for the whole circle. Invariance and continuity force the representative to be constant in $\arg z$ on that circle, whereas (F51.4) has the same two opposite values for arbitrarily small w. This is a contradiction. All isotropy groups in this example are trivial; its dense circle orbits make the groupoid nonproper. Properness, not merely a bound on stabilizers, is material.

3. The intrinsic covering comparison can fail as well. Let $\mathbb{Z}$ act on $\mathbb{P}^1$ by

$$g(z) = \frac{z}{1 + z}, \qquad D = \{0\}. \tag{F51.5}$$

The action is understood projectively at the pole and at infinity. In the coordinate $w = 1/z$, its restriction to $\mathbb{P}^1 \setminus D = \mathbb{C}_w$ is $w \mapsto w + 1$. Every such orbit accumulates at D in $\mathbb{P}^1$. Therefore an invariant open neighbourhood of D is necessarily all of $\mathbb{P}^1$: it meets every orbit in an end of its sequence, and invariance then contains that entire orbit. The intrinsic punctured neighbourhood pro-topos is consequently constant, with value

$$\mathrm{Sh}([\mathbb{C}_w/\mathbb{Z}]) \simeq \mathrm{Sh}(\mathbb{C}^*), \tag{F51.6}$$

where the quotient map is $w \mapsto e^{2\pi i w}$. Its fundamental group is $\mathbb{Z}$.

On the other hand $g'(0) = 1$, so the action on the normal line at D is trivial. Its punctured normal stack is $[\mathbb{C}^*/\mathbb{Z}_{\mathrm{trivial}}]$, with classifying homotopy type $S^1 \times B\mathbb{Z}$ and fundamental group $\mathbb{Z}^2$. The tangential comparison sends the normal meridian to zero: the small circle $z = \epsilon e^{i\theta}$ becomes a closed circle in the simply connected plane $\mathbb{C}_w$. The operator generator is represented by the path $w \to w + 1$, which becomes the standard loop of the quotient. Thus on these fundamental groups the comparison is

$$\mathbb{Z} \oplus \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad (a, b) \longmapsto b. \tag{F51.7}$$

It is not an equivalence, even on finite covering categories: a nontrivial finite normal-meridian action does not descend through (F51.7). Hence F.19's intrinsic-stack equivalence requires its properness or common-stage hypothesis. Its coherent chart-germ construction is a different assertion and is not refuted by this example. The linear unipotent example of F.20 still proves exactly the stated failure of an invariant differentiable tubular embedding; that failure alone is not an argument for or against full-topos germ descent.

4. The common-stage replacement for every stack. Retain an actual open substack $\mathcal{U} \supset Z$ and its full atlas nerve $X_\bullet \times_{\mathcal{X}} \mathcal{U}$. At this one stage ordinary etale descent gives $\mathrm{Sh}(\mathcal{U})$ as the bicategorical colimit of the sheaf topoi of the nerve. Now let *this entire diagram* vary with $\mathcal{U}$. Mapping to a fixed target topos uses the inverse-image functor on degree zero, an invertible comparison on degree one, and its cocycle equality on degree two. Each item is represented at one actual $\mathcal{U}$; their finitely many stage indices have a common refinement. The same argument treats natural transformations and their equalities. Thus filtered two-colimits of these mapping categories commute with this descent pseudolimit. For a pro-target take the further limit over target stages. This proves the colimit assertion for the synchronized pro-diagram. It does

not claim that its degree-q term is cofinal among all independent neighbourhoods of Z_q in X_q; Parts 1–2 exhibit exactly the failure of that claim.

Likewise, at each actual stage $\mathcal{U}$, the opens $\mathcal{X} \setminus Z$ and $\mathcal{U}$ cover $\mathcal{X}$. Their two-open descent is a finite pseudolimit on mapping categories. The same common-refinement argument proves

$$\mathrm{Sh}(\mathcal{X}) \simeq \mathrm{Sh}(\mathcal{X} \setminus Z) \amalg_{\mathfrak{V}^*_{Z,\mathcal{X}}} \mathfrak{V}_{Z,\mathcal{X}} \tag{F51.8}$$

in pro-topoi, with the left side constant, without a properness assumption. All neighbourhoods in this formula are actual open substacks throughout.

5. Why proper etale presentations admit the chartwise repair. A proper etale groupoid admits local charts by finite isotropy groups, without an effectiveness assumption; see Moerdijk (2002, §3.4) (Moerdijk, 2002). Choose a locally finite relatively compact shrinking of these charts on the paracompact locally compact orbit space. Over a pair or triple of compact chart closures, properness makes the space of relevant arrows or composable arrows compact. Etaleness therefore covers it by finitely many local bisections. Thus the chart datum, overlap isomorphisms, cocycle equalities, and comparisons of two such data impose only finitely many radius bounds near any point of this locally finite system.

First take the minimum over the finite isotropy group in each chart. Next shrink the compact chart cores so every representative bound is positive there. A partition of unity on the orbit space supplies one positive continuous minorant of these locally finite bounds. Its pullbacks give invariant compatible tube radii; the finite control identities for incident branches are incorporated by the successive shrinking already proved in F.13. Descent on the resulting common neighbourhood realizes the full datum. Applying the same procedure to comparison maps and their equalities proves full faithfulness as well as essential surjectivity. This proves the proper-stack version of F.13 and identifies its independently formed chart germs with the intrinsic construction, including all flag data. The argument retains the finite stabilizers; it does not replace a quotient stack by its coarse space.

This procedure does not assert that an arbitrary neighbourhood in an unbounded atlas contains an invariant neighbourhood. Even for the proper translation action of $\mathbb{Z}$ on $\mathbb{R}_x \times \mathbb{R}_y$, the neighbourhood $|y| < e^{-x^2}$ of $y = 0$ has no invariant refinement. One first restricts the germ datum to the relatively compact quotient charts, performs the finite synchronizations there, and glues a stack neighbourhood. Comparison with the original atlas datum is then by its individual common refinements. These comparisons are precisely the morphisms in the atlas-germ descent category; no uniform lower bound along an entire noncompact orbit is used.

For a nonproper presentation one may instead require exactly this common-stage realization property as part of its controlled atlas data. The synchronized statement of Part 4 then applies. No arbitrary interchange of a filtered germ colimit with an infinite atlas descent diagram is included in either replacement.

The exact certificates in `code/foundations/F51_checks.py` check the finite rotation divisibilities, tuple compatibilities, and parabolic composition identities used above; their actual output is attached. □

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